

Diversity and Distribution of Floral Scent

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Abstract

A list of 1719 chemical compounds identified from headspace samples of floral scent is presented. The list has been compiled from some 270 published papers, including analyses of 991 species of flowering plants and a few gymnosperms, a sample including seed plants from 90 families and 38 orders. The compounds belong to seven major compound classes, of which the aliphatics, the benzenoids and phenylpropanoids, and, among the terpenes, the mono- and sesquiterpenes, occur in most orders of seeds plants. C5-branched compounds, irregular terpenes, nitrogen-containing compounds, and a class of miscellaneous cyclic compounds have been recorded in about two-thirds of the orders. Sulfur-containing compounds occur in a third of the orders, whereas diterpenes have been reported from three orders only. The most common single compounds in floral scent are the monoterpenes limonene, (*E*)- β -ocimene, myrcene, linalool, α - and β -pinene, and the benzenoids benzaldehyde, methyl 2-hydroxybenzoate (methyl salicylate), benzyl alcohol, and 2-phenyl ethanol, which occur in 54–71% of the families investigated so far. The sesquiterpene caryophyllene and the irregular terpene 6-methyl-5-hepten-2-one are also common and occur in more than 50% of the families. Orchidaceae are by far the best investigated family, followed by several families known to have many species with strongly scented flowers, such as Araceae, Arecaceae, Magnoliaceae, and Rosaceae. However, the majority of angiosperm families are still poorly investigated. Relationships between floral scent and pollination, chemistry, evolution, and phylogeny are briefly discussed. It is concluded that floral scent chemistry is of little use for phylogenetic estimates above the genus level, whereas the distribution and combinations of floral scent compounds at species and subspecific levels is a promising field of investigation for the understanding of adaptations and evolutionary processes in angiosperms.

Introduction

In 1993, one of us published a checklist of floral scent constituents collected with headspace techniques (Knudsen et al., 1993). Since then a large number of papers dealing with floral odors in many different groups of seed plants have been published. The number of known compounds present in floral scents has increased from 730 to 1719, and an updated checklist has been requested for a long time. The present review is intended to meet this demand. However, apart from adding to the list of known compounds, we also felt the need to summarize the present knowledge of floral scent in a broader context. Advances in phylogenetic research during the last decade have produced a much better understanding of relationships of angiosperms at the family level and below (e.g., Soltis et al., 1999; Hilu et al., 2003). Consequently, the distribution of various scent compounds is here compared to phylogenies of angiosperms orders based on recent analyses of DNA sequence data. In addition, new techniques and research have provided much needed insight into the evolution of floral scent at lower taxonomic levels and its role in evolutionary processes at the population level, and the chemistry of floral scent in general. A general scheme with references to different kinds of investigations with regard to floral scent is presented in Table I.

The present review is based on a total of 268 papers including data on floral scent composition. Many of these are cited by Knudsen et al. (1993); others were found

Table I

Broad categories of biological and chemical aspects in studies of floral scent (numbers refer to the references in Appendix III).

Environmental conditions, influence on floral scent production
89, 104, 161, 164, 172, 186, 215
Diurnal and nocturnal changes, and circadian rhythmicity in floral scent emission
1, 2, 8, 64, 69, 89, 91, 93, 95, 96, 102, 104, 105, 115, 116, 121, 132, 145, 146, 161, 166, 167, 168, 172, 173, 174, 175, 186, 187, 204, 215, 224
Flower age and post-pollination effects on floral scent emission
8, 64, 84, 152, 161, 164, 168, 172, 174, 196, 198, 199, 205, 213, 214, 219, 222, 223, 224, 231, 241, 244
Variation in floral scent composition within flowers/inflorescences
19, 59, 60, 61, 62, 69, 74, 86, 140, 172, 180, 184, 188, 199, 205, 222, 262
Attraction and behavior of pollinators
22, 23, 24, 27, 30, 59, 68, 83, 90, 91, 94, 129, 133, 145, 147, 149, 185, 187, 188, 197, 202, 205, 210, 213, 214, 223, 224, 226, 229
Attraction of pests/host selection, rust fungi, and defense
3, 4, 25, 55, 69, 70, 75, 152, 156, 182, 185, 217, 220, 226
Pollination syndromes and floral scent
butterflies, 5, 117; bats, 21, 94, 125, 142; moths, 113, 115, 116, 117, 125, 127, 131, 141, 169, 175; flies, 26, 48, 116, 117, 129, 130, 133, 134; bees, 116, 117; beetles, 135, 211, 225; hummingbirds, Knudsen et al. (2004); male euglossine bees, 7, 77, 78, 79, 80, 95, 116, 117, 119, 145, 222, 252, 253, 254, 255, 256, 257, 258, 259, 260; wind, 211
Mutualisms and floral scent
18, 82, 83, 84, 205
New natural products for flavors, perfumes, or medicines
6, 14, 32, 33, 34, 35, 36, 37, 38, 39, 40, 42, 43, 44, 45, 47, 57, 58, 99, 102, 107, 108, 109, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 125, 150, 153, 154, 155, 170, 171, 179, 200, 201, 203, 234, 238, 249, 250, 266, 268
Attached vs. detached flowers, effect on scent
41, 91, 93, 176, 177, 178, 186, 235
Community or environmental studies of floral scent
10, 119
Sampling and analytical techniques
16, 48, 51, 87, 100, 115, 128, 150, 176, 178, 221, 227, 228, 230, 232, 233
Natural history of pollination
65, 66, 92, 129, 138, 169, 189, 190, 191, 192, 193, 194, 195, 206, 207
Characterization of whole composition
46, 71, 106, 126, 248, 251, 265, 267
Reports of selected floral scent compounds
110, 113, 115, 118, 119, 121, 124, 125, 148, 149, 150, 158, 162, 163, 164, 183, 252, 254
Flower color and scent
73, 150, 178, 197
Within species variation in floral scent composition
Magnoliaceae, 9; Orobanchaceae, 20; Vitaceae, 42, 47; Fumariaceae, 56, 197; Oleaceae, 57, 58; Rosaceae, 72, 128; Orchidaceae, 13, 30, 80, 116, 117, 118, 180, 204, 243, 253, 257, 258, 259, 260; Araceae, 129; Ericaceae, 136; Arecaceae, 145; Nyctaginaceae, 151; Onagraceae, 218; Apiaceae, 242; Salicaceae, 245; Lamiaceae, 261; Caryophyllaceae, Dötterl et al. (2005)
Among species variation in floral scent composition
Magnoliaceae, 11, 240, 264; Orchidaceae, 13, 15, 17, 23, 27, 28, 29, 30, 53, 76, 80, 98, 117, 119, 181, 243, 253, 255; Apiaceae, 31, 246; Fumariaceae, 56; Amaryllidaceae, 63; Sterculiaceae, 67; Arecaceae, 68, 137, 143; Moraceae, 82; Ranunculaceae, 85, 208, 209; Myrsinaceae, 103; Araceae, 133, 146, 262; Ericaceae, 136; Theophrastaceae, 139; Lecythidaceae, 144; Nyctaginaceae, 151; Passifloraceae, 157; Winteraceae, 239; Fabaceae, 212; Salicaceae, 245; Rutaceae, 247; Caryophyllaceae, 112; Annonaceae, 111

Table I (continued)

Phenetic and phylogenetic analysis using floral scent
13, 56, 63, 80, 112, 157, 240, Levin et al. (2003), Barkman (2001), Azuma et al. (1999), Williams & Whitten (1999)
Leaf volatiles
11, 49, 50, 70, 87, 105, 131, 156, 208, 219, 235, 261
Horticulture and agriculture
12, 49, 50, 52, 54, 72, 75, 81, 87, 88, 159, 160, 161, 162, 163, 164, 165, 213, 214, 216, 220, 221, 235, 236, 237, 263
Biosynthesis and excretory pathways of floral scent components
64, 229
Heredity of floral scent composition
103, 160, 194, 195, 218

through searches in Biological Abstracts (BA; <http://www.biosis.org>) and Chemical Abstracts (CA; <http://www.cas.org>). The last searches in BA and CA were made in October 2002. Some studies that have come to our attention at a later date have also been cited, but are not included in the Appendices.

Only studies in which headspace techniques in a broad sense were used to collect the floral scent are included in the present review. This delimitation was made in order to better reflect the world of scent that flower visitors and pollinators actually encounter, for example, during their search for food or a site for breeding, mating, egg laying, or floral scent collecting. This means that studies using other methods, such as solvent extraction or steam distillation, are not included because the compounds detected may not necessarily be released from intact flowers under natural conditions.

A relational database, SCENTbase, was developed in FileMaker Pro 5.5 to handle all compound, taxon, and reference data, and to generate the lists. In addition, a separate database was created to analyze various properties of the data at different chemical and taxonomic levels, and to export matrices for further phylogenetic analyses. A limited version of SCENTbase is searchable online at <http://www2.botany.gu.se/SCENTbase.html>.

Collection Methods and Materials

The preferred method to collect floral scent is through dynamic headspace adsorption onto various artificial porous polymers and activated charcoal. This method is usually easy to apply in the field by enclosing flowers or floral structures in glass vials or polyacetate bags, and then creating a stream of air with a battery-operated pump. The samples are extracted by a solvent in the field, or stored until extracted in the laboratory, and then analyzed by coupled gas chromatography/mass spectrometry. Thermal desorption, in which the sample is desorbed directly onto a gas chromatographic column using heat, is less commonly employed. Recently, static headspace adsorption or solid-phase microextraction (SPME) with direct thermal desorption has been employed in studies of floral scents either alone or as a supplement to traditional dynamic headspace sampling. Detailed reviews of collection methods have been published by Raguso and Pellmyr (1998) and Agelopoulos and Pickett (1998). It is important to keep in mind that methodological differences among the reviewed studies may be responsible for inconsistencies in the compiled information. If present, they should only affect the accuracy of interspecific comparisons, which are not the focus of this review.

The total number of investigated species and subspecific taxa is 991, representing 90

Table II

The 10 most collected families ordered according to number of species analyzed.

	No. of species and subspecies	No. genera
Orchidaceae	417	104
Araceae	55	10
Arecaceae	40	18
Magnoliaceae	26	3
Rosaceae	24	8
Cactaceae	21	12
Rutaceae	21	3
Solanaceae	21	8
Caryophyllaceae	20	3
Nyctaginaceae	20	3

families and 38 orders of seeds plants. Two of the orders are gymnosperms, with a total of eight species investigated. Although representatives of most of the orders recognized by the APG II (2003) have been analyzed, the usually broad circumscriptions of these orders and the estimated total number of angiosperm species clearly indicate that the sample is still very small. In addition, the sample is very uneven. Orchidaceae is by far the best-collected family, followed by several families (e.g., Araceae, Arecaceae, Cactaceae, and Nyctaginaceae) with a reasonably large proportion of taxa sampled (Table II). Most families, however, remain poorly sampled or still not investigated with regard to floral scent chemistry.

Chemical Classification

In this checklist, our goal has been to classify floral scent compounds on the basis of their mode of biosynthesis, because this should be of greatest value for making comparisons with phylogenetic hypotheses. Although the broad outlines of biosynthesis are known for most large groups of floral volatiles, the actual routes by which most individual compounds are formed are still uncertain. Thus, it is especially difficult to arrange compounds biosynthetically at lower levels of classification. We divide floral scent constituents into seven major compound classes based on inspection of their structures and knowledge of the major pathways of plant secondary metabolism. These classes are aliphatics, benzenoids and phenylpropanoids, C5-branched compounds, terpenoids (including monoterpenes, sesquiterpenes, diterpenes, and irregular terpenes), nitrogen-containing compounds, sulfur-containing compounds, and a class of miscellaneous cyclic compounds. Within classes, subdivisions have been made primarily on the basis of chain length or skeletal type and secondarily according to functional groups (for details, see the legend of Appendix I). While this classification may not accurately reflect biosynthetic considerations, it does align compounds by their chemical properties and help in locating them in the list.

The accuracy and thoroughness with which individual chemical compounds have been identified varies among studies. Sometimes the precise isomers present were not determined, and often the enantiomeric composition was not resolved. The specific information on isomers and enantiomers reported in the original literature has been preserved in the checklist. However, when these details are not specified, the information is also missing here. It is important to be aware of this problem when comparing distributions of individual compounds among taxa. For example, "lilac alcohol," which is listed under monoterpenes/acyclic/alcohols, may include one or more of the four possible isomers A, B, C, and D.

Plant Names and Classification

No taxonomic evaluations of the usage of names have been made by us. Consequently, a certain taxon may occur under two or more different names. However, since we focus on the distribution of chemical compounds or compound classes at the level of plant family and plant order, such mistakes do not interfere with the general patterns observed.

We follow a recent classification of the angiosperms (APG II, 2003). However, to retain as much information as possible, we have adopted the narrower circumscription options in that work, recognizing Agavaceae, Amaryllidaceae, Fumariaceae, Hemerocallidaceae, Hyacinthaceae, Linnaeaceae, Rhizophoraceae, Ruscaceae, and Valerianaceae as separate families.

The phylogenetic trees are based on those presented by APG II (2003).

Floral Scent at Different Taxonomic Levels

POPULATION-LEVEL VARIATION

Studies of floral scent variation at individual or population levels including more than just a few individuals are still scarce. However, some studies have been made on selected species of Apiaceae (Tollsten & Øvstedal, 1994), Arecaceae (Knudsen, 2002), Caryophyllaceae (Dötterl et al., 2005), Magnoliaceae (Azuma et al., 2001), Mimosaceae (Pettersson & Knudsen, 2001), Orchidaceae (Whitten & Williams, 1992; Moya & Ackerman, 1993; Tollsten & Bergström, 1993), Fumariaceae (Olesen & Knudsen, 1994) and Pyrolaceae (Knudsen, 1994). In general, species relying mainly on one class of pollinator have higher similarity among populations than more generalized species pollinated by several orders of insects (Knudsen, unpubl.). Furthermore, the variation in floral scent composition may be clinal, with similarity among populations being negatively correlated with distance (Knudsen, 2002). However, this picture may be obscured by species with deceptive pollination systems in which variation both within and between populations is extreme, disrupting the associative learning of visiting insects (Moya & Ackerman, 1993). This may also be the case if floral scent types rarely or never influence pollinator behavior and reproductive success. Scent types caused by random genetic drift could then be maintained in natural populations (Azuma et al., 2001).

SPECIES- AND GENUS-LEVEL VARIATION

Many studies have shown that the floral scent composition usually differs among closely related species. This has been shown in works on *Nicotiana* (Raguso et al., 2003), *Narcissus* (Dobson et al., 1997), and *Silene* (Jürgens et al., 2002), as well as on various genera of Arecaceae (Knudsen, 1999a; Knudsen et al., 2001), Lecythidaceae (Knudsen & Mori, 1996), Magnoliaceae (Thien et al., 1975), Nyctaginaceae (Levin et al., 2001), and Orchidaceae (Gregg, 1983; Whitten & Williams, 1992; Kaiser, 1993; Barkman et al., 1997). In addition, floral scent composition may vary as much among genera within a family as among species of a given genus. Thus, unless monospecific, taxa above the species level usually cannot be characterized by a distinct floral scent profile.

FAMILY- AND ORDER-LEVEL VARIATION

Most compound classes are present in most orders of flowering plants (Figs. 1, 2), suggesting that the distribution of floral scent compounds is not phylogenetically constrained, at least not at this broad scale. Monoterpenes are found in all orders, and three,

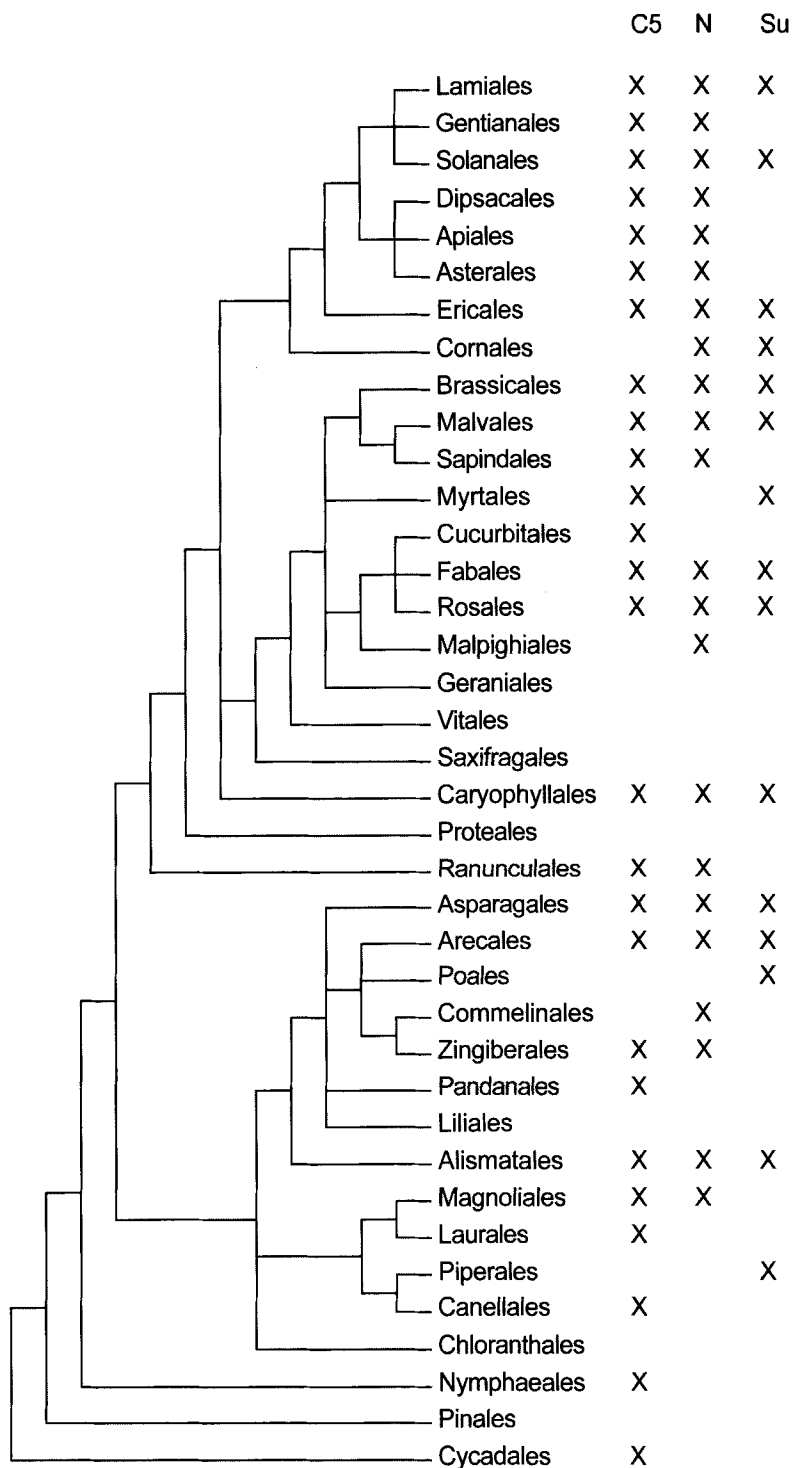


Fig. 1. Order-level distribution of major classes of floral scent compounds. C5, C5-branched chain compounds; N, nitrogen-containing compounds; Su, sulfur-containing compounds.

	A	B	S	D	IT
Lamiales	X	X	X	X	X
Gentianales	X	X	X		X
Solanales	X	X	X	X	X
Dipsacales	X	X	X		X
Apiales	X	X	X		X
Asterales	X	X	X		X
Ericales	X	X	X		X
Cornales		X			X
Brassicales	X	X	X		X
Malvales	X	X	X		X
Sapindales	X	X	X		X
Myrtales	X	X	X		X
Cucurbitales	X	X	X		
Fabales	X	X	X		X
Rosales	X	X	X		X
Malpighiales	X	X	X		X
Geraniales	X		X		
Vitales	X	X	X		X
Saxifragales					
Caryophyllales	X	X	X		X
Proteales	X	X	X		
Ranunculales	X	X	X		X
Asparagales	X	X	X		X
Arecales	X	X	X		X
Poales	X	X	X		
Commelinales			X		X
Zingiberales	X	X	X		X
Pandanales	X	X			X
Liliales	X	X	X		X
Alismatales	X	X	X		X
Magnoliales	X	X	X		X
Laurales	X	X	X		X
Piperales	X	X			X
Canellales	X	X	X		X
Chloranthales	X		X	X	
Nymphaeales	X	X	X		X
Pinales	X	X			
Cycadales	X	X	X		X

Fig. 2. Order-level distribution of major classes of floral scent compounds. A, aliphatics; B, benzenoids and phenyl propanoids; S, sesquiterpenes; D, diterpenes; IT, irregular terpenes. Monoterpenes occur in all orders and are not listed.

Table III

Floral scent compounds occurring in more than half of the families of seed plants.

Limonene	71%
(<i>E</i>)-Ocimene	71%
Myrcene	70%
Linalool	70%
α -Pinene	67%
Benzaldehyde	64%
β -Pinene	59%
Methyl 2-hydroxybenzoate	57%
Benzyl alcohol	56%
2-Phenyl ethanol	54%
Caryophyllene	52%
6-Methyl-5-hepten-2-one	52%

four and six orders out of 38 lack aliphatics, benzenoids and sesquiterpenes respectively. However, in most of the orders lacking these classes only one genus has been investigated and the result thus likely reflects lack of sampling. Diterpenes are rare in floral scents and are only reported from four species belonging to three plant orders, reflecting their low volatility.

C5-branched compounds are lacking in 12 orders, based on information from one or a few genera from each order. The same patterns were found for irregular terpenes and miscellaneous compounds, suggesting low sampling frequency rather than true absence.

Nitrogen- and sulfur-containing compounds are present in 63% and 39% of the plant orders, respectively. However, orders lacking nitrogen-containing compounds are all sampled at low frequency, while orders sampled at both high and low frequency lack sulfur-containing compounds. The latter indicates that the occurrence of volatile sulfur-containing compounds is probably restricted within seed plants.

Twelve compounds occur in more than 50% of the families investigated (Table III), and these should probably, even with investigation of additional taxa, be regarded as the most common compounds in floral scents. Most of these are monoterpenes, that is, limonene (71%), (*E*)- β -ocimene (71%), myrcene (70%), linalool (70%), α -pinene (67%), β -pinene (59%); followed by benzenoids, that is, benzaldehyde (64%), methyl 2-hydroxybenzoate (57%), benzyl alcohol (56%), 2-phenyl ethanol (54%); a sesquiterpene, caryophyllene (52%); and an irregular terpene, 6-methyl-5-hepten-2-one (52%). The widespread distribution of these substances suggests that they may have other roles in flowers in addition to pollinator attraction (see Floral Scent and Evolution).

Floral Scent and Pollination Biology

During the exploration of the function of floral scents in pollination, it is necessary to sort out which chemicals constitute the actual signals and which are historical and/or biosynthetic artifacts (Raguso, 2001). Historical and biosynthetic artifacts may contain important phylogenetic information, either alone or in combination. However, some compounds, such as thio- and isothiocyanates, may be both a signal and a historical artifact.

Sulfur-containing compounds are found mainly in plants pollinated by bats and carrion flies. The latter group of pollinators is restricted mainly to the Araceae and a few

other families of basal angiosperms. Thus, the presence of sulfur-containing compounds may be constrained phylogenetically, or these compounds may have developed across plant orders as a way of attracting a group of pollinators with similar scent preferences, as has been suggested for microbats in the New World tropics (Knudsen & Tollsten, 1995; Helversen et al., 2000).

Floral Scent Chemistry and Biochemistry

The constituents of floral scent comprise a large variety of generally lipophilic plant products with molecular masses less than 300. By definition, all constituents have high enough vapor pressures at atmospheric pressure and normal growth temperatures to allow significant rates of release into the air. The main classes of floral volatiles are the same as those reported to be released from other parts of the plant. Vegetative organs also release many different volatile constituents, especially after herbivore damage (Pichersky & Gershenzon, 2002). However, the overall diversity of vegetative volatiles is less than that found in floral scent. When headspace collections are made from intact plants, it is in fact difficult to readily distinguish between volatiles released from flowers and those released from nearby vegetative organs. Nevertheless, the sites of synthesis and storage of floral volatiles seem to be consistently associated with the floral organs themselves. Several genes and enzymes involved in the biosynthesis of floral scent compounds are expressed locally in the tissues of the petals, stigma, and style (Dudareva & Pichersky, 2000; Kolosova et al., 2001).

In the last few years, the biochemistry and molecular biology of floral scent formation have become major topics of research in several laboratories. Studies have centered around three species, *Antirrhinum majus* (Scrophulariaceae), *Clarkia breweri* (Onagraceae), and *Rosa hybrida* (Rosaceae), whose floral scents are dominated by terpenoid and benzenoid compounds. The biosynthesis of other classes of floral scent compounds has not yet been rigorously examined at the gene and enzyme levels, but the general routes can be inferred from investigations on similar or identical compounds in vegetative tissues.

Among the major categories of floral volatiles listed in Appendix I, the aliphatics are probably biosynthesized predominantly from fatty acids. For example, the abundant C6 and C9 aldehydes and alcohols are formed via lipoxygenase-catalyzed degradations of linolenic and linoleic acids (Hatanaka, 1999). Other compounds arise by oxidation of the double bonds or chain-shortening via β -oxidation.

The class of benzenoids and phenylpropanoids is formed starting from the phenylpropanoid pathway, which begins with the deamination of phenylalanine. Biosynthesis of the benzenoids (C6–C1), the most widespread members of this group in floral scent, thus requires the loss of one or two carbon atoms from a phenylpropanoid precursor (C6–C3). However, the steps of this process are not yet clear (Jarvis et al., 2000). Alternatively, methyl 2-hydroxybenzoate (methyl salicylate) and similar benzenoids could arise from an intermediate of the shikimate pathway prior to phenylalanine (Wildermuth et al., 2001). Several late steps in the formation of benzenoid esters and ethers have been well characterized (Murfitt et al., 2000; D'Auria et al., 2002; Lavid et al., 2002).

The C5-branched compounds are probably derived from the amino acids valine, leucine, and isoleucine, but there is only a little direct evidence to date (Rowan et al., 1996). Although the basic isopentane carbon skeleton of this group is the same as that of the terpenoids, the oxidation patterns present make it very unlikely that any are terpenoid-derived.

The terpenes themselves are formed from C5 isopentanoid building blocks synthe-

sized by the mevalonate or methylerythritol 4-phosphate (MEP) pathways (Gershenzon & Kreis, 1999; Rodriguez-Concepcion & Boronat, 2002). After assembly of the C5 units into prenyl diphosphate precursors, enzymes known as terpene synthases catalyze the formation of the basic terpene skeletons of monoterpenes (C10), sesquiterpenes (C15), or diterpenes (C20) (Bohlmann et al., 1998). The most well-studied enzyme of floral scent biosynthesis, linalool synthase, is a terpene synthase that converts geranyl diphosphate to linalool (Dudareva & Pichersky, 2000). The initial terpene synthase products, such as linalool, can be further modified to form other floral volatiles (Burkhardt & Mosandl, 2003; Shalit et al., 2003). The irregular terpenoids include cleavage products of carotenoids, such as the widespread ionones, and derivatives of smaller terpenoids (Winterhalter & Rouseff, 2002). For example, one member of the latter group, the widely-occurring C11 volatile, (*E*)-4,8-dimethyl-1,3,7-nonatriene, is believed to be formed by oxidative cleavage of a sesquiterpene (C15) precursor (Boland & Gäbler, 1989).

Both the nitrogen- and sulfur-containing floral scent compounds are derived from amino acid metabolism. Indole, the most widely distributed member of these groups, is also one of the few whose biosynthesis is understood in plants. It is formed by direct cleavage of the tryptophan precursor, indole-3-glycerol phosphate. (Frey et al., 2000). The miscellaneous cyclic compounds are collectively of uncertain biosynthetic origin, although some are undoubtedly derivatives of fatty acids or amino acids.

In summary, the constituents of floral scent are drawn from nearly all of the major pathways of plant secondary metabolism. As these pathways are present in all plants, where they make a wide range of pigments, membrane constituents, cell wall components, hormones, and other signaling compounds, it is perhaps not surprising that the principal classes of floral scent compounds are so widely distributed in seed plants.

Floral Scent and Evolution

The primary function of floral scent in flowering plants is to attract and guide pollinators (Dobson, 1994; Raguso, 2001; Metcalf, 1987; Robacker et al., 1988; Williams, 1983). However, additional functions may be ascribed to the presence of volatile chemicals in flowers (reviewed by Pichersky & Gershenzon, 2002), including defense and protection against abiotic stresses. These additional functions may help explain some of the abundance and variety of different constituents detected. The possibility that flowers are chemically well defended against herbivores and pathogens is not surprising. By producing pollen and ovules for the next generation, flowers have a very high fitness value to the plant and must be protected accordingly. In addition, the attraction of insects for pollination could increase the risk of herbivory on floral structures, and floral tissues may therefore require relatively more protection from enemies. Representatives of all of the major classes of floral volatiles have been shown to have toxic or deterrent activity against microbes and herbivores (e.g., De Moraes et al., 2001; Friedman et al., 2002; Hammer et al., 2003). Certain floral volatiles could also have a physiological role of providing resistance to abiotic stress. For example, some of the monoterpenes found in abundance in flowers have also been shown to ameliorate high temperatures and reduce damage caused by oxidative stress (Delfine et al., 2000; Loreto et al., 2004). The most abundant floral sesquiterpene, caryophyllene, is very reactive with ozone (Bonn & Moortgat, 2003).

The defensive and physiological functions of floral volatiles may well predate the origin of the angiosperms (Pellmyr & Thien, 1986; Thien et al., 2000), and floral scent, especially that of pollen, may therefore constitute an ancient trait, already present in

preangiosperms (Dobson & Bergström, 2000). A likely scenario suggests that some insects overcame the repellence of the floral chemicals and that pollination in early angiosperms was based on a meshing of the sexual life cycle of insects with that of plants, in which volatile floral chemicals served as mediating cues for rendezvous and mating sites (the flower) and food (primarily microspores) for pollinating insects. This hypothesis is supported by several observations: (1) floral scent is present in most extant basal angiosperms and in some nonangiospermous seed plants of Gnetales, Cycadales, and Pinales; (2) floral scent compounds similar to many general herbivore repellents are present; (3) all groups of ancient insects involved in pollination share phytophagy on nonangiosperms as an ancestral condition; and (4) a high percentage of flowers function as mating sites in extant basal angiosperms (Thien et al., 2000, 2003; Bernhardt et al., 2003).

In addition, thermogenesis may also have existed in preangiosperms, since this phenomenon occurs in cycads and is particularly common in basal angiosperms (Thien et al., 2000, 2003). Most thermogenic plants are beetle- or fly-pollinated. In beetles, heat produced by the plant may help to regulate body temperature. Because of this, Seymour and Schultze-Motel (1997) suggested that beetle pollination coevolved with thermogenesis and floral scent production in plants. Recently, Seymour et al. (2003) showed that heat is a reward to *Cyclocephala colasi* (Scarabaeidae: Cyclocephalini) beetles visiting *Philodendron solimoesense* (Araceae), the energy requirements being from 2.0 to 4.8 times lower inside the heated floral chamber than outside.

The correlation of high angiosperm diversity and the adoption of biotic pollination is undisputed. However, the mechanism or mechanisms responsible for this diversity are still not understood. Most likely a number of mechanisms, for example, coevolution of insect herbivores and plant chemical defenses and coevolution of seed dispersing animals and plants, have worked concomitantly to produce the diversity of animals and plants that we see today (see Gorelick, 2001, for a review and additional hypotheses).

Pollinators show different scent and color preferences, which may result in a certain degree of flower constancy. In combination with divergence in floral traits related to pollination (scent, color, flowering phenology, reward), these factors have led to reproductive isolation in present-day angiosperms (e.g., Gregg, 1983; Groth et al., 1987; Whitten & Williams, 1992; Dobson et al., 1997; Knudsen 1999a). The selective pressures probably have been especially strong in sympatric, coflowering species (Knudsen, 1999b; Schiestl & Ayasse, 2002). On the other hand, floral scents have converged in chemical composition even across plant orders in species sharing a suite of morphological and phenological characters adapting them to pollination by one particular group of pollinators (pollination syndromes), for example, by moths or bats (Miyake et al., 1998; Levin et al., 2001; Kaiser, 1993; Kaiser & Tollsten, 1995; Knudsen & Tollsten, 1993, 1995; Bestmann et al., 1997; Helversen et al., 2000; Raguso et al., 2003), or production has ceased, as in hummingbird-pollinated species in the neotropics (Knudsen et al., 2004). This contradiction suggests that floral scent evolution is influenced by several factors and that floral scent is best defined as a mosaic product of biosynthetic pathway dynamics, phylogenetic constraints, and balancing selection due to pollinator and florivore attraction (Raguso, 2001).

Floral Scent and Phylogeny

Seed plants are still poorly sampled with regard to floral scent compounds. In addition, the available information is also very uneven, with some groups being well sampled (e.g., 42% of all taxa investigated so far are orchids) and others poorly sampled or

not at all (e.g., Ebenaceae, Marantaceae). Consequently, the present information on the distribution of floral scent compounds is of limited use for broad phylogenetic reconstructions. Nonetheless, during the course of this work a few analyses of orders and families with compound classes classified into broad as well as less-restricted groups were made (with 30 and 68 characters, respectively). For example, aliphatics were classified into groups with 8 or 25 characters based on the number of carbon atoms present. Both analyses resulted in the >25,000 shortest trees producing largely unresolved strict consensus cladograms, corroborating the supposition that floral scent data are of little value for phylogenetic estimates at high taxonomic levels. It is also evident that the few groups that do appear in the consensus cladograms are mere artifacts of the uneven data set. Order and family-level analyses of the same group characters using the APG tree (APG II, 2003) as a constraint exposed high levels of homoplasy, with a majority of the characters having consistency indices between 0.05 and 0.15.

A few studies have used the distribution of floral scent compounds for reconstructing plant phylogenies and to infer the evolutionary history of ecological relationships (Azuma et al., 1997, 1999; Williams & Whitten, 1999; Lindberg et al., 2000; Barkman, 2001; Levin et al., 2003). The outcome of most of these studies is that only the outermost branching pattern is consistent with phylogenetic trees obtained using either morphological or DNA sequence data. This outcome indicates that, in general, floral scent chemicals may be too evolutionarily labile to be useful for phylogenetic inference (Williams & Whitten, 1999; Barkman, 2001). However, this does not exclude the possibility that some floral scent chemicals are patterned phylogenetically at lower taxonomic levels (Levin et al., 2003; Barkman, 2001). Furthermore, all these studies have been based on all identified compounds in the floral scent blends, and it is possible that a selection of compounds may produce alternative interpretations.

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Literature Cited

- Agelopoulos, N.G. & J.A. Pickett.** 1998. Headspace analysis in chemical ecology: effects of different sampling methods on the ratios of volatile compounds present in headspace samples. *J. Chem. Ecol.* 24: 1161–1172.
- APG II.** 2003. An update of the Angiosperm Phylogeny Group classification for orders and families of flowering plants: APG II. *Bot. J. Linn. Soc.* 141: 399–436.
- Azuma, H., L.B. Thien & S. Kawano.** 1999. Molecular phylogeny of *Magnolia* (Magnoliaceae) inferred from cpDNA sequences and evolutionary divergence of the floral scents. *J. Pl. Res.* 112: 291–306.
- , **Y. Toyota, Y. Asakawa, R. Yamaoka, J.G. Garcia-Franco, G. Dieringer, L.B. Thien & S. Kawano.** 1997. Chemical divergence in floral scents of *Magnolia* and allied genera (Magnoliaceae). *Pl. Spec. Biol.* 12: 69–83.
- , & **Y. Asakawa.** 2001. Intraspecific variation of floral scent chemistry in *Magnolia kobus* DC. (Magnoliaceae). *J. Pl. Res.* 114: 411–422.
- Barkman, T.J.** 2001. Character coding of secondary chemical variation for use in phylogenetic analyses. *Biochem. Syst. Ecol.* 29: 1–20.
- , **J.H. Beaman & D.A. Gage.** 1997. Floral fragrance variation in *Cypripedium*: implications for evolutionary and ecological studies. *Phytochemistry* 44: 875–882.
- Bernhardt, P., T. Sage, P. Weston, H. Azuma, M. Lam, L.B. Thien & J. Bruhl.** 2003. The pollination

- of *Trimenia moorei* (Trimeniaceae): floral volatiles, insect/wind pollen vectors and stigmatic self-incompatibility in a basal angiosperm. *Ann. Bot. (Oxford)* 92: 1–14.
- Bestmann, H. G., L. Winkler, & O. von Helversen.** 1997. Headspace analysis of volatile flower scent constituents of bat-pollinated plants. *Phytochemistry* 46: 1169–1172.
- Bohlmann, J., G. Meyer-Gauen & R. Croteau.** 1998. Plant terpenoid synthases: molecular biology and phylogenetic analysis. *Proc. Natl. Acad. Sci. U.S.A.* 95: 4126–4133.
- Boland, W. & A. Gäbler.** 1989. Biosynthesis of homoterpenes in higher plants. *Helv. Chim. Acta* 72: 247–253.
- Burkhardt, D. & A. Mosandl.** 2003. Biogenetic studies in *Syringa vulgaris* L.: bioconversion of 18O(2H)-labeled precursors into lilac aldehydes and lilac alcohols. *J. Agric. Food Chem.* 51: 7391–7395.
- Bonn B. & G. K. Moortgat.** 2003. Sesquiterpene ozonolysis: origin of atmospheric new particle formation from biogenic hydrocarbons. *Geophys. Res. Lett.* 30: 1585.
- D'Auria, J. C., F. Chen & E. Pichersky.** 2002. Characterization of an acyltransferase capable of synthesizing benzylbenzoate and other volatile esters in flowers and damaged leaves of *Clarkia breweri*. *Pl. Physiol.* 130: 466–476.
- Delfine, S., O. Csiky, G. Seufert & F. Loreto.** 2000. Fumigation with exogenous monoterpenes of a non-isoprenoid-emitting oak (*Quercus suber*): monoterpene acquisition, translocation, and effect on the photosynthetic properties at high temperatures. *New Phytol.* 146: 27–36.
- De Moraes C. M., M. C. Mescheer & J. H. Tumlinson.** 2001. Caterpillar-induced nocturnal plant volatiles repel nonspecific females. *Nature* 410: 577–580.
- Dobson, H. E. M.** 1994. Floral volatiles in insect biology. Pp. 47–81 in E. A. Bernays (ed.), *Insect-plant interactions*, Vol. V, CRC Press, Boca Raton, Florida.
- & G. Bergström. 2000. The ecology and evolution of pollen odors. *Pl. Syst. Evol.* 222: 63–87.
- , J. Arroyo, G. Bergström & I. Groth. 1997. Interspecific variation in floral fragrances within the genus *Narcissus* (Amaryllidaceae). *Biochem. Syst. Ecol.* 25: 685–706.
- Dudareva, N. & E. Pichersky.** 2000. Biochemical and molecular genetic aspects of floral scents. *Pl. Physiol.* 122: 627–633.
- Dötterl, S., L. M. Wolfe & A. Jürgens.** 2005. Qualitative and quantitative analyses of floral scent in *Silene latifolia*. *Phytochemistry* 66: 195–205.
- Frey, M., C. Stettner, P. W. Pare, E. A. Schmelz, J. H. Tumlinson & A. Gierl.** 2000. An herbivore elicitor activates the gene for indole emission in maize. *Proc. Natl. Acad. Sci. U.S.A.* 97: 14801–14806.
- Friedman M., P. R. Henika & R. E. Mandrell.** 2002. Bactericidal activities of plant essential oils and some of their isolated constituents against *Campylobacter jejuni*, *Escherichia coli*, *Listeria monocytogenes*, and *Salmonella enterica*. *J. Food Protect.* 65: 1545–1560.
- Gershenzon, J. & W. Kreis.** 1999. Biochemistry of terpenoids: monoterpenes, sesquiterpenes, diterpenes, sterols, cardiac glycosides and steroid saponins. Pp. 222–299 in M. Wink (ed.), *Biochemistry of plant secondary metabolism*. Sheffield Academic Press, Sheffield.
- Gorelick, R.** 2001. Did insect pollination cause increased seed plant diversity? *Biol. J. Linn. Soc.* 74: 407–427.
- Gregg, K. B.** 1983. Variation in floral fragrances and morphology: incipient speciation in *Cynoches*? *Bot. Gaz.* 144: 566–576.
- Groth, I., Bergström, G. & O. Pellmyr.** 1987. Floral fragrances in *Cimicifuga*: chemical polymorphism and incipient speciation in *Cimicifuga simplex*. *Biochem. Syst. Ecol.* 15: 441–444.
- Hammer, K. A., C. F. Carson & T. V. Riley.** 2003. Antifungal activity of the components of *Melaleuca alternifolia* (tea tree) oil. *J. Appl. Microbiol.* 95: 853–860.
- Hatanaka, A.** 1999. Biosynthesis of so-called “green odor” emitted by green leaves. Pp. 83–117 in U. Sankawa (ed.), *Comprehensive natural products chemistry*, vol. 1, Polyketides and other secondary metabolites. Elsevier, Amsterdam.
- Helversen, O. v., L. Winkler & H. J. Bestmann.** 2000. Sulphur containing “perfumes” attract flower-visiting bats. *J. Comp. Physiol. A* 186: 143–153.
- Hilu, K. W., T. Borsch, K. Muller, D. E. Soltis, P. S. Soltis, V. Savolainen, M. W. Chase, M. P. Powell, L. A. Alice, R. Evans, H. Sauquet, C. Neinhuis, T. A. B. Slot, J. G. Rohwer, C. S. Campbell, L. W. Chatrou.** 2003. Angiosperm phylogeny based on matK sequence information. *Amer. J. Bot.* 90: 1758–1776.

- Jarvis, A. P., O. Schaaf & N. J. Oldham.** 2000. 3-Hydroxy-3-phenylpropanoic acid is an intermediate in the biosynthesis of benzoic acid and salicylic acid but benzaldehyde is not. *Planta* 212: 119–126.
- Jürgens, A., T. Witt & G. Gottsberger.** 2002. Flower scent composition in night-flowering *Silene* species (Caryophyllaceae). *Biochem. Syst. Ecol.* 30: 383–397.
- Kaiser, R.** 1993. The scents of orchids. Elsevier, Amsterdam.
- & **L. Tollsten.** 1995. An introduction to the scent of cacti. *Flav. Fragr. J.* 10: 153–164.
- Knudsen, J. T.** 1994. Floral scent variation in the *Pyrola rotundifolia* complex in Scandinavia and western Greenland. *Nordic J. Bot.* 14: 277–282.
- 1999a. Floral scent chemistry in geonoid palms (Palmae: Geonomeae) and its importance in maintaining reproductive isolation. *Mem. New York Bot. Gard.* 83: 141–168.
- 1999b. Floral scent differentiation among coflowering, sympatric species of *Geonoma* (Arecaceae). *Pl. Spec. Biol.* 2: 137–142.
- 2002. Variation in floral scent composition within and between populations of *Geonoma macrostachys* (Arecaceae) in the western Amazon. *Amer. J. Bot.* 89: 1772–1778.
- & **S. A. Mori.** 1996. Floral scent in neotropical Lecythidaceae. *Biotropica* 28: 42–60.
- & **L. Tollsten.** 1993. Trends in floral scent chemistry in pollination syndromes: floral scent composition in moth taxa. *Bot. J. Linn. Soc.* 113: 263–284.
- & ———. 1995. Floral scent in bat-pollinated plants: a case of convergent evolution. *Bot. J. Linn. Soc.* 119: 45–57.
- , ——— & **G. Bergström.** 1993. A review: Floral scents—a check list of volatile compounds isolated by head-space techniques. *Phytochemistry* 33: 253–280.
- , ——— & **F. Ervik.** 2001. Flower scent and pollination in selected neotropical palms. *Pl. Biol.* 3: 1–12.
- , ———, **I. Groth, G. Bergström & R. A. Raguso.** 2004. Trends in floral scent chemistry in pollination syndromes: floral scent composition in hummingbird-pollinated taxa. *Bot. J. Linn. Soc.* 146: 191–199.
- Kolosova, N., D. Sherman, D. Karlson & N. Dudareva.** 2001. Cellular and subcellular localization of S-adenosyl-l-methionine:benzoic acid carboxyl methyltransferase, the enzyme responsible for biosynthesis of the volatile ester methylbenzoate in snapdragon flowers. *Pl. Physiol.* 126: 956–964.
- Lavid N., J. H. Wang, M. Shalit, I. Guterman, E. Bar, T. Beuerle, N. Menda, S. Shafir, D. Zamir, Z. Adam, A. Vainstein & D. Weiss.** 2002. O-Methyltransferases involved in the biosynthesis of volatile phenolic derivatives in rose petals. *Pl. Physiol.* 129: 1899–1907.
- Levin, R. A., R. A. Raguso & L. A. McDade.** 2001. Fragrance chemistry and pollinator affinities in Nyctaginaceae. *Phytochemistry* 58: 429–440.
- , **L. A. McDade & R. A. Raguso.** 2003. The systematic utility of floral and vegetative fragrance in two genera of Nyctaginaceae. *Syst. Biol.* 52: 334–351.
- Lindberg, A. B., J. T. Knudsen & J. M. Olesen.** 2000. Independence of floral morphology and scent chemistry as trait groups in a set of *Passiflora* species. Pp. 91–111 in Ö. Totland (ed.), The Scandinavian Association for Pollination Ecology honours Knut Fægri. Det Norske Videnskaps-Akademi. I. Mat.-Naturv. Klasse, Ny Ser., 39.
- Loreto, F., P. Pinelli, F. Manes & H. Kollist.** 2004. Impact of ozone on monoterpene emissions and evidence for an isoprene-like antioxidant action of monoterpenes emitted by *Quercus ilex* leaves. *Tree Physiol.* 24: 361–367.
- Metcalfe, R. L.** 1987. Plant volatiles as insect attractants. *Crit. Rev. Pl. Sci.* 45: 251–301
- Miyake, T., R. Yamaoka & T. Yahara.** 1998. Floral scents of hawkmoth pollinated flowers in Japan. *J. Pl. Res.* 111: 199–205.
- Moya, S. & J. D. Ackerman.** 1993. Variation in the floral fragrance of *Epidendrum ciliare* (Orchidaceae). *Nordic J. Bot.* 13: 41–47.
- Murfitt, L. M., N. Kolosova, C. J. Mann & N. Dudareva.** 2000. Purification and characterization of S-adenosyl-l-methionine:benzoic acid carboxyl methyltransferase, the enzyme responsible for biosynthesis of the volatile ester, methyl benzoate in flowers of *Antirrhinum majus*. *Arch. Biochem. Biophys.* 382: 145–151.
- Olesen, J. M. & J. T. Knudsen.** 1994. Scent profiles of flower colour morphs of *Corydalis cava* (Fumariaceae) in relation to foraging behaviour of bumblebee queens (*Bombus terrestris*). *Biochem. Syst. Ecol.* 22: 231–237.

- Pellmyr, O. & L. B. Thien.** 1986. Insect reproduction and floral fragrances: keys to the evolution of the angiosperms? *Taxon* 35: 76–85.
- Pettersson, S. & J. T. Knudsen.** 2001. Floral scent and nectar production in *Parkia biglobosa* Jacq. (Leguminosae: Mimosoideae). *Bot. J. Linn. Soc.* 135: 97–106.
- Pichersky, E. & J. Gershenzon.** 2002. The formation and function of plant volatiles: perfumes for pollinator attraction and defence. *Curr. Opin. Pl. Biol.* 5: 237–243.
- Raguso, R. A.** 2001. Floral scent, olfaction, and scent driven foraging behavior. Pp. 83–105 in L. Chittka & J. D. Thomson (eds.), *Cognitive ecology of pollination; Animal behavior and floral evolution*. Cambridge University Press, Cambridge.
- & **O. Pellmyr.** 1998. Dynamic headspace analysis of floral volatiles: a comparison of methods. *Oikos* 81: 238–254.
- , **R. A. Levin, S. E. Foose, M. W. Holmberg & L. A. McDade.** 2003. Fragrance chemistry, nocturnal rhythms and pollination “syndromes” in *Nicotiana*. *Phytochemistry* 63: 265–284.
- Robacker, D. C., B. J. D. Meeuse & E. H. Erickson.** 1988. Floral aromas. How far will plants go to attract pollinators? *BioScience* 38: 390–398.
- Rodriguez-Concepcion, M. & A. Boronat.** 2002. Elucidation of the methylerythritol phosphate pathway for isoprenoid biosynthesis in bacteria and plastids. A metabolic milestone achieved through genomics. *Pl. Physiol.* 130: 1079–1089.
- Rowan, D. D., H. P. Lane, J. M. Allen, S. Fielder & M. B. Hunt.** 1996. Biosynthesis of 2-methylbutyl, 2-methyl-2-butenyl, and 2-methylbutanoate esters in Red Delicious and Granny Smith apples using deuterium-labeled substrates. *J. Agric. Food Chem.* 44: 3276–3285.
- Schiestl, F. P. & M. Ayasse.** 2002. Do changes in floral odor cause speciation in sexually deceptive orchids? *Pl. Syst. Evol.* 234: 111–119.
- Seymour, R. S. & P. Schultze-Motel.** 1997. Heat-producing flowers. *Endeavour* 21: 125–129.
- , **C. R. White & M. Gibernau.** 2003. Heat reward for insect pollinators. *Nature* 426: 243–244.
- Shalit, M., I. Guterman, H. Volpin, E. Bar, T. Tamari, N. Menda, Z. Adam, D. Zamir, A. Vainstein, D. Weiss, E. Pichersky & E. Lewinsohn.** 2003. Volatile ester formation in roses. Identification of an acetyl-coenzyme A. Geraniol/citronellol acetyltransferase in developing rose petals. *Pl. Physiol.* 131: 1868–1876.
- Soltis P. S., D. E. Soltis & M. W. Chase.** 1999. Angiosperm phylogeny inferred from multiple genes as a tool for comparative biology. *Nature* 402: 402–404.
- Thien, L. B., H. Azuma & S. Kawano.** 2000. New perspectives on the pollination biology of basal angiosperms. *Int. J. Pl. Sci.* 161: 225–235.
- , **Heimermann, W. H. & R. T. Holman.** 1975. Floral odors and quantitative taxonomy of *Magnolia* and *Liriodendron*. *Taxon* 24: 557–568.
- , **T. L. Sage, T. Jaffré, P. Bernhardt, V. Pontieri, P. H. Weston, D. Malloch, H. Azuma, S. W. Graham, M. A. McPherson, H. S. Rai, R. F. Sage & J.-L. Dupre.** 2003. The population structure and floral biology of *Amborella trichopoda* (Amborellaceae). *Ann. Missouri Bot. Gard.* 90: 466–490.
- Tollsten, L. & L. G. Bergström.** 1993. Fragrance chemotypes in *Platanthera* (Orchidaceae)—the result of adaptation to pollinating moths? *Nordic J. Bot.* 13: 607–613.
- & **D. O. Øvstedal.** 1994. Differentiation in floral scent chemistry among populations of *Conopodium majus* (Apiaceae). *Nordic J. Bot.* 14: 361–367.
- Whitten, W. M. & N. H. Williams.** 1992. Floral fragrances of *Stanhopea* (Orchidaceae). *Lindleyana* 7: 130–153.
- Wildermuth, M. C., J. Dewdney, G. Wu & F. M. Ausubel.** 2001. Isochorismate synthase is required to synthesize salicylic acid for plant defence. *Nature* 414: 562–565.
- Williams, N. H.** 1983. Floral fragrances as cues in animal behavior. Pp. 50–72 in C. E. Jones & R. J. Little (eds.), *Handbook of experimental pollination biology*, Van Nostrand Reinhold, New York.
- & **W. M. Whitten.** 1999. A study of *Stanhopea*. *Pl. Spec. Biol.* 14: 129–136.
- Winterhalter, P. & R. Rouseff.** 2002. Carotenoid-derived aroma compounds: an introduction. Pp. 1–17 in P. Winterhalter & R. Rouseff (eds.), *Carotenoid-derived Aroma Compounds*. ACS Symposium Series No. 802, American Chemical Society, Washington, DC.

Appendix I: Distribution of Floral Scent Compounds at the Level of Plant Family

Floral scent constituents are ordered according to their inferred biosynthetic origin into seven large groups: aliphatics, benzenoids and phenylpropanoids, C5-branched chain compounds, terpenes (including monoterpenes, sesquiterpenes, diterpenes, and irregular terpenes), nitrogen-containing compounds, sulfur-containing compounds, and miscellaneous cyclic compounds. Each large class is then divided into subclasses according to skeleton type or chain length, and the subclasses are then divided according to functional groups if appropriate. Numbers within square brackets refer to reference numbers in Appendix III.

Aliphatics are straight-chain compounds that have been divided into groups by ascending chain length. Within each length the compounds are ordered into functional groups: alkanes, alkenes, acids, aldehydes, ketones, alcohols, esters (including lactones), and ethers. Among the esters, the parent acid determines which chain length the compounds are listed by.

Benzenoids and phenylpropanoids are classified together with all other compounds that have a benzene ring in their skeleton. They are ordered into groups reflecting the longest carbon chain (ranging from no chain to C7) attached to the benzene ring. Within each skeletal group, compounds are ordered into hydrocarbons, acids, aldehydes, ketones, alcohols, esters, ethers, chlorinated compounds, benzofurans, benzopyrans, and nitrogen-containing compounds.

C5-branched chain compounds have an iso-pentenoid carbon chain as their principal structural element and are subdivided into saturated and unsaturated compounds. These categories are further divided into hydrocarbons, acids, aldehydes, ketones, alcohols, and esters.

Terpenes include all compounds with more than five carbon atoms that appear to have an isoprenoid carbon skeleton. **Monoterpenes** are divided into the following main groups: acyclic, irregular, and cyclic skeletons. The latter group is subdivided into: *p*-menthane, bicyclo[2.2.1], bicyclo[3.1.0], bicyclo[3.1.1], bicyclo[4.1.0], and tricyclic skeletons. When appropriate, each of these groups is further divided into hydrocarbons, aldehydes, ketones, alcohols, esters, and ethers. **Sesquiterpenes** are divided into acyclic and cyclic skeletons, and within these two groups into hydrocarbons, aldehydes, ketones, alcohols, esters, and ethers. **Diterpenes** are divided in acyclic and cyclic skeletons. The **irregular terpenes** contain compounds with apocarotenoid skeletons and those with an apparent isoprenoid skeleton but an irregular number of carbon atoms, including C8, C9, C10, C11, C12, C13, C14, C16, and C18. Each of these groups is further subdivided into aldehydes, ketones, alcohols, esters, and ethers, when present.

Nitrogen-containing compounds are ordered according to chain length and, within these groups, into amines, amides, nitro compounds, nitriles, oximes, esters, and cyclic compounds, including indole, imidazole, pyrazine, pyrazole, pyridine, and triazine.

Sulfur-containing compounds are ordered into acyclic, including sulfoxides, sulfides, thiols, thioesters, thiocyanates, and isothiocyanates, and cyclic compounds, including thiofurans and thiazoles.

Miscellaneous cyclic compounds are divided into carbocyclic and heterocyclic compounds. The former group is further divided into monocyclic and bicyclic compounds (bicyclo[3.1.0], bicyclo[3.1.1], bicyclo[4.1.0], bicyclo[4.4.0] (naphthalenes), bicyclo[5.3.0] (azulenes)) and the latter into furans, pyrans, bicyclo[3.2.1], bicyclo[5.1.0], and spiro compounds. When appropriate, each of these groups is ordered by functional group.

Within each functional group compounds are ordered alphabetically disregarding numbers and the following prefixes or combinations of them: (*E*)-, (*Z*)-, *cis*-, *trans*-, α , β , γ , δ , *endo*-, *epi*-, *exo*-, *allo*-, *neo*-, *neoallo*-, *m*-, *o*-, *p*-, *di-tert*-, *ar*-, *1H*-, *N*-, *N,N*-, *S*-, *1S*-, *T*- and *Bis*-.

ALIPHATICS

C1

Aldehydes

Formaldehyde

Brassicaceae [221]

Alcohols

Methanol

Brassicaceae [221], Rosaceae [220]

Esters

Cyclohexyl formate

Oleaceae [267]

Diethyl carbonate

- Orchidaceae [117]
- (Z)-3-Hexenyl formate
- Orchidaceae [117, 118]
- C2
- Acids
- Acetic acid
- Araceae [86, 133, 134], Asteraceae [65], Brassicaceae [221], Calycanthaceae [266], Cyclanthaceae [225], Fabaceae [120], Nelumbonaceae [184], Orchidaceae [8, 77, 106, 117, 203], Rosaceae [12, 61], Rutaceae [247], Winteraceae [239], Zingiberaceae [174]
- Aldehydes
- Acetaldehyde
- Araceae [86, 262], Brassicaceae [221], Hydnoraceae [48], Orchidaceae [75, 201], Rosaceae [220]
- Alcohols
- Ethanol
- Araceae [86, 133, 229], Brassicaceae [221], Magnoliaceae [238], Orchidaceae [27, 75, 201], Rosaceae [219, 220, 228], Rubiaceae [249], Vitaceae [44]
- 2-(2-Methoxyethoxy)ethanol
- Oleaceae [177]
- Esters
- Butyl acetate
- Amaryllidaceae [6], Annonaceae [170], Arecaceae [68, 137], Asteraceae [132], Bromeliaceae [21], Fabaceae [175, 179, 212], Gentianaceae [45], Musaceae [21], Oleaceae [39, 267], Orchidaceae [77, 117], Rosaceae [33, 35, 40, 43, 128], Sapindaceae [46], Sapotaceae [263], Theophrastaceae [139], Winteraceae [210, 239]
- Cyclohexyl acetate
- Rosaceae [43]
- (E,E)-2,4-Decadienyl acetate
- Orchidaceae [77, 116, 117]
- (E,Z)-2,4-Decadienyl acetate
- Orchidaceae [77, 116, 117]
- (Z,Z)-3,6-Decadienyl acetate
- Orchidaceae [117]
- Decenyl acetate
- Orchidaceae [29]
- (E)-4-Decenyl acetate
- Orchidaceae [117]
- (Z)-4-Decenyl acetate
- Orchidaceae [77, 117, 120]
- Decyl acetate
- Caprifoliaceae [102], Fabaceae [49, 156], Orchidaceae [13, 17, 23, 27, 28, 29, 77, 117, 147, 188, 253]
- Dodecyl acetate
- Fabaceae [49, 156], Orchidaceae [13, 17, 23, 29, 117, 147, 253], Rutaceae [172]
- (E)-3,4-Epoxy-hexyl acetate
- Orchidaceae [118]
- (Z)-3,4-Epoxy-hexyl acetate
- Oleaceae [114], Orchidaceae [118]
- Ethyl acetate
- Actinidiaceae [237], Annonaceae [170], Araceae [86, 133, 134], Arecaceae [137], Cycadaceae [211], Magnoliaceae [238], Oleaceae [39, 57, 58, 267], Orchidaceae [75, 117], Rosaceae [12, 219, 220], Rubiaceae [249], Winteraceae [210, 239], Vitaceae [44, 47], Zingiberaceae [268]
- Heptyl acetate
- Orchidaceae [17, 27, 29, 77], Passifloraceae [157]
- 2-Heptyl acetate
- Orchidaceae [75, 117]
- Hexadecyl acetate
- Orchidaceae [13, 17, 23, 27, 117, 147, 253], Rosaceae [60, 61]
- Hexenyl acetate
- Dipsacaceae [185]

(E)-2-Hexenyl acetate

Actinidiaceae [237], Lamiaceae [5], Nyctaginaceae [151], Oleaceae [52], Rosaceae [33, 115], Sapotaceae [263], Valerianaceae [37]

3-Hexenyl acetate

Asteraceae [59], Oleaceae [171, 267], Ranunculaceae [207, 209], Rosaceae [12, 43, 61], Ruscaceae [171]

(E)-3-Hexenyl acetate

Lauraceae [74], Orchidaceae [118], Rosaceae [35]

(Z)-3-Hexenyl acetate

Actinidiaceae [237], Annonaceae [111], Apiaceae [242], Asteraceae [5, 34, 37, 65], Berberidaceae [182], Brassicaceae [70, 217], Caprifoliaceae [102, 178], Caryophyllaceae [5, 54, 66, 112], Combretaceae [10], Dipsacaceae [5], Ericaceae [136], Fabaceae [49, 50, 107, 120, 156, 179], Hemerocallidaceae [33], Hyacinthaceae [123], Lamiaceae [5], Lecythidaceae [144], Malvaceae [142], Moraceae [82], Nyctaginaceae [151], Oleaceae [33, 39, 52, 57, 58, 107, 119, 176, 177, 178], Onagraceae [218], Orchidaceae [17, 117, 118, 120, 169], Orobanchaceae [20], Passifloraceae [157], Primulaceae [36], Ranunculaceae [19], Rosaceae [33, 35, 37, 40, 60, 62, 72, 87, 88, 115, 166, 176, 177, 178, 219, 220, 232], Rubiaceae [249], Ruscaceae [232, 233], Rutaceae [247], Salicaceae [245], Sapotaceae [263], Scrophulariaceae [5], Solanaceae [165], Theophrastaceae [139], Valerianaceae [37], Winteraceae [210]

(Z)-4-Hexenyl acetate

Caryophyllaceae [5], Oleaceae [51]

Hexyl acetate

Actinidiaceae [237], Amaryllidaceae [6], Apiaceae [31], Asteraceae [37, 132], Berberidaceae [182], Cactaceae [21], Caprifoliaceae [178], Fabaceae [179], Lecythidaceae [144], Oleaceae [119], Orchidaceae [17, 27, 77, 117, 120, 188], Passifloraceae [157], Pinaceae [25], Ranunculaceae [209], Rosaceae [33, 35, 40, 43, 51, 60, 61, 62, 72, 87, 88, 93, 115, 128, 176, 177, 178], Rubiaceae [5, 249], Scrophulariaceae [5], Solanaceae [94], Valerianaceae [37], Winteraceae [210, 239]

Isobutyl acetate, see 2-Methylpropyl acetate

Isopropyl acetate

Passifloraceae [157]

Methyl acetate

Actinidiaceae [237], Oleaceae [39], Orchidaceae [75, 106], Rosaceae [219, 220]

(E)-2-Methylcyclopentanol acetate

Rosaceae [51]

2-Methyl-5-(1-methylethenyl)-2-cyclohexenyl acetate

Oleaceae [57, 58]

3-Methylpentyl acetate

Asteraceae [37]

2-Methylpropen-2-yl acetate

Oleaceae [267]

1-Methylpropyl acetate

Araceae [26]

2-Methylpropyl acetate (Isobutyl acetate)

Amaryllidaceae [6], Annonaceae [111], Apiaceae [31], Arecaceae [137], Asteraceae [37], Lecythidaceae [144], Magnoliaceae [11, 238], Oleaceae [52], Orchidaceae [77, 117, 201], Winteraceae [210, 239]

(Z,Z)-3,6-Nonadienyl acetate

Orchidaceae [117]

(Z)-3-Nonenyl acetate

Orchidaceae [117]

Nonyl acetate

Orchidaceae [17, 27, 29, 117]

2-Nonyl acetate

Orchidaceae [75]

Octadecyl acetate

Orchidaceae [13]

3-Octenyl acetate, see 3-Octyl acetate

Octenyl acetate

- Orchidaceae [27, 28, 29]
- 1-Octen-3-yl acetate
 - Nyctaginaceae [151]
- (Z)-3-Octenyl acetate
 - Orchidaceae [120]
- (Z)-5-Octenyl acetate
 - Orchidaceae [117]
- Octyl acetate
 - Apiaceae [31], Hemerocallidaceae [33], Hyacinthaceae [123], Orchidaceae [17, 23, 27, 28, 29, 77, 117, 118, 147, 188], Rosaceae [33, 35], Valerianaceae [37]
- 3-Octyl acetate
 - Araceae [86]
- 2-Pentadecyl acetate
 - Orchidaceae [117]
- Pentyl acetate
 - Actinidiaceae [237], Annonaceae [170], Orchidaceae [17], Rosaceae [33, 35, 40], Theophrastaceae [139]
- 2-Pentyl acetate
 - Amaryllidaceae [21], Bignoniaceae [21], Cycadaceae [211], Orchidaceae [117]
- Propyl acetate
 - Annonaceae [170], Lecythidaceae [144], Magnoliaceae [238], Orchidaceae [117], Winteraceae [210, 239]
- 2-Propyl acetate
 - Arecaceae [68], Orchidaceae [75]
- Tetradecyl acetate
 - Orchidaceae [13, 17, 23, 117, 147, 253], Rosaceae [60, 61]
- 2-Tridecyl acetate
 - Amaryllidaceae [63], Orchidaceae [75, 117]
- Undecyl acetate
 - Orchidaceae [17]
- 2-Undecyl acetate
 - Amaryllidaceae [63], Orchidaceae [75, 117]
- Ethers and epoxides
 - 1,1-Diethoxyethane
 - Calycanthaceae [266], Rosaceae [12]
 - Diethoxymethane
 - Rosaceae [12]
 - Diethyl ether
 - Araceae [229]
- Chloro
 - Tetrachloroethylene
 - Rosaceae [12]
- C3
 - Acids
 - Dimethylpropanoic acid
 - Brassicaceae [221]
 - Propanoic acid
 - Araceae [26, 229], Orchidaceae [27, 117]
 - Aldehydes
 - 2-Methylpropanal
 - Rosaceae [12, 220]
 - 2-Methyl-2-propenal
 - Araceae [229], Hydnoraceae [48], Orchidaceae [8, 201], Rutaceae [247], Zingiberaceae [174]
 - Propanal
 - Brassicaceae [221], Hydnoraceae [48]
 - Ketones
 - Acetone
 - Araceae [86, 229], Brassicaceae [221], Orchidaceae [75, 201], Rosaceae [12, 72, 220], Vitaceae [44, 47]
 - 2,2-Dimethoxypropanone
 - Orchidaceae [106]

1-Methoxy-2-propanone

Araceae [86]

Alcohols

Isobutanol, see 2-Methylpropanol

2-Methylpropanol (Isobutanol)

Actinidiaceae [237], Araceae [86, 262], Asteraceae [37], Lecythidaceae [144], Magnoliaceae [238], Oleaceae [39], Orchidaceae [77, 117], Rosaceae [12, 220], Winteraceae [210, 239]

Propanol

Araceae [133], Rosaceae [12]

2-Propanol

Orchidaceae [201]

Esters

Butyl acrylate, see Butyl 2-propenoate

Butyl 2-methylpropanoate

Annonaceae [111]

Butyl propanoate

Passifloraceae [157]

Butyl 2-propenoate (Butyl acrylate)

Zingiberaceae [268]

Ethyl propanoate

Magnoliaceae [238], Orchidaceae [117]

Ethyl 2-methylpropanoate

Caryophyllaceae [54], Magnoliaceae [11, 238], Rubiaceae [249]

Ethyl 2-methyl-2-propenoate

Magnoliaceae [238]

(Z)-3-Hexenyl 2-methylpropanoate

Oleaceae [107], Rubiaceae [249]

(Z)-3-Hexenyl propanoate

Annonaceae [111], Asteraceae [37], Nyctaginaceae [151], Oleaceae [107], Rubiaceae [249]

Hexyl 2-methylpropanoate

Rubiaceae [249]

Hexyl propanoate

Asteraceae [37], Passifloraceae [157]

Isobutyl acetate, see 2-Methylpropyl acetate

Isopropyl propanoate

Annonaceae [111]

Methyl 2-hydroxy-2-methylpropanoate

Cactaceae [113]

Methyl isobutanoate, see Methyl 2-methylpropanoate

Methyl 2-methylpropanoate (Methyl isobutanoate)

Araceae [133], Brassicaceae [221], Cycadaceae [211], Nymphaeaceae [14, 135], Orchidaceae [117]

Methyl 2-methylpropenoate

Nymphaeaceae [14, 135]

3-Methylpentyl 2-methylpropanoate

Asteraceae [37]

3-Methylpentyl 2-methyl-2-propenoate

Asteraceae [37]

Methyl propanoate

Rubiaceae [228]

2-Methyl-2-propenyl 2-methyl-2-propenoate

Asteraceae [37]

2-Methylpropyl 2-methylpropanoate

Annonaceae [111], Asteraceae [37], Magnoliaceae [238]

2-Methylpropyl 2-methyl-2-propenoate

Asteraceae [37]

2-Pentyl propanoate

Cycadaceae [211]

Ethers

2-Methylpropyl ethenyl ether

Orchidaceae [201]

C4

Alkanes

Ethylcyclobutane

Rosaceae [12]

Alkenes

2,3-Dimethyl-1,3-butadiene

Araceae [86]

2,3-Dimethyl-2-butene

Rosaceae [40]

Acids

Butanoic acid

Araceae [86], Asteraceae [65], Orchidaceae [77, 116, 117]

Aldehydes

Butanal

Araceae [86], Brassicaceae [221], Hydnoraceae [48], Rosaceae [12]

2-Butenal

Araceae [229], Rosaceae [12]

(E)-2-Butenal

Hydnoraceae [48]

Ketones

2,3-Butanedione

Araceae [86], Arecaceae [137], Brassicaceae [221], Zingiberaceae [174]

2-Butanone

Actinidiaceae [237], Araceae [86, 229], Brassicaceae [221], Fabaceae [179], Oleaceae [39], Orchidaceae [201, 236], Rosaceae [12, 219, 220]

1-Buten-3-one, see 3-Buten-2-one

3-Buten-2-one (1-Buten-3-one)

Araceae [229], Calycanthaceae [266], Liliaceae [92], Rutaceae [247], Zingiberaceae [174]

3-Hydroxy-2-butanone

Actinidiaceae [237], Araceae [86], Arecaceae [10, 137], Asteraceae [5], Acanthaceae [10], Brassicaceae [221], Fabaceae [50, 120, 156], Malvaceae [142], Oleaceae [57, 58], Orchidaceae [117], Rhizophoraceae [10], Rosaceae [219, 220], Sapotaceae [263], Scrophulariaceae [5], Theophrastaceae [139], Thymelaeaceae [5]

Alcohols

2,3-Butanediol, see 2,3-Dihydroxybutane

Butanol

Araceae [86], Gentianaceae [45], Magnoliaceae [238], Oleaceae [39], Orchidaceae [77, 117], Rosaceae [12, 33, 35], Vitaceae [44, 47]

2-Butanol

Apiaceae [246], Araceae [86], Brassicaceae [221], Rosaceae [12, 219, 220]

2,3-Dihydroxybutane (2,3-Butanediol)

Acanthaceae [10], Fabaceae [50, 156], Orchidaceae [27], Rhizophoraceae [10]

Esters

Butyl butanoate

Arecaceae [137], Asteraceae [5], Orchidaceae [117], Violaceae [248]

2-Butyl butanoate

Moraceae [121]

Ethyl butanoate

Araceae [86], Arecaceae [137], Lythraceae [10], Magnoliaceae [11, 238], Orchidaceae [117], Pinaceae [25], Rubiaceae [249]

Ethyl 2-butenate

Magnoliaceae [238]

Ethyl (E)-2-butenate

Arecaceae [137], Orchidaceae [117]

Heptyl butanoate

Lythraceae [10]

2-Heptyl butanoate

Orchidaceae [117]

(Z)-3,5-Hexadienyl butanoate

Orchidaceae [117]

2-Hexenyl butanoate

- Lythraceae [10], Nyctaginaceae [151]
- (*E*)-2-Hexenyl butanoate
 - Oleaceae [52]
- 3-Hexenyl butanoate
 - Lythraceae [10], Nyctaginaceae [151], Oleaceae [57, 267]
- (*Z*)-3-Hexenyl butanoate
 - Actinidiaceae [237], Berberidaceae [182], Caprifoliaceae [102], Iridaceae [178], Lecythidaceae [144], Oleaceae [39, 52, 58, 107, 119, 177], Orchidaceae [117, 201], Rubiaceae [249]
- Hexyl butanoate
 - Actinidiaceae [237], Apiaceae [31], Lecythidaceae [144], Moraceae [121], Nyctaginaceae [151], Oleaceae [57, 58], Orchidaceae [117], Passifloraceae [157], Rubiaceae [249]
- Isobutyl butanoate, see 2-Methylpropyl butanoate
- Isopropyl butanoate
 - Arecaceae [68], Oleaceae [52], Scrophulariaceae [226]
- Methyl butanoate
 - Araceae [86, 129, 133], Fabaceae [216], Lythraceae [10], Moraceae [121], Nymphaeaceae [14, 135], Orchidaceae [106, 117], Pinaceae [25], Rosaceae [43], Rubiaceae [249]
- Methyl 2-butanoate
 - Nymphaeaceae [135], Oleaceae [39]
- Methyl (*E*)-2-butanoate (Methyl crotonate)
 - Arecaceae [137], Nymphaeaceae [14], Orchidaceae [117]
- Methyl crotonate, see Methyl (*E*)-2-butanoate
- 2-Methylhexyl butanoate
 - Lythraceae [10]
- Methyl 3-hydroxybutanoate
 - Orchidaceae [77]
- Methyl 4-methoxybutanoate
 - Orchidaceae [106]
- 2-Methylpropyl butanoate (Isobutyl butanoate)
 - Asteraceae [37], Magnoliaceae [238], Orchidaceae [117]
- Octyl butanoate
 - Apiaceae [31], Orchidaceae [117]
- Pentyl butanoate
 - Passifloraceae [157]
- 2-Pentyl butanoate
 - Cycadaceae [211], Orchidaceae [117]
- Propyl butanoate
 - Annonaceae [111]
- Ethers
 - 2-Butoxyethanol
 - Violaceae [248]
 - 2-Methoxybutane
 - Oleaceae [39]
- C5
 - Alkanes
 - Pentane
 - Brassicaceae [221], Orchidaceae [201], Rosaceae [12, 220]
 - 2-Methylpentane
 - Brassicaceae [221], Rosaceae [12, 220]
 - 3-Methylpentane
 - Brassicaceae [221], Rosaceae [12]
 - 2,2',3,4-Tetramethylpentane
 - Oleaceae [267]
 - 2,2,4-Trimethylpentane
 - Rosaceae [12]
 - Alkenes
 - (*E*)-3-Methyl-1,2-pentadiene
 - Brassicaceae [221]
 - 3-Methyl-1,3-pentadiene
 - Rosaceae [220]

2-Methyl-1-pentene
Araceae [86], Rosaceae [220]

Acids

4-Methylpentanoic acid
Araceae [134]
(*E*)-3-Methyl-2-pentenoic acid
Asteraceae [65]
(*Z*)-3-Methyl-2-pentenoic acid
Asteraceae [65]

Pentanoic acid
Asteraceae [65], Brassicaceae [217], Orchidaceae [27, 117]

Aldehydes

2-Methyl-4-pentenal
Actinidiaceae [237]

Pentanal

Araceae [229], Bignoniaceae [142], Brassicaceae [221], Meliaceae [250], Orchidaceae [201],
Rosaceae [12, 40, 220], Rutaceae [247]

2-Pentenal

Amaryllidaceae [177], Araceae [229]

(*E*)-2-Pentenal

Araceae [86], Orchidaceae [117]

(*Z*)-2-Pentenal

Amaryllidaceae [6]

Ketones

4-Hydroxy-4-methyl-2-pentanone
Calycanthaceae [266], Oleaceae [177]

3-Hydroxy-2-pentanone
Orchidaceae [117]

2-Hydroxy-3-pentanone
Orchidaceae [117]

1-Methoxy-3-methylene-2-pentanone
Actinidiaceae [237], Oleaceae [39]

Methyl isobutenyl ketone, see 4-Methyl-3-penten-2-one

3-Methyl-2-pentanone
Araceae [86, 129, 133, 229], Brassicaceae [221]

4-Methyl-2-pentanone
Cyclanthaceae [225], Orchidaceae [201]

4-Methyl-3-penten-2-one (Methyl isobutenyl ketone)
Annonaceae [111], Fabaceae [121], Liliaceae [92], Orchidaceae [192]

2,3-Pentanedione
Orchidaceae [117]

2-Pentanone

Actinidiaceae [237], Araceae [86, 229], Brassicaceae [221], Cycadaceae [211], Oleaceae [39],
Orchidaceae [75], Rosaceae [219]

3-Pentanone

Actinidiaceae [237], Arecaceae [143], Brassicaceae [221], Chloranthaceae [251], Oleaceae
[39], Rosaceae [12, 219, 220]

3-Penten-2-one

Araceae [86]

4-Penten-2-one

Fabaceae [153]

Alcohols

3-Methylpentanol
Asteraceae [37], Solanaceae [165, 166]

3-Methyl-2-pentanol
Cycadaceae [211]

4-Methylpentanol
Araceae [86], Solanaceae [165]

4-Methyl-2-pentanol
Liliaceae [92]

3-Methyl-4-pentanol

- Araceae [229]
- Pentanol
 - Araceae [86], Cactaceae [125], Orchidaceae [30, 77, 117], Rosaceae [33, 35, 61, 219], Winteraceae [210]
- 2-Pentanol
 - Brassicaceae [221], Cycadaceae [211], Orchidaceae [27, 30, 147]
- 3-Pentanol
 - Arecaceae [143], Rosaceae [219, 220]
- 1-Penten-2-ol
 - Gentianaceae [45]
- 1-Penten-3-ol
 - Actinidiaceae [237], Chloranthaceae [251], Gentianaceae [45], Meliaceae [250], Oleaceae [39], Orchidaceae [201]
- (Z)-3-Pentenol
 - Orchidaceae [117]
- 3-Penten-2-ol
 - Zingiberaceae [174]
- 4-Pentenol
 - Araceae [86]
- Esters
 - Butyl pentanoate
 - Orchidaceae [117]
 - Dimethyl pentanedioate
 - Lauraceae [74]
 - Hexyl pentanoate
 - Orchidaceae [117]
 - Methyl 2-acetoxy-3-methylpentanoate
 - Eupomatiaceae [18]
 - Methyl 2-acetoxy-4-methylpentanoate
 - Eupomatiaceae [18]
 - Methyl 2-hydroxy-3-methylpentanoate
 - Eupomatiaceae [18], Nymphaeaceae [14], Orchidaceae [117]
 - Methyl 2-hydroxy-4-methylpentanoate
 - Eupomatiaceae [18], Orchidaceae [117]
 - Methyl 2-hydroxy-3-methylpentenoate
 - Nymphaeaceae [135]
 - Methyl hydroxypentanoate
 - Nymphaeaceae [14, 135]
 - Methyl 2-hydroxypentanoate
 - Orchidaceae [117]
 - Methyl 3-methylpentanoate
 - Solanaceae [165]
 - Methyl 2-oxo-3-methylpentanoate
 - Orchidaceae [117]
 - Methyl pentanoate
 - Nymphaeaceae [14, 135], Orchidaceae [27, 117]
 - Methyl 2-pentenoate
 - Orchidaceae [203]
 - Methyl 3-pentenoate
 - Amaryllidaceae [63]
 - 2-Pentyl pentanoate
 - Cycadaceae [211]
 - Propyl pentanoate
 - Arecaceae [68]
- C6
 - Alkanes
 - 3,3-Dimethylhexane
 - Oleaceae [57, 58]
 - Hexane
 - Actinidiaceae [237], Araceae [86], Brassicaceae [221], Rosaceae [12]
 - 2-Methylhexane

- Oleaceae [267]
- 2,3,4-Trimethylhexane
- Asteraceae [213]
- Alkenes
 - 4,5-Dimethyl-1-hexene
 - Cucurbitaceae [71]
 - 1-Hexene
 - Rosaceae [12], Vitaceae [44]
- Acids
 - 2-Ethylhexanoic acid
 - Malvaceae [67]
 - Hexanoic acid
 - Araceae [86], Fabaceae [120], Orchidaceae [8, 23, 27, 77, 147], Passifloraceae [157], Rosaceae [12], Rutaceae [247]
- Aldehydes
 - 2-Ethylhexanal
 - Moraceae [82]
 - 2,4-Hexadienal
 - Araceae [86]
 - Hexanal
 - Actinidiaceae [237], Apiaceae [242], Araceae [229], Arecaceae [68, 143], Brassicaceae [221], Cactaceae [125], Caprifoliaceae [102], Caryophyllaceae [54], Fabaceae [212], Hemerocallidaceae [33], Hydnoraceae [48], Malvaceae [142], Meliaceae [250], Oleaceae [52], Orchidaceae [8, 23, 117, 147, 201, 224, 227], Rhizophoraceae [10], Rosaceae [12, 33, 35, 72, 219, 220], Scrophulariaceae [5], Valerianaceae [37], Verbenaceae [5]
 - Hexenal
 - Amaryllidaceae [6], Orchidaceae [23]
 - 2-Hexenal
 - Nyctaginaceae [151], Rosaceae [33, 35, 219]
 - (*E*)-2-Hexenal
 - Actinidiaceae [237], Cactaceae [21], Fabaceae [49, 50, 120, 156, 179], Malvaceae [94], Oleaceae [52, 177], Ranunculaceae [209], Verbenaceae [5]
 - (*Z*)-2-Hexenal
 - Caprifoliaceae [102], Orchidaceae [117], Rosaceae [72]
 - (*Z*)-3-Hexenal
 - Araceae [86], Cactaceae [125], Fabaceae [179], Orchidaceae [117]
- Ketones
 - 2,5-Hexanedione
 - Orchidaceae [117]
 - 3,4-Hexanedione
 - Orchidaceae [117]
 - 2-Hexanone
 - Araceae [229], Cactaceae [125], Caprifoliaceae [102], Fabaceae [50, 156], Orchidaceae [117]
 - 3-Hexanone
 - Amaryllidaceae [21], Araceae [229], Arecaceae [21, 143], Bignoniaceae [21], Bromeliaceae [21], Cactaceae [125], Orchidaceae [117]
 - 3-Hexen-2-one
 - Annonaceae [111]
 - 5-Hexen-2-one
 - Oleaceae [57, 58]
 - 4-Hydroxy-3-hexanone
 - Orchidaceae [117]
 - 4-Methyl-3-hexanone
 - Oleaceae [51]
 - 5-Methyl-2-hexanone
 - Zingiberaceae [174]
 - 5-Methyl-4-hydroxy-2-hexanone
 - Lauraceae [74]
- Alcohols
 - (*Z*)-3,4-Epoxy-hexanol
 - Orchidaceae [118]

2-Ethylhexanol

Amaryllidaceae [6], Araceae [229], Fabaceae [104], Malvaceae [67], Moraceae [82], Nelumbonaceae [184], Rosaceae [12, 128], Violaceae [248]

Hexanol

Actinidiaceae [237], Alliaceae [36], Amaryllidaceae [6], Apiaceae [31, 242], Apocynaceae [178], Araceae [86], Arecaceae [68, 137, 143], Berberidaceae [182], Cactaceae [113, 125], Caryophyllaceae [66], Dipsacaceae [5], Fabaceae [50, 62, 120, 156, 179], Hyacinthaceae [123], Lecythidaceae [144], Malvaceae [67, 142], Myrsinaceae [103], Nyctaginaceae [151], Oleaceae [119], Orchidaceae [17, 23, 27, 28, 29, 30, 77, 116, 117, 118, 120, 147], Papaveraceae [62], Passifloraceae [157], Pinaceae [25], Ranunculaceae [209], Rosaceae [12, 33, 35, 40, 43, 51, 61, 62, 72, 87, 88, 128, 166, 228], Rubiaceae [5], Rutaceae [119], Scrophulariaceae [5], Solanaceae [165], Theophrastaceae [139], Thymelaeaceae [5], Valerianaceae [5, 37], Winteraceae [210], Vitaceae [44]

2-Hexanol

Cactaceae [125], Caprifoliaceae [102], Cycadaceae [211], Orchidaceae [30], Solanaceae [165]

3-Hexanol

Cactaceae [125]

Hexenol

Oleaceae [267]

1-Hexen-3-ol

Vitaceae [44, 47]

2-Hexenol

Ranunculaceae [209]

(E)-2-Hexenol

Actinidiaceae [237], Araceae [86], Asteraceae [37], Cactaceae [125], Dipsacaceae [5], Fabaceae [50, 120], Nyctaginaceae [151], Orchidaceae [117], Rosaceae [33], Valerianaceae [37]

(Z)-2-Hexenol

Orchidaceae [117]

3-Hexenol

Annonaceae [111], Asteraceae [59], Calycanthaceae [266], Caryophyllaceae [112], Moraceae [83], Oleaceae [57], Ranunculaceae [209], Rosaceae [43, 61], Ruscaceae [171], Winteraceae [210]

(E)-3-Hexenol

Actinidiaceae [237], Arecaceae [137], Fabaceae [156, 179], Lauraceae [74], Orchidaceae [117, 203], Rosaceae [35]

(Z)-3-Hexenol

Actinidiaceae [237], Alliaceae [36], Apocynaceae [141], Arecaceae [143], Asteraceae [5, 34, 37, 65], Acanthaceae [10], Berberidaceae [182], Brassicaceae [70, 221], Cactaceae [122, 125], Caprifoliaceae [102], Caryophyllaceae [5, 66, 141], Combretaceae [10], Dipsacaceae [5], Ericaceae [136], Fabaceae [49, 50, 120, 156, 160, 179], Gentianaceae [45], Hemerocallidaceae [33], Hyacinthaceae [123], Lecythidaceae [144], Magnoliaceae [264], Malvaceae [67, 142], Myrsinaceae [103], Nyctaginaceae [151], Oleaceae [33, 39, 52, 58, 107, 114, 119, 177], Orchidaceae [17, 23, 77, 116, 117, 118, 120, 141, 169], Orobanchaceae [20], Papaveraceae [62], Passifloraceae [157], Primulaceae [36], Ranunculaceae [19], Rhizophoraceae [10], Rosaceae [33, 35, 37, 62, 87, 88, 128, 166, 219, 220], Rubiaceae [5, 141, 249], Rutaceae [119], Salicaceae [245], Scrophulariaceae [5], Solanaceae [141, 165, 166], Theophrastaceae [139], Valerianaceae [37], Verbenaceae [5], Zingiberaceae [141]

4-Methylhexanol

Solanaceae [165]

Esters**Butyl hexanoate**

Lecythidaceae [144], Orchidaceae [117], Passifloraceae [157]

Butyl (Z)-3-hexenoate

Orchidaceae [117]

Dimethyl hexanedioate

Lauraceae [74]

Ethyl hexanoate

Annonaceae [111], Fabaceae [126], Magnoliaceae [11, 238], Orchidaceae [117, 203]

Ethyl 3-hexanoate

- Annonaceae [111]
- 2-Heptyl hexanoate
 - Orchidaceae [117]
- 3-Hexenyl hexanoate
 - Nyctaginaceae [151]
- (*E*)-3-Hexenyl hexanoate
 - Passifloraceae [157]
- (*Z*)-3-Hexenyl hexanoate
 - Lecythidaceae [144], Oleaceae [51], Orchidaceae [117]
- (*Z*)-3-Hexenyl (*Z*)-3-hexenoate
 - Oleaceae [119], Orchidaceae [117]
- Hexyl hexanoate
 - Lecythidaceae [144], Orchidaceae [117], Passifloraceae [157]
- Methyl 2-ethylhexanoate
 - Arecaceae [137], Cucurbitaceae [71]
- Methyl hexanoate
 - Fabaceae [216], Lecythidaceae [144], Lythraceae [10], Magnoliaceae [11], Nymphaeaceae [14, 135], Orchidaceae [117], Passifloraceae [157]
- Methyl hexenoate
 - Magnoliaceae [11]
- Methyl (*E*)-2-hexenoate
 - Nymphaeaceae [14, 135]
- Methyl (*Z*)-2-hexenoate
 - Nymphaeaceae [14, 135]
- Methyl 3-hexenoate
 - Oleaceae [39]
- Methyl 4-methylhexanoate
 - Nymphaeaceae [14, 135]
- 2-Methylpropyl hexanoate
 - Annonaceae [111]
- Pentyl hexanoate
 - Orchidaceae [117]
- 2-Pentyl hexanoate
 - Cycadaceae [211]
- Propyl hexanoate
 - Orchidaceae [117]
- 2-Propyl hexanoate
 - Arecaceae [68]
- Ethers
 - 1,6-Diacetoxylhexane
 - Oleaceae [57, 58]

C7

- Alkanes
 - 3,4-Dimethylheptane
 - Zingiberaceae [268]
 - Heptane
 - Actinidiaceae [237], Oleaceae [267], Orchidaceae [201], Rosaceae [12]
 - 3-Methylheptane
 - Oleaceae [267]
- Alkenes
 - 2,4-Dimethyl-2,4-heptadiene
 - Araceae [86]
 - 1-Heptene
 - Rosaceae [12]
 - Trimethylheptadiene
 - Araceae [133]
 - 2,3,6-Trimethyl-1,5-heptadiene
 - Araceae [129]
- Acids
 - 2-Ethylheptanoic acid
 - Onagraceae [265]

- Heptanoic acid
 - Asteraceae [65], Orchidaceae [23, 27]
- Aldehydes
 - 2,3-Dimethyl-5-heptenal
 - Sapindaceae [46]
 - 2,6-Dimethyl-5-heptenal
 - Oleaceae [267]
- Heptanal
 - Amaryllidaceae [6], Arecaceae [10, 143], Berberidaceae [215], Brassicaceae [221], Cactaceae [125], Caprifoliaceae [102], Caryophyllaceae [112], Hydnoraceae [48], Lauraceae [74], Malvaceae [142], Meliaceae [250], Orchidaceae [23, 28, 117, 147, 201, 224], Rosaceae [12, 219, 220]
- 4-Heptenal
 - Araceae [229]
- Ketones
 - 2,3-Heptandione
 - Scrophulariaceae [5]
 - 2-Heptanone
 - Araceae [129, 133, 134], Arecaceae [137, 143], Cactaceae [125], Caprifoliaceae [102], Cycadaceae [211], Ericaceae [136], Fabaceae [50, 138, 156], Lecythidaceae [144], Lythraceae [10], Orchidaceae [28, 29, 75, 116, 117], Rosaceae [12, 60]
 - 3-Heptanone
 - Araceae [229], Orchidaceae [116], Rosaceae [12], Violaceae [248], Zingiberaceae [268]
 - 2-Hydroxy-3-heptanone
 - Scrophulariaceae [5]
 - 3-Hydroxy-2-heptanone
 - Scrophulariaceae [5]
 - 3-Methyl-2-heptanone
 - Rosaceae [220]
 - 6-Methyl-2-heptanone
 - Oleaceae [267]
 - 4-Methyl-3-heptanone
 - Orchidaceae [203]
 - 6-Methyl-3-heptanone
 - Gentianaceae [45]
 - Methylheptenone
 - Orchidaceae [201]
- Alcohols
 - (*E,E*)-2,4-Heptadienol
 - Araceae [86]
 - Heptanol
 - Cactaceae [125], Hyacinthaceae [123], Orchidaceae [23, 27, 28, 29, 30, 117, 118, 147], Ranunculaceae [19, 59, 209]
 - 2-Heptanol
 - Arecaceae [143], Cactaceae [125], Caprifoliaceae [102], Cycadaceae [211], Fabaceae [138], Orchidaceae [23, 28, 29, 117], Rosaceae [60]
 - 3-Heptanol
 - Orchidaceae [28]
 - (*Z*)-4-Heptenol
 - Orchidaceae [117]
 - 2-Methylheptanol
 - Cactaceae [125]
 - 4-Methyl-2-heptanol
 - Oleaceae [57, 58]
 - 6-Methylheptanol
 - Calycanthaceae [266], Solanaceae [165]
- Esters
 - 3-Hexenyl heptanoate
 - Nyctaginaceae [151]
 - Methyl heptanoate
 - Orchidaceae [117]

C8

Alkanes

Octane

Actinidiaceae [237], Araceae [262], Caprifoliaceae [102], Orchidaceae [27, 201], Rosaceae [12, 40, 219]

Alkenes

3,4-Dimethyl-2,4,6-octatriene

Cucurbitaceae [71]

2,7-Methyloctadiene

Nyctaginaceae [151]

Octadiene

Nyctaginaceae [151]

1,3-Octadiene

Araceae [86, 229], Zamiaceae [211]

1,3,6-Octatriene

Orchidaceae [180]

1,3,7-Octatriene

Araceae [229]

1-Octene

Brassicaceae [221], Rosaceae [12]

Acids

Octanoic acid

Asteraceae [65], Fabaceae [120], Orchidaceae [8, 23, 27, 28, 77, 147, 192], Rosaceae [12]

Aldehydes

Octanal

Amaryllidaceae [6], Apiaceae [242], Araceae [229, 262], Brassicaceae [221], Cactaceae [122, 125], Caprifoliaceae [102], Caryophyllaceae [5, 54, 112], Gentianaceae [5], Hydnoraceae [48], Lauraceae [74], Meliaceae [250], Orchidaceae [8, 23, 28, 77, 117, 118, 120, 147, 176, 201, 224], Papaveraceae [62], Rosaceae [12, 219, 220], Violaceae [248]

Octenal

Araceae [205]

(E)-2-Octenal

Orchidaceae [117]

(Z)-2-Octenal

Araceae [86]

Ketones

3,5-Octadien-2-one

Araceae [86]

2-Octanone

Fabaceae [138], Malvaceae [142], Orchidaceae [27, 117]

3-Octanone

Araceae [86, 229], Arecaceae [137], Bignoniaceae [142], Fabaceae [49, 156], Malvaceae [142], Nymphaeaceae [14, 135], Orchidaceae [116, 117]

1-Octen-3-one

Araceae [229], Bignoniaceae [142], Malvaceae [142], Orchidaceae [116, 117]

(E)-3-Octen-2-one

Moraceae [82]

Alcohols

Octanol

Apiaceae [31], Araceae [229], Berberidaceae [182, 215], Cactaceae [125], Caprifoliaceae [102], Caryophyllaceae [5], Hyacinthaceae [38], Orchidaceae [17, 23, 27, 28, 29, 30, 117, 118, 147, 148], Ranunculaceae [19, 59, 209], Rosaceae [12], Salicaceae [245], Theophrastaceae [139], Valerianaceae [37], Vitaceae [44]

2-Octanol

Arecaceae [143], Fabaceae [138], Orchidaceae [28]

3-Octanol

Araceae [86, 229], Nymphaeaceae [14, 135], Orchidaceae [77, 116, 117]

1-Octenol

Orchidaceae [28, 29]

(E)-2-Octenol

Orchidaceae [117]

- (Z)-2-Octenol
 - Orchidaceae [117]
- (Z)-5-Octenol
 - Orchidaceae [117, 118]
- 7-Octenol
 - Malvaceae [67]
- Octen-3-ol
 - Fabaceae [126]
- 1-Octen-3-ol
 - Araceae [86, 229], Bignoniaceae [21, 142], Cactaceae [21, 122], Caryophyllaceae [5], Dipsacaceae [5], Fabaceae [49, 50, 107, 156], Gentianaceae [45], Hyacinthaceae [38, 123], Malvaceae [142], Orchidaceae [116, 117], Polemoniaceae [5, 21], Scrophulariaceae [5], Verbenaceae [5]
- 7-Octen-4-ol
 - Nyctaginaceae [151]
- Esters
 - Butyl octanoate
 - Orchidaceae [117]
 - Butyl octenoate
 - Orchidaceae [117]
 - Ethyl octanoate
 - Fabaceae [126], Orchidaceae [13, 27, 117]
 - Ethyl 3-octenoate
 - Orchidaceae [13]
 - Hexyl octanoate
 - Orchidaceae [117]
 - Methyl 3-methyloctanoate
 - Orchidaceae [116, 117]
 - Methyl octanoate
 - Araceae [129], Eupomatiaceae [18], Magnoliaceae [11, 240, 264], Myrsinaceae [103], Orchidaceae [13, 117], Passifloraceae [157]
 - Methyl (*E*)-2-octenoate
 - Nymphaeaceae [14, 135]
 - Methyl 4-octenoate
 - Cucurbitaceae [71], Orchidaceae [13]
 - 2-Pentyl octanoate
 - Cycadaceae [211]
- C9
 - Alkanes
 - Nonane
 - Apiaceae [31], Araceae [262], Brassicaceae [105], Caprifoliaceae [102], Gentianaceae [45], Meliaceae [250], Orchidaceae [8, 201, 224], Pittosporaceae [107], Rosaceae [12, 40, 128, 219]
 - Alkenes
 - 1-Nonene
 - Asteraceae [36]
 - Acids
 - Nonanoic acid
 - Araceae [86, 229], Asteraceae [65], Fabaceae [120], Malvaceae [67], Onagraceae [265], Orchidaceae [23]
 - Aldehydes
 - 2,6-Nonadienal
 - Nyctaginaceae [151]
 - (*E,Z*)-2,6-Nonadienal
 - Orchidaceae [117]
 - Nonanal
 - Amaryllidaceae [6], Apiaceae [242], Araceae [129, 205, 229, 262], Arecaceae [143], Asteraceae [65], Berberidaceae [215], Brassicaceae [221], Cactaceae [122, 125], Caprifoliaceae [102], Caryophyllaceae [54, 66, 112], Cucurbitaceae [71], Dipsacaceae [5], Fabaceae [120], Gentianaceae [5], Hydnoraceae [48], Lauraceae [74], Malvaceae [67], Meliaceae [250], Myrsinaceae [103], Orchidaceae [8, 17, 23, 27, 28, 77, 116, 117, 118,

- 147, 175, 176, 192, 201, 224], Papaveraceae [62], Primulaceae [5], Ranunculaceae [209], Rosaceae [12, 72, 219, 220], Rutaceae [247], Sapindaceae [46], Sapotaceae [263], Solanaceae [69], Valerianaceae [37], Verbenaceae [5], Violaceae [248], Vitaceae [42, 44, 47]
- 2-Nonenal
 Nyctaginaceae [151]
 (*E*)-2-Nonenal
 Cactaceae [113, 125], Orchidaceae [8, 117]
- Ketones
 Nonanone
 Myrsinaceae [103]
 2-Nonanone
 Araceae [129, 133], Arecaceae [143], Berberidaceae [215], Brassicaceae [22], Fabaceae [138], Malvaceae [41], Orchidaceae [27, 28, 29, 75, 117, 224], Rosaceae [60]
 3-Nonanone
 Araceae [229]
- Alcohols
 Nonanol
 Araceae [262], Brassicaceae [36], Orchidaceae [23, 27, 29, 30, 117, 118, 147]
 2-Nonanol
 Araceae [129, 205], Fabaceae [138], Orchidaceae [27, 28, 29, 30, 224]
 4-Nonenol
 Araceae [86]
 (*Z*)-3-Nonenol
 Orchidaceae [117], Solanaceae [121]
 (*E*)-6-Nonenol
 Malvaceae [67]
 (*Z*)-6-Nonenol
 Solanaceae [121]
 (*Z,Z*)-3,6-Nonadienol
 Caprifoliaceae [102], Orchidaceae [117], Solanaceae [121]
- Esters
 Methyl nonanoate
 Orchidaceae [13, 117]
- C10
 Alkanes
 Decane
 Apiaceae [31], Araceae [26, 146, 231], Arecaceae [143], Brassicaceae [105], Caprifoliaceae [102], Caryophyllaceae [54], Fabaceae [179, 235], Malvaceae [67], Meliaceae [250], Myrsinaceae [103], Oleaceae [57, 58], Orchidaceae [8, 23, 27, 28, 147, 201, 224], Ranunculaceae [209], Rosaceae [12, 40, 43, 128, 219], Zingiberaceae [268]
 2,6,8-Trimethyldecane
 Oleaceae [51]
- Alkenes
 1-Decene
 Araceae [133], Asteraceae [36]
- Acids
 Decanoic acid
 Araceae [229], Calycanthaceae [266], Caprifoliaceae [102], Orchidaceae [8]
 Decatrienoic acid
 Orchidaceae [77]
 (*Z*)-4-Decenoic acid
 Orchidaceae [117]
 (*E*)-3-Methyl-4-decenoic acid
 Orchidaceae [117]
 (*Z*)-3-Methyl-3-decenoic acid
 Fabaceae [120]
- Aldehydes
 (*E,E*)-2,4-Decadienal
 Amaryllidaceae [6], Orchidaceae [77, 116, 117, 121, 227]
 (*E,Z*)-2,4-Decadienal

- Orchidaceae [77, 116, 117, 121, 227]
- 2,5-Decadienal
 - Asteraceae [214]
- (Z,Z)-4,7-Decadienal
 - Apocynaceae [121], Orchidaceae [118]
- Decanal
 - Apiaceae [242], Araceae [129, 229, 262], Asteraceae [5], Berberidaceae [215], Brassicaceae [221], Cactaceae [122, 125], Caprifoliaceae [102, 175], Caryophyllaceae [5, 54, 66, 112], Dipsacaceae [5], Fabaceae [216], Gentianaceae [5, 45], Hydnoraceae [48], Lauraceae [74], Liliaceae [92], Malvaceae [67], Myrsinaceae [103], Nymphaeaceae [14, 135], Orchidaceae [5, 8, 23, 28, 77, 116, 117, 118, 176, 192, 201, 224], Papaveraceae [62], Primulaceae [5], Rosaceae [12, 43, 72, 219, 220], Rutaceae [247], Solanaceae [69], Violaceae [73]
- (E,Z,Z)-2,4,7-Decatrienal
 - Apocynaceae [121], Orchidaceae [121]
- Decenal
 - Araceae [86]
- (E)-2 Decenal
 - Amaryllidaceae [6]
- (Z)-4-Decenal
 - Orchidaceae [117]
- (E)-3-Methyl-4-decenal
 - Orchidaceae [117]
- Ketones
 - 2-Decanone
 - Arecaceae [137]
 - 4-Decanone
 - Calycanthaceae [266]
 - 3-Methyl-3-decen-2-one
 - Oleaceae [51]
- Alcohols
 - (E,E)-2,4-Decadienol
 - Orchidaceae [77, 117, 121]
 - (E,Z)-2,4-Decadienol
 - Orchidaceae [77, 117, 120, 121]
 - (Z,Z)-4,7-Decadienol
 - Apocynaceae [121], Orchidaceae [118]
- Decanol
 - Malvaceae [67], Orchidaceae [17, 28, 118], Sapindaceae [46]
- 2-Decanol
 - Orchidaceae [29]
- (E,Z,Z)-2,4,7-Decatrienol
 - Apocynaceae [121], Orchidaceae [121]
- (Z)-3-Decenol
 - Orchidaceae [117]
- (Z)-4-Decenol
 - Orchidaceae [117, 118]
- (Z)-7-Decenol
 - Orchidaceae [118]
- (E)-3-Methyl-4-decenol
 - Orchidaceae [117]
- Esters
 - Ethyl (E,E)-2,4-decadienoate
 - Orchidaceae [117]
 - Ethyl (Z,E)-2,4-decadienoate
 - Orchidaceae [117]
 - Ethyl decanoate
 - Orchidaceae [13, 117]
 - Ethyl 3-decenoate
 - Orchidaceae [13]
 - Ethyl (Z)-4-decenoate

- Orchidaceae [117]
 - Methyl caprate, see Methyl decanoate
 - Methyl (Z)-cascarillate
 - Fabaceae [120], Orchidaceae [120]
 - Methyl decadienoate
 - Magnoliaceae [240]
 - Methyl 2,4-decadienoate
 - Nymphaeaceae [14, 135]
 - Methyl (E,E)-2,4-decadienoate
 - Orchidaceae [116, 117]
 - Methyl (E,Z)-2,4-decadienoate
 - Orchidaceae [116, 117, 120]
 - Methyl decanoate (Methyl caprate)
 - Asteraceae [214], Gentianaceae [5], Magnoliaceae [11, 240], Myrsinaceae [103], Nyctaginaceae [151], Orchidaceae [13, 117]
 - Methyl decatrienoate
 - Orchidaceae [117]
 - Methyl decenoate
 - Magnoliaceae [11, 240]
 - Methyl 4-decenoate
 - Orchidaceae [13]
 - Methyl (Z)-4-decenoate
 - Orchidaceae [116, 117]
 - Methyl (Z)-3-methyl-3-decenoate
 - Fabaceae [120]
 - Methyl (E)-3-methyl-4-decenoate
 - Fabaceae [120], Orchidaceae [120]
 - 2-Pentyl decanoate
 - Cycadaceae [211]
- C11
- Alkanes
 - Undecane
 - Actinidiaceae [237], Araceae [26, 146, 231], Arecaceae [68, 137, 143], Asteraceae [5], Brassicaceae [105], Caprifoliaceae [102], Fabaceae [216], Malvaceae [67], Myrsinaceae [103], Oleaceae [51], Orchidaceae [8, 23, 27, 28, 30, 147, 175, 201, 224], Pittosporaceae [107], Ranunculaceae [209], Rosaceae [12, 33, 35, 51, 220], Rutaceae [172, 247], Sapotaceae [263], Violaceae [248]
 - Alkenes
 - Methylundecene
 - Araceae [206]
 - (E,E)-1,3,5-Undecatriene
 - Cactaceae [21], Orchidaceae [78, 120]
 - (E,Z)-1,3,5-Undecatriene
 - Cactaceae [21], Orchidaceae [78, 117, 120, 121]
 - Undecene
 - Apiaceae [31], Orchidaceae [23, 27]
 - 1-Undecene
 - Asteraceae [36], Brassicaceae [22]
 - 2-Undecene
 - Araceae [86]
 - Acids
 - Undecanoic acid
 - Hydnoraceae [48]
 - Aldehydes
 - Undecanal
 - Amaryllidaceae [6], Caprifoliaceae [102], Caryophyllaceae [54], Hydnoraceae [48], Lauraceae [74], Orchidaceae [23, 117, 201, 224]
 - Ketones
 - 2-Undecanone
 - Araceae [26], Myrsinaceae [103], Orchidaceae [23, 28, 29, 75, 117], Rosaceae [51, 60, 61, 128]

- 3-Undecanone
 - Orchidaceae [27]
- 6-Undecanone
 - Oleaceae [57, 58], Onagraceae [265]
- Alcohols
 - 2-Undecanol
 - Orchidaceae [28]
- Esters
 - Ethyl undecanoate
 - Orchidaceae [117]
 - Methyl undecanoate
 - Orchidaceae [77]
- Ethers and epoxides
 - (*Z*)-5(6)-Epoxy-(*E*)-1,3-undecadiene
 - Orchidaceae [121]
- C12
 - Alkanes
 - Dodecane
 - Actinidiaceae [237], Araceae [26, 86, 231, 262], Arecaceae [143], Brassicaceae [105], Cactaceae [113], Calycanthaceae [120], Caprifoliaceae [102], Caryophyllaceae [5], Malvaceae [67], Myrsinaceae [103], Nymphaeaceae [14, 135], Oleaceae [51], Orchidaceae [8, 23, 27, 28, 147, 175, 201, 224], Papaveraceae [62], Ranunculaceae [207, 209], Rosaceae [43, 51, 219], Sapotaceae [263], Valerianaceae [37]
 - 6-Methyldodecane
 - Oleaceae [51]
 - 2,7,10-Trimethyldodecane
 - Araceae [262]
 - Alkenes
 - 1-Dodecene
 - Araceae [86], Brassicaceae [22]
 - Acids
 - Dodecanoic acid
 - Araceae [152], Fabaceae [120], Hydnoraceae [48], Orchidaceae [117]
 - Aldehydes
 - Dodecanal
 - Caprifoliaceae [102], Gentianaceae [5], Orchidaceae [28, 117, 224]
 - Ketones
 - Dimethyldodecenone
 - Araceae [205, 206]
 - 2-Dodecanone (Methyl undecyl ketone)
 - Orchidaceae [77]
 - Methyl undecyl ketone, see 2-Dodecanone
 - Alcohols
 - (*Z,Z*)-3,6-Dodecadienol
 - Orchidaceae [118]
 - Dodecanol
 - Oleaceae [51], Orchidaceae [8, 13, 17, 30, 117, 118], Rutaceae [172]
 - (*Z,Z,Z*)-3,6,9-Dodecatrienol
 - Orchidaceae [117, 118]
 - 2-Dodecenol
 - Sapindaceae [46]
 - (*Z*)-6-Dodecenol
 - Orchidaceae [118]
 - Esters
 - Butyl dodecanoate
 - Orchidaceae [117]
 - Ethyl dodecanoates
 - Orchidaceae [13, 117]
 - Methyl dodecadienoate
 - Magnoliaceae [240]
 - Methyl 3,6-dodecadienoate

- Orchidaceae [13]
- Methyl dodecanoate
 - Caprifoliaceae [102], Magnoliaceae [11, 240, 264], Orchidaceae [13, 117]
- Methyl dodecenoate
 - Magnoliaceae [11, 240]
- C13
 - Alkanes
 - 2,5-Dimethyltridecane
 - Oleaceae [51], Rosaceae [51]
 - Tridecane
 - Actinidiaceae [237], Araceae [26, 86, 231], Arecaceae [137, 143], Berberidaceae [215], Brassicaceae [105], Cactaceae [113], Caprifoliaceae [102], Dipsacaceae [185], Fabaceae [120, 216], Magnoliaceae [11, 240], Malvaceae [67], Myrsinaceae [103], Nelumbonaceae [184], Orchidaceae [8, 23, 27, 29, 30, 76, 147, 201, 224], Papaveraceae [62], Polemoniaceae [5], Ranunculaceae [209], Rosaceae [51, 60, 128, 219], Sapotaceae [263], Violaceae [73], Zamiaceae [211]
 - Alkenes
 - Tridecadiene
 - Orchidaceae [28, 29]
 - Tridecene
 - Araceae [26], Orchidaceae [23, 27, 28, 29, 30, 147]
 - 1-Tridecene
 - Araceae [86], Lauraceae [74], Nymphaeaceae [14, 135], Vitaceae [44]
 - 2-Tridecene
 - Lauraceae [74]
 - 6-Tridecene
 - Malvaceae [67]
 - Acids
 - Tridecanoic acid
 - Hydnoraceae [48]
 - Aldehydes
 - (Z,Z)-4,7-Tridecadial
 - Orchidaceae [117]
 - Tridecanal
 - Araceae [262], Orchidaceae [117, 224]
 - Ketones
 - Tridecanone
 - Orchidaceae [227], Vitaceae [47]
 - 2-Tridecanone
 - Arecaceae [137], Asteraceae [214], Orchidaceae [28, 75, 117], Rosaceae [51, 60, 61, 128], Vitaceae [44]
 - Alcohols
 - Tridecanol
 - Orchidaceae [227]
 - 2-Tridecanol
 - Sapindaceae [46]
- C14
 - Alkanes
 - 2-Methyltetradecane
 - Malvaceae [67]
 - Tetradecane
 - Actinidiaceae [237], Amaryllidaceae [6], Annonaceae [111], Araceae [26, 86, 262], Arecaceae [137, 143], Brassicaceae [105], Cactaceae [113, 125], Calycanthaceae [120], Caryophyllaceae [5], Fabaceae [120], Gentianaceae [45], Lauraceae [74], Magnoliaceae [11, 240, 264], Malvaceae [67], Nelumbonaceae [184, 198, 199], Nymphaeaceae [14, 135], Orchidaceae [23, 27, 28, 30, 76, 117, 147, 175, 201, 224], Papaveraceae [62], Polemoniaceae [5], Ranunculaceae [209], Rosaceae [62, 128, 219], Rutaceae [119], Sapotaceae [263], Scrophulariaceae [5], Valerianaceae [33], Violaceae [73], Zamiaceae [211]
 - Alkenes
 - Tetradecene
 - Arecaceae [137], Magnoliaceae [11, 240], Malvaceae [67], Orchidaceae [27, 28, 30]

- 1-Tetradecene
 - Araceae [86], Musaceae [21], Nymphaeaceae [14, 135]
- Acids
 - Myristic acid, see Tetradecanoic acid
 - Tetradecanoic acid (Myristic acid)
 - Araceae [152, 229], Fabaceae [120], Hydnoraceae [48], Orchidaceae [227], Rutaceae [247]
- Aldehydes
 - Tetradecanal
 - Araceae [262], Berberidaceae [215], Caprifoliaceae [102], Orchidaceae [117, 224, 253], Rosaceae [60, 61]
- Alcohols
 - 1,14-Tetradecanediol
 - Caryophyllaceae [5]
 - Tetradecanol
 - Orchidaceae [30, 117], Rosaceae [128]
 - (Z)-7-Tetradecen-1-ol
 - Nelumbonaceae [184]
- Esters
 - Ethyl tetradecanoate
 - Caprifoliaceae [102], Orchidaceae [117]
 - Isopropyl tetradecanoate, see 2-Propyl tetradecanoate
 - Methyl tetradecadienoate
 - Magnoliaceae [240]
 - Methyl tetradecanoate
 - Cactaceae [125], Caprifoliaceae [102], Magnoliaceae [11, 240], Orchidaceae [30, 117, 227]
 - Methyl tetradecenoate
 - Magnoliaceae [11, 240]
 - Propyl tetradecanoate
 - Araceae [152], Orchidaceae [23, 27]
 - 2-Propyl tetradecanoate (Isopropyl tetradecanoate)
 - Asteraceae [5], Dipsacaceae [5], Orchidaceae [195]
- C15
 - Alkanes
 - Methylpentadecane
 - Papaveraceae [62]
 - Pentadecane
 - Actinidiaceae [237], Amaryllidaceae [6], Apiaceae [242, 246], Araceae [26, 86, 262], Arecaceae [137, 143], Asteraceae [5], Caprifoliaceae [102], Caryophyllaceae [5], Fabaceae [126, 212, 216], Gentianaceae [45], Geraniaceae [32], Magnoliaceae [11, 240, 264], Malvaceae [67], Myrsinaceae [103], Nelumbonaceae [178, 184, 198, 199], Nymphaeaceae [14, 135, 178], Oleaceae [107], Orchidaceae [16, 23, 27, 28, 29, 30, 117, 147, 201, 224, 253], Papaveraceae [62], Polemoniaceae [5], Ranunculaceae [209], Rosaceae [33, 35, 60, 61, 62, 220], Rubiaceae [5], Rutaceae [119, 172, 247], Sapotaceae [263], Thymelaeaceae [5], Valerianaceae [37], Verbenaceae [5, 101], Violaceae [73], Zamiaceae [211]
 - Alkenes
 - Pentadecadiene
 - Orchidaceae [28, 29]
 - (Z)-1,8-Pentadecadiene
 - Orchidaceae [117]
 - Pentadecene
 - Arecaceae [137, 143], Cucurbitaceae [4], Magnoliaceae [11], Nelumbonaceae [198, 199], Orchidaceae [23, 27, 28, 29, 30, 147], Papaveraceae [62]
 - 1-Pentadecene
 - Araceae [86], Malvaceae [67], Orchidaceae [117], Vitaceae [42, 44, 47]
 - Acids
 - Pentadecanoic acid
 - Araceae [152], Hydnoraceae [48], Onagraceae [175], Orchidaceae [227]
 - Ketones
 - 2-Pentadecanone
 - Arecaceae [137], Caprifoliaceae [102], Fumariaceae [56], Orchidaceae [75, 117], Rosaceae [61, 128]

Alcohols

Pentadecanol

Caprifoliaceae [102], Rosaceae [128]

Esters

Methyl pentadecanoate

Caprifoliaceae [102], Orchidaceae [30, 117]

Methyl 13-methylpentadecanoate

Orchidaceae [180]

Methyl 14-methylpentadecanoate

Orchidaceae [227]

C16

Alkanes

Hexadecane

Actinidiaceae [237], Annonaceae [111], Apiaceae [242, 246], Araceae [26, 86, 129, 262], Arecaceae [137, 143], Berberidaceae [215], Cactaceae [125], Caprifoliaceae [102], Caryophyllaceae [5], Fabaceae [120, 212], Gentianaceae [5], Magnoliaceae [240], Malvaceae [67], Meliaceae [250], Nymphaeaceae [14, 135], Oleaceae [52], Orchidaceae [8, 23, 27, 28, 29, 30, 117, 147, 201, 224], Papaveraceae [62], Polemoniaceae [5], Primulaceae [5], Ranunculaceae [209], Rosaceae [33, 35, 43, 62, 128], Rubiaceae [5], Sapotaceae [263], Thymelaeaceae [5], Valerianaceae [5], Verbenaceae [5], Zamiaceae [211]

Alkenes

Hexadecadiene

Arecaceae [143]

Hexadecene

Araceae [129], Arecaceae [143], Magnoliaceae [240], Malvaceae [67], Orchidaceae [27, 30], Papaveraceae [62]

1-Hexadecene

Musaceae [21], Nymphaeaceae [14, 135]

Acids

Hexadecanoic acid

Araceae [152, 229], Fabaceae [120], Hydnoraceae [48], Liliaceae [107], Oleaceae [57, 58], Onagraceae [175], Orchidaceae [227], Rosaceae [43], Rutaceae [247], Vitaceae [44]

9-Hexadecanoic acid

Orchidaceae [227]

(Z)-9-Hexadecenoic acid

Hydnoraceae [48]

Aldehydes

Hexadecanal

Caprifoliaceae [102], Orchidaceae [117], Rosaceae [60, 61]

Alcohols

Hexadecanol

Hydnoraceae [48], Orchidaceae [27, 30, 117], Rosaceae [128]

Esters

Ethyl hexadecanoate

Orchidaceae [117, 227]

Methyl hexadecanoate

Cactaceae [125], Fabaceae [107], Liliaceae [107], Magnoliaceae [11], Orchidaceae [30, 117, 141], Rosaceae [37], Vitaceae [44, 47]

Methyl hexadecenoate

Orchidaceae [30]

Propyl hexadecanoate

Orchidaceae [27]

2-Propyl hexadecanoate

Orchidaceae [191]

Ethers

2-Dodecen-1-yl-succinic anhydride

Rosaceae [128]

C17

Alkanes

Heptadecane

Actinidiaceae [237], Araceae [26], Arecaceae [137, 143], Cactaceae [125], Caprifoliaceae [102], Caryophyllaceae [5], Fabaceae [212, 216], Gentianaceae [5], Liliaceae [107], Myrsinaceae [103], Nelumbonaceae [198, 199], Oleaceae [107], Orchidaceae [16, 23, 27, 28, 29, 30, 117, 147, 201, 224], Papaveraceae [62], Ranunculaceae [207, 209], Rosaceae [33, 35, 51, 62], Rubiaceae [5], Rutaceae [119, 172, 247]

Heptadecane isomer

Caryophyllaceae [5]

Methylheptadecane

Papaveraceae [62]

Alkenes

Heptadecadiene

Arecaceae [137, 143], Magnoliaceae [11, 240], Nymphaeaceae [178], Orchidaceae [27], Papaveraceae [62]

Heptadecene

Arecaceae [137, 143], Liliaceae [107], Magnoliaceae [11, 240], Nelumbonaceae [198, 199], Oleaceae [107, 119], Orchidaceae [23, 27, 30, 147], Papaveraceae [62], Rutaceae [119, 247]

1-Heptadecene

Geraniaceae [32], Nelumbonaceae [184], Orchidaceae [117]

8-Heptadecene

Nelumbonaceae [184]

(Z)-8-Heptadecene

Rutaceae [172]

Acids

Heptadecanoic acid

Araceae [152], Hydnoraceae [48]

Aldehydes

(Z,Z)-Heptadeca-8,11-dienal

Caprifoliaceae [102]

(Z,Z,Z)-Heptadeca-8,11,14-trienal

Caprifoliaceae [102]

Ketones

2-Heptadecanone

Fumariaceae [56], Moraceae [82], Orchidaceae [28, 117], Rosaceae [62]

C18

Alkanes

Octadecane

Actinidiaceae [237], Araceae [26], Arecaceae [137], Caryophyllaceae [5], Orchidaceae [23, 27], Papaveraceae [62], Rubiaceae [5]

Alkenes

Octadecene

Araceae [129], Orchidaceae [27], Papaveraceae [62]

Acids

Octadecanoic acid

Araceae [152], Hydnoraceae [48], Lauraceae [74]

Octadecenoic acid

Araceae [152]

(Z)-9-Octadecenoic acid

Lauraceae [74]

Alcohols

Octadecanol

Orchidaceae [30]

Esters

Methyl (Z,Z)-9,12-octadecadienoate

Orchidaceae [117]

Methyl octadecenoate

Orchidaceae [30]

C19

Alkanes

Methylnonadecane

Papaveraceae [62]

- Nonadecane
 - Actinidiaceae [237], Araceae [26, 86], Orchidaceae [23, 27, 30, 117, 148, 201], Papaveraceae [62], Ranunculaceae [207, 208, 209], Rosaceae [33, 35], Rutaceae [247], Vitaceae [44]
- Alkenes
 - Nonadecadiene
 - Papaveraceae [62]
 - Nonadecatriene
 - Ranunculaceae [209]
 - Nonadecene
 - Arecaceae [137], Orchidaceae [23, 27, 30], Papaveraceae [62], Ranunculaceae [207, 209], Rosaceae [33, 35]
 - 1-Nonadecene
 - Orchidaceae [117], Vitaceae [44]
- Ketones
 - 2-Nonadecanone
 - Orchidaceae [117]
- C20
 - Alkanes
 - Eicosane
 - Actinidiaceae [237], Orchidaceae [23, 117, 201], Papaveraceae [62], Ranunculaceae [209], Rosaceae [33, 35]
 - Alkenes
 - Eicosene
 - Papaveraceae [62]
- C21
 - Alkanes
 - Heneicosane
 - Actinidiaceae [237], Oleaceae [52], Orchidaceae [23, 27, 148], Papaveraceae [62], Ranunculaceae [209], Rosaceae [33, 35, 60]
 - Alkenes
 - Heneicosene
 - Papaveraceae [62]
- C22
 - Alkanes
 - Docosane
 - Oleaceae [52]
- C23
 - Alkanes
 - Tricosane
 - Myrsinaceae [103], Oleaceae [52], Orchidaceae [27, 148]
 - Alkenes
 - Tricosene
 - Orchidaceae [23]
- C24
 - Alkanes
 - Tetracosane
 - Myrsinaceae [103]
- C25
 - Alkanes
 - Pentacosane
 - Myrsinaceae [103], Oleaceae [52]

BENZENOIDS AND PHENYL PROPANOIDS

- C6-C0
 - Hydrocarbons
 - Benzene
 - Orchidaceae [23, 28, 147], Rosaceae [12]
 - Alcohols
 - 1,2-Benzenediol, see 1,2-Dihydroxybenzene
 - 1,2-Dihydroxybenzene (1,2-Benzenediol)
 - Orchidaceae [227]

2,6-Dimethoxyphenol, see 1-Hydroxy-2,6-dimethoxybenzene

Guaiacol, see 1-Hydroxy-2-methoxybenzene

Hydroxybenzene (Phenol)

Araceae [152, 229], Caryophyllaceae [54], Myrsinaceae [103], Orchidaceae [106]

1-Hydroxy-2,6-dimethoxybenzene (2,6-Dimethoxyphenol)

Orchidaceae [227]

1-Hydroxy-2-methoxybenzene (Methoxyphenol, Guaiacol)

Araceae [137], Caryophyllaceae [112], Eupomatiaceae [18], Orchidaceae [117, 227], Rhizophoraceae [10], Rubiaceae [141]

1-Hydroxy-4-methoxybenzene (4-Methoxyphenol)

Cactaceae [125]

2-Methoxyphenol, see 1-Hydroxy-2-methoxybenzene

4-Methoxyphenol, see 1-Hydroxy-4-methoxybenzene

Phenol, see Hydroxybenzene

2-Phenoxyethanol

Amaryllidaceae [63], Fabaceae [81, 216], Nelumbonaceae [184], Rosaceae [62]

Esters

Phenyl acetate

Caryophyllaceae [112]

Ethers

Anisole, see Methoxybenzene

Dimethoxybenzene

Orchidaceae [29]

1,2-Dimethoxybenzene (Veratrole)

Araceae [134], Arecaceae [137], Caryophyllaceae [112], Fumariaceae [56], Hyacinthaceae [123], Lecythidaceae [144], Magnoliaceae [9, 11], Oleaceae [107, 108], Orchidaceae [77, 117, 120], Primulaceae [5], Ranunculaceae [85], Rhizophoraceae [10], Rubiaceae [5], Zingiberaceae [268]

1,3-Dimethoxybenzene

Theophrastaceae [139]

1,4-Dimethoxybenzene (Hydroquinone dimethyl ether)

Amaryllidaceae [63, 232, 233], Arecaceae [137, 143], Cactaceae [125], Cucurbitaceae [4], Ericaceae [136, 140], Gesneriaceae [77], Hyacinthaceae [38, 123], Nelumbonaceae [178, 184, 198, 199], Oleaceae [33, 51, 107, 150, 171, 178, 232, 233], Orchidaceae [8, 13, 17, 28, 76, 77, 78, 116, 117, 120, 188, 201, 253, 258], Salicaceae [245], Theophrastaceae [139], Violaceae [248]

Diphenyl ether

Oleaceae [107]

Hydroquinone dimethyl ether, see 1,4-Dimethoxybenzene

Methoxybenzene (Anisole)

Araceae [26], Arecaceae [137], Magnoliaceae [9], Nyctaginaceae [151], Orchidaceae [117], Ranunculaceae [19], Rosaceae [40, 62, 72, 219]

4-Methoxybenzene

Zingiberaceae [174]

1,2,3,5-Tetramethoxybenzene

Orchidaceae [117]

Trimethoxybenzene

Magnoliaceae [11]

1,2,3-Trimethoxybenzene

Araceae [137]

1,2,4-Trimethoxybenzene

Cucurbitaceae [4], Ericaceae [136, 140], Hyacinthaceae [38, 123], Malvaceae [142], Oleaceae [107, 150], Orchidaceae [8, 117, 201]

1,3,5-Trimethoxybenzene

Cactaceae [125], Ericaceae [136, 140], Oleaceae [107, 108], Orchidaceae [13, 17, 77, 117], Passifloraceae [157]

Veratrole, see 1,2-Dimethoxybenzene

Chloro

Dichlorobenzene

Cucurbitaceae [71]

N-compounds

- 2-Methoxybenzeneamine
 - Fumariaceae [56]
- 1-Methoxy-2-nitrobenzene
 - Fumariaceae [56]
- C6-C1
- Hydrocarbons
 - Dimethylbenzene
 - Asteraceae [213], Winteraceae [239]
 - 1,2-Dimethylbenzene
 - Annonaceae [111], Caryophyllaceae [112], Meliaceae [250], Orchidaceae [23, 27, 28, 75, 147], Rosaceae [12]
 - 1,3-Dimethylbenzene
 - Actinidiaceae [237], Meliaceae [250], Orchidaceae [23, 27, 28, 147], Rosaceae [12]
 - 1,4-Dimethylbenzene
 - Annonaceae [111], Orchidaceae [23, 27, 28, 75, 147], Rosaceae [12]
 - Methylbenzene (Toluene)
 - Actinidiaceae [237], Araceae [86, 146], Meliaceae [250], Orchidaceae [23, 27, 28, 75, 147], Rosaceae [12]
 - Tetramethylbenzene
 - Lauraceae [74]
 - Toluene, see Methylbenzene
 - Trimethylbenzene
 - Annonaceae [111], Cucurbitaceae [71], Orchidaceae [147]
 - 1,2,3-Trimethylbenzene
 - Caryophyllaceae [112]
 - 1,2,4-Trimethylbenzene
 - Lauraceae [74], Meliaceae [250], Rosaceae [12]
 - 1,3,5-Trimethylbenzene
 - Lauraceae [74], Rosaceae [12]
- Acids
 - Benzoic acid
 - Araceae [86], Calycanthaceae [266], Nelumbonaceae [184], Oleaceae [57, 58], Orchidaceae [8], Rutaceae [247]
 - 2-Hydroxymethylbenzoic acid
 - Oleaceae [58]
- Aldehydes
 - Benzaldehyde
 - Actinidiaceae [237], Alliaceae [36], Amaryllidaceae [6, 63], Annonaceae [111], Apiaceae [31, 242, 246], Apocynaceae [141, 178, 232, 233], Araceae [86, 133, 146, 258, 262], Arecaceae [137], Asteraceae [5, 37, 55, 59, 65, 132], Berberidaceae [182, 215], Brassicaceae [22, 36, 70, 202, 217, 221], Bromeliaceae [77], Cactaceae [21, 113, 115, 122, 125], Calycanthaceae [120], Caprifoliaceae [102], Caryophyllaceae [5, 66, 112, 141], Cucurbitaceae [175], Dipsacaceae [5], Ericaceae [136, 140], Fabaceae [49, 62, 81, 104, 120, 126, 212, 216], Fumariaceae [56, 197], Gentianaceae [5, 45], Hemerocallidaceae [33], Hyacinthaceae [38, 123], Hydnoraceae [48], Lamiaceae [5], Lecythidaceae [144], Linnaeaceae [90], Magnoliaceae [9, 11, 264], Malvaceae [142], Meliaceae [250], Moraceae [82], Myrsinaceae [103], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [33, 39, 51, 171, 267], Orchidaceae [2, 5, 8, 13, 17, 27, 28, 29, 77, 78, 115, 116, 117, 118, 120, 141, 169, 176, 180, 195, 201, 204, 236, 241, 243, 244, 253, 254, 258], Papaveraceae [62], Passifloraceae [157], Polemoniaceae [5, 232, 233, 234], Primulaceae [5, 36], Ranunculaceae [208], Rosaceae [12, 33, 35, 37, 43, 62, 72, 87, 88, 128, 219, 220], Rubiaceae [5, 141], Ruscaceae [141, 171], Rutaceae [119, 247], Sapindaceae [46], Sapotaceae [263], Scrophulariaceae [5, 226], Solanaceae [69, 91, 131, 141, 154, 165, 166, 167, 168], Theaceae [200], Theophrastaceae [139], Thymelaeaceae [5, 24], Valerianaceae [37], Verbenaceae [5, 101, 175], Violaceae [73, 248], Zamiaceae [211], Zingiberaceae [141, 174]
 - 1,4-Dicarboxaldehydebenzene
 - Oleaceae [57]
 - 3,4-Dimethoxybenzaldehyde (Veratraldehyde)
 - Hydrangaceae [108], Onagraceae [218], Orchidaceae [117]
 - 2-Hydroxybenzaldehyde

- Apiaceae [246], Cucurbitaceae [71], Fumariaceae [56], Magnoliaceae [9], Nyctaginaceae [151], Orchidaceae [117, 204], Rosaceae [37], Salicaceae [245]
- 4-Hydroxy-3-methoxybenzaldehyde (Vanillin)
Asteraceae [214], Cactaceae [125], Caryophyllaceae [54], Nyctaginaceae [151], Onagraceae [218], Orchidaceae [77, 117, 120, 141, 244, 253], Passifloraceae [157], Rosaceae [43], Zingiberaceae [141, 174]
- Methoxybenzaldehyde
Magnoliaceae [9]
- 4-Methoxybenzaldehyde
Arecaceae [68], Asteraceae [5], Brassicaceae [70], Caryophyllaceae [5], Fumariaceae [56], Nymphaeaceae [178], Oleaceae [33, 51, 107, 171, 232, 233], Orchidaceae [13, 17, 28, 115, 116, 117, 118], Primulaceae [36], Rosaceae [33, 35, 37, 87, 88, 220], Solanaceae [121], Theophrastaceae [139]
- 4-Methylbenzaldehyde
Moraceae [83]
- 3,4-Methylenedioxybenzaldehyde (Piperonal)
Malvaceae [41], Oleaceae [150]
- Piperonal, see 3,4-Methylenedioxybenzaldehyde
- 3,4,5-Trimethoxybenzaldehyde
Orchidaceae [117]
- Vanillin, see 4-Hydroxy-3-methoxybenzaldehyde
- Veratraldehyde, see 3,4-Dimethoxybenzaldehyde
- Ketones
Benzophenone
Araceae [26, 262]
- 2,6-Dimethyl-3-methoxymethyl-4-benzoquinone
Orchidaceae [227]
- Alcohols
Anisyl alcohol, see Methoxybenzyl alcohol
- Benzyl alcohol (Phenylmethanol)
Amaryllidaceae [6, 107], Annonaceae [111], Apocynaceae [121, 141, 173, 232, 233], Araceae [86, 129, 133, 146, 229], Arecaceae [10, 68, 137, 143], Asteraceae [5], Berberidaceae [182, 215], Brassicaceae [22, 186, 202], Cactaceae [113, 115, 122, 125], Calycanthaceae [120, 266], Caprifoliaceae [102, 178], Caryophyllaceae [112, 141], Cucurbitaceae [4], Cyclanthaceae [225], Ericaceae [136, 140], Fabaceae [49, 81, 104, 120, 138, 156, 212, 216], Fumariaceae [56, 197], Gentianaceae [45], Hemerocallidaceae [33], Hyacinthaceae [38, 123], Lecythidaceae [144], Linnaeaceae [90], Magnoliaceae [9, 11, 240, 264], Malvaceae [142], Moraceae [82, 84], Myrsinaceae [103], Nyctaginaceae [151], Nymphaeaceae [14, 135, 178], Oleaceae [33, 39, 52, 114, 176, 178], Orchidaceae [8, 13, 17, 77, 78, 115, 116, 117, 118, 120, 141, 169, 175, 195, 201, 203, 204, 241, 243, 244, 253, 258], Passifloraceae [157], Pittosporaceae [107], Polemoniaceae [232, 233], Primulaceae [36], Ranunculaceae [207, 208, 209], Rosaceae [12, 33, 35, 37, 43, 51, 60, 61, 87, 88, 166, 219, 220], Rubiaceae [121, 141, 249], Ruscaceae [141, 171, 232, 233], Rutaceae [119], Salicaceae [245], Sapotaceae [263], Scrophulariaceae [226], Solanaceae [141, 154, 165, 166, 167, 168, 222], Theaceae [200], Theophrastaceae [139], Thymelaeaceae [24, 234], Valerianaceae [33], Vitaceae [42, 44, 47], Zamiaceae [211], Zingiberaceae [141, 174]
- 5-Benzyloxy-1-pentanol
Oleaceae [58]
- m*-Cresol, see 3-Methyl-1-hydroxybenzene
- o*-Cresol, see 2-Methyl-1-hydroxybenzene
- p*-Cresol, see 4-Methyl-1-hydroxybenzene
- A,α-Dimethyl benzyl alcohol, see 2,6-Dimethylbenzyl alcohol
- 2,6-Dimethylbenzyl alcohol (A,α-Dimethyl benzyl alcohol)
Moraceae [82]
- 1-Hydroxy-2,3,5,6-tetramethylbenzene (2,3,5,6-Tetramethylphenol)
Orchidaceae [227]
- Methoxybenzyl alcohol (Anisyl alcohol)
Orchidaceae [116]
- 4-Methoxybenzyl alcohol

- Arecaceae [68], Cucurbitaceae [4], Orchidaceae [13, 117, 118], Polemoniaceae [5], Primulaceae [5], Theophrastaceae [139]
- 2-Methoxy-4-methyl-1-hydroxybenzene (2-Methoxy-4-methylphenol, 4-Methyl-2-methoxy-1-hydroxybenzene)
- Amaryllidaceae [177], Araceae [86], Asteraceae [132], Oleaceae [52], Orchidaceae [8]
- 2-Methoxy-4-methylphenol, see 2-Methoxy-4-methyl-1-hydroxybenzene
- Methylhydroxybenzene (Methylphenol)
- Asteraceae [132], Caryophyllaceae [54], Oleaceae [171], Orchidaceae [23, 117]
- 2-Methyl-1-hydroxybenzene (2-Methylphenol, *o*-Cresol)
- Apiaceae [31], Araceae [86, 229]
- 3-Methyl-1-hydroxybenzene (*m*-Cresol)
- Araceae [86]
- 4-Methyl-1-hydroxybenzene (4-Methylphenol, *p*-Cresol)
- Amaryllidaceae [63, 177], Apiaceae [246], Araceae [86, 129, 133, 229], Arecaceae [68], Caryophyllaceae [112], Hydnoraceae [48], Oleaceae [52, 114], Orchidaceae [28, 77, 116, 117, 203, 253, 258], Sapotaceae [263], Theophrastaceae [139], Violaceae [248]
- 4-Methyl-2-methoxy-1-hydroxybenzene, see 2-Methoxy-4-methyl-1-hydroxybenzene
- Methylphenol, see Methylhydroxybenzene
- 2-Methylphenol, see 2-Methyl-1-hydroxybenzene
- 4-Methylphenol, see 4-Methyl-1-hydroxybenzene
- Phenylmethanol, see Benzyl alcohol
- 2,3,5,6-Tetramethylphenol, see 1-Hydroxy-2,3,5,6-tetramethylbenzene
- Esters**
- Benzyl acetate**
- Amaryllidaceae [6, 63, 177], Annonaceae [111, 170], Araceae [77, 146, 152, 229, 258], Arecaceae [137], Asteraceae [132], Berberidaceae [215], Brassicaceae [186], Bromeliaceae [77], Cactaceae [113, 122, 125], Calycanthaceae [120, 266], Caryophyllaceae [54, 112, 141], Euphorbiaceae [7], Fabaceae [104, 107, 120, 212], Fumariaceae [197], Hemerocallidaceae [33], Hyacinthaceae [38, 123], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [240], Myrsinaceae [103], Nyctaginaceae [151], Nymphaeaceae [14, 135, 178], Oleaceae [52, 58, 114, 171, 176, 178, 267], Onagraceae [218], Orchidaceae [8, 17, 77, 78, 95, 97, 98, 115, 116, 117, 118, 120, 141, 169, 181, 195, 236, 243, 244, 253, 254, 256, 258, 259], Passifloraceae [157], Pittosporaceae [107], Rosaceae [33, 35, 43, 128, 166], Rubiaceae [121, 141, 249], Ruscaceae [141, 171, 232, 233], Sapotaceae [263], Scrophulariaceae [5], Solanaceae [91, 131, 154, 165, 166, 167, 222], Theophrastaceae [139], Thymelaeaceae [5], Verbenaceae [175], Violaceae [73], Zingiberaceae [141, 174]
- Benzyl angelate, see Benzyl (*Z*)-2-methyl-2-butenate
- Benzyl benzoate**
- Amaryllidaceae [6, 63], Asteraceae [5, 65], Berberidaceae [215], Brassicaceae [36], Cactaceae [125], Caprifoliaceae [102], Caryophyllaceae [54, 66, 112, 141], Fabaceae [120], Hemerocallidaceae [33], Hyacinthaceae [38, 123, 150], Lecythidaceae [144], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [114], Onagraceae [218], Orchidaceae [13, 77, 116, 117, 118, 141, 175, 180, 187, 195, 241, 243, 244, 253, 254, [258], Passifloraceae [157], Primulaceae [5], Rosaceae [33, 35, 37], Rubiaceae [5, 141], Ruscaceae [141], Sapotaceae [263], Solanaceae [131, 141, 167], Thymelaeaceae [5], Vitaceae [42, 44, 47], Zingiberaceae [141, 174, 268]
- Benzyl butanoate**
- Arecaceae [137], Berberidaceae [215], Nyctaginaceae [151], Nymphaeaceae [14, 135], Orchidaceae [77, 116, 117], Passifloraceae [157], Rubiaceae [249], Sapotaceae [263], Zingiberaceae [174]
- Benzyl 2-butenate (Benzyl crotonate)**
- Nymphaeaceae [14, 135]
- Benzyl crotonate, see Benzyl 2-butenate
- Benzyl formate**
- Berberidaceae [215], Nymphaeaceae [14, 135], Orchidaceae [116, 117], Sapindaceae [46]
- Benzyl heptanoate**
- Orchidaceae [117]
- Benzyl hexanoate**
- Nymphaeaceae [14, 135], Orchidaceae [117], Passifloraceae [157]
- Benzyl 2-hexenoate**

- Nymphaeaceae [14, 135]
 Benzyl (*E*)-2-hexenoate
 Orchidaceae [117]
 Benzyl (*Z*)-3-hexenoate
 Orchidaceae [117]
 Benzyl 2-hydroxybenzoate (Benzyl salicylate)
 Cactaceae [125], Caryophyllaceae [54], Fabaceae [120], Orchidaceae [77, 116, 117], Rubiaceae [121, 141], Solanaceae [131, 167]
 Benzyl 2-methylbutanoate
 Berberidaceae [215], Nymphaeaceae [14, 135], Orchidaceae [117], Thymelaeaceae [5], Zingiberaceae [141, 174]
 Benzyl 3-methylbutanoate
 Cactaceae [113, 125], Caryophyllaceae [5, 66], Fabaceae [216], Lecythidaceae [144], Myrsinaceae [103], Nyctaginaceae [151], Nymphaeaceae [14, 135], Orchidaceae [116, 117], Rubiaceae [5, 249], Solanaceae [165], Zingiberaceae [141, 268]
 Benzyl (*Z*)-2-methyl-2-butenolate (Benzyl angelate)
 Nymphaeaceae [14, 135]
 Benzyl 4-methylhexanoate
 Nymphaeaceae [14, 135]
 Benzyl 2-methylpropanoate
 Nymphaeaceae [14, 135], Orchidaceae [115, 117], Rubiaceae [249], Thymelaeaceae [5], Zingiberaceae [174]
 Benzyl 2-methylpropenoate
 Nymphaeaceae [14, 135]
 Benzyl octanoate
 Orchidaceae [117]
 Benzyl pentanoate
 Nymphaeaceae [14, 135], Orchidaceae [117], Solanaceae [165]
 Benzyl propanoate
 Nymphaeaceae [14, 135], Orchidaceae [117], Passifloraceae [157], Rubiaceae [249], Zingiberaceae [141]
 Benzyl salicylate, see Benzyl 2-hydroxybenzoate
 Benzyl tiglate
 Cactaceae [125], Hyacinthaceae [38, 123], Lauraceae [74], Lecythidaceae [144], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [150], Orchidaceae [116, 117], Rubiaceae [121, 249], Sapotaceae [263], Zingiberaceae [141, 174]
 Butyl benzoate
 Araceae [229], Caryophyllaceae [54], Magnoliaceae [9], Nyctaginaceae [151], Orchidaceae [117, 180]
 Cyclohexyl benzoate
 Nyctaginaceae [151], Oleaceae [267]
 Dibutyl 1,2-benzenedicarboxylate
 Onagraceae [265]
 Dibutyl-*o*-phthalate
 Oleaceae [57]
 Diethyl-*o*-phthalate
 Oleaceae [57]
 (*Z*)-3,4-Epoxy-hexyl benzoate
 Oleaceae [114], Orchidaceae [118]
 Ethyl benzoate (α -Oxo-ethyl benzoate)
 Agavaceae [142], Araceae [86], Arecaceae [68], Berberidaceae [215], Cactaceae [113, 115, 125], Caprifoliaceae [102], Caryophyllaceae [5, 54], Fumariaceae [197], Hyacinthaceae [38, 123], Lecythidaceae [144], Magnoliaceae [9, 11], Nyctaginaceae [151], Oleaceae [39, 57, 58], Orchidaceae [13, 77, 115, 116, 117, 118], Pinaceae [25], Rhizophoraceae [10], Rosaceae [33, 35, 37, 40], Rubiaceae [249], Theophrastaceae [139]
 Ethyl 2-hydroxybenzoate (Ethyl salicylate)
 Cactaceae [113, 125], Caryophyllaceae [54], Orchidaceae [117], Pinaceae [25]
 Ethyl 2-methoxybenzoate
 Hyacinthaceae [123], Orchidaceae [117]
 Ethyl 4-methoxybenzoate
 Fumariaceae [197]

Ethyl salicylate, see Ethyl 2-hydroxybenzoate

Hexenyl benzoate

Oleaceae [114]

(*E*)-3-Hexenyl benzoate

Caryophyllaceae [54]

(*Z*)-3-Hexenyl benzoate

Cactaceae [122, 125], Caprifoliaceae [102, 107], Caryophyllaceae [54, 141], Fabaceae [107, 119], Hydrangeaceae [107], Lecythidaceae [144], Nyctaginaceae [151], Oleaceae [33, 150], Orchidaceae [116, 117, 118], Passifloraceae [157], Rubiaceae [5], Solanaceae [167], Zingiberaceae [141]

Hexyl benzoate

Brassicaceae [36], Caryophyllaceae [54], Nyctaginaceae [151], Oleaceae [33], Orchidaceae [117], Rosaceae [33, 35, 37]

Hexyl 2-hydroxybenzoate

Caryophyllaceae [54]

Isoprenyl benzoate, see 3-Methyl-3-butenyl benzoate

Isopropyl benzoate

Caryophyllaceae [54]

Isopropyl 2-hydroxybenzoate

Nyctaginaceae [151]

2-Methoxybenzyl acetate

Amaryllidaceae [6], Hyacinthaceae [123]

4-Methoxybenzyl acetate

Araceae [77, 152, 258], Nymphaeaceae [178], Orchidaceae [77, 78, 116, 117, 118, 258], Theophrastaceae [139]

4-Methoxybenzyl butanoate

Orchidaceae [117]

4-Methoxybenzyl hexanoate

Orchidaceae [117]

Methyl anisate, see Methyl methoxybenzoate

Methyl benzoate

Amaryllidaceae [6, 63, 175], Annonaceae [111, 170], Apocynaceae [2, 141, 173, 178], Araceae [77, 133, 146, 229], Arecaceae [10, 137, 143], Asteraceae [5, 37], Berberidaceae [215], Brassicaceae [36], Cactaceae [113, 115, 122, 125], Caprifoliaceae [102], Caryophyllaceae [5, 54, 112, 141], Ericaceae [140], Eupomatiaceae [18], Fabaceae [107, 119, 126, 216], Fumariaceae [56, 197], Hemerocallidaceae [33], Hyacinthaceae [38, 123, 150], Lecythidaceae [121, 144], Lythraceae [10], Magnoliaceae [9, 11, 264], Myrsinaceae [103], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [39, 52, 267], Onagraceae [175], Orchidaceae [8, 13, 76, 77, 78, 96, 97, 98, 100, 115, 116, 117, 118, 120, 121, 141, 180, 181, 187, 194, 195, 241, 243, 244, 253, 255, 258, 259], Passifloraceae [157], Pinaceae [25], Polemoniaceae [5], Ranunculaceae [85], Rhizophoraceae [10], Rosaceae [33, 35, 37, 128, 196, 220], Rubiaceae [5, 121, 141, 249], Ruscaceae [141], Rutaceae [119], Sapotaceae [263], Scrophulariaceae [64], Solanaceae [121, 131, 141, 154, 165, 166, 167, 168], Theaceae [200], Valerianaceae [37], Verbenaceae [5, 101, 175], Zingiberaceae [141, 174, 268]

2-Methylbenzyl acetate

Myrsinaceae [103]

(*E*)-2-Methyl-2-butenyl benzoate

Orchidaceae [117]

3-Methyl-2-butenyl benzoate (Prenyl benzoate)

Amaryllidaceae [63], Orchidaceae [117]

3-Methyl-3-butenyl benzoate (Isoprenyl benzoate)

Amaryllidaceae [63], Orchidaceae [117]

2-Methylbutyl benzoate

Orchidaceae [117], Zingiberaceae [174]

3-Methylbutyl benzoate

Cactaceae [125], Caryophyllaceae [54], Magnoliaceae [11], Nymphaeaceae [14, 135], Orchidaceae [117], Primulaceae [5], Rubiaceae [5], Zingiberaceae [141, 174, 268]

3-Methylbutyl 2-hydroxybenzoate

Magnoliaceae [11], Orchidaceae [117]

Methyl 3,5-dimethylbenzoate

Oleaceae [51]

Methyl 2-hydroxybenzoate (Methyl salicylate)

Agavaceae [178], Amaryllidaceae [175], Apiaceae [31, 242, 246], Apocynaceae [1, 141, 232, 233], Araceae [77, 129], Arecaceae [137, 143], Asteraceae [5, 55], Berberidaceae [182, 215], Brassicaceae [22], Bromeliaceae [77], Cactaceae [113, 115, 122, 125], Calycanthaceae [120, 266], Caprifoliaceae [102], Caryophyllaceae [5, 54, 66, 112, 141], Combretaceae [10], Dipsacaceae [5, 185], Eupomatiaceae [18], Fabaceae [49, 120, 156, 216], Fumariaceae [56, 197], Gesneriaceae [77], Hemerocallidaceae [33], Hyacinthaceae [38, 123], Lauraceae [74], Lecythidaceae [121, 144], Magnoliaceae [9, 11], Malvaceae [142], Moraceae [83], Myrsinaceae [103], Nyctaginaceae [151], Oleaceae [39, 57, 267], Onagraceae [218], Orchidaceae [13, 76, 77, 78, 97, 98, 116, 117, 120, 141, 181, 195, 236, 241, 243, 244, 253, 255, 256, 258, 259], Passifloraceae [157], Pinaceae [25], Polemoniaceae [5], Primulaceae [36], Ranunculaceae [19, 85], Rhizophoraceae [10], Rosaceae [12, 33, 35, 87, 88, 128, 220], Rubiaceae [5, 121, 141, 249], Ruscaceae [141], Rutaceae [119, 247], Salicaceae [245], Sapotaceae [263], Solanaceae [131, 141, 165, 167, 168], Theaceae [200], Theophrastaceae [139], Verbenaceae [5, 175], Violaceae [73], Zamiaceae [211], Zingiberaceae [141, 174]

Methyl 3-hydroxybenzoate

Orchidaceae [192]

Methyl 4-hydroxybenzoate

Orchidaceae [117, 227]

Methyl 4-hydroxy-3-methoxybenzoate

Orchidaceae [117, 227]

Methyl methoxybenzoate (Methyl anisate)

Myrsinaceae [103], Sapotaceae [263]

Methyl 2-methoxybenzoate

Asteraceae [5, 55], Caryophyllaceae [141], Hyacinthaceae [38, 123], Lecythidaceae [144], Oleaceae [108], Orchidaceae [117], Rubiaceae [5], Zingiberaceae [141]

Methyl 4-methoxybenzoate

Ericaceae [140], Fumariaceae [197], Orchidaceae [117], Rosaceae [220], Theophrastaceae [139]

Methyl 2-methylbenzoate

Orchidaceae [117]

2-Methylpropyl benzoate

Caryophyllaceae [54], Magnoliaceae [11], Nyctaginaceae [151], Orchidaceae [117], Zingiberaceae [268]

2-Methylpropyl 2-hydroxybenzoate

Magnoliaceae [11]

Methyl salicylate, see Methyl 2-hydroxybenzoate

 α -Oxo-ethyl benzoate, see Ethyl benzoate

Pentyl benzoate

Asteraceae [214], Caryophyllaceae [54], Nyctaginaceae [151], Rosaceae [220]

Pentyl 2-hydroxybenzoate

Caryophyllaceae [54]

Phenylethyl 2-hydroxybenzoate

Orchidaceae [117]

2-Phenylethyl 2-hydroxybenzoate

Orchidaceae [117]

Prenyl benzoate, see 3-Methyl-2-butenyl benzoate

Propyl benzoate

Magnoliaceae [11], Nyctaginaceae [151], Orchidaceae [117]

Ethers

Benzyl methyl ether

Arecaceae [68, 143], Cactaceae [125], Hyacinthaceae [123], Nyctaginaceae [151], Oleaceae [107, 178, 232, 233], Orchidaceae [117], Rosaceae [72]

1,2-Dimethoxy (methyl) benzene

Oleaceae [107]

1,2-Dimethoxy-4-methylbenzene (3,4-Dimethoxytoluene)

Malvaceae [142], Orchidaceae [8], Valerianaceae [37]

1,3-Dimethoxy-5-methylbenzene (3,5-Dimethoxy-1-methylbenzene, 3,5-Dimethoxytoluene)

Amaryllidaceae [63, 177, 232, 233], Berberidaceae [215], Lecythidaceae [144], Orchidaceae

- [117, 201], Rosaceae [33, 35, 72, 93, 176, 177, 178, 183], Solanaceae [121], Theophrastaceae [139]
- 1,4-Dimethoxy-5-methylbenzene (2,5-Dimethoxytoluene, 2,5-Dimethoxy-1-methylbenzene)
Ericaceae [136]
- 2,6-Dimethoxy-1-methylbenzene (2,6-Dimethoxytoluene)
Rosaceae [128]
- 3,4-Dimethoxytoluene, see 1,2-Dimethoxy-4-methylbenzene
- 3,5-Dimethoxy-1-methylbenzene, see 1,3-Dimethoxy-5-methylbenzene
- 2,5-Dimethoxytoluene, see 1,4-Dimethoxy-5-methylbenzene
- 2,6-Dimethoxytoluene, see 2,6-Dimethoxy-1-methylbenzene
- 3,5-Dimethoxytoluene, see 1,3-Dimethoxy-5-methylbenzene
- 4-Hydroxy-3-methoxybenzyl ethyl ether
Orchidaceae [117]
- 2-Methoxybenzyl methyl ether
Oleaceae [150]
- 4-Methoxybenzyl methyl ether
Orchidaceae [117]
- Methoxymethylbenzene (Methylanisol)
Apiaceae [31], Oleaceae [171], Rosaceae [40], Rubiaceae [5]
- 1-Methoxy-4-methylbenzene (4-Methylanisole, 4-Methylphenyl methyl ether)
Amaryllidaceae [63, 177], Annonaceae [170], Apocynaceae [141], Araceae [133], Arecaceae [68, 143], Caryophyllaceae [54], Hyacinthaceae [38, 123], Lecythidaceae [144], Magnoliaceae [9], Malvaceae [142], Oleaceae [33, 150], Orchidaceae [13, 17, 27, 77, 106, 117, 120, 258], Rosaceae [72], Rubiaceae [5], Sapotaceae [263], Valerianaceae [37], Violaceae [248]
- Methylanisol, see Methoxymethylbenzene
- 4-Methylanisole, see 1-Methoxy-4-methylbenzene
- 4-Methylphenyl methyl ether, see 1-Methoxy-4-methylbenzene
- 1,2,3-Trimethoxy-5-methylbenzene
Amaryllidaceae [63]
- N-compounds
- Aminobenzaldehyde
Orchidaceae [13]
- 2-Aminobenzaldehyde
Agavaceae [108], Apiaceae [31], Cactaceae [125], Cucurbitaceae [71], Fabaceae [107, 108, 126], Fumariaceae [56, 197], Hydrangeaceae [107, 108], Liliaceae [108], Magnoliaceae [9], Orchidaceae [117, 120, 141], Pittosporaceae [107, 108], Ranunculaceae [85], Rutaceae [108, 119, 247]
- Benzyl cyanide
Magnoliaceae [9]
- Carbamoylbenzoate
Oleaceae [267]
- Diethyltoluamide
Arecaceae [137]
- 2-*N*-Dimethylaminobenzaldehyde
Orchidaceae [120]
- N*-(1,1-Dimethylethyl)-4-methylbenzamide
Malvaceae [67]
- Ethyl 2-aminobenzoate (Ethyl anthranilate)
Orchidaceae [13]
- 2-Formamidobenzaldehyde
Hydrangeaceae [108]
- Methoxy methyl benzeneamine isomer
Fumariaceae [56]
- 2-*N*-Methylaminobenzaldehyde
Fumariaceae [197], Orchidaceae [13, 76, 77, 78, 117, 120]
- Methyl 2-aminobenzoate (Methyl anthranilate)
Agavaceae [108, 178], Apiaceae [31], Arecaceae [137], Asteraceae [5], Berberidaceae [215], Cactaceae [125], Cucurbitaceae [71], Fabaceae [107, 108, 119, 126], Fumariaceae [197], Lecythidaceae [144], Nyctaginaceae [151], Oleaceae [39, 114, 267], Orchidaceae [115, 116, 117, 120, 141], Passifloraceae [157], Ranunculaceae [85], Rhizophoraceae [10], Rubiaceae [141], Rutaceae [2, 108, 119, 247], Solanaceae [121]

Methyl 2-amino-3-methoxybenzoate

Arecaceae [137]

N-Methylaniline

Annonaceae [111]

Methyl anthranilate, see Methyl 2-aminobenzoate

Methyl *N*-formylanthranilate, see Methyl *N*-methylaminobenzoate

Methyl *N*-methylaminobenzoate (Methyl *N*-formylanthranilate, Methyl *N*-methylantranilate)

Orchidaceae [117], Rutaceae [247]

Methyl 2-*N*-methylaminobenzoate

Hyacinthaceae [38], Orchidaceae [120]

Methyl *N*-methylantranilate, see Methyl *N*-methylaminobenzoate

1-Methyl-4-nitrobenzene

Orchidaceae [180]

2-Nitro-4-methylphenol

Araceae [133]

N-Phenyl formamide

Dipsacaceae [5]

C6-C2

Hydrocarbons

1,2-Diethylbenzene

Oleaceae [51]

1,3-Diethylbenzene

Oleaceae [51]

1,4-Diethylbenzene

Annonaceae [111]

Dimethyl styrene

Orchidaceae [254]

α -*p*-Dimethylstyrene

Orchidaceae [243]

Ethenylbenzene

Arecaceae [137], Caryophyllaceae [112], Ericaceae [136]

1-Ethenyl-4-ethylbenzene

Rosaceae [40]

Ethylbenzene

Actinidiaceae [237], Annonaceae [111], Meliaceae [250], Orchidaceae [23, 27, 75], Rosaceae [12]

Ethylmethylbenzene

Meliaceae [250], Orchidaceae [147]

1-Ethyl-2-methylbenzene

Annonaceae [111]

1-Ethyl-3-methylbenzene

Rosaceae [12]

1-Ethyl-4-methylbenzene

Rosaceae [12]

Isopropylbenzene

Orchidaceae [201]

(1-Methylethenyl)benzene

Moraceae [82]

α -Methylstyrene

Rosaceae [12]

Styrene

Caryophyllaceae [54], Moraceae [82], Orchidaceae [117], Rosaceae [12], Scrophulariaceae [5], Violaceae [248]

Styrene dimer

Orchidaceae [180]

Triethylbenzene

Asteraceae [213]

Acids

Phenylacetic acid

Asteraceae [65]

Aldehydes

Ethylbenzaldehyde

- Araceae [26], Oleaceae [57, 58]
 2-Hydroxy-2-phenylacetaldehyde (Phenylglyoxal)
 Orchidaceae [77, 116, 117]
 Phenylacetaldehyde
 Amaryllidaceae [6, 63], Apiaceae [246], Arecaceae [137], Asteraceae [5, 55], Berberidaceae [215], Brassicaceae [22, 36, 70, 186, 202, 221], Cactaceae [125], Caprifoliaceae [102], Caryophyllaceae [5, 112], Cucurbitaceae [175], Dipsacaceae [5], Ericaceae [140], Fabaceae [119, 212, 216], Fumariaceae [56], Hyacinthaceae [38], Lamiaceae [5], Liliaceae [107], Linnaeaceae [90], Lythraceae [10], Magnoliaceae [11], Nyctaginaceae [151], Oleaceae [33, 51, 108, 150], Orchidaceae [5, 13, 23, 116, 117, 141, 147, 236], Orobanchaceae [20], Passifloraceae [157], Polemoniaceae [5, 232, 233, 234], Rosaceae [62], Scrophulariaceae [5], Solanaceae [91, 154]
 Phenylglyoxal, see 2-Hydroxy-2-phenylacetaldehyde
- Ketones**
 Acetophenone
 Araceae [26, 86, 134], Asteraceae [5], Calycanthaceae [120], Caprifoliaceae [102], Fabaceae [50, 156], Hydrangeaceae [107, 108], Magnoliaceae [9, 11, 240], Oleaceae [33], Orchidaceae [8, 29, 117, 192], Primulaceae [36], Solanaceae [69], Theaceae [200], Violaceae [248], Zingiberaceae [174]
 2,4-Dimethylacetophenone (2,4-Dimethylphenylethanone)
 Oleaceae [57, 58]
 2,4-Dimethylphenylethanone, see 2,4-Dimethylacetophenone
 4-Ethylacetophenone (1-(4-Ethylphenyl)ethanone)
 Fabaceae [104], Rosaceae [220]
 1-(Ethylphenyl)ethanone
 Orchidaceae [27]
 1-(4-Ethylphenyl)ethanone, see 4-Ethylacetophenone
 α -Hydroxyacetophenone
 Orchidaceae [77, 116, 117]
 4-Hydroxyacetophenone
 Araceae [26]
 4-Methoxyacetophenone
 Ericaceae [136], Orchidaceae [117]
 4-Methylacetophenone
 Oleaceae [150], Orchidaceae [117]
 1,1(1,4-Phenylene)-bisethanone
 Oleaceae [58]
- Alcohols**
p-Cymol
 Ruscaceae [171]
ar-Ethylbenzyl alcohol
 Orchidaceae [180]
 2-Methoxy- β -phenylpropanol
 Amaryllidaceae [6]
 2-Methoxy-2-phenylethanol (*o*-Methoxy- β -phenylethyl alcohol)
 Amaryllidaceae [6]
 3-Methoxy-2-phenylethanol (3-Methoxyphenylethyl alcohol)
 Oleaceae [177]
o-Methoxy- β -phenylethyl alcohol, see 2-Methoxy-2-phenylethanol
 3-Methoxyphenylethyl alcohol, see 3-Methoxy-2-phenylethanol
 4-Methoxyphenylethyl alcohol, see 4-Methoxy-2-phenylethanol
 4-Methoxyphenylethanol, see 4-Methoxy-2-phenylethanol
 4-Methoxy-2-phenylethanol (4-Methoxyphenylethanol, 4-Methoxyphenylethyl alcohol, 2-(4-Methoxyphenyl) ethanol)
 Annonaceae [111], Araceae [134], Hyacinthaceae [38, 123], Orchidaceae [13, 117, 120]
 Phenylethanol
 Rosaceae [12], Rutaceae [247]
 1-Phenylethanol
 Fabaceae [50, 156]
 2-Phenylethanol
 Actinidiaceae [237], Amaryllidaceae [6, 63], Annonaceae [111], Apiaceae [31, 246], Apocy-

naceae [232, 233], Araceae [26, 86, 134, 146], Arecaceae [10, 137, 143], Asteraceae [5, 55, 65], Acanthaceae [10], Berberidaceae [182, 215], Brassicaceae [22, 36, 70, 186, 202, 217], Cactaceae [122, 125], Caprifoliaceae [102], Caryophyllaceae [5, 112, 141], Cucurbitaceae [4], Dipsacaceae [5], Fabaceae [49, 50, 104, 107, 119, 126, 138, 156, 212], Fumariaceae [197], Gentianaceae [5], Hyacinthaceae [38, 123], Lamiaceae [5], Lauraceae [74], Lecythidaceae [121, 144], Liliaceae [107], Linnaeaceae [90], Lythraceae [10], Magnoliaceae [11], Malvaceae [41, 142], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [33, 51, 52], Orchidaceae [5, 8, 13, 17, 29, 77, 78, 98, 106, 115, 116, 117, 120, 141, 169, 180, 204, 253, 258], Passifloraceae [157], Polemoniaceae [5, 232, 233, 234], Primulaceae [36], Ranunculaceae [19, 59, 85], Rosaceae [33, 35, 43, 51, 60, 61, 62, 72, 87, 88, 93, 115, 128, 155, 166, 176, 177, 178, 183, 196, 228, 232], Rubiaceae [5, 141], Rutaceae [119], Salicaceae [245], Sapotaceae [263], Scrophulariaceae [5], Solanaceae [131, 154, 165, 167], Theaceae [200], Theophrastaceae [139], Thymelaeaceae [5, 234], Violaceae [73], Vitaceae [42, 44, 47], Zingiberaceae [141, 174]

Esters

2-Methoxyphenylethyl acetate

Amaryllidaceae [6]

2-Methoxyphenylisopropyl acetate (2-Methoxy- β -phenylpropyl acetate)

Amaryllidaceae [6]

2-Methoxy- β -phenylpropyl acetate, see 2-Methoxyphenylisopropyl acetate

2-Methylpropyl phenylacetate

Berberidaceae [215]

3-Methylbutyl 2-phenylacetate

Passifloraceae [157]

4-Methoxyphenylethyl acetate

Orchidaceae [78, 117]

Methoxyphenylethyl acetate

Orchidaceae [77]

Methyl 2-hydroxy-2-phenylacetate (Methyl 2-hydroxy-2-phenylethanoate)

Eupomatiaceae [18]

Methyl 2-hydroxy-2-phenylethanoate, see Methyl 2-hydroxy-2-phenylacetate

Methyl 4-methoxyphenylacetate (Methyl 4-methoxyphenylethanoate)

Orchidaceae [117]

Methyl 4-methoxyphenylethanoate, see Methyl 4-methoxyphenylacetate

Methyl phenylacetate (Methyl phenylethanoate)

Brassicaceae [36], Magnoliaceae [11], Orchidaceae [116, 117, 120], Primulaceae [36]

Methyl 2-phenylacetate

Passifloraceae [157]

Methyl phenylethanoate, see Methyl phenylacetate

Phenylethyl acetate

Actinidiaceae [237], Amaryllidaceae [6], Brassicaceae [186], Bromeliaceae [77], Fabaceae [104, 107], Hyacinthaceae [123, 150], Nyctaginaceae [151], Oleaceae [52], Orchidaceae [13, 77, 78, 117, 120, 253], Rosaceae [33, 40, 72, 177, 178, 196]

1-Phenylethyl acetate

Araceae [134]

2-Phenylethyl acetate

Amaryllidaceae [63], Annonaceae [111], Arecaceae [68, 137, 143], Brassicaceae [36], Caryophyllaceae [54], Dipsacaceae [5], Fabaceae [126, 212], Gentianaceae [45], Hyacinthaceae [38], Lamiaceae [5], Lecythidaceae [144], Magnoliaceae [11], Malvaceae [41], Orchidaceae [8, 17, 98, 106, 116, 117, 141, 181, 188, 236, 258], Passifloraceae [157], Rosaceae [35, 43, 60, 61, 62, 93, 128, 232], Ruscaceae [141], Sapotaceae [263]

Phenylethyl benzoate

Hyacinthaceae [123, 150], Nyctaginaceae [151], Orchidaceae [116, 117]

2-Phenylethyl benzoate

Hyacinthaceae [38], Malvaceae [41], Orchidaceae [116, 117], Rubiaceae [141], Sapotaceae [263], Zingiberaceae [141]

Phenylethyl butanoate

Fabaceae [107, 119]

2-Phenylethyl butanoate

Lecythidaceae [144], Orchidaceae [117], Sapotaceae [263]

Phenylethyl formate

- Orchidaceae [77, 117]
- 2-Phenylethyl formate
 - Arecaceae [137], Hyacinthaceae [38], Orchidaceae [117], Passifloraceae [157]
- 2-Phenylethyl hexanoate
 - Lecythidaceae [144]
- Phenylethyl 2-methylbutanoate
 - Nymphaeaceae [14, 135]
- Phenylethyl 2-methylpropanoate
 - Rosaceae [33]
- 2-Phenylethyl 2-methylpropanoate
 - Caryophyllaceae [5], Passifloraceae [157], Rosaceae [35], Sapotaceae [263]
- Phenylethyl 3-methylbutanoate
 - Orchidaceae [117]
- 2-Phenylethyl 3-methylbutanoate
 - Caryophyllaceae [5], Polemoniaceae [5], Sapotaceae [263]
- 2-Phenylethyl pentanoate
 - Lecythidaceae [144]
- Phenylethyl tiglate
 - Orchidaceae [117]
- 2-Phenylethyl tiglate
 - Sapotaceae [263]
- Ethers
 - 4-Ethenylanisole, see 4-Methoxy-1-ethenylbenzene
 - 4-Ethenyl-1-methoxybenzene, see 4-Methoxy-1-ethenylbenzene
 - 1-Ethoxyethylbenzene
 - Oleaceae [57, 58]
 - 4-Methoxy-1-ethenylbenzene (4-Ethenylanisole, 4-Ethenyl-1-methoxybenzene)
 - Araceae [152], Arecaceae [143], Orchidaceae [117, 118, 120], Passifloraceae [157], Rosaceae [33, 35, 72]
 - 2-Phenylethyl methyl ether
 - Rosaceae [62], Salicaceae [245]
- Benzofurans
 - 7,7-Dimethyl-4,5,6,7-tetrahydro-1(3H)-isobenzofuranone
 - Orchidaceae [120]
- N-compounds
 - Benzonitrile
 - Fabaceae [179], Hemerocallidaceae [33], Rosaceae [37]
 - 1-Nitro-2-phenylethane (2-Phenylnitroethane)
 - Apocynaceae [2, 173, 178], Brassicaceae [36], Cactaceae [122, 125], Caprifoliaceae [107], Fabaceae [107, 212], Hemerocallidaceae [33], Lecythidaceae [121], Orchidaceae [117], Primulaceae [36], Rutaceae [119]
 - Phenylacetaldoxime
 - Brassicaceae [36], Cactaceae [125], Caprifoliaceae [102, 107], Fabaceae [107], Magnoliaceae [115], Orchidaceae [115, 116, 117], Rubiaceae [115], Ruscaceae [115, 232, 233], Rutaceae [115, 119, 247], Zingiberaceae [115, 141]
 - Phenylacetaldoxime-*O*-methyl ether
 - Cactaceae [125], Rubiaceae [109], Rutaceae [119]
 - Phenylacetonitrile, see 2-Phenylacetonitrile
 - 2-Phenylacetonitrile (Phenylacetonitrile)
 - Amaryllidaceae [6, 177], Apiaceae [31], Brassicaceae [36, 202], Cactaceae [113, 125], Caprifoliaceae [102, 107], Fabaceae [107, 212], Hemerocallidaceae [33], Lecythidaceae [121, 144], Magnoliaceae [264], Malvaceae [142], Nyctaginaceae [151], Orchidaceae [77, 115, 116, 117, 169], Orobanchaceae [20], Primulaceae [36], Rubiaceae [141], Ruscaceae [171, 232, 233], Rutaceae [119, 247], Zingiberaceae [141, 174]
 - Phenylethylamine hydrochloride
 - Orchidaceae [180]
 - 2-Phenylnitroethane, see 1-Nitro-2-phenylethane
- C6-C3
 - Hydrocarbons
 - Anethole, see 1-Methoxy-4-(prop-1-enyl)benzene

1-Isopropenyl-4-methylbenzene (4-Isopropenyl-1-methylbenzene)

Araceae [229], Asteraceae [34], Dipsacaceae [185], Malvaceae [41], Orchidaceae [117, 258], Primulaceae [36], Rosaceae [33]

4-Isopropenyl-1-methylbenzene, see 1-Isopropenyl-4-methylbenzene

1-Methyl-4-(1-methylethyl)benzene

Cucurbitaceae [71]

4-Propenylanisole, see 1-Methoxy-4-(prop-1-enyl)benzene

Aldehydes

Cinnamic aldehyde (Zimtaldehyde)

Apiaceae [246], Apocynaceae [141], Caryophyllaceae [141], Fabaceae [81, 120], Magnoliaceae [264], Myrsinaceae [103], Nymphaeaceae [14, 135], Oleaceae [57, 58], Orchidaceae [77, 96, 117, 175, 195, 241, 243, 244, 258], Rosaceae [43]

(E)-Cinnamic aldehyde

Berberidaceae [215], Cactaceae [122], Caryophyllaceae [112], Ericaceae [136, 140], Oleaceae [33], Orchidaceae [13, 117, 253], Polemoniaceae [5], Rubiaceae [5], Scrophulariaceae [5], Solanaceae [167], Theophrastaceae [139], Thymelaeaceae [5]

(Z)-Cinnamic aldehyde

Berberidaceae [215], Ericaceae [136, 140], Orchidaceae [13, 117, 253], Rubiaceae [5], Scrophulariaceae [5], Theophrastaceae [139], Thymelaeaceae [5]

Cumin aldehyde, see 4-Isopropylbenzaldehyde

4-Isopropenylbenzaldehyde

Rosaceae [33, 35]

4-Isopropylbenzaldehyde (Cumin aldehyde)

Orchidaceae [117]

4-Methoxycinnamic aldehyde

Theophrastaceae [139]

Phenylpropanal

Berberidaceae [215], Ericaceae [136]

3-Phenylpropanal

Orchidaceae [117]

Zimtaldehyde, see Cinnamic aldehyde

Ketones

1-Phenyl-1-hydroxypropan-2-one

Oleaceae [108]

1-Phenyl-1,2-propandione

Orchidaceae [117]

Phenyl-2-propanone

Oleaceae [267]

Propiophenone

Berberidaceae [215]

Propiovanillone

Asteraceae [214]

p-Tolyl-2-propanone

Araceae [86]

Vanillyl methylketone

Nyctaginaceae [151]

Alcohols

Allyl benzoate, see 3-Prop-1-enyl benzoate

Chavicol, see 3-(4-Hydroxyphenyl)prop-1-ene

Cinnamic alcohol (Styralyl alcohol)

Amaryllidaceae [6], Apocynaceae [141], Araceae [86], Caryophyllaceae [141], Ericaceae [140], Fabaceae [81], Hemerocallidaceae [33], Magnoliaceae [240], Orchidaceae [53, 77, 115, 116, 141, 195, 241, 243, 244, 253, 258], Polemoniaceae [5], Primulaceae [5], Rosaceae [43], Rubiaceae [5], Ruscaceae [141, 232, 233], Scrophulariaceae [5], Thymelaeaceae [5, 234]

(E)-Cinnamic alcohol

Berberidaceae [215], Cactaceae [125], Calycanthaceae [120], Cucurbitaceae [4], Ericaceae [136], Fabaceae [120], Hyacinthaceae [38, 123], Magnoliaceae [264], Myrsinaceae [103], Oleaceae [33], Orchidaceae [77, 116, 117, 118], Primulaceae [36], Sapotaceae [263], Solanaceae [167], Theophrastaceae [139]

- (*Z*)-Cinnamic alcohol
 Ericaceae [136], Hyacinthaceae [38, 123], Magnoliaceae [264], Myrsinaceae [103], Orchidaceae [117]
- Cuminol, see 4-Isopropylbenzyl alcohol
- Cuminy alcohol, see 4-Isopropylbenzyl alcohol
- 2,6-Dimethoxy-4-(2-propenyl)-1-hydroxybenzene (2,6-Dimethoxy-4-(2-propenyl)-phenol)
 Rhizophoraceae [10]
- 2,6-Dimethoxy-4-(2-propenyl)-phenol, see 2,6-Dimethoxy-4-(2-propenyl)-1-hydroxybenzene
- 2,6-Dimethyl-4-(2-propenyl)-1-hydroxybenzene (2,6-Dimethyl-4-(2-propenyl)-phenol)
 Oleaceae [51]
- 2,6-Dimethyl-4-(2-propenyl)-phenol, see 2,6-Dimethyl-4-(2-propenyl)-1-hydroxybenzene
- 1-Hydroxy-2,6-dimethoxy-4-(1-propenyl)benzene
 Sapotaceae [263]
- 2-Hydroxy-1-isopropyl-4-methylbenzene
 Orchidaceae [117]
- 3-(4-Hydroxyphenyl)prop-1-ene (Chavicol)
 Fabaceae [107], Orchidaceae [77, 78, 117, 120]
- Isochavicol
 Orchidaceae [117]
- 4-Isopropylbenzyl alcohol (Cuminol, Cuminy alcohol)
 Araceae [229], Oleaceae [33]
- 4-Methoxycinnamic alcohol
 Orchidaceae [116, 117]
- 4-Methoxyphenylpropanol
 Ericaceae [136], Theophrastaceae [139]
- 1-Methoxy-*p*-tolyl-2-propanol
 Araceae [86]
- 2-Methoxy-*p*-tolyl-1-propanol
 Araceae [86]
- Phenylpropanol
 Polemoniaceae [5]
- 2-Phenyl-2-propanol
 Orchidaceae [23]
- 3-Phenylpropanol
 Amaryllidaceae [6], Berberidaceae [215], Caryophyllaceae [141], Ericaceae [136, 140], Fabaceae [120], Hemerocallidaceae [33], Hyacinthaceae [38, 123], Lecythidaceae [144], Myrsinaceae [103], Orchidaceae [117, 244, 253], Solanaceae [167], Theophrastaceae [139]
- (*E*)-4-(1-Propenyl)-1-hydroxybenzene ((*E*)-4-(1-Propen-1-yl)phenol)
 Apocynaceae [108], Fabaceae [107], Oleaceae [108]
- (*Z*)-4-(1-Propenyl)-1-hydroxybenzene ((*Z*)-4-(1-Propen-1-yl)phenol)
 Apocynaceae [108], Fabaceae [107], Oleaceae [108]
- (*E*)-4-(1-Propen-1-yl)phenol, see (*E*)-4-(1-Propenyl)-1-hydroxybenzene
- (*Z*)-4-(1-Propen-1-yl)phenol, see (*Z*)-4-(1-Propenyl)-1-hydroxybenzene
- Styralyl alcohol, see Cinnamic alcohol
- p*-Tolyl-1,2-propanediol
 Araceae [86]
- Esters
- Cinnamic acetate
 Amaryllidaceae [6], Caryophyllaceae [141], Ericaceae [140], Fabaceae [81], Hemerocallidaceae [33], Hyacinthaceae [38, 123], Orchidaceae [77, 78, 116, 195, 236, 241, 243, 244, 253, 258], Ruscaceae [141]
- (*E*)-Cinnamic acetate
 Calycanthaceae [120], Ericaceae [136], Fabaceae [120], Orchidaceae [77, 116, 117, 120], Sapotaceae [263], Solanaceae [167], Theophrastaceae [139]
- (*Z*)-Cinnamic acetate
 Ericaceae [136], Orchidaceae [117], Rubiaceae [5]
- Cinnamic butanoate
 Orchidaceae [117]
- Coumarin
 Malvaceae [41], Orchidaceae [117]

- Ethyl 3-phenyl-2-propenoate, see Ethyl cinnamate
- Ethyl 3-phenylpropanoate
Araceae [86], Fumariaceae [197]
- Ethyl cinnamate
Arecaceae [137], Fumariaceae [197], Rosaceae [51]
- Ethyl (*E*)-cinnamate
Orchidaceae [117], Theophrastaceae [139]
- Ethyl (*Z*)-cinnamate
Theophrastaceae [139]
- Eugenyl acetate
Caryophyllaceae [54], Nyctaginaceae [151]
- 4-Methoxycinnamic acetate
Orchidaceae [117, 118, 120]
- Methyl cinnamate
Amaryllidaceae [6, 63], Berberidaceae [215], Fabaceae [50, 156], Fumariaceae [197], Orchidaceae [28, 97, 98, 254, 256, 257, 258]
- Methyl (*E*)-cinnamate
Fabaceae [120], Oleaceae [52], Orchidaceae [77, 78, 116, 117, 120, 253], Passifloraceae [157], Sapotaceae [263], Theophrastaceae [139]
- Methyl (*Z*)-cinnamate
Orchidaceae [78, 117, 253], Passifloraceae [157], Theophrastaceae [139]
- Methyl 4-methoxycinnamate
Orchidaceae [258, 260]
- Methyl (*E*)-4-methoxycinnamate
Orchidaceae [78, 117, 120]
- Methyl (*Z*)-4-methoxycinnamate
Orchidaceae [78, 117, 120]
- Methyl 4-methoxyphenylpropanoate
Orchidaceae [117, 120]
- Methyl 3-phenyl-2-hydroxypropanoate
Fumariaceae [197], Orchidaceae [117], Passifloraceae [157]
- Methyl 3-phenylpropanoate
Fumariaceae [197], Orchidaceae [117]
- Methyl 3-phenyl-2-propenoate, see Methyl cinnamate
- Phenylpropyl acetate
Ericaceae [140], Hyacinthaceae [123], Orchidaceae [258], Rubiaceae [5]
- γ -Phenylpropyl acetate, see 3-Phenylpropyl acetate
- 3-Phenylpropyl acetate (γ -Phenylpropyl acetate)
Amaryllidaceae [6, 63], Berberidaceae [215], Caryophyllaceae [141], Ericaceae [136], Fabaceae [120], Orchidaceae [117], Ruscaceae [141], Theophrastaceae [139]
- Phenylpropyl butanoate
Orchidaceae [117]
- 3-Prop-1-enyl benzoate (Allyl benzoate)
Nyctaginaceae [151]
- Ethers
- 4-Allyl-1,2-dimethoxybenzene, see Methyleugenol
- Anethole, see 1-Methoxy-4-(prop-1-enyl)benzene
- (*E*)-Anethole, see (*E*)-1-Methoxy-4-(prop-1-enyl)benzene
- (*Z*)-Anethole, see (*Z*)-1-Methoxy-4-(prop-1-enyl)benzene
- Dillapiole
Apiaceae [242]
- 1,2-Dimethoxy-4-(2-propenyl)-benzene, see Methyleugenol
- Elemicin
Arecaceae [68, 143], Hyacinthaceae [38, 123], Lecythidaceae [144], Oleaceae [33, 232, 233], Orchidaceae [53, 116, 117], Passifloraceae [157]
- (*E*)-Elemicin
Orchidaceae [77]
- Estragole, see 1-Methoxy-4-(prop-2-enyl)benzene
- Eugenol (2-Methoxy-4-(2-propenyl)-1-hydroxybenzene)
Apocynaceae [178], Araceae [77, 133, 258], Arecaceae [137, 145], Asteraceae [214], Acanthaceae [10], Caprifoliaceae [102], Caryophyllaceae [54], Fabaceae [81, 119], Hy-

- acinthaceae [38, 123], Lauraceae [74], Lecythidaceae [121, 144], Magnoliaceae [264], Malvaceae [142], Nyctaginaceae [151], Oleaceae [52, 114], Onagraceae [218], Orchidaceae [8, 13, 77, 78, 97, 116, 117, 120, 141, 195, 241, 243, 244, 253, 258], Passifloraceae [157], Polemoniaceae [5], Rhizophoraceae [10], Rosaceae [43, 51, 60, 61, 128], Rubiaceae [121, 141], Sapotaceae [263], Solanaceae [165], Theophrastaceae [139], Zingiberaceae [141, 174]
- (*E*)-Isoelemicin
Orchidaceae [117], Passifloraceae [157]
- (*Z*)-Isoelemicin
Passifloraceae [157]
- (*E*)-Isoeugenol
Caryophyllaceae [54], Lecythidaceae [121], Oleaceae [52], Orchidaceae [78, 117], Zingiberaceae [174]
- Methoxyeugenol, see Methyleugenol
- 6-Methoxyeugenol
Orchidaceae [117]
- 1-Methoxy-4-(prop-1-enyl)benzene (4-Propenylanisole, Anethole)
Araceae [86], Fabaceae [81], Magnoliaceae [11, 264], Orchidaceae [77, 117]
- (*E*)-1-Methoxy-4-(prop-1-enyl)benzene ((*E*)-Anethole)
Araceae [152], Fabaceae [107], Ranunculaceae [85]
- (*Z*)-1-Methoxy-4-(prop-1-enyl)benzene ((*Z*)-Anethole)
Araceae [152], Fabaceae [107], Ranunculaceae [85]
- 1-Methoxy-4-(prop-2-enyl)benzene (Estragole, Methylchavicol)
Amaryllidaceae [63], Araceae [77], Cactaceae [125], Fabaceae [81], Hyacinthaceae [123], Lamiaceae [261], Magnoliaceae [9, 11], Oleaceae [33, 150], Orchidaceae [77, 117, 120], Rosaceae [72], Sapotaceae [263]
- Methylchavicol, see 1-Methoxy-4-(prop-2-enyl)benzene
- (*E*)-Methylisoeugenol
Arecaceae [137], Fabaceae [107], Lecythidaceae [144], Passifloraceae [157], Zingiberaceae [141, 174]
- Isoelemicin
Orchidaceae [258]
- Isoeugenol
Fabaceae [107, 108], Nyctaginaceae [151], Onagraceae [218], Orchidaceae [141, 195, 243, 244], Passifloraceae [157], Ranunculaceae [85], Zingiberaceae [141]
- (*Z*)-Isoeugenol
Caryophyllaceae [54], Oleaceae [52], Orchidaceae [117], Zingiberaceae [174]
- Methoxyeugenol, see Methyleugenol
- Methyleugenol (1,2-Dimethoxy-4-(2-propenyl)-benzene, 4-Allyl-1,2-dimethoxybenzene, Methoxyeugenol)
Amaryllidaceae [6, 63], Araceae [152], Arecaceae [137, 143, 145], Fabaceae [81, 104], Hyacinthaceae [38, 123], Lamiaceae [261], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [9], Malvaceae [142], Oleaceae [33, 150], Onagraceae [218], Orchidaceae [116, 117, 258], Passifloraceae [157], Polemoniaceae [233], Primulaceae [36], Rhizophoraceae [10], Rosaceae [51, 60, 61, 72, 128], Valerianaceae [37], Zingiberaceae [268]
- Methylisoeugenol
Amaryllidaceae [6], Arecaceae [143, 145], Fabaceae [81], Onagraceae [218], Orchidaceae [77, 117], Polemoniaceae [233]
- (*Z*)-Methylisoeugenol
Arecaceae [137], Passifloraceae [157], Zingiberaceae [174]
- Benzopyrans
6,7-Dimethoxy-4*H*-1-benzopyran
Fabaceae [110]
6-Methoxy-4*H*-1-benzopyran-7-ol
Fabaceae [110]
6-Pentyl- α -pyrone
Orchidaceae [120]
- C6-C4
Hydrocarbons
Butylbenzene
Moraceae [82]

- 4-Isoprenyl-1-methylbenzene, see 4-(3-Methyl-3-butenyl)-1-methylbenzene
 4-(3-Methyl-3-butenyl)-1-methylbenzene (4-Isoprenyl-1-methylbenzene)
 Rosaceae [35]

Ketones

- Gingerone, see 3-Methoxy-4-hydroxyphenyl-2-butanone
 3-Hydroxy-4-phenyl-2-butanone
 Orchidaceae [117], Sapotaceae [263]
 4-Hydroxyphenyl-2-butanone
 Orchidaceae [117]
 3-Methoxy-4-hydroxyphenyl-2-butanone (Gingerone)
 Orchidaceae [117]
 1-Phenyl-2,3-butanedione (1-Phenylbutane-2,3-dione)
 Oleaceae [108], Sapotaceae [263]
 1-Phenylbutane-2,3-dione, see 1-Phenyl-2,3-butanedione
 4-Phenyl-2-butanone
 Orchidaceae [75, 116, 117], Solanaceae [69]
 4-Phenyl-3-buten-2-one
 Orchidaceae [13]

Alcohols

- 2,6-di-*tert*-Butyl-4-methyl-1-hydroxybenzene (2,6-di-*tert*-Butyl-4-methylphenol)
 2,6-di-*tert*-Butyl-4-methylphenol, see 2,6-di-*tert*-Butyl-4-methyl-1-hydroxybenzene
 Calycanthaceae [266], Malvaceae [67], Onagraceae [265]

Ethers

- 2,4-di-*tert*-Butylanisole, see 2,4-di-*tert*-Butyl-1-methoxybenzene
 2,4-di-*tert*-Butyl-1-methoxybenzene (2,4-di-*tert*-Butylanisole)
 Oleaceae [51]

C6-C5

Alcohols

- 5-Phenylmethoxypentanol
 Oleaceae [57]

C6-C7

Hydrocarbons

- 1-Heptenylbenzene
 Asteraceae [213]

C5-BRANCHED CHAIN COMPOUNDS

Saturated

Hydrocarbons

- 2-Methylbutane
 Rosaceae [220]

Acids

- 2-Methylbutanoic acid
 Orchidaceae [116, 117]
 3-Methylbutanoic acid
 Araceae [86], Asteraceae [65, 132], Cyclanthaceae [225], Orchidaceae [77, 116], Theophrastaceae [139]

Aldehydes

- 2-Methylbutanal
 Brassicaceae [221], Meliaceae [250], Orchidaceae [117], Rosaceae [12, 219, 220], Zingiberaceae [268]
 3-Methylbutanal
 Brassicaceae [221], Cactaceae [125], Fabaceae [179, 212, 216], Meliaceae [250], Orchidaceae [116, 117], Rosaceae [12, 219, 220], Sapindaceae [46], Zingiberaceae [174]

Ketones

- 3-Methyl-2-butanone
 Araceae [86, 129, 133, 229], Brassicaceae [221], Oleaceae [57, 58], Rosaceae [219, 220]

Alcohols

- 2-Methylbutanol
 Araceae [262], Arecaceae [10, 137], Fabaceae [49, 179], Hemerocallidaceae [33], Orchidaceae [117], Rosaceae [219, 220], Winteraceae [210, 239]
 3-Methylbutanol

Actinidiaceae [237], Apocynaceae [1, 233], Araceae [262], Arecaceae [10, 137], Cactaceae [125], Calycanthaceae [266], Cucurbitaceae [71], Cycadaceae [211], Fabaceae [49, 153, 156, 179, 212, 216], Lecythidaceae [144], Magnoliaceae [238], Malvaceae [142], Nymphaeaceae [14, 135], Oleaceae [39], Orchidaceae [77, 116, 117], Orobanchaceae [141], Polemoniaceae [5], Ranunculaceae [207, 208, 209], Rosaceae [12, 33, 35, 37, 61, 62, 219, 220], Scrophulariaceae [5], Solanaceae [141], Theophrastaceae [139], Thymelaeaceae [24], Winteraceae [210], Zingiberaceae [174]

Esters

Butyl 2-methylbutanoate

Orchidaceae [117]

Butyl 3-methylbutanoate

Asteraceae [37]

Ethyl 2-methylbutanoate

Annonaceae [111], Apiaceae [31], Arecaceae [137], Asteraceae [37], Cycadaceae [211], Magnoliaceae [238], Orchidaceae [77, 117], Rubiaceae [249]

Ethyl 3-methylbutanoate

Annonaceae [111], Araceae [86], Arecaceae [137], Asteraceae [37], Cycadaceae [211], Magnoliaceae [238]

Hexenyl 2-methylbutanoate

Lythraceae [10]

(Z)-3-Hexenyl 2-methylbutanoate

Fabaceae [120], Nyctaginaceae [151], Orchidaceae [117]

Hexenyl 3-methylbutanoate

Hemerocallidaceae [33]

(Z)-3-Hexenyl 3-methylbutanoate

Asteraceae [34], Brassicaceae [217], Cactaceae [125], Nyctaginaceae [151], Oleaceae [107], Ranunculaceae [19], Rubiaceae [249]

Hexyl 2-methylbutanoate

Lecythidaceae [144], Orchidaceae [117]

Hexyl 3-methylbutanoate

Rubiaceae [249]

Methyl (2R)-2-acetoxy-3-methylbutanoate

Eupomatiaceae [18]

2-Methylbutyl acetate

Fabaceae [179], Orchidaceae [117], Winteraceae [210, 239]

3-Methylbutyl acetate

Actinidiaceae [237], Amaryllidaceae [6, 63], Annonaceae [111], Apiaceae [31], Araceae [134, 146], Arecaceae [137], Asteraceae [37, 132], Caryophyllaceae [141], Fabaceae [175, 179, 212, 216], Lecythidaceae [144], Magnoliaceae [238], Orchidaceae [77, 116, 117], Rosaceae [33, 35, 72], Theophrastaceae [139], Winteraceae [210], Zingiberaceae [174, 268]

2-Methylbutyl butanoate

Arecaceae [68], Moraceae [121], Orchidaceae [117]

3-Methylbutyl butanoate

Nymphaeaceae [14, 135], Orchidaceae [117]

2-Methylbutyl hexanoate

Orchidaceae [117]

3-Methylbutyl hexanoate

Nymphaeaceae [14, 135]

2-Methylbutyl 2-methylbutanoate

Asteraceae [37]

3-Methylbutyl 3-methylbutanoate

Annonaceae [111], Cactaceae [125], Caryophyllaceae [5], Cycadaceae [211], Hemerocallidaceae [33], Orchidaceae [117], Valerianaceae [37]

2-Methylbutyl 2-methylpropanoate

Asteraceae [37]

3-Methylbutyl 2-methylpropanoate

Asteraceae [37]

3-Methylbutyl 2-methylprop-2-enoate

Asteraceae [37]

3-Methylbutyl pentanoate

- Rosaceae [33, 35]
- 3-Methylbutyl propanoate
 - Annonaceae [111]
- Methyl 2-hydroxy-3-methylbutanoate
 - Cactaceae [125], Eupomatiaceae [18]
- Methyl 2-methylbutanoate
 - Amaryllidaceae [63], Araceae [129, 133], Arecaceae [137], Brassicaceae [221], Cycadaceae [211], Eupomatiaceae [18], Fabaceae [212], Lythraceae [10], Magnoliaceae [11], Nymphaeaceae [14, 135], Oleaceae [39], Orchidaceae [117, 203], Rubiaceae [249]
- Methyl 3-methylbutanoate
 - Cactaceae [125], Cycadaceae [211], Nymphaeaceae [14, 135], Orchidaceae [117], Rubiaceae [249], Zingiberaceae [268]
- 3-Methylpentyl 3-methylbutanoate
 - Asteraceae [37]
- 2-Methylpropyl 2-methylbutanoate
 - Annonaceae [111], Asteraceae [37]
- 2-Methylpropyl 3-methylbutanoate
 - Annonaceae [111]
- Propyl 2-methylbutanoate
 - Asteraceae [37]
- 2-Propyl 3-methylbutanoate
 - Arecaceae [68]
- Unsaturated
 - Hydrocarbons
 - 2-Methyl-1,3-butadiene
 - Araceae [86], Brassicaceae [221], Rosaceae [12]
 - Aldehydes
 - 2-Methylbutenal
 - Sapindaceae [46]
 - 2-Methyl-2-butenal
 - Brassicaceae [221], Hemerocallidaceae [33]
 - (*E*)-2-Methyl-2-butenal
 - Cactaceae [125], Orchidaceae [117]
 - 3-Methyl-2-butenal (Prenal)
 - Arecaceae [143], Rutaceae [247], Zingiberaceae [174]
 - Prenal, see 3-Methyl-2-butenal
 - Ketones
 - 3-Methyl-3-buten-2-one
 - Zingiberaceae [174]
 - Alcohols
 - Isoprenol, see 3-Methyl-3-butenol
 - 2-Methyl-2-butenol
 - Amaryllidaceae [6], Oleaceae [39]
 - (*E*)-2-Methyl-2-butenol
 - Orchidaceae [77, 117]
 - 2-Methyl-3-buten-2-ol
 - Amaryllidaceae [63], Araceae [229], Brassicaceae [221], Fabaceae [212], Oleaceae [39], Orchidaceae [117], Zingiberaceae [174]
 - 3-Methyl-2-butenol (Prenal)
 - Araceae [86], Arecaceae [143], Berberidaceae [182], Cactaceae [125], Fabaceae [216], Orchidaceae [117]
 - 3-Methyl-3-butenol (Isoprenol)
 - Amaryllidaceae [63], Cactaceae [125], Fabaceae [212, 216], Oleaceae [39], Orchidaceae [117]
 - Prenal, see 3-Methyl-2-butenol
- Esters
 - Butyl (*Z*)-2-methyl-2-butenolate
 - Asteraceae [37]
 - Butyl tiglate
 - Orchidaceae [117]
 - (*E*)-2(3)-Epoxy-2-methylbutyl tiglate
 - Orchidaceae [117]

- Ethyl (Z)-2-methyl-2-butenolate
Asteraceae [37]
- Ethyl tiglate
Annonaceae [111], Arecaceae [137], Nymphaeaceae [14, 135], Rubiaceae [249]
- Geranyl tiglate
Orchidaceae [117]
- Heptyl tiglate
Orchidaceae [117]
- (Z)-3-Hexenyl (Z)-2-methyl-2-butenolate
Orchidaceae [117]
- Hexenyl tiglate
Caprifoliaceae [175]
- (Z)-3-Hexenyl tiglate
Caprifoliaceae [102, 107], Nyctaginaceae [151], Orchidaceae [117, 203], Rubiaceae [249]
- Hexyl tiglate
Orchidaceae [117], Rubiaceae [249]
- Isoprenyl acetate, see 3-Methyl-3-butenyl acetate
- Isoprenyl hexanoate, see 3-Methyl-3-butenyl hexanoate
- Methyl angelate, see Methyl (Z)-2-methyl-2-butenolate
- 2-Methyl-2-butenyl acetate
Asteraceae [37], Hyacinthaceae [123]
- (E)-2-Methyl-2-butenyl acetate
Orchidaceae [77, 117]
- 3-Methyl-2-butenyl acetate (Prenyl acetate)
Annonaceae [111], Apiaceae [31], Berberidaceae [182], Fabaceae [216], Orchidaceae [117]
- 3-Methyl-3-butenyl acetate (Isoprenyl acetate)
Amaryllidaceae [63], Orchidaceae [117]
- (E)-2-Methyl-2-butenyl butanoate
Orchidaceae [117]
- (E)-2-Methyl-2-butenyl hexanoate
Orchidaceae [117]
- 3-Methyl-2-butenyl hexanoate (Prenyl hexanoate)
Orchidaceae [117]
- 3-Methyl-3-butenyl hexanoate (Isoprenyl hexanoate)
Orchidaceae [117]
- (E)-2-Methyl-2-butenyl isobutanoate
Orchidaceae [77]
- (E)-2-Methyl-2-butenyl 2-methylbutanoate
Orchidaceae [77, 117]
- 3-Methyl-2-butenyl 3-methylbutanoate (Prenyl 3-methylbutanoate)
Orchidaceae [117]
- 3-Methyl-2-butenyl octanoate (Prenyl octanoate)
Orchidaceae [117]
- (E)-2-Methyl-2-butenyl propanoate
Orchidaceae [117]
- (E)-2-Methyl-2-butenyl tiglate
Caprifoliaceae [107], Orchidaceae [77, 117]
- 2-Methylbutyl (Z)-2-methyl-2-butenolate
Asteraceae [37]
- 3-Methylbutyl (Z)-2-methyl-2-butenolate
Asteraceae [37]
- 2-Methylbutyl tiglate
Orchidaceae [117]
- 3-Methylbutyl tiglate
Hyacinthaceae [123], Nymphaeaceae [14, 135], Orchidaceae [117]
- Methyl (Z)-2-methyl-2-butenolate (Methyl angelate)
Magnoliaceae [11], Nymphaeaceae [14, 135]
- Methyl 3-methyl-2-butenolate
Amaryllidaceae [63], Arecaceae [137], Asteraceae [213], Cycadaceae [211]
- 3-Methylpentyl (Z)-2-methyl-2-butenolate
Asteraceae [37]

- 2-Methyl-2-propenyl (Z)-2-methyl-2-butenolate
Asteraceae [37]
- 2-Methyl-2-propenyl tiglate
Asteraceae [37]
- 2-Methylpropyl (Z)-2-methyl-2-butenolate
Asteraceae [37]
- 2-Methylpropyl tiglate
Nyctaginaceae [151]
- Methyl tiglate
Asteraceae [37], Caryophyllaceae [54], Magnoliaceae [11], Nymphaeaceae [14, 135], Orchidaceae [117], Rubiaceae [249]
- Pentyl 2-methyl-2-butenolate
Orchidaceae [203]
- Pentyl (Z)-2-methyl-2-butenolate
Asteraceae [37]
- Pentyl tiglate
Orchidaceae [117]
- Prenyl acetate, see 3-Methyl-2-butenyl acetate
- Prenyl hexanoate, see 3-Methyl-2-butenyl hexanoate
- Prenyl 3-methylbutanoate, see 3-Methyl-2-butenyl 3-methylbutanoate
- Prenyl octanoate, see 3-Methyl-2-butenyl octanoate
- Propyl (Z)-2-methyl-2-butenolate
Asteraceae [37]
- Propyl tiglate
Orchidaceae [117]

TERPENES

MONOTERPENES

Acyclic

Hydrocarbons

β -Citronellene

Araceae [86, 129]

Cosmene, see (*E,Z*)-2,6-Dimethyl-1,3,5,7-octatetraene

Dehydroocimene

Amaryllidaceae [232, 233]

2,6-Dimethyl-2(3)epoxyocta-5(*E*),7-dien-4-one, see (*E*)-2(3)-Epoxy-2,6-dimethyl-5,7-octadien-4-one

Dimethyloctadiene

Araceae [133]

2,7-Dimethyl-3,5-octadiene

Oleaceae [57, 58]

3,7-Dimethyl-1,6-octadiene

Araceae [229]

3,7-Dimethyl-1,3,6-octatriene, see Ocimene

(*E*)-2,6-Dimethyl-1,3,5,7-octatetraene

Lauraceae [74]

(*E,E*)-2,6-Dimethyl-1,3,5,7-octatetraene

Asteraceae [37], Caryophyllaceae [66], Hemerocallidaceae [33], Hyacinthaceae [38], Orchidaceae [77, 116, 117]

(*E,Z*)-2,6-Dimethyl-1,3,5,7-octatetraene (Cosmene)

Asteraceae [37], Hemerocallidaceae [33], Hyacinthaceae [38], Orchidaceae [77, 117, 253]

(*Z,E*)-2,6-Dimethyl-1,3,5,7-octatetraene

Cactaceae [122], Hyacinthaceae [123, 150]

3,7-Dimethyloctatriene

Orchidaceae [204], Winteraceae [239]

3,7-Dimethyl-1,3,7-octatriene

Rosaceae [12]

Isocitronellene

Nyctaginaceae [151]

2-Methyl-6-methylene-1,3,7-octatriene

Arecaceae [143]

Myrcene

Actinidiaceae [237], Alliaceae [193], Amaryllidaceae [6, 21, 63, 175], Annonaceae [111], Apiaceae [31, 217, 242, 246], Apocynaceae [141], Araceae [77, 86, 129, 133, 146, 229, 252], Arecaceae [68, 137, 143, 145], Asteraceae [34, 36, 37, 59, 65, 132], Berberidaceae [182], Bignoniaceae [21, 142], Brassicaceae [70, 105, 186, 221], Bromeliaceae [21], Cactaceae [125, 142], Calycanthaceae [120, 266], Caprifoliaceae [102], Caricaceae [141], Caryophyllaceae [54, 66, 112], Chloranthaceae [251], Cucurbitaceae [71], Cyclanthaceae [225], Dipsacaceae [5, 185], Euphorbiaceae [7], Fabaceae [21, 49, 62, 81, 138, 156, 159, 160, 179, 193, 212, 216, 235], Gesneriaceae [77], Grossulariaceae [89], Hemerocallidaceae [33], Hyacinthaceae [38, 123], Lamiaceae [261], Lauraceae [74], Lecythidaceae [121, 144], Liliaceae [92], Magnoliaceae [9, 11, 240, 264], Malvaceae [41, 142], Moraceae [82], Musaceae [21], Myrsinaceae [103], Nelumbonaceae [184, 198, 199], Nyctaginaceae [151], Oleaceae [33, 51, 57, 58, 119], Onagraceae [218], Orchidaceae [17, 23, 27, 28, 29, 76, 77, 78, 80, 96, 98, 116, 117, 120, 147, 169, 189, 190, 191, 192, 193, 195, 201, 203, 204, 236, 252, 253, 255, 258, 259], Passifloraceae [157], Pinaceae [25], Pittosporaceae [107], Polemoniaceae [5, 21, 217], Primulaceae [5, 36], Ranunculaceae [85, 207, 208, 209], Rosaceae [12, 33, 35, 40, 43, 51, 72, 128, 196, 219, 220, 228], Rubiaceae [141, 249], Ruscaceae [171], Rutaceae [119, 159, 172, 247], Salicaceae [245], Scrophulariaceae [64], Solanaceae [141, 165, 166, 167, 168, 222], Theophrastaceae [139], Thymelaeaceae [5], Valerianaceae [37], Verbenaceae [5], Winteraceae [210, 239], Violaceae [73], Zamiaceae [211], Zingiberaceae [141, 174]

 α -Ocimene

Calycanthaceae [266], Fabaceae [153], Rubiaceae [249]

allo-Ocimene

Apiaceae [31], Araceae [229], Arecaceae [137], Fabaceae [179], Lauraceae [74], Moraceae [82], Nyctaginaceae [151], Orchidaceae [17, 180], Rubiaceae [5], Rutaceae [247], Verbenaceae [5]

(E)-allo-Ocimene

Arecaceae [143], Cactaceae [125], Zingiberaceae [174]

(Z)-allo-Ocimene

Zingiberaceae [174]

(E,Z)-allo-Ocimene

Berberidaceae [215]

neoallo-Ocimene

Araceae [229], Lauraceae [74]

(E,E)-neoallo-Ocimene (see also neoallo-Ocimene)

Berberidaceae [215]

Ocimene (3,7-Dimethyl-1,3,6-octatriene)

Amaryllidaceae [175], Araceae [134], Fabaceae [160, 161, 162, 163, 164, 175], Moraceae [82, 84], Oleaceae [171, 267], Orchidaceae [15, 76, 80, 95, 96, 98, 254, 255, 258, 260], Ranunculaceae [190], Rubiaceae [249], Ruscaceae [171], Rutaceae [247], Solanaceae [79], Verbenaceae [101]

(E)-Ocimene ((E)- β -Ocimene, see also Ocimene)

Actinidiaceae [237], Alliaceae [193], Amaryllidaceae [6, 63, 177, 232, 233], Apiaceae [31, 217, 246], Apocynaceae [141], Araceae [229], Arecaceae [68, 137, 143, 145], Asteraceae [5, 59, 132], Acanthaceae [10], Berberidaceae [215], Bignoniaceae [21, 142], Brassicaceae [70, 186, 221], Bromeliaceae [77], Cactaceae [113, 115, 122, 125], Calycanthaceae [120], Caprifoliaceae [102], Caricaceae [141], Caryophyllaceae [5, 66, 112, 141], Chloranthaceae [251], Combretaceae [10], Commelinaceae [121], Cucurbitaceae [71], Dipsacaceae [5], Fabaceae [21, 49, 50, 62, 81, 107, 120, 121, 138, 156, 179, 193, 212, 216, 235], Fumariaceae [56], Gentianaceae [45], Geraniaceae [32], Grossulariaceae [89], Hemerocallidaceae [33], Hyacinthaceae [38, 123, 150], Lamiaceae [5], Lauraceae [74], Lecythidaceae [121, 144], Lythraceae [10], Magnoliaceae [9, 11, 264], Malvaceae [41, 142], Moraceae [83], Nyctaginaceae [151], Oleaceae [33, 57, 58, 107, 150, 176, 178, 232, 233], Onagraceae [218], Orchidaceae [5, 8, 13, 17, 23, 27, 28, 77, 78, 116, 117, 118, 120, 121, 169, 191, 192, 193, 195, 201, 203, 236, 241, 243, 244, 253], Papaveraceae [62], Passifloraceae [157], Pittosporaceae [107], Polemoniaceae [5, 21], Primulaceae [36], Ranunculaceae [19, 59, 85, 207, 208, 209, 217], Rhizophoraceae [10], Rosaceae [40, 61, 72, 87, 88, 219, 220, 228], Rubiaceae [5, 121, 141], Ruscaceae [141, 232, 233], Rutaceae [119], Salicaceae [245], Sapotaceae [263], Scrophulariaceae [5, 64, 226], Solanaceae [121, 131,

141, 165, 166, 222], Throphastaceae [139], Thymelaeaceae [5, 24, 234], Valerianaceae [5], Verbenaceae [5], Winteraceae [210, 239], Violaceae [73], Zamiaceae [211], Zingiberaceae [141, 174]

(Z)-Ocimene ((Z)- β -Ocimene)

Actinidiaceae [237], Alliaceae [193], Amaryllidaceae [6, 63, 232, 233], Apiaceae [31, 217, 246], Apocynaceae [141], Araceae [26, 86, 229], Arecaceae [68, 137, 143, 145], Asteraceae [5, 132], Acanthaceae [10], Berberidaceae [182, 215], Bignoniaceae [142], Brassicaceae [186], Cactaceae [125], Calycanthaceae [120], Caricaceae [141], Caryophyllaceae [141], Chloranthaceae [251], Cucurbitaceae [71], Dipsacaceae [5], Fabaceae [21, 50, 62, 81, 138, 156, 179, 193, 212, 216, 235], Grossulariaceae [89], Hemerocallidaceae [33], Hyacinthaceae [38, 123, 150], Lecythidaceae [121, 144], Magnoliaceae [9, 11, 264], Malvaceae [142], Nyctaginaceae [151], Oleaceae [33, 57, 58, 150, 176, 178], Orchidaceae [13, 17, 23, 27, 77, 117, 118, 120, 191, 192, 193, 195, 201, 203, 236, 244, 253], Passifloraceae [157], Polemoniaceae [5, 21], Ranunculaceae [19, 85, 208, 217], Rosaceae [87, 88, 219, 220], Rubiaceae [5, 141], Rutaceae [172], Salicaceae [245], Scrophulariaceae [5, 226], Solanaceae [121, 141], Thymelaeaceae [5, 24], Valerianaceae [5], Verbenaceae [5], Winteraceae [239], Zamiaceae [211], Zingiberaceae [141, 174]

(Z)- β -Ocimene, see (Z)-Ocimene

Aldehydes

(E)-Citral, see Geranial

(Z)-Citral, see Neral

Citronellal

Arecaceae [137], Berberidaceae [215], Caprifoliaceae [102], Fabaceae [216], Gentianaceae [45], Myrsinaceae [103], Orchidaceae [23, 29, 77, 80, 117, 141, 147, 181, 255, 256, 258, 259], Passifloraceae [157], Rosaceae [51, 228], Rubiaceae [141], Ruscaceae [171], Rutaceae [119], Solanaceae [141], Violaceae [248], Zingiberaceae [268]

6(7)-Epoxyneral

Orchidaceae [117]

Geranial ((E)-Citral)

Apiaceae [246], Araceae [86, 133], Arecaceae [137], Cactaceae [125], Caprifoliaceae [102], Commelinaceae [121], Ericaceae [140], Lamiaceae [261], Lecythidaceae [121, 144], Magnoliaceae [11], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [178], Orchidaceae [28, 29, 77, 116, 117, 141, 180, 195, 241, 243, 244], Passifloraceae [157], Ranunculaceae [208, 209], Rosaceae [33, 35, 40, 43, 51, 60, 61, 72, 93, 128, 228], Rubiaceae [5], Rutaceae [247], Solanaceae [141], Thymelaeaceae [5]

Lilac aldehydes

Amaryllidaceae [63, 107], Apiaceae [246], Araceae [86], Caryophyllaceae [141], Oleaceae [33, 107, 178, 232, 233], Orchidaceae [13, 116, 117, 204, 241, 243, 244], Salicaceae [245]

Lilac aldehyde A (see also Lilac aldehydes)

Apocynaceae [121], Asteraceae [5], Caryophyllaceae [112], Lamiaceae [5], Orchidaceae [117], Polemoniaceae [5], Rubiaceae [5], Violaceae [73]

Lilac aldehyde B (see also Lilac aldehydes)

Apocynaceae [121], Berberidaceae [182], Caryophyllaceae [112], Lamiaceae [5], Orchidaceae [117], Polemoniaceae [5], Primulaceae [5], Rubiaceae [5], Violaceae [73]

Lilac aldehyde C (see also Lilac aldehydes)

Apocynaceae [121], Asteraceae [5], Berberidaceae [182], Lamiaceae [5], Orchidaceae [117], Polemoniaceae [5], Rubiaceae [5]

Lilac aldehyde D (see also Lilac aldehydes)

Apocynaceae [121], Asteraceae [5], Caryophyllaceae [112], Lamiaceae [5], Orchidaceae [117], Polemoniaceae [5], Rubiaceae [5]

Neral (*cis*-Citral)

Araceae [86], Cactaceae [125], Caprifoliaceae [102], Commelinaceae [121], Ericaceae [140], Fabaceae [216], Lamiaceae [261], Lecythidaceae [121, 144], Magnoliaceae [11], Nyctaginaceae [151], Orchidaceae [28, 77, 116, 117, 141, 180, 195, 241, 243, 244], Passifloraceae [157], Ranunculaceae [208, 209], Rosaceae [33, 35, 43, 60, 61, 93, 128], Rutaceae [247], Solanaceae [141], Verbenaceae [5]

Ketones

(E)-2(3)-Epoxy-2,6-dimethyl-5,7-octadien-4-one (2,6-Dimethyl-2(3)epoxyocta-5(E),7-dien-4-one)

Orchidaceae [117, 120]

- Ipsdienone (2-methyl-6-methylene-2,7-octadien-4-one)
 Amaryllidaceae [63], Arecaceae [137, 143, 145], Caryophyllaceae [5], Orchidaceae [78, 117, 120]
- Ipsenone (2-methyl-6-methylene-7-octen-4-one)
 Arecaceae [137, 145]
- 5-Ketolinalool
 Adoxaceae [108], Fabaceae [108], Oleaceae [108], Simaroubaceae [108]
- 2-Methyl-6-methylene-1,7-octadien-2-one
 Actinidiaceae [237]
- 2-Methyl-6-methylene-1,7-octadien-3-one
 Asteraceae [213], Chloranthaceae [251], Cucurbitaceae [71], Moraceae [83], Rutaceae [247]
- 2-methyl-6-methylene-2,7-octadien-4-one, see Ipsdienone
- 2-methyl-6-methylene-7-octen-4-one, see Ipsenone
- Ocimenone
 Arecaceae [137]
- Alcohols
- Amitinol (2-Methyl-6-methylene-3,7-octadien-2-ol)
 Arecaceae [145], Rutaceae [247]
- β -Citronellol, see Citronellol
- Citronellol (β -Citronellol)
 Araceae [86, 129, 133, 229], Berberidaceae [215], Cactaceae [125], Commelianaceae [121], Ericaceae [140], Fabaceae [216], Gentianaceae [45], Lecythydaceae [144], Myrsinaceae [103], Oleaceae [57, 58], Orchidaceae [8, 13, 23, 29, 76, 77, 97, 116, 117, 141, 147, 243, 255, 256, 258], Passifloraceae [157], Ranunculaceae [208], Rosaceae [33, 35, 40, 51, 60, 61, 72, 93, 115, 128, 155, 178, 183, 196, 228, 232], Rubiaceae [141], Ruscaceae [171], Rutaceae [119, 247], Solanaceae [141]
- 2,6-Dimethyl-1,7-octadien-3,6-diol
 Arecaceae [137], Rutaceae [247], Thymelaeaceae [24], Zingiberaceae [174]
- (*E*)-2,6-Dimethyl-5,7-octadien-2,3-diol
 Orchidaceae [117]
- 2,6-Dimethyl-3,7-octadien-2,6-diol
 Amaryllidaceae [63], Arecaceae [137], Berberidaceae [182], Fabaceae [121], Passifloraceae [157], Rubiaceae [5], Rutaceae [247], Simaroubaceae [108], Thymelaeaceae [5, 24], Zingiberaceae [174]
- 2,6-Dimethyl-1,7-octadien-3-ol
 Araceae [86, 229]
- 2,6-Dimethyl-3,7-octadien-2-ol
 Araceae [229]
- 3,7-Dimethyl-1,6-octadien-3,4-diol
 Orchidaceae [117]
- 3,7-Dimethyl-1,6-octadien-3-ol, see Linalool
- (*E*)-3,7-Dimethyl-2,6-octadien-1-ol, see Geraniol
- (*Z*)-3,7-Dimethyl-2,6-octadien-1-ol, see Nerol
- 2,6-Dimethyl-1,3,7-octatrien-6-ol, see Hotrienol
- 2,6-Dimethyl-1,5,7-octatrien-3-ol
 Rutaceae [247], Zingiberaceae [174]
- 2,6-Dimethyl-3,5,7-octatrien-2-ol, see Ocimenol
- (*E*)-2,6-Dimethyl-3,5,7-octatrien-2-ol, see (*E*)-Ocimenol
- (*Z*)-2,6-Dimethyl-3,5,7-octatrien-2-ol, see (*Z*)-Ocimenol
- 3,7-Dimethyl-1,5,7-octatrien-3-ol
 Zingiberaceae [174]
- 2(3)-Epoxygeraniol
 Orchidaceae [120]
- Geraniol ((*E*)-Geraniol, (*E*)-3,7-Dimethyl-2,6-octadien-1-ol)
 Amaryllidaceae [6], Araceae [26, 86, 133], Arecaceae [137], Berberidaceae [215], Cactaceae [125], Caprifoliaceae [102], Dipsacaceae [185], Ericaceae [140], Fabaceae [81, 126, 216], Gentianaceae [45], Iridaceae [178], Lecythydaceae [121, 144], Magnoliaceae [11, 264], Malvaceae [41], Myrsinaceae [103], Oleaceae [39, 177, 178], Orchidaceae [8, 28, 29, 53, 77, 78, 115, 116, 117, 118, 141, 169, 195, 241, 243, 244, 254, 256, 258, 259], Passifloraceae [157], Primulaceae [36], Ranunculaceae [207, 208, 209], Rosaceae [33, 35, 40, 43, 51, 60, 61, 72, 93, 115, 128, 155, 178, 183, 196, 228, 232], Rubiaceae [141], Ruscaceae

- [171, 232, 233], Rutaceae [119, 247], Sapindaceae [46], Scrophulariaceae [226], Solanaceae [141], Thymelaeaceae [5], Verbenaceae [5], Vitaceae [42, 44, 47], Zingiberaceae [174]
- Hotrienol (2,6-Dimethyl-1,3,7-octatrien-6-ol)
Adoxaceae [108], Arecaceae [137], Oleaceae [108], Orchidaceae [77, 117], Rutaceae [172, 247], Simaroubaceae [108]
- 5-Hydroxylinalool
Adoxaceae [108], Fabaceae [107], Oleaceae [108], Orchidaceae [117], Simaroubaceae [108]
- Ipsdienol (2-methyl-6-methylene-2,7-octadien-4-ol)
Amaryllidaceae [63], Araceae [252], Arecaceae [137, 143, 145], Orchidaceae [77, 78, 117, 120, 252, 253, 254, 258], Solanaceae [222]
- Ipsenol (2-methyl-6-methylene-7-octen-4-ol)
Arecaceae [137]
- (*E*)-Isogeraniol
Cactaceae [125]
- (*Z*)-Isogeraniol
Cactaceae [125]
- γ -Isogeraniol
Cactaceae [125]
- Lilac alcohols
Amaryllidaceae [63, 107], Apiaceae [246], Caryophyllaceae [141], Oleaceae [33, 107, 171, 178, 232, 233], Orchidaceae [13, 116, 117, 194, 195, 204, 241, 243, 244], Polemoniaceae [233, 234], Salicaceae [245]
- Lilac alcohol A (see also Lilac alcohols)
Apocynaceae [121], Berberidaceae [182], Orchidaceae [117], Polemoniaceae [5], Rubiaceae [5]
- Lilac alcohol B (see also Lilac alcohols)
Apocynaceae [121], Berberidaceae [182], Orchidaceae [117], Polemoniaceae [5], Rubiaceae [5]
- Lilac alcohol C (see also Lilac alcohols)
Apocynaceae [121], Berberidaceae [182], Orchidaceae [117], Polemoniaceae [5], Rubiaceae [5]
- Lilac alcohol D (see also Lilac alcohols)
Apocynaceae [121], Berberidaceae [182], Orchidaceae [117], Polemoniaceae [5], Rubiaceae [5]
- Linalool (3,7-Dimethyl-1,6-octadien-3-ol)
Alliaceae [193], Amaryllidaceae [6, 21, 63, 175, 177], Annonaceae [170], Apiaceae [31, 246], Apocynaceae [1, 2, 121, 141, 173, 232, 233], Araceae [86, 146, 205], Arecaceae [10, 68, 137, 143, 145], Asteraceae [5, 65, 132], Berberidaceae [182, 215], Brassicaceae [22, 36, 70, 105, 142, 186, 217, 221], Cactaceae [21, 113, 115, 122, 125], Calycanthaceae [120, 266], Caprifoliaceae [102, 107, 175, 178], Caricaceae [141], Caryophyllaceae [5, 54, 112], Combretaceae [10], Commelianaceae [121], Cucurbitaceae [71, 175], Dipsacaceae [5], Fabaceae [62, 81, 107, 119, 120, 121, 126, 138, 153, 159, 175, 179, 193, 212, 216], Fumariaceae [197], Gentianaceae [45], Hemerocallidaceae [33], Hyacinthaceae [123, 150], Iridaceae [150, 178], Lamiaceae [261], Lauraceae [74], Lecythidaceae [121, 144], Liliaceae [92, 107], Lythraceae [10], Magnoliaceae [9, 11, 240, 264], Malvaceae [41, 94, 142], Moraceae [82, 83, 84], Myrsinaceae [103], Nelumbonaceae [184, 198, 199], Nyctaginaceae [151], Oleaceae [33, 39, 51, 52, 57, 58, 108, 114, 150, 176, 178, 267], Onagraceae [127, 175, 218, 265], Orchidaceae [5, 8, 13, 17, 28, 29, 30, 53, 75, 77, 78, 80, 97, 98, 100, 115, 116, 117, 118, 120, 121, 141, 169, 181, 188, 191, 192, 193, 194, 195, 201, 204, 241, 243, 244, 253, 255, 258, 259, 260], Passifloraceae [157], Pittosporaceae [107], Polemoniaceae [5, 217, 232, 233, 234], Primulaceae [36, 190], Ranunculaceae [19, 85, 207, 208, 209], Rhizophoraceae [10], Rosaceae [12, 33, 35, 37, 40, 43, 51, 128, 166, 219, 228], Rubiaceae [5, 141, 249], Ruscaceae [171], Rutaceae [2, 119, 159, 173, 247], Salicaceae [245], Sapotaceae [263], Simaroubaceae [108], Solanaceae [94, 141, 154, 165, 166], Theaceae [200], Theophrastaceae [139], Thymelaeaceae [5, 24, 234], Valerianaceae [37], Verbenaceae [5, 101, 175], Winteraceae [210, 239], Violaceae [73], Vitaceae [42, 44, 47], Zamiaceae [211], Zingiberaceae [141, 174, 268]
- 3-Methylene-7-methyl-1,6-octadien-4-ol, see 2-Methyl-6-methylene-2,7-octadien-5-ol
- 3-Methylene-7-methyl-1,7-octadien-6-ol, see 2-Methyl-6-methylene-1,7-octadien-3-ol
- 2-Methyl-6-methylene-1,7-octadien-3-ol (3-Methylene-7-methyl-1,7-octadien-6-ol)

- Araceae [86], Rutaceae [247]
 2-methyl-6-methylene-2,7-octadien-4-ol, see Ipsdienol
 2-methyl-6-methylene-7-octen-4-ol, see Ipsenol
 2-Methyl-6-methylene-2,7-octadien-5-ol (3-Methylene-7-methyl-1,6-octadien-4-ol)
 Zingiberaceae [174]
 2-Methyl-6-methylene-3,7-octadien-2-ol, see Amitinol
 Myrcenol
 Zingiberaceae [268]
 Nerol ((Z)-Geraniol, (Z)-3,7-Dimethyl-2,6-octadien-1-ol)
 Annonaceae [111], Araceae [146], Arecaceae [137], Brassicaceae [217], Cactaceae [125],
 Commelinaceae [121], Ericaceae [140], Fabaceae [81, 216], Iridaceae [178], Lecythi-
 daceae [121, 144], Magnoliaceae [11, 264], Malvaceae [41], Myrsinaceae [103], Nyctag-
 inaceae [151], Oleaceae [39], Orchidaceae [8, 13, 28, 76, 77, 98, 117, 141, 195, 243, 254,
 258], Papaveraceae [62], Passifloraceae [157], Primulaceae [36], Ranunculaceae [207,
 208, 209], Rosaceae [33, 35, 40, 51, 60, 61, 72, 93, 178, 228], Rubiaceae [141], Rus-
 caceae [171], Rutaceae [247], Solanaceae [141], Thymelaeaceae [5], Verbenaceae [5]
 Ocimenols (2,6-Dimethyl-3,5,7-octatrien-2-ol)
 Adoxaceae [115], Agavaceae [115], Amaryllidaceae [232, 233], Apocynaceae [115], Araceae
 [86], Cactaceae [115], Ericaceae [115], Fabaceae [115, 138], Hyacinthaceae [115, 150],
 Iridaceae [115, 150], Liliaceae [115], Magnoliaceae [108, 115], Malvaceae [115], Myrsi-
 naceae [115], Oleaceae [115], Orchidaceae [115, 253], Passifloraceae [157], Resedaceae
 [115], Rutaceae [115], Solanaceae [115], Thymelaeaceae [115], Zingiberaceae [115]
 (*E*)-Ocimenol (*E*)-2,6-Dimethyl-3,5,7-octatrien-2-ol, see also Ocimenols)
 Amaryllidaceae [63], Apocynaceae [141], Arecaceae [68, 137, 143, 145], Asteraceae [37],
 Cactaceae [113, 115, 122], Caryophyllaceae [66, 141], Fabaceae [62], Hemerocallidaceae
 [33], Hyacinthaceae [38, 123], Lecythidaceae [144], Malvaceae [142], Orchidaceae [77,
 116, 117], Rubiaceae [5, 141], Solanaceae [141], Verbenaceae [5], Zingiberaceae [141]
 (*Z*)-Ocimenol (*Z*)-2,6-Dimethyl-3,5,7-octatrien-2-ol, see also Ocimenols)
 Amaryllidaceae [63], Apocynaceae [141], Arecaceae [137, 143], Asteraceae [37], Cactaceae
 [113, 115], Caryophyllaceae [66, 141], Hemerocallidaceae [33], Hyacinthaceae [38, 123],
 Lecythidaceae [144], Malvaceae [142], Orchidaceae [77, 117], Rubiaceae [5], Solanaceae
 [141], Thymelaeaceae [5], Zingiberaceae [141]
- Esters
- Citronellyl acetate
 Amaryllidaceae [6], Berberidaceae [215], Cactaceae [125], Myrsinaceae [103], Orchidaceae
 [8, 13, 29, 117], Passifloraceae [157], Ranunculaceae [208, 209], Rosaceae [33, 35, 40,
 51, 60, 61, 72, 128, 155, 228, 232], Ruscaceae [171, 232, 233]
- Citronellyl butanoate
 Orchidaceae [117]
- Citronellyl formate
 Berberidaceae [215], Rosaceae [228]
- Citronellyl hexanoate
 Orchidaceae [117]
- Citronellyl pentanoate
 Orchidaceae [117]
- 3,7-Dimethyl-2,3-dihydroxy-6-octen-1-yl acetate
 Orchidaceae [120]
- 3,7-Dimethyl-7-hydroxyocta-1,5-dien-3-yl acetate
 Rutaceae [247]
- 3,7-Dimethyl-2-oxo-6-octen-1-yl acetate
 Orchidaceae [120]
- 2(3)-Epoxygeranyl acetate (6(7)-Epoxygeranyl acetate)
 Orchidaceae [77, 78, 117, 120]
- 6(7)-Epoxygeranyl acetate, see 2(3)-Epoxygeranyl acetate
- Ethyl 3,7-dimethyl-6-octenoate
 Orchidaceae [117]
- Geranyl acetate
 Amaryllidaceae [6], Apiaceae [31], Cactaceae [125], Fabaceae [81, 216], Lecythidaceae [121,
 144], Lythraceae [10], Magnoliaceae [11], Malvaceae [41], Oleaceae [33], Orchidaceae
 [8, 77, 78, 116, 117, 120, 141], Passifloraceae [157], Ranunculaceae [207, 208, 209],
 Rosaceae [33, 35, 40, 43, 51, 60, 61, 72, 128, 228, 232], Ruscaceae [171], Solanaceae
 [141]

- Geranyl butanoate
 - Orchidaceae [77], Passifloraceae [157]
- Geranyl formate
 - Passifloraceae [157], Rosaceae [51, 228]
- Geranyl propanoate
 - Asteraceae [5], Rosaceae [228]
- Ipsdienyl acetate
 - Arecaceae [145], Orchidaceae [117, 120]
- Lilac acetates
 - Amaryllidaceae [63], Caryophyllaceae [141], Orchidaceae [117, 241, 243]
- Lilac acid methyl ester
 - Orchidaceae [117]
- Linalyl acetate
 - Amaryllidaceae [6], Araceae [86], Cactaceae [113, 125], Caryophyllaceae [5, 54], Lauraceae [74], Musaceae [21], Orchidaceae [117], Passifloraceae [157], Polemoniaceae [21], Rosaceae [40], Rutaceae [247], Violaceae [248]
- Methyl citronellate
 - Myrsinaceae [103], Orchidaceae [13]
- Methyl 2,7-dimethyl-6-octenoate
 - Orchidaceae [117]
- Methyl geranate
 - Araceae [86], Arecaceae [137], Berberidaceae [215], Ericaceae [140], Orchidaceae [77, 117, 120, 180], Passifloraceae [157], Ranunculaceae [207, 208, 209], Rosaceae [40, 93], Verbenaceae [5], Zamiaceae [211]
- Methyl (*E*)-geranate
 - Rosaceae [33, 35]
- Methyl (*Z*)-geranate, see Methyl nerylolate
- Methyl nerylolate (Methyl (*Z*)-geranate)
 - Berberidaceae [215], Magnoliaceae [11], Rosaceae [33, 35]
- Myrcenyl acetate
 - Arecaceae [145]
- Neryl acetate
 - Cactaceae [125], Cucurbitaceae [71], Fabaceae [81, 216], Lecythidaceae [121, 144], Orchidaceae [8, 77, 117], Ranunculaceae [207, 208, 209], Rosaceae [33, 35, 40, 60, 61, 72, 128], Ruscaceae [171]
- Neryl 2-methylbutanoate
 - Fabaceae [49, 156]
- Ethers and epoxides
 - Acetaldehyde citronellylethyl acetal
 - Orchidaceae [117]
 - 2(3)-Epoxy-citral
 - 50 Orchidaceae [117]
 - 3(4)-Epoxy-3,7-dimethyl-1,6-octadiene (5(6)-Epoxy-2,6-dimethyl-2,7-octadiene)
 - Solanaceae [121]
 - (*E*)-3(4)-Epoxy-3,7-dimethyl-1,6-octadiene ((*E*)-5(6)-Epoxy-2,6-dimethyl-2,7-octadiene)
 - Cactaceae [125], Calycanthaceae [120], Lecythidaceae [121], Orchidaceae [77, 117]
 - (*Z*)-3(4)-Epoxy-3,7-dimethyl-1,6-octadiene
 - Calycanthaceae [120]
 - (*E*)-5(6)-Epoxy-2,6-dimethyl-2,7-octadiene, see (*E*)-3(4)-Epoxy-3,7-dimethyl-1,6-octadiene
 - 5(6)-Epoxy-2,6-dimethyl-2,7-octadiene, see 3(4)-Epoxy-3,7-dimethyl-1,6-octadiene
 - Epoxylinolool
 - Cucurbitaceae [71]
 - Epoxy rose furan
 - Arecaceae [143], Cactaceae [125]
 - Epoxy tagetone
 - Solanaceae [165]
 - Geranyl methyl ether
 - Magnoliaceae [11]
 - cis*-Linalool-6,7-epoxide, see (*Z*)-6(7)-Linalool epoxide
 - 6(7)-Linalool epoxide (Linalool-6,7-oxides)
 - Rutaceae [247]
 - (*E*)-6(7)-Linalool epoxide (see also 6(7)-Linalool epoxide)

- Zingiberaceae [174]
 (Z)-6(7)-Linalool epoxide (see also 6(7)-Linalool epoxide)
 Zingiberaceae [174]
 Linalool oxides
 Asteraceae [59], Brassicaceae [186], Calycanthaceae [266], Fabaceae [175, 179], Moraceae [82, 84], Orchidaceae [13, 27], Ranunculaceae [59, 208], Ruscaceae [171]
 (E)-Linalool oxide (see also Linalool oxides)
 Amaryllidaceae [6], Cucurbitaceae [71], Orchidaceae [13]
 (Z)-Linalool oxide (see also Linalool oxides)
 Amaryllidaceae [6], Caryophyllaceae [112], Orchidaceae [13]
 Linalool oxide (furanoid) (Linalool-3,6-oxides, see also Linalool oxides)
 Amaryllidaceae [63], Apiaceae [246], Araceae [26], Asteraceae [132], Cactaceae [113], Caprifoliaceae [102, 150], Caricaceae [141], Caryophyllaceae [141], Hydrangeaceae [108], Iridaceae [178], Lecythidaceae [144], Liliaceae [92], Magnoliaceae [9, 11], Malvaceae [67], Nyctaginaceae [151], Oleaceae [58, 177], Onagraceae [265], Orchidaceae [28, 29, 244], Polemoniaceae [217], Rubiaceae [141], Rutaceae [247], Solanaceae [141], Theaceae [200], Winteraceae [210], Zingiberaceae [141]
 (E)-Linalool oxide (furanoid) (see also Linalool oxides and linalool oxide (furanoid))
 Apiaceae [31], Apocynaceae [121], Araceae [86], Arecaceae [137, 143], Asteraceae [5], Berberidaceae [215], Cactaceae [125], Calycanthaceae [120], Dipsacaceae [5], Fabaceae [120, 121, 138, 153, 212], Gentianaceae [45], Grossulariaceae [89], Malvaceae [142], Moraceae [83], Oleaceae [39, 57, 267], Orchidaceae [30, 77, 117, 169, 191, 192, 243], Passifloraceae [157], Polemoniaceae [5], Ranunculaceae [19], Rubiaceae [5], Thymelaeaceae [5, 24], Verbenaceae [5], Zamiaceae [211], Zingiberaceae [174]
 (Z)-Linalool oxide (furanoid) (see also Linalool oxides and linalool oxide (furanoid))
 Apiaceae [31], Apocynaceae [121], Araceae [86], Arecaceae [137, 143], Asteraceae [5], Berberidaceae [182, 215], Cactaceae [115, 125], Calycanthaceae [120], Dipsacaceae [5], Fabaceae [121, 138, 212], Grossulariaceae [89], Lauraceae [74], Malvaceae [142], Oleaceae [39, 57], Onagraceae [218], Orchidaceae [77, 117, 243], Passifloraceae [157], Polemoniaceae [5], Ranunculaceae [19], Rubiaceae [5], Thymelaeaceae [5, 24], Verbenaceae [5], Zamiaceae [211], Zingiberaceae [174]
 Linalool-3,6-oxide, see Linalool oxide (furanoid)
 Linalool oxide (pyranoid) (see also Linalool oxides)
 Araceae [86], Arecaceae [137, 143], Asteraceae [132], Brassicaceae [217], Cactaceae [113], Caprifoliaceae [102, 150], Hydrangeaceae [108], Iridaceae [178], Magnoliaceae [11], Nyctaginaceae [151], Oleaceae [39, 58], Orchidaceae [115], Rubiaceae [141], Theaceae [200], Winteraceae [210]
 (E)-Linalool oxide (pyranoid) (see also Linalool oxides and linalool oxide (pyranoid))
 Apocynaceae [121], Asteraceae [5], Cactaceae [125], Calycanthaceae [120], Fabaceae [212], Lauraceae [74], Malvaceae [142], Oleaceae [57], Onagraceae [218], Orchidaceae [77, 117], Rubiaceae [5], Thymelaeaceae [24], Verbenaceae [5]
 (Z)-Linalool oxide (pyranoid) (see also Linalool oxides and linalool oxide (pyranoid))
 Asteraceae [5], Cactaceae [115, 125], Calycanthaceae [120], Fabaceae [121, 153, 212], Lauraceae [74], Malvaceae [142], Oleaceae [57], Orchidaceae [77, 116, 117], Primulaceae [5], Rubiaceae [5], Thymelaeaceae [24], Verbenaceae [5]
 Linalool oxide acetate
 Rosaceae [228]
 Linalool oxide acetate (pyranoid) (2,6,6-Trimethyl-2-vinyl-4-acetoxytetrahydropyran, see also Linalool oxide acetate)
 Asteraceae [132], Cactaceae [113]
 (E)-Linalool oxide acetate (pyranoid) ((E)-2,6,6-Trimethyl-2-vinyl-4-acetoxytetrahydropyran, see also Linalool oxide acetate and Linalool oxide acetate (pyranoid))
 Orchidaceae [117]
 (Z)-Linalool oxide acetate (pyranoid) (see also Linalool oxide acetate and Linalool oxide acetate (pyranoid))
 Cactaceae [125]
 Linalool-6,7-oxides, see 6(7)-Linalool epoxide
 Linalool oxide (pyranoid: alcohol)
 Magnoliaceae [9]
 Linalool oxide (pyranoid: ketone)
 Magnoliaceae [9, 11]

- Linalyl methyl ether
 - Cactaceae [113, 125]
- 4-Methyl-6-(2'-methyl-1'-propenyl)-3,6-dihydro-2*H*-pyran, see Rose oxide
- Myrcene epoxide, see Myrcene oxide
- Myrcene oxide (Myrcene epoxide)
 - Amaryllidaceae [63], Arecaceae [137, 143, 145], Orchidaceae [77, 78, 117], Rutaceae [247], Solanaceae [222]
- Nerol oxide
 - Orchidaceae [117], Rosaceae [33, 35]
- Neryl methyl ether
 - Magnoliaceae [11]
- Ocimene epoxide, see Ocimene oxide
- (*E*)-Ocimene epoxide, see (*E*)-Ocimene oxide
- 2(3)-Ocimene epoxide, see Ocimene oxide
- (*E*)-2(3)-Ocimene epoxide, see (*E*)-Ocimene oxide
- Ocimene oxide (2(3)-Ocimene epoxide, Ocimene epoxide)
 - Fabaceae [62], Rutaceae [247]
- (*E*)-Ocimene oxide ((*E*)-2(3)-Ocimene epoxide, (*E*)-Ocimene epoxide, (*E*)-6,7-Ocimene oxide, see also Ocimene oxide)
 - Amaryllidaceae [63, 232, 233], Apiaceae [246], Apocynaceae [141], Arecaceae [137, 143], Cactaceae [125], Caryophyllaceae [66, 141], Hemerocallidaceae [33], Lecythidaceae [144], Orchidaceae [77, 116, 117, 120, 121], Polemoniaceae [5], Ranunculaceae [19], Solanaceae [121], Valerianaceae [5], Verbenaceae [5], Zingiberaceae [141, 174]
- (*Z*)-Ocimene oxide ((*Z*)-Ocimene epoxide, (*Z*)-6,7-Ocimene oxide, see also Ocimene oxide)
 - Arecaceae [137, 143], Hemerocallidaceae [33], Valerianaceae [5], Zingiberaceae [174]
- (*Z*)-6,7-Ocimene oxide, see (*Z*)-Ocimene oxide
- (*E*)-6,7-Ocimene oxide, see (*E*)-Ocimene oxide
- Rose furan
 - Cactaceae [125]
- Rose oxide (4-Methyl-6-(2'-methyl-1'-propenyl)-3,6-dihydro-2*H*-pyran)
 - Cactaceae [125], Orchidaceae [117], Rosaceae [72, 178, 196, 228, 232], Ruscaceae [171]
- (*E*)-Rose oxide ((*E*)-4-Methyl-6-(2'-methyl-1'-propenyl)-3,6-dihydro-2*H*-pyran, see also Rose oxide)
 - Berberidaceae [215], Gentianaceae [45], Malvaceae [41], Rosaceae [33, 35, 51]
- (*Z*)-Rose oxide ((*Z*)-4-Methyl-6-(2'-methyl-1'-propenyl)-3,6-dihydro-2*H*-pyran, see also rose oxide)
 - Berberidaceae [215], Cactaceae [113], Gentianaceae [45], Myrsinaceae [103], Rosaceae [33, 35, 51]
- 2,6,6-Trimethyl-2-vinyl-4-acetoxytetrahydropyran, see Linalool oxide acetate (pyranoid)
- Irregular
 - Ketones
 - Artemisia ketone
 - Asteraceae [34, 37]
- Alcohols
 - Artemisia alcohol
 - Asteraceae [34, 37]
 - (*E*)-Chrysanthemol
 - Asteraceae [34]
 - Lavandulol
 - Asteraceae [37], Lecythidaceae [144], Orchidaceae [117], Polemoniaceae [5], Sapotaceae [263], Scrophulariaceae [5], Theophrastaceae [139], Thymelaeaceae [5], Valerianaceae [37]
 - Yomogi alcohol
 - Asteraceae [34]
- Esters
 - Lavandulyl acetate
 - Asteraceae [5, 37], Lecythidaceae [144], Malvaceae [142], Orchidaceae [117], Primulaceae [5], Theophrastaceae [139]
 - Lavandulyl 2-methylbutanoate
 - Valerianaceae [37]
 - Lavandulyl 3-methylbutanoate

Valerianaceae [37]
 Lavandulyl pentanoate
 Valerianaceae [37]
 Lavandulyl propanoate
 Asteraceae [5]
 Methyl chrysantemate
 Orchidaceae [23]

p-menthane skeleton

Hydrocarbons

Cymene

Apiaceae [217], Moraceae [82]

o-Cymene

Araceae [129], Asteraceae [132]

p-Cymene

Annonaceae [111], Apiaceae [31, 242], Apocynaceae [141], Araceae [86, 229], Arecaceae [143], Asteraceae [5, 34, 37], Berberidaceae [182], Brassicaceae [22], Cactaceae [113, 125], Calycanthaceae [266], Caryophyllaceae [5, 54], Cycadaceae [211], Dipsacaceae [185], Fabaceae [126, 212], Gentianaceae [45], Geraniaceae [32], Hemerocallidaceae [33], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [264], Malvaceae [41], Meliaceae [250], Musaceae [21], Nelumbonaceae [184], Oleaceae [177], Orchidaceae [23, 27, 77, 117, 120, 201, 254, 258], Primulaceae [5, 36], Rosaceae [33, 35, 37, 43, 228], Rutaceae [247], Solanaceae [141], Theophrastaceae [139], Zamiaceae [211]

p-Cymenene

Araceae [86], Lauraceae [74]

Limonene

Actinidiaceae [237], Agavaceae [142, 178], Alliaceae [36], Amaryllidaceae [6, 21, 63], Annonaceae [111], Apiaceae [31, 242, 246], Apocynaceae [141], Araceae [77, 86, 129, 133, 134, 146, 205, 229], Arecaceae [21, 68, 137, 143], Asteraceae [34, 36, 37, 59, 65, 132, 213], Berberidaceae [182, 215], Bignoniaceae [21, 142], Brassicaceae [22, 70, 105, 186, 217, 221], Bromeliaceae [21, 77], Cactaceae [21, 113, 115, 122, 125], Calycanthaceae [120, 266], Caprifoliaceae [102], Caryophyllaceae [5, 54, 66, 112, 141], Chloranthaceae [251], Cucurbitaceae [71], Cyclanthaceae [225], Dipsacaceae [5], Euphorbiaceae [7, 254], Fabaceae [21, 49, 62, 138, 156, 159, 160, 193, 212, 235], Fumariaceae [197], Gentianaceae [45], Geraniaceae [32], Gesneriaceae [77], Grossulariaceae [89], Hemerocallidaceae [33], Hyacinthaceae [38], Lamiaceae [261], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [9, 11, 240, 264], Malvaceae [41, 94, 142], Moraceae [82, 84], Musaceae [21], Myrsinaceae [103], Nelumbonaceae [184, 198, 199], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [33, 51, 57, 58, 119, 150, 267], Onagraceae [218], Orchidaceae [8, 13, 17, 23, 27, 28, 29, 75, 76, 77, 78, 80, 95, 96, 106, 116, 117, 120, 141, 147, 169, 180, 189, 190, 191, 192, 193, 195, 201, 204, 224, 253, 254, 258], Pinaceae [25], Polemoniaceae [5, 21, 217], Primulaceae [36, 190], Ranunculaceae [85, 208, 209, 217], Rosaceae [12, 33, 35, 37, 40, 43, 51, 72, 87, 88, 128, 196, 219, 220, 228], Rubiaceae [5, 141, 249], Ruscaceae [141, 171], Rutaceae [2, 119, 159, 172, 247], Salicaceae [245], Sapindaceae [46], Solanaceae [94, 121, 131, 141, 165, 166, 167, 168, 222], Theophrastaceae [139], Valerianaceae [37], Verbenaceae [5], Winteraceae [210, 239], Violaceae [73, 248], Vitaceae [42, 44, 47], Zamiaceae [211], Zingiberaceae [141, 174]

Menthatriene

Nyctaginaceae [151], Rubiaceae [5], Thymelaeaceae [5]

p-Menthatriene

Fabaceae [81]

p-Mentha-1,3,8-triene

Araceae [86]

p-Mentha-1(7),2,8-triene

Araceae [86], Arecaceae [143]

p-Mentha-1(7),3,8-triene

Araceae [86]

Perillene

Amaryllidaceae [21], Araceae [86], Bignoniaceae [21], Brassicaceae [70], Cactaceae [125], Fabaceae [21, 104], Magnoliaceae [11, 264], Orchidaceae [117, 191, 201], Rosaceae [33, 35, 43, 166], Zingiberaceae [268]

Phellandrene

Nyctaginaceae [151]

 α -Phellandrene (see also Phellandrene)

Apiaceae [242], Araceae [86, 229], Asteraceae [36, 132], Caryophyllaceae [5, 112, 141], Chloranthaceae [251], Fabaceae [212], Geraniaceae [32], Lauraceae [74], Lecythidaceae [144], Malvaceae [41], Orchidaceae [77, 78, 95, 117, 181, 203, 254, 258], Rosaceae [219], Ruscaceae [171], Verbenaceae [5], Winteraceae [239]

 β -Phellandrene (see also Phellandrene)

Apiaceae [31, 242, 246], Araceae [86, 229], Arecaceae [143], Asteraceae [34, 59], Brassicaceae [142], Caryophyllaceae [5], Chloranthaceae [251], Fabaceae [212], Geraniaceae [32], Grossulariaceae [89], Lecythidaceae [144], Magnoliaceae [11, 240], Malvaceae [41, 142], Moraceae [82], Orchidaceae [17, 23, 28, 77, 78, 117, 191, 192, 201, 203], Pinaceae [25], Polemoniaceae [5], Primulaceae [5], Rosaceae [33, 35], Ruscaceae [171], Rutaceae [247], Solanaceae [121], Verbenaceae [5], Winteraceae [239], Zamiaceae [211]

 α -Terpinene

Araceae [26, 86, 229], Arecaceae [137, 143], Asteraceae [36], Brassicaceae [22], Fabaceae [138, 212], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [264], Nelumbonaceae [184, 198, 199], Nyctaginaceae [151], Orchidaceae [28, 77, 100, 117, 180, 191, 258], Rosaceae [51, 228], Ruscaceae [171], Zamiaceae [211]

 β -Terpinene

Cucurbitaceae [71], Nyctaginaceae [151]

 γ -Terpinene

Amaryllidaceae [21], Apiaceae [31], Araceae [86, 146], Arecaceae [143], Berberidaceae [215], Bignoniaceae [21], Cactaceae [125], Calycanthaceae [120], Caprifoliaceae [102], Cucurbitaceae [71], Dipsacaceae [185], Fabaceae [21, 212], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [240], Malvaceae [41], Moraceae [82], Musaceae [21], Nelumbonaceae [184, 198, 199], Orchidaceae [117, 180, 189, 191, 203], Rosaceae [12], Ruscaceae [171], Rutaceae [2, 119], Theophrastaceae [139], Zamiaceae [211]

Terpinolene

Amaryllidaceae [6], Araceae [129, 133, 229], Arecaceae [137, 143], Asteraceae [34, 36], Bignoniaceae [142], Calycanthaceae [120], Ericaceae [136], Fabaceae [21, 138, 212], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [11, 240], Malvaceae [41], Nelumbonaceae [184, 198, 199], Orchidaceae [77, 117, 120, 258], Pinaceae [25], Primulaceae [5], Rosaceae [12, 43, 228], Rubiaceae [5], Ruscaceae [171], Solanaceae [131], Thymelaeaceae [5], Winteraceae [239], Violaceae [73], Zamiaceae [211], Zingiberaceae [174]

 α -Terpinolene

Apiaceae [31], Araceae [86, 146], Nyctaginaceae [151], Polemoniaceae [217]

Aldehydes

p-Mentha-1,3-diene-7-al

Araceae [229]

p-Menth-1-en-7-al

Orchidaceae [117]

Perillaldehyde

Araceae [229]

Phellandral

Araceae [229]

Ketones

Carvenone, see Carvone

Carvone (Carvenone)

Araceae [146, 229], Caryophyllaceae [54, 112], Euphorbiaceae [7], Gesneriaceae [77], Oleaceae [57, 58], Orchidaceae [77, 97, 98, 117, 254, 256, 258]

Carvotanacetone

Orchidaceae [117]

Dihydrocarvone

Araceae [86, 146], Orchidaceae [77, 117, 254, 258]

(E)-Dihydrocarvone (see also Dihydrocarvone)

Araceae [146], Cucurbitaceae [71]

(Z)-Dihydrocarvone (see also Dihydrocarvone)

Araceae [146]

Dill ether

- Araceae [86], Calycanthaceae [120]
- Eucarvone
 - Nyctaginaceae [151]
- 5-Hydroxy-*p*-mentha-6-ene-2-one
 - Araceae [229]
- 4-Hydroxy-3-methyl-6-(1-methylethyl)-2-cyclohexen-1-one
 - Araceae [229]
- Isomenthone
 - Orchidaceae [117]
- Menthone
 - Araceae [229], Caryophyllaceae [54], Gentianaceae [45], Oleaceae [57, 58], Orchidaceae [117]
- 3-Methyl-6-(-1-methylethyl)-2-cyclohexen-1-one, see Piperitone
- Piperitenone, see Piperitone
- Piperitone (Piperitenone, 3-Methyl-6-(-1-methylethyl)-2-cyclohexen-1-one)
 - Araceae [86, 229], Lauraceae [74], Nyctaginaceae [151], Orchidaceae [117], Polemoniaceae [5]
- (+)-Pulegone
 - Lamiaceae [261], Malvaceae [41]
- Alcohols
 - Carvacrol
 - Araceae [229], Caryophyllaceae [5], Lauraceae [74], Orchidaceae [117], Primulaceae [5]
 - Carveol
 - Nyctaginaceae [151]
 - (*E*)-Carveol (see also Carveol)
 - Asteraceae [214], Malvaceae [41]
 - L*-Carveol (see also Carveol)
 - Annonaceae [111]
 - (*Z*)-Carveol (see also Carveol)
 - Araceae [86], Caryophyllaceae [112]
 - p*-Cymenol-(8), see *p*-Cymen-8-ol
 - p*-Cymen-8-ol (*p*-Cymenol-(8))
 - Araceae [86, 229], Oleaceae [33], Orchidaceae [117, 243], Zamiaceae [211]
 - 5-Hydroxycineole
 - Araceae [146]
 - endo*-2-Hydroxycineole
 - Apocynaceae [232, 233]
 - exo*-2-Hydroxycineole
 - Fabaceae [212]
 - (*E*)-2-Hydroxy-1,8-cineole
 - Orchidaceae [117]
 - Isomenthol
 - Orchidaceae [117]
 - p*-Mentha-1,8-dien-4-ol
 - Araceae [229]
 - p*-Mentha-1,3-dien-8-ol
 - Orchidaceae [117]
 - p*-Mentha-1,5-dienol-(8)
 - Hemerocallidaceae [33]
 - p*-Mentha-1(7),2-dienol-(8)
 - Hemerocallidaceae [33]
 - (*E*)-*p*-Mentha-1,7-dien-3-ol
 - Orchidaceae [117]
 - (*E*)-*p*-Mentha-2,8-dien-1-ol
 - Araceae [86]
 - (*Z*)-*p*-Mentha-2,8-dien-1-ol
 - Araceae [86]
 - p*-Menth-6-en-3,8-diol
 - Zingiberaceae [174]
 - (*E*)-Menthenol
 - Arecaceae [143]

- p*-Menth-1(7)-en-8-ol
 - Zingiberaceae [174]
- Menthol
 - Gentianaceae [45], Lauraceae [74], Magnoliaceae [264], Myrsinaceae [103], Oleaceae [57, 58], Orchidaceae [117], Passifloraceae [157], Rosaceae [43]
- Perilla alcohol
 - Orchidaceae [117]
- α -Phellandren-8-ol
 - Araceae [86]
- β -Phellandren-8-ol
 - Araceae [86]
- (*E*)-Piperitol
 - Araceae [229]
- (*Z*)-Piperitol
 - Araceae [86, 229]
- Terpinen-4-ol (4-Terpineol)
 - Amaryllidaceae [6], Araceae [86, 129], Caprifoliaceae [102], Dipsacaceae [185], Fabaceae [138, 212], Lauraceae [74], Lecythidaceae [144], Nelumbonaceae [178, 184, 198, 199], Nyctaginaceae [151], Orchidaceae [77, 116, 117, 191, 258], Rosaceae [33, 35, 228], Rutaceae [119], Zingiberaceae [268]
- α -Terpineol
 - Agavaceae [178], Amaryllidaceae [6, 107], Araceae [77, 86, 146], Arecaceae [137], Asteraceae [37], Berberidaceae [215], Brassicaceae [186], Cactaceae [122, 125], Calycanthaceae [120], Caprifoliaceae [102], Caricaceae [141], Caryophyllaceae [5, 54, 112, 141], Cucurbitaceae [71], Dipsacaceae [5], Ericaceae [136], Fabaceae [126, 138, 212], Fumariaceae [56], Iridaceae [150, 178], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [11, 240], Myrsinaceae [103], Nelumbonaceae [178, 184, 198, 199], Oleaceae [33, 39, 57, 58], Orchidaceae [77, 96, 116, 117, 141, 192, 201, 253, 255], Primulaceae [36], Rosaceae [228], Rutaceae [2, 119, 247], Solanaceae [131, 141, 165, 166, 167], Theophrastaceae [139], Violaceae [73], Zamiaceae [211], Zingiberaceae [141, 174]
- δ -Terpineol
 - Orchidaceae [117]
- γ -Terpineol
 - Amaryllidaceae [6]
- 4-Terpineol, see Terpinen-4-ol
- Terpin-1-ol hydrate
 - Oleaceae [39]
- Thymol
 - Araceae [229], Lauraceae [74], Oleaceae [33, 51], Rosaceae [43]
- Esters
 - Perillyl acetate
 - Asteraceae [214]
 - Terpinen-4-yl acetate, see 4-Terpinenyl acetate
 - 4-Terpinenyl acetate (Terpinen-4-yl acetate)
 - Orchidaceae [117]
 - α -Terpinyl acetate
 - Caprifoliaceae [102], Caryophyllaceae [5], Lauraceae [74]
- Ethers and epoxides
 - Ascaridol
 - Araceae [86], Asteraceae [214]
 - Carvacrol methyl ether
 - Lauraceae [74]
 - (*E*)-(*E*)-Carveol epoxide
 - Orchidaceae [117, 120]
 - Carvone epoxide, see Carvone oxide
 - Carvone oxide (Carvone epoxide)
 - Euphorbiaceae [7]
 - (*E*)-Carvone oxide (*trans*-Carvone epoxide, see also Carvone oxide)
 - Euphorbiaceae [7, 254], Gesneriaceae [77], Orchidaceae [77, 117, 120, 158, 254]
 - (*Z*)-Carvone epoxide (see also Carvone oxide)
 - Orchidaceae [117, 120]

Cineole

Rosaceae [128]

1,4-Cineole (see also Cineole)

Caryophyllaceae [54], Oleaceae [33]

1,8-Cineole (see also Cineole)

Actinidiaceae [237], Amaryllidaceae [6, 21, 63], Annonaceae [111], Apocynaceae [1, 121, 141], Araceae [77, 146, 152], Arecaceae [68, 137, 143], Asteraceae [34, 37], Bignoniaceae [21], Brassicaceae [22, 70, 105, 186], Bromeliaceae [21, 77], Calycanthaceae [120], Caprifoliaceae [102], Caryophyllaceae [54, 112], Cucurbitaceae [71], Dipsacaceae [5, 185], Euphorbiaceae [254], Fabaceae [212], Gentianaceae [5, 45], Geraniaceae [32], Gesneriaceae [77], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [240, 264], Malvaceae [142], Meliaceae [250], Moraceae [82, 83], Nelumbonaceae [184, 198, 199], Oleaceae [33], Orchidaceae [76, 77, 78, 95, 96, 97, 98, 106, 116, 117, 120, 181, 192, 201, 224, 253, 254, 255, 256, 258, 259], Polemoniaceae [217], Rosaceae [33, 35], Rutaceae [119, 247], Salicaceae [245], Sapindaceae [46], Solanaceae [131, 141, 165, 166, 167, 168, 222], Valerianaceae [5], Violaceae [73], Zingiberaceae [141, 174]

Limonene diepoxide

Nyctaginaceae [151]

(E)-Limonene epoxide, see *(E)*-Limonene oxide*(Z)*-Limonene epoxide, see *(Z)*-Limonene oxide

Limonene oxides

Araceae [229], Orchidaceae [77, 78]

(E)-Limonene oxide (*(E)*-Limonene epoxide, *(E)*-Limonene-1,2-oxide, see also Limonene oxides)

Araceae [86], Euphorbiaceae [95], Orchidaceae [95, 117, 253], Violaceae [248]

(Z)-Limonene oxide (*(Z)*-Limonene epoxide, *(Z)*-Limonene-1,2-oxide, see also Limonene oxides)

Araceae [86], Orchidaceae [95, 117], Violaceae [248]

(E)-Limonene-1,2-oxide, see *(E)*-Limonene oxide*(Z)*-Limonene-1,2-oxide, see *(Z)*-Limonene oxide

Menthofuran

Araceae [86]

Piperitone oxide

Araceae [86]

Terpinolene epoxide

Araceae [229]

 α -Terpinyl methyl ether

Berberidaceae [215]

Thymol methyl ether

Lauraceae [74]

Bicyclo[2.2.1]

Hydrocarbons

Camphene

Amaryllidaceae [63], Apiaceae [31, 242, 246], Araceae [86, 229], Asteraceae [34, 37, 99], Berberidaceae [215], Caryophyllaceae [54], Chloranthaceae [251], Euphorbiaceae [7, 254], Fabaceae [21], Gentianaceae [45], Lauraceae [74], Malvaceae [41, 142], Moraceae [82], Musaceae [21], Nyctaginaceae [151], Oleaceae [33, 51, 150], Orchidaceae [117, 254, 258], Ranunculaceae [85], Rosaceae [12, 33, 35, 40, 128, 219, 220, 228], Rutaceae [172], Salicaceae [245], Sapindaceae [46], Winteraceae [210, 239], Zingiberaceae [174]

 α -Camphene (see also Camphene)

Lamiaceae [261]

 α -Fenchene

Apiaceae [31], Araceae [86], Oleaceae [33], Orchidaceae [180]

Santene

Nyctaginaceae [151]

Ketones

Camphor

Amaryllidaceae [21, 63], Arecaceae [21], Asteraceae [34, 99], Bromeliaceae [21], Caryophyllaceae [54, 112], Gentianaceae [45], Hydnoraceae [48], Lauraceae [74], Malvaceae [41], Meliaceae [250], Musaceae [21], Oleaceae [150], Orchidaceae [224], Polemoniaceae [217], Rosaceae [12, 35, 37, 40], Zingiberaceae [268]

- Fenchone
 - Araceae [86, 229], Magnoliaceae [264], Orchidaceae [117]
- Alcohols
 - Borneol
 - Amaryllidaceae [63], Araceae [86], Asteraceae [34], Calycanthaceae [120], Lauraceae [74], Oleaceae [33], Orchidaceae [27], Pinaceae [25], Primulaceae [36], Rosaceae [43]
 - Fenchol
 - Araceae [86], Fabaceae [212], Myrsinaceae [103]
 - Isoborneol
 - Amaryllidaceae [63]
- Esters
 - Bornyl acetate
 - Amaryllidaceae [6], Araceae [86], Asteraceae [214], Bignoniaceae [21], Calycanthaceae [120], Gentianaceae [45], Lamiaceae [261], Musaceae [21], Oleaceae [171], Orchidaceae [30, 117], Pinaceae [25], Rosaceae [43]
 - Fenchyl acetate
 - Araceae [86, 229], Caryophyllaceae [112]
 - Isobornyl acetate
 - Araceae [229], Caryophyllaceae [54], Lauraceae [74], Orchidaceae [117]
- Ethers and epoxides
 - Camphene epoxide 1
 - Amaryllidaceae [63]
 - Camphene epoxide 2
 - Amaryllidaceae [63]
- Bicyclo[3.1.0]
 - Hydrocarbons
 - Sabinene
 - Agavaceae [142], Amaryllidaceae [6, 21], Annonaceae [111], Apiaceae [31, 217, 242, 246], Apocynaceae [121, 141, 232, 233], Araceae [86, 129, 133, 146, 229], Arecaceae [137, 143], Asteraceae [34, 36, 37], Berberidaceae [215], Bignoniaceae [142], Brassicaceae [70, 105, 142, 186], Calycanthaceae [120], Caprifoliaceae [102], Chloranthaceae [251], Cucurbitaceae [71], Cycadaceae [211], Dipsacaceae [5], Fabaceae [62, 81, 138, 212, 216], Geraniaceae [32], Gesneriaceae [77], Grossulariaceae [89], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [9, 11, 264], Malvaceae [41, 142], Moraceae [82, 83], Myrsinaceae [103], Nelumbonaceae [178, 184, 198, 199], Nyctaginaceae [151], Oleaceae [33, 119], Orchidaceae [17, 77, 78, 117, 120, 147, 192, 195, 201, 253], Primulaceae [5], Ranunculaceae [85], Rosaceae [33, 35, 228], Rubiaceae [141], Rutaceae [119, 247], Salicaceae [245], Solanaceae [141, 165, 166, 167, 168], Theophrastaceae [139], Verbenaceae [5], Violaceae [73], Zingiberaceae [141, 174]
 - α -Thujene
 - Agavaceae [142], Araceae [77, 86], Arecaceae [137, 143], Berberidaceae [215], Bignoniaceae [21, 142], Brassicaceae [105, 142, 221], Calycanthaceae [120], Chloranthaceae [251], Cucurbitaceae [71], Fabaceae [21, 81, 126, 212, 216], Lauraceae [74], Lecythidaceae [144], Malvaceae [142], Moraceae [82], Musaceae [21], Orchidaceae [77, 117, 201, 253], Primulaceae [5], Solanaceae [131, 165, 166], Verbenaceae [5], Zingiberaceae [268]
 - β -Thujene
 - Calycanthaceae [266], Oleaceae [51]
- Ketones
 - Sabina ketone
 - Calycanthaceae [266]
 - 3-Thujen-2-one
 - Lauraceae [74]
 - α -Thujone
 - Cucurbitaceae [71], Fabaceae [126]
- Alcohols
 - Sabinene hydrate
 - Asteraceae [213], Orchidaceae [116], Rosaceae [51]
 - (*E*)-Sabinene hydrate (*trans*-4-Thujanol, *trans*-Thujanol, see also Sabinene hydrate)
 - Araceae [86, 129], Arecaceae [137], Calycanthaceae [120], Caprifoliaceae [102], Lauraceae [74], Orchidaceae [30, 117], Passifloraceae [157], Rosaceae [33, 35], Rutaceae [119], Solanaceae [131], Theophrastaceae [139], Verbenaceae [5]

(Z)-Sabinene hydrate (*cis*-4-Thujanol, see also Sabinene hydrate)

Araceae [86, 146], Arecaceae [137], Cucurbitaceae [71], Lauraceae [74], Theophrastaceae [139], Violaceae [73]

Sabinol

Araceae [86]

trans-Thujanol, see (*E*)-Sabinene hydrate

cis-4-Thujanol, see (Z)-Sabinene hydrate

trans-4-Thujanol, see (*E*)-Sabinene hydrate

Thujol

Nyctaginaceae [151]

Thujylalcohol

Orchidaceae [254]

Bicyclo[3.1.1]

Hydrocarbons

α -Pinene

Agavaceae [142], Alliaceae [36], Amaryllidaceae [6, 21, 63], Annonaceae [111], Apiaceae [31, 242, 246], Apocynaceae [1], Araceae [86, 129, 133, 146, 205, 229], Arecaceae [21, 137, 143], Asteraceae [5, 34, 36, 37, 59, 65, 99], Berberidaceae [182], Bignoniaceae [21, 142], Brassicaceae [22, 70, 105, 142, 186, 217, 221], Bromeliaceae [21], Calycanthaceae [120], Caprifoliaceae [102], Caryophyllaceae [5, 54, 66, 112], Chloranthaceae [251], Cucurbitaceae [71], Dipsacaceae [5, 185], Euphorbiaceae [7, 254], Fabaceae [21, 81, 126, 138, 212, 216, 235], Fumariaceae [197], Gentianaceae [45], Geraniaceae [32], Gesneriaceae [77], Grossulariaceae [89], Hemerocallidaceae [33], Lamiaceae [261], Lauraceae [74], Lecythidaceae [144], Liliaceae [92], Magnoliaceae [9, 11, 240], Malvaceae [94, 142], Meliaceae [250], Moraceae [82, 83], Musaceae [21], Nelumbonaceae [184], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [33, 51, 119, 150], Onagraceae [218], Orchidaceae [5, 17, 23, 27, 28, 29, 76, 77, 78, 80, 95, 96, 97, 98, 116, 117, 120, 141, 147, 169, 180, 181, 189, 190, 192, 195, 201, 204, 224, 244, 253, 254, 255, 256, 258, 259], Orobanchaceae [20], Passifloraceae [157], Pinaceae [25], Pittosporaceae [107], Polemoniaceae [21, 217], Primulaceae [36], Ranunculaceae [85, 209], Rosaceae [12, 33, 35, 40, 43, 128, 219, 220, 228], Rutaceae [2, 119, 172], Salicaceae [245], Sapindaceae [46], Scrophulariaceae [226], Solanaceae [94, 131, 222], Valerianaceae [37], Verbenaceae [5], Winteraceae [210, 239], Violaceae [73], Zamiaceae [211]

β -Pinene

Agavaceae [142], Alliaceae [36], Amaryllidaceae [6, 63], Annonaceae [111], Apiaceae [31, 217, 242, 246], Apocynaceae [1], Araceae [77, 86, 129, 133, 146, 205, 229], Arecaceae [137, 143], Asteraceae [34, 36, 37, 59, 65], Bignoniaceae [21, 142], Brassicaceae [70, 105, 142, 186], Bromeliaceae [21], Calycanthaceae [120], Caryophyllaceae [54, 112], Chloranthaceae [251], Cucurbitaceae [71], Dipsacaceae [5], Euphorbiaceae [7], Fabaceae [21, 138, 212], Gentianaceae [45], Geraniaceae [32], Gesneriaceae [77], Grossulariaceae [89], Hemerocallidaceae [33], Lauraceae [74], Lecythidaceae [144], Liliaceae [92], Magnoliaceae [11, 238, 240, 264], Malvaceae [142], Meliaceae [250], Moraceae [82], Musaceae [21], Myrsinaceae [103], Nelumbonaceae [184], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [33, 119, 150, 267], Orchidaceae [13, 17, 23, 27, 28, 29, 76, 77, 78, 80, 95, 96, 97, 98, 116, 117, 120, 147, 180, 192, 201, 203, 204, 253, 255, 256, 258, 259], Pinaceae [25], Polemoniaceae [5, 21], Primulaceae [5, 36], Ranunculaceae [85, 209], Rosaceae [12, 33, 35, 40, 72, 128, 196], Rutaceae [2, 119, 172], Salicaceae [245], Sapindaceae [46], Solanaceae [131, 165, 222], Theophrastaceae [139], Valerianaceae [37], Verbenaceae [5], Winteraceae [210, 239], Violaceae [73], Zamiaceae [211]

Verbenene

Araceae [86]

Aldehydes

(*E*)-Myrtanal

Hydrangeaceae [108]

(Z)-Myrtanal

Hydrangeaceae [108], Orchidaceae [117]

Myrtanal

Hydrangeaceae [108], Magnoliaceae [108], Orchidaceae [117]

Ketones

Chrysanthenone

- Dipsacaceae [5], Magnoliaceae [108]
- Isopinocampnone
 - Chloranthaceae [251], Lauraceae [74], Magnoliaceae [11, 108, 240]
- 3-Methylnopinone
 - Dipsacaceae [185]
- Nopinone
 - Hydrangeaceae [108]
- Pinocampnone
 - Calycanthaceae [266]
- Pinocarvone
 - Amaryllidaceae [63], Araceae [229], Asteraceae [37], Oleaceae [33]
- 3,6,6-Trimethyl-bicyclo[3.1.1]heptan-2-one
 - Dipsacaceae [5], Primulaceae [5]
- 4,6,6-Trimethyl-bicyclo[3.1.1]hept-3-enone, see Verbenone
- (Z)-Verbanone
 - Magnoliaceae [108]
- Verbenone (4,6,6-Trimethyl-bicyclo[3.1.1]hept-3-enone)
 - Araceae [262], Asteraceae [214], Dipsacaceae [5, 185], Lecythidaceae [144], Magnoliaceae [11, 108, 240, 264], Nyctaginaceae [151], Oleaceae [57, 58], Orchidaceae [5, 27, 116, 117, 204], Polemoniaceae [217], Scrophulariaceae [5]
- Alcohols
 - (E)-Myrtanol
 - Orchidaceae [117]
 - (Z)-Myrtanol
 - Orchidaceae [117]
 - Myrtenol
 - Araceae [229], Asteraceae [37], Hydrangeaceae [108], Lecythidaceae [144], Magnoliaceae [108], Oleaceae [33], Orchidaceae [117]
 - 3-Pinen-2-ol
 - Orchidaceae [28]
 - (E)-Pinocarveol
 - Asteraceae [37], Lecythidaceae [144], Orchidaceae [117]
 - (Z)-Pinocarveol
 - Orchidaceae [117]
 - (E)-Verbenol
 - Araceae [86], Magnoliaceae [108], Orchidaceae [117]
 - (Z)-Verbenol
 - Orchidaceae [117]
- Esters
 - Myrtenyl 2-methylbutanoate
 - Asteraceae [37]
 - Myrtenyl 2-methylpropanoate
 - Asteraceae [37]
- Ethers and epoxides
 - α -Pinene epoxides, see α -Pinene oxide
 - α -Pinene oxide (α -Pinene epoxide)
 - Nyctaginaceae [151], Orchidaceae [253]
 - (E)-Verbenone epoxide
 - Orchidaceae [117]
- Bicyclo[4.1.0]
 - Hydrocarbons
 - δ -Carene, see 3-Carene
 - δ -3-Carene, see 3-Carene
 - 2-Carene
 - Nyctaginaceae [151]
 - 3-Carene (δ -3-Carene, δ -Carene)
 - Amaryllidaceae [21], Apiaceae [31, 217, 242, 246], Araceae [26, 86, 205], Asteraceae [59], Brassicaceae [105, 221], Caryophyllaceae [5, 54], Fabaceae [126], Gentianaceae [45], Grossulariaceae [89], Lauraceae [74], Lecythidaceae [144], Malvaceae [41], Moraceae [82], Orchidaceae [17, 23, 27, 77, 180, 191, 192, 195, 258], Pinaceae [25], Primulaceae [5], Ranunculaceae [208, 209], Rosaceae [43, 220], Winteraceae [210, 239], Zingiber-

aceae [268], Amaryllidaceae [21], Araceae [86], Brassicaceae [221], Caryophyllaceae [54], Malvaceae [41], Moraceae [82]

Alcohols

(+)-(E)-3,7,7-Trimethylbicyclo[4.1.0]heptan-5-ol
Oleaceae [51]

Tricyclic

Hydrocarbons

Tricyclene

Fabaceae [21], Orchidaceae [192], Violaceae [73]

SESQUITERPENES

Acyclic

Hydrocarbons

Farnesene

Rosaceae [128]

α -Farnesene (see also Farnesene)

Actinidiaceae [237], Amaryllidaceae [63], Apiaceae [31, 246], Apocynaceae [141, 173, 178], Araceae [77, 133], Acanthaceae [10], Brassicaceae [70, 105], Cactaceae [122], Caprifoliaceae [102, 107], Caricaceae [141], Caryophyllaceae [112, 141], Chloranthaceae [251], Combretaceae [10], Fabaceae [62, 81, 107, 179], Geraniaceae [32], Hyacinthaceae [123, 150], Lecythidaceae [144], Liliaceae [92], Lythraceae [10], Magnoliaceae [9, 11, 240], Myrsinaceae [103], Nyctaginaceae [151], Oleaceae [51, 114], Orchidaceae [13, 17, 23, 27, 28, 30, 77, 141, 180, 188, 191, 203, 236, 253], Orobanchaceae [141], Primulaceae [190], Ranunculaceae [19, 207, 208, 209], Rhizophoraceae [10], Rosaceae [60, 62, 176, 177, 178, 219, 220], Rubiaceae [141, 249], Rutaceae [247], Salicaceae [245], Solanaceae [131, 154], Winteraceae [239], Vitaceae [42, 44, 47], Zingiberaceae [141]

(E)- α -Farnesene (see also α -Farnesene)

Lauraceae [74], Meliaceae [250], Orchidaceae [77, 117], Passifloraceae [157]

(Z)- α -Farnesene (see also α -Farnesene)

Arecaceae [68, 143], Lauraceae [74], Meliaceae [250], Orchidaceae [77, 117], Passifloraceae [157]

(E,E)- α -Farnesene (see also α -Farnesene)

Amaryllidaceae [232, 233], Arecaceae [137, 143, 145], Asteraceae [5], Berberidaceae [182], Brassicaceae [22, 36], Cactaceae [113, 125], Hyacinthaceae [38], Lauraceae [74], Lecythidaceae [144], Moraceae [83], Orchidaceae [77, 116, 117, 120, 201], Passifloraceae [157], Primulaceae [36], Rubiaceae [5], Rutaceae [119], Sapotaceae [263], Scrophulariaceae [5, 226], Solanaceae [167], Thymelaeaceae [5], Valerianaceae [37], Verbenaceae [5], Zingiberaceae [174]

(E,Z)- α -Farnesene (see also α -Farnesene)

Arecaceae [137]

(Z,E)- α -Farnesene (see also α -Farnesene)

Arecaceae [145], Asteraceae [5], Brassicaceae [36], Cactaceae [113, 115, 125], Hyacinthaceae [38], Orchidaceae [116], Rubiaceae [5], Scrophulariaceae [5], Thymelaeaceae [5], Valerianaceae [37]

(Z,Z)- α -Farnesene (see also α -Farnesene)

Fabaceae [126]

β -Farnesene

Araceae [86], Cactaceae [113, 122], Fabaceae [49], Malvaceae [41], Myrsinaceae [103], Nyctaginaceae [151], Oleaceae [51, 267], Orchidaceae [77, 191, 201], Rosaceae [40], Rubiaceae [249], Rutaceae [247], Vitaceae [42, 44, 47]

(E)- β -Farnesene (see also β -Farnesene)

Apiaceae [31], Arecaceae [137, 143, 145], Asteraceae [5, 37], Berberidaceae [215], Cactaceae [125], Caprifoliaceae [102], Fabaceae [62, 120, 156], Geraniaceae [32], Hyacinthaceae [38], Lecythidaceae [144], Orchidaceae [17, 23, 27, 77, 116, 117, 118, 120, 192], Passifloraceae [157], Polemoniaceae [5], Solanaceae [165, 167], Valerianaceae [37], Verbenaceae [5], Violaceae [73]

(Z)- β -Farnesene (see also β -Farnesene)

Araceae [86], Chloranthaceae [251], Fabaceae [126], Geraniaceae [32], Lecythidaceae [144], Meliaceae [250], Orchidaceae [78]

3,7,11-Trimethyl-1,3,6,11-dodecatetraene

Zingiberaceae [268]

- 3,7,11-Trimethyl-1,6,10-dodecatriene
Araceae [86]
- Aldehydes
(*E*)-2(3)-Dihydrofarnesal
Arecaceae [119], Commelianaceae [121], Orchidaceae [119], Rutaceae [119]
Farnesal
Zingiberaceae [174]
(*E,E*)-Farnesal (see also Farnesal)
Cactaceae [125], Fabaceae [120], Orchidaceae [116, 117], Rutaceae [119]
(*Z,E*)-Farnesal (see also Farnesal)
Cactaceae [125], Orchidaceae [117]
(*E*)- β -Sinensal
Arecaceae [120]
(*Z*)- β -Sinensal
Arecaceae [120]
- Alcohols
Dihydrofarnesol
Orchidaceae [23]
2,3-Dihydrofarnesol (see also Dihydrofarnesol)
Myrsinaceae [103], Ruscaceae [232, 233]
(*E*)-2(3)-Dihydrofarnesol
Commelianaceae [121], Orchidaceae [117, 120], Rutaceae [119]
Farnesol
Malvaceae [41], Nyctaginaceae [151], Orchidaceae [120], Rosaceae [128], Ruscaceae [232, 233]
(*E,E*)-Farnesol (see also Farnesol)
Apocynaceae [141], Cactaceae [125], Caprifoliaceae [102], Fabaceae [120], Lecythidaceae [121, 144], Myrsinaceae [103], Oleaceae [52, 267], Orchidaceae [29, 77, 116, 117, 118, 141, 169, 201], Orobanchaceae [141], Passifloraceae [157], Polemoniaceae [5], Primulaceae [5], Rubiaceae [141], Rutaceae [119], Scrophulariaceae [5], Solanaceae [131], Theophrastaceae [139]
(*E,Z*)-Farnesol (see also Farnesol)
Solanaceae [131]
(*Z,E*)-Farnesol (see also Farnesol)
Cactaceae [125], Lecythidaceae [144], Myrsinaceae [103], Orchidaceae [8, 77, 117, 141], Rutaceae [119], Solanaceae [131]
(*Z,Z*)-Farnesol (see also Farnesol)
Apocynaceae [141], Lecythidaceae [144], Orchidaceae [141], Orobanchaceae [141], Theophrastaceae [139]
- Nerolidol
Amaryllidaceae [175], Apocynaceae [141], Cactaceae [113, 115], Caprifoliaceae [107, 175], Chloranthaceae [251], Cucurbitaceae [4], Fabaceae [107, 119], Gentianaceae [45], Lecythidaceae [144], Malvaceae [41], Nyctaginaceae [151], Orchidaceae [13, 78, 116, 201], Orobanchaceae [141], Rubiaceae [141], Rutaceae [2, 119, 173, 247], Sapotaceae [263], Solanaceae [141, 165], Vitaceae [42, 44, 47], Zingiberaceae [141]
(*E*)-Nerolidol (see also Nerolidol)
Araceae [86], Arecaceae [137, 145], Berberidaceae [182], Cactaceae [125], Caprifoliaceae [102], Fabaceae [126], Hemerocallidaceae [33], Orchidaceae [8, 77, 116, 117, 120, 169], Passifloraceae [157], Polemoniaceae [5], Rubiaceae [5], Solanaceae [131], Verbenaceae [5]
(*Z*)-Nerolidol (see also Nerolidol)
Polemoniaceae [5]
(*E*)- β -Sinensol
Arecaceae [120]
- Esters
(*E*)-2(3)-Dihydrofarnesyl acetate
Orchidaceae [117]
Farnesyl acetate
Lecythidaceae [144], Malvaceae [41]
(*E*)-Farnesyl acetate (see also Farnesyl acetate)
Caprifoliaceae [102]

- (*E,E*)-Farnesyl acetate (see also Farnesyl acetate)
 Fabaceae [120], Oleaceae [52], Orchidaceae [117, 118, 120], Passifloraceae [157]
- (*Z,E*)-Farnesyl acetate (see also Farnesyl acetate)
 Orchidaceae [117]
- Farnesyl hexanoate
 Orchidaceae [223]
- Farnesyl octanoate
 Orchidaceae [29]
- Methyl (*E,E*)-3,7,11-trimethyl-2,6,10-dodecatrienoate
 Orchidaceae [116, 117]
- Ethers and epoxides
- Dendrolasin
 Calycanthaceae [120], Moraceae [83], Orchidaceae [191], Zingiberaceae [268]
- (*E,E*)-2(3)-Epoxy-2,6,10-trimethyl-6,9,11-dodecatriene, see (*E,E*)- α -Farnesene epoxide
- (*E,E*)- α -Farnesene epoxide ((*E,E*)-2(3)-Epoxy-2,6,10-trimethyl-6,9,11-dodecatriene)
 Orchidaceae [78, 117, 120]
- (*Z*)- α -Farnesene epoxide
 Orchidaceae [120]
- (*E*)- β -Farnesene epoxide
 Arecaceae [145]
- Nerolidyl methyl ether
 Cactaceae [113, 115, 125]
- Cyclic
- Hydrocarbons
- Albene
 Asteraceae [36]
- α -Amorphene
 Araceae [86]
- Aristolene
 Araceae [86], Nyctaginaceae [151]
- Aromadendrene
 Araceae [86, 129], Geraniaceae [32]
- allo*-Aromadendrene
 Araceae [86, 129, 229], Arecaceae [68, 137, 143], Asteraceae [34], Chloranthaceae [251],
 Fabaceae [62, 104], Geraniaceae [32], Lecythidaceae [144], Malvaceae [142], Meliaceae
 [250], Moraceae [83], Orchidaceae [77, 117], Polemoniaceae [5], Rosaceae [51]
- α -Bergamotene
 Apiaceae [31], Geraniaceae [32], Nyctaginaceae [151], Orchidaceae [77, 117, 190, 191, 192,
 201], Ranunculaceae [59]
- (*E*)- α -Bergamotene (see also α -Bergamotene)
 Araceae [86], Arecaceae [137], Cactaceae [125], Fabaceae [62, 138, 216], Lecythidaceae
 [144], Moraceae [83], Orchidaceae [77, 78, 117, 120], Ranunculaceae [19], Rubiaceae
 [5], Solanaceae [131], Violaceae [73]
- (*Z*)- α -Bergamotene (see also α -Bergamotene)
 Fabaceae [216], Geraniaceae [32], Lecythidaceae [144], Orchidaceae [253], Ranunculaceae
 [19], Verbenaceae [5]
- α -(*E*)- β -Bergamotene
 Geraniaceae [32]
- (*Z*)- α -*cis*-Bergamotene
 Araceae [129]
- Bicyclogermacrene
 Araceae [86, 133], Arecaceae [143], Asteraceae [34, 37], Cactaceae [125], Geraniaceae [32],
 Lauraceae [74], Magnoliaceae [11], Moraceae [83], Oleaceae [114], Orchidaceae [117],
 Rutaceae [172]
- Bicyclosquiphellandrene
 Fabaceae [216]
- β -Bisabolene
 Amaryllidaceae [63], Arecaceae [137], Oleaceae [51], Orchidaceae [77, 116, 117, 120, 180,
 190, 201, 253], Vitaceae [42, 44, 47]
- Bourbonene
 Rosaceae [72]

β -Bourbonene

Apiaceae [242], Arecaceae [137], Asteraceae [5], Caprifoliaceae [102], Caryophyllaceae [112], Dipsacaceae [5], Fabaceae [104], Gentianaceae [5], Geraniaceae [32], Lecythidaceae [144], Malvaceae [41], Nyctaginaceae [151], Oleaceae [150], Orchidaceae [117], Passifloraceae [157], Rosaceae [219, 220], Rubiaceae [5], Verbenaceae [5]

 β -Bourbonene isomer

Rubiaceae [5]

Cadalene

Orchidaceae [201], Rosaceae [37]

Cadina-1,4-diene

Araceae [86], Meliaceae [250]

Cadinene

Moraceae [82], Orchidaceae [148, 149], Ranunculaceae [59], Rosaceae [219, 220]

 α -Cadinene (see also Cadinene)

Araceae [86]

 β -Cadinene (see also Cadinene)

Nyctaginaceae [151]

 δ -Cadinene (see also Cadinene)

Annonaceae [111], Araceae [86, 129], Arecaceae [68, 143], Bignoniaceae [21], Caprifoliaceae [102], Caryophyllaceae [112], Fabaceae [104, 138], Gentianaceae [5], Geraniaceae [32], Lamiaceae [261], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [240], Moraceae [83], Nelumbonaceae [184], Orchidaceae [117], Passifloraceae [157], Polemoniaceae [5], Rosaceae [51], Rubiaceae [5], Verbenaceae [5]

 γ -Cadinene (see also Cadinene)

Annonaceae [111], Apiaceae [31], Araceae [86], Arecaceae [137], Caryophyllaceae [112], Geraniaceae [32], Magnoliaceae [240, 264], Malvaceae [41], Nyctaginaceae [151], Oleaceae [267], Orchidaceae [23, 27, 28, 147], Rosaceae [40], Verbenaceae [5], Vitaceae [42, 44, 47]

(E)- γ -Cadinene (see also Cadinene)

Lauraceae [74]

(Z)- γ -Cadinene (see also Cadinene)

Lauraceae [74]

 α -Calacorene

Araceae [86], Fabaceae [138]

Calamenene

Araceae [86, 229], Nyctaginaceae [151]

1*S*-*cis*-Calamenene (see also Calamenene)

Arecaceae [137], Lecythidaceae [144], Rubiaceae [5], Verbenaceae [5]

Calarene, see β -GurjuneneCaryophyllene (β -Caryophyllene)

Annonaceae [111], Apiaceae [31, 242], Apocynaceae [141], Araceae [26, 86, 129, 229], Arecaceae [10, 68, 137, 143], Asteraceae [5, 37, 59, 132], Bignoniaceae [21], Brassicaceae [36, 70, 217], Cactaceae [125], Calycanthaceae [120, 266], Caricaceae [141], Caryophyllaceae [5, 54, 66, 112], Chloranthaceae [251], Ericaceae [136], Fabaceae [21, 49, 50, 62, 81, 104, 107, 121, 126, 156, 193, 216], Gentianaceae [45], Geraniaceae [32], Hyacinthaceae [38], Lamiaceae [261], Lauraceae [74], Lecythidaceae [121, 144], Magnoliaceae [11, 264], Malvaceae [142], Moraceae [82, 83], Myrsinaceae [103], Nelumbonaceae [198, 199], Nyctaginaceae [151], Oleaceae [33, 52, 267], Orchidaceae [2, 13, 17, 23, 27, 28, 29, 77, 80, 100, 116, 117, 147, 187, 190, 192, 193, 195, 201, 244, 253, 258], Papaveraceae [62], Passifloraceae [157], Polemoniaceae [5, 21, 142], Primulaceae [36], Ranunculaceae [19, 85], Rosaceae [33, 35, 40, 51, 62, 72, 128, 219, 220], Rubiaceae [5, 141, 228], Rutaceae [119, 172, 247], Salicaceae [245], Sapindaceae [46], Solanaceae [165, 166, 167, 168], Theophrastaceae [139], Valerianaceae [5], Verbenaceae [5], Winteraceae [239], Vitaceae [42, 44, 47], Zamiaceae [211], Zingiberaceae [141]

 α -Caryophyllene, see Humulene β -Caryophyllene, see Caryophyllene*(E)*-Caryophyllene (*(E)*- β -Caryophyllene)

Nelumbonaceae [184], Rosaceae [93]

(Z)-Caryophyllene

Moraceae [82], Oleaceae [267]

(E)- β -Caryophyllene, see *(E)*-Caryophyllene

α -Cedrene

Brassicaceae [70], Caryophyllaceae [54], Orchidaceae [192], Verbenaceae [5], Zingiberaceae [268]

 β -Cedrene

Arecaceae [137], Asteraceae [5], Caryophyllaceae [112], Magnoliaceae [240], Rubiaceae [5], Solanaceae [131]

Chamigrene

Nyctaginaceae [151]

Clovene

Moraceae [82]

Copaene

Rubiaceae [141]

 α -Copaene (see also Copaene)

Apiaceae [31, 246], Araceae [86, 129, 229], Arecaceae [21, 68, 137, 143], Asteraceae [5], Bignoniaceae [21], Bromeliaceae [21], Cactaceae [125], Caprifoliaceae [102], Caryophyllaceae [112], Chloranthaceae [251], Fabaceae [49, 62, 138, 156], Gentianaceae [5], Lauraceae [74], Lecythidaceae [144], Malvaceae [41, 142], Moraceae [82, 83], Musaceae [21], Myrsinaceae [103], Nyctaginaceae [151], Orchidaceae [17, 23, 27, 29, 77, 116, 117, 147, 148, 149], Passifloraceae [157], Polemoniaceae [5], Rosaceae [33, 35, 40, 62, 219, 220], Rubiaceae [5], Verbenaceae [5]

 β -Copaene (see also Copaene)

Annonaceae [111], Calycanthaceae [266], Moraceae [83], Orchidaceae [117]

Cubebene

Moraceae [82]

 α -Cubebene (see also Cubebene)

Apocynaceae [121], Araceae [229], Arecaceae [68, 143], Cactaceae [125], Fabaceae [62, 120], Magnoliaceae [264], Moraceae [83], Myrsinaceae [103], Nyctaginaceae [151], Orchidaceae [117], Polemoniaceae [5], Rosaceae [40, 72], Theophrastaceae [139], Verbenaceae [5, 175]

 β -Cubebene (see also Cubebene)

Apiaceae [31], Araceae [229], Arecaceae [68, 137, 143], Cactaceae [125], Fabaceae [62, 104, 216], Geraniaceae [32], Lauraceae [74], Lecythidaceae [144], Magnoliaceae [240], Moraceae [83], Nelumbonaceae [184], Nyctaginaceae [151], Orchidaceae [117], Passifloraceae [157], Polemoniaceae [5], Rosaceae [40, 93, 220], Verbenaceae [5]

ar-Curcumen

Asteraceae [34], Geraniaceae [32], Orchidaceae [77], Verbenaceae [5]

 α -Curcumen

Apiaceae [246], Fabaceae [62], Orchidaceae [201]

T-Curcumen

Asteraceae [34]

Cyclosativene

Apiaceae [31], Moraceae [82, 83], Orchidaceae [23, 27, 28, 29, 30]

Cyperene

Arecaceae [137]

 α -Elemene

Rosaceae [176, 177, 178]

 β -Elemene

Apiaceae [31], Araceae [86], Arecaceae [137], Asteraceae [37], Brassicaceae [70], Cactaceae [125], Calycanthaceae [120], Caprifoliaceae [102], Fabaceae [62, 107], Lauraceae [74], Magnoliaceae [240, 264], Malvaceae [41], Moraceae [82], Myrsinaceae [103], Nyctaginaceae [151], Orchidaceae [117], Rutaceae [247], Solanaceae [131], Verbenaceae [5]

 γ -Elemene

Apiaceae [31], Chloranthaceae [251], Fabaceae [62, 138], Magnoliaceae [264], Polemoniaceae [5], Rubiaceae [5], Verbenaceae [5]

 δ -Elemene

Araceae [86], Geraniaceae [32], Lauraceae [74], Nyctaginaceae [151], Verbenaceae [5]

Germacrene A

Magnoliaceae [264], Moraceae [83], Orchidaceae [116, 117]

Germacrene B

Araceae [129], Arecaceae [137], Geraniaceae [32], Lecythidaceae [144], Nelumbonaceae [184], Passifloraceae [157]

Germacrene D

- Apiaceae [242], Apocynaceae [121], Araceae [86, 129, 229], Arecaceae [68, 137, 143], Asteraceae [5, 34, 36], Cactaceae [125], Calycanthaceae [120], Caprifoliaceae [102, 107, 150, 175], Caricaceae [141], Caryophyllaceae [5], Fabaceae [62, 138], Geraniaceae [32], Lauraceae [74], Lecythidaceae [121, 144], Magnoliaceae [264], Malvaceae [41], Moraceae [83], Nelumbonaceae [184], Nyctaginaceae [151], Oleaceae [114], Orchidaceae [77, 116, 117, 121], Passifloraceae [157], Polemoniaceae [5, 233], Primulaceae [36], Ranunculaceae [19], Rosaceae [33, 35, 72, 87, 88, 115, 228], Rubiaceae [5, 121, 141], Scrophulariaceae [5], Solanaceae [222], Verbenaceae [5]
- 6,9-Guaiadiene
Geraniaceae [32]
- α -Guaiene
Araceae [86, 129], Chloranthaceae [251], Lauraceae [74], Rubiaceae [249], Vitaceae [42, 44, 47]
- δ -Guaiene
Nyctaginaceae [151], Vitaceae [42, 44, 47]
- α -Gurjunene
Araceae [86, 129], Arecaceae [137], Asteraceae [5], Fabaceae [179], Lecythidaceae [144], Nyctaginaceae [151], Orchidaceae [117]
- β -Gurjunene (Calarene)
Araceae [86], Caprifoliaceae [102], Chloranthaceae [251], Meliaceae [250], Passifloraceae [157], Polemoniaceae [5], Solanaceae [165]
- γ -Gurjunene
Annonaceae [111], Arecaceae [137], Asteraceae [5], Magnoliaceae [240]
- β -Himachalene
Orchidaceae [192]
- Humulene (α -Caryophyllene, α -Humulene)
Annonaceae [111], Apiaceae [242], Araceae [26, 86, 129, 229], Arecaceae [10, 68, 143], Asteraceae [37], Cactaceae [125], Calycanthaceae [120], Caricaceae [141], Caryophyllaceae [54, 66, 112], Chloranthaceae [251], Cucurbitaceae [71], Fabaceae [21, 62, 81, 121, 216], Geraniaceae [32], Lauraceae [74], Lecythidaceae [121, 144], Magnoliaceae [11, 264], Moraceae [82, 83], Myrsinaceae [103], Nelumbonaceae [184], Nyctaginaceae [151], Oleaceae [267], Orchidaceae [13, 17, 28, 77, 116, 117, 147], Papaveraceae [62], Passifloraceae [157], Polemoniaceae [5], Ranunculaceae [85], Rosaceae [33, 35, 43, 176, 178], Rubiaceae [5], Rutaceae [172], Solanaceae [165], Theophrastaceae [139], Verbenaceae [5], Winteraceae [239], Vitaceae [42, 44, 47], Zamiaceae [211]
- α -Humulene, see Humulene
- Isocaryophyllene
Araceae [86], Lecythidaceae [121], Orchidaceae [77, 117], Rosaceae [177]
- Junipene
Araceae [86], Nyctaginaceae [151]
- Ledene, see Viridiflorene
- Lepidozene
Moraceae [82]
- Longicyclene
Orchidaceae [148, 149]
- Longifolene
Amaryllidaceae [21, 63], Araceae [129], Musaceae [21], Rosaceae [51]
- Longipinene
Asteraceae [34]
- (Z)-Muurolo-4(14),5-diene
Lauraceae [74]
- Muurolole isomer
Orchidaceae [23, 147]
- α -Muurolole
Annonaceae [111], Apiaceae [31], Araceae [86], Caryophyllaceae [112], Geraniaceae [32], Lecythidaceae [144], Moraceae [83], Nyctaginaceae [151], Rosaceae [33, 35, 87, 219, 220], Rubiaceae [5], Verbenaceae [5]
- γ -Muurolole
Annonaceae [111], Araceae [86, 229], Arecaceae [137], Caryophyllaceae [112], Fabaceae [49, 156], Geraniaceae [32], Lauraceae [74], Lecythidaceae [144], Rosaceae [72], Rubiaceae [5, 249]
- β -Patchoulene

- Lauraceae [74]
- α -Santalene
 - Asteraceae [36], Caryophyllaceae [54], Chloranthaceae [251], Orchidaceae [23, 27, 120]
- β -Santalene
 - Asteraceae [36], Orchidaceae [27, 28, 29, 120]
- epi*- β -Santalene
 - Asteraceae [36]
- α -Selinene
 - Arecaceae [137], Orchidaceae [117]
- β -Selinene
 - Araceae [86], Arecaceae [137], Asteraceae [37], Nyctaginaceae [151], Orchidaceae [190], Polemoniaceae [5], Ranunculaceae [19], Theophrastaceae [139], Vitaceae [42, 44, 47]
- δ -Selinene
 - Vitaceae [42, 44, 47]
- β -Sesquiphellandrene
 - Arecaceae [137, 145], Fabaceae [138, 216], Geraniaceae [32]
- Thujopsene
 - Fabaceae [104], Orchidaceae [201], Verbenaceae [5]
- Valencene
 - Annonaceae [111], Orchidaceae [77], Vitaceae [42, 44, 47]
- Viridiflorene (Ledene)
 - Araceae [86, 129], Lecythidaceae [144]
- α -Ylangene
 - Arecaceae [137], Fabaceae [62, 138], Lauraceae [74], Lecythidaceae [144]
- Zingiberene
 - Orchidaceae [201]
- α -Zingiberene (see also Zingiberene)
 - Geraniaceae [32], Verbenaceae [5]
- Aldehydes
 - (*Z*)- α -*trans*-Bergamotenal
 - Orchidaceae [117]
- Ketones
 - Germacrone
 - Nyctaginaceae [151]
 - Longiverbenone
 - Asteraceae [34]
- Alcohols
 - (*Z*)- α -*trans*-Bergamotol
 - Orchidaceae [117]
 - β -Bisabolol
 - Fumariaceae [56]
 - α -Cadinol
 - Oleaceae [114], Zingiberaceae [174]
 - δ -Cadinol
 - Asteraceae [214], Nyctaginaceae [151]
 - T-Cadinol
 - Oleaceae [114], Primulaceae [5], Verbenaceae [5]
 - Carotol
 - Nyctaginaceae [151]
 - Caryophylladienol
 - Lecythidaceae [121]
 - Caryophyll-5-en-2- α -ol
 - Lecythidaceae [121], Orchidaceae [116, 117]
 - Cubenol
 - Orchidaceae [23, 28, 147]
 - Elemol
 - Caprifoliaceae [102], Myrsinaceae [103], Passifloraceae [157], Rosaceae [51]
 - α,β -Eudesmol
 - Nyctaginaceae [151]
 - β -Eudesmol
 - Myrsinaceae [103]

- Germacra-1,6-dien-5-ol
 - Asteraceae [36], Lecythidaceae [144], Oleaceae [114], Passifloraceae [157]
- Germacra-1(10),5-dien-4-ol, see Germacrene-D-4-ol
- Germacrene-D-4-ol (Germacra-1(10),5-dien-4-ol)
 - Asteraceae [65], Cactaceae [125], Lauraceae [74], Orchidaceae [77, 117], Polemoniaceae [5]
- Guaiol
 - Nyctaginaceae [151]
- T-Murolol
 - Fabaceae [138], Oleaceae [114], Orchidaceae [23, 28, 147]
- Sesqui-1,8-cineole
 - Orchidaceae [120]
- Spathulenol
 - Nyctaginaceae [151], Orchidaceae [117]
- Teresantalol
 - Magnoliaceae [240]
- Esters
 - Nerolidol epoxyacetate
 - Nyctaginaceae [151]
- Ethers and epoxides
 - Aromadendrene epoxide
 - Nyctaginaceae [151]
 - Bisabolol oxide A
 - Asteraceae [37]
 - Bisabolol oxide B
 - Asteraceae [37]
 - Caryophyllan-2,6- α -oxide
 - Lecythidaceae [121]
 - Caryophyllene β -epoxide, see Caryophyllene oxide
 - Caryophyllene epoxide, see Caryophyllene oxide
 - Caryophyllene oxide (Caryophyllene β -epoxide, Caryophyllene epoxide)
 - Apiaceae [242], Araceae [86, 229], Arecaceae [137, 143], Asteraceae [34], Caryophyllaceae [54, 66], Fabaceae [62, 121], Gentianaceae [45], Lauraceae [74], Lecythidaceae [121, 144], Magnoliaceae [11], Malvaceae [142], Nelumbonaceae [184], Nyctaginaceae [151], Orchidaceae [28, 77, 116, 117], Rubiaceae [5], Rutaceae [247], Solanaceae [165], Verbenaceae [5], Vitaceae [42, 44, 47]
 - Humulene epoxide
 - Arecaceae [143], Orchidaceae [116, 117]

DITERPENES

- Acyclic
 - Hydrocarbons
 - neo*-Phytadiene
 - Solanaceae [165]
 - Alcohols
 - Geranyl linalool
 - Oleaceae [114]
 - Isophytol
 - Oleaceae [114]
 - Phytol
 - Oleaceae [114]
 - Cyclic
 - Hydrocarbons
 - Cembrene
 - Solanaceae [165]
 - Sandaracopimaradiene
 - Chloranthaceae [251]

IRREGULAR TERPENES

Monoterpenes, sesquiterpenes, and diterpenes are listed under these specific headings. The following class contains terpenes whose skeletons fall outside of these three main groups.

Apocarotenoid

Aldehydes

β -Cyclocitral, see 2,6,6-Trimethyl-1-cyclohexene-1-carboxaldehyde

Safranal, see 2,6,6-Trimethyl-1,3-cyclohexadiene-1-carboxaldehyde

2,6,6-Trimethyl-1,3-cyclohexadiene-1-carboxaldehyde (Safranal)

Iridaceae [150], Scrophulariaceae [5]

1,3,4-Trimethyl-3-cyclohexene-1-carboxaldehyde

Theophrastaceae [139]

2,6,6-Trimethyl-1-cyclohexene-1-carboxaldehyde (β -Cyclocitral)

Araceae [86], Iridaceae [150, 178], Scrophulariaceae [226], Theophrastaceae [139]

Ketones

Cyclic- β -Ionone

Lecythidaceae [121], Orchidaceae [117], Rutaceae [172]

Damascenone

Orchidaceae [117]

β -Damascenone

Oleaceae [177, 178], Rosaceae [232]

(*E*)- β -Damascone

Cyclanthaceae [225]

(*Z*)- β -Damascone

Cyclanthaceae [225]

8,9-Dehydro-4,5-dihydrotheaspirone

Asteraceae [214]

2,3-Dihydro-3,5-dihydroxy-6-methyl-4*H*-pyrene-4-one

Orchidaceae [227]

Dihydro- β -ionone

Arecaceae [10, 143], Cactaceae [125], Commelianaceae [121], Iridaceae [178], Lecythidaceae [121], Oleaceae [57, 58, 177], Orchidaceae [17, 77, 116, 117, 120], Rosaceae [33, 35, 93, 176, 178], Verbenaceae [5], Violaceae [248]

Dihydrooxoisophorone

Orchidaceae [5], Polemoniaceae [5], Scrophulariaceae [5]

7(11)-Epoxy megastigma-5(7),6(11)-dien-9-one

Orchidaceae [117, 120]

7(11)-Epoxy megastigma-5(6)-en-9-one

Orchidaceae [15, 77, 117, 120]

5(6)-Epoxy-2,2,6-trimethylcyclohexan-1,4-dione

Orchidaceae [117]

3-Hydroxy-7(*E*)-megastigmen-9-one

Orchidaceae [77, 116]

3-Hydroxy-2-methyl-4-pyrone (Maltol)

Orchidaceae [227]

α -Ionone

Arecaceae [10], Cactaceae [125], Fabaceae [120], Malvaceae [142], Nyctaginaceae [151], Oleaceae [57, 58], Orchidaceae [117], Violaceae [248]

β -Ionone

Amaryllidaceae [63], Araceae [86, 229], Arecaceae [10], Cactaceae [125], Fabaceae [120], Iridaceae [178], Lecythidaceae [121], Malvaceae [142], Oleaceae [57, 58, 119, 177, 178], Orchidaceae [17, 27, 77, 78, 115, 116, 117, 120], Passifloraceae [157], Rosaceae [33, 35, 43, 72], Rutaceae [172], Scrophulariaceae [5]

β -Ionone epoxide

Orchidaceae [116, 117]

β -Ionone-5(6)-epoxide

Orchidaceae [120]

Isophorone

Primulaceae [5], Rosaceae [128]

α -Isophorone (3,5,5-Trimethyl-2-cyclohexen-1-one)

Iridaceae [150], Primulaceae [5]

β -Isophorone

Iridaceae [150]

Maltol, see 3-Hydroxy-2-methyl-4-pyrone

7(*E*)-Megastigmen-3,9-dione (3-Oxo-7(*E*)-megastigmen-9-one)

- Orchidaceae [77, 116]
- 4-Oxo- β -ionone
 - Iridaceae [178], Oleaceae [57, 58, 177, 178]
- Oxoisophorone
 - Combretaceae [10], Scrophulariaceae [226]
- 4-Oxoisophorone (2,6,6-Trimethyl-2-cyclohexen-1,4-dione)
 - Apiaceae [242, 246], Asteraceae [5], Dipsacaceae [5], Iridaceae [150], Orchidaceae [5, 117], Polemoniaceae [5], Scrophulariaceae [5], Theophrastaceae [139]
- Oxoisophorone oxide
 - Scrophulariaceae [5]
- 3-Oxo-7(*E*)-megastigmen-9-one, see 7(*E*)-Megastigmen-3,9-dione
- (*E,E*)-Pseudoionone
 - Orchidaceae [117]
- (*E,Z*)-Pseudoionone
 - Orchidaceae [117]
- retro- γ -Ionone
 - Orchidaceae [117]
- (*Z*)-retro- γ -Ionone
 - Lecythidaceae [121]
- 2,2,6-Trimethyl-1,4-cyclohexanedione
 - Orchidaceae [117], Scrophulariaceae [226], Theophrastaceae [139]
- 2,6,6-Trimethyl-2-cyclohexen-1,4-dione, see 4-Oxoisophorone
- 3,5,5-Trimethyl-2-cyclohexen-1-one, see α -Isophorone
- 2,2,6-Trimethyl-6-hydroxycyclohexanone
 - Oleaceae [171]
- 6,10,14-Trimethyl-2-pentadecanone
 - Orchidaceae [117]
- Alcohols
 - Dihydro- β -ionol
 - Arecaceae [10], Oleaceae [177, 178], Rosaceae [33, 35, 177, 178]
 - 6,10-Dimethyl-5,9-undecadien-2-ol
 - Orchidaceae [117]
 - (*E*)-6,10-Dimethyl-5,9-undecadien-2-ol
 - Commelinaceae [121], Winteraceae [239]
 - (*Z*)-6,10-Dimethyl-5,9-undecadien-2-ol
 - Winteraceae [239]
- Ionol
 - Orchidaceae [77]
- α -Ionol
 - Violaceae [248]
- β -Ionol
 - Violaceae [248]
- 4-Oxo- β -ionol
 - Iridaceae [178]
- Esters
 - Dihydroactinidiolide
 - Amaryllidaceae [107], Arecaceae [10], Onagraceae [265], Orchidaceae [116, 117]
 - 6,10-Dimethyl-5,9-undecadien-2-yl acetate
 - Orchidaceae [117]
 - (*E*)-6,10-Dimethyl-5,9-undecadien-2-yl acetate
 - Winteraceae [239]
 - (*Z*)-6,10-Dimethyl-5,9-undecadien-2-yl acetate
 - Winteraceae [239]
- Ethers
 - cis*-Theaspirane
 - Rosaceae [33, 35]
 - trans*-Theaspirane
 - Rosaceae [33, 35]
- Unknown
 - Edulan II
 - Orchidaceae [117]

C8

Ketones

6-Methyl-3,5-heptadien-2-one

Oleaceae [57, 58], Orchidaceae [117], Rutaceae [247], Zingiberaceae [174]

6-Methylheptan-2,4-dione

Cyclanthaceae [225]

5-Methyl-5-hepten-2-one

Araceae [229]

6-Methyl-5-hepten-2-one

Actinidiaceae [237], Amaryllidaceae [21], Apiaceae [242, 246], Apocynaceae [141], Araceae [86, 129, 133, 229], Arecaceae [10, 137, 143], Asteraceae [37], Berberidaceae [215], Bignoniaceae [142], Cactaceae [122, 125], Caprifoliaceae [102], Caricaceae [141], Caryophyllaceae [5, 54, 66, 112, 141], Commelinaceae [121], Ericaceae [136, 140], Fabaceae [62, 120, 212], Gentianaceae [5, 45], Hemerocallidaceae [33], Hydnoraceae [48], Lauraceae [74], Lecythidaceae [144], Malvaceae [67, 142], Myrsinaceae [103], Nyctaginaceae [151], Oleaceae [33, 39, 51, 52, 57, 58, 176, 178, 267], Orchidaceae [8, 23, 30, 77, 117, 120, 141, 147, 203, 224, 241, 243, 244], Papaveraceae [62], Passifloraceae [157], Polemoniaceae [5, 21], Primulaceae [5, 36], Ranunculaceae [208, 209], Rosaceae [12, 33, 35, 60, 61, 219, 220], Rubiaceae [5, 141], Ruscaceae [141], Rutaceae [247], Salicaceae [245], Scrophulariaceae [5], Solanaceae [141], Theophrastaceae [139], Thymelaeaceae [5], Valerianaceae [5], Verbenaceae [5], Winteraceae [210, 239], Violaceae [248], Vitaceae [42, 44, 47], Zamiaceae [211], Zingiberaceae [141, 174]

Alcohols

6-Methyl-5-hepten-2-ol

Arecaceae [137], Cactaceae [125], Malvaceae [142], Orchidaceae [117], Winteraceae [210, 239]

6-Methyl-6-hepten-3-ol

Araceae [262]

Esters

6-Methyl-5-hepten-2-yl acetate

Orchidaceae [117], Winteraceae [210, 239]

C9

Ketones

Crypton (4-Isopropyl-2-cyclohexen-1-one)

Araceae [229]

3,3-Dimethylbicyclo[2.2.1]heptan-2-one

Fabaceae [212]

4-Isopropyl-2-cyclohexen-1-one, see Crypton

C10

Ketones

4,7(*endo*)-Dimethylbicyclo[3.2.1]oct-3-en-6-one

Calycanthaceae [120]

4,7(*exo*)-Dimethylbicyclo[3.2.1]oct-3-en-6-one

Calycanthaceae [120]

4-Methylene-7(*endo*)-methylbicyclo[3.2.1]octan-6-one

Calycanthaceae [120]

4-Methylene-7(*exo*)-methylbicyclo[3.2.1]octan-6-one

Calycanthaceae [120]

Alcohols

4,7(*endo*)-Dimethylbicyclo[3.2.1]oct-3-en-6(*endo*)-ol

Calycanthaceae [120]

4,7(*endo*)-Dimethylbicyclo[3.2.1]oct-3-en-6(*exo*)-ol

Calycanthaceae [120]

4,7(*exo*)-Dimethylbicyclo[3.2.1]oct-3-en-6(*endo*)-ol

Calycanthaceae [120]

4,7(*exo*)-Dimethylbicyclo[3.2.1]oct-3-en-6(*exo*)-ol

Calycanthaceae [120]

C11

Hydrocarbons

4,8-Dimethyl-1,3,7-nonatriene

Nyctaginaceae [151], Rosaceae [33, 35]

(E)-4,8-Dimethyl-1,3,7-nonatriene

Apocynaceae [115, 141], Arecaceae [68, 137, 143], Asteraceae [5], Berberidaceae [182], Cactaceae [21, 113, 115, 125], Caryophyllaceae [5], Combretaceae [10], Cyclanthaceae [225], Fabaceae [62, 115, 120, 179], Hydrangeaceae [115], Lamiaceae [5], Lecythidaceae [144], Liliaceae [115], Lythraceae [10], Magnoliaceae [9, 11, 115], Malvaceae [142], Oleaceae [115, 119], Orchidaceae [115, 117, 120], Orobanchaceae [141], Passifloraceae [157], Polemoniaceae [5], Rhizophoraceae [10], Rosaceae [72, 219, 220], Rubiaceae [5, 141], Ruscaceae [115], Rutaceae [119], Scrophulariaceae [5], Solanaceae [121, 141], Theophrastaceae [139], Thymelaeaceae [115], Verbenaceae [5, 101], Zingiberaceae [141]

(Z)-4,8-Dimethyl-1,3,7-nonatriene

Arecaceae [143], Asteraceae [5], Cactaceae [113, 115, 125], Magnoliaceae [9, 11], Orchidaceae [117], Rubiaceae [5], Theophrastaceae [139]

Ketones

(E)-Cyclanthone (*(E)*-2,6-Dimethyl-6,8-nonadien-4-one)

Cyclanthaceae [225]

(Z)-Cyclanthone

Cyclanthaceae [225]

(E)-6(7)-Epoxy-2,6-dimethyl-8-nonen-4-one

Cyclanthaceae [225]

(Z)-6(7)-Epoxy-2,6-dimethyl-8-nonen-4-one

Cyclanthaceae [225]

Alcohols

(E)-2,6-Dimethyl-6,8-nonadien-4-ol

Cyclanthaceae [225]

(E)-2,6-Dimethyl-6,8-nonadien-4-one, see *(E)*-Cyclanthone*(E)*-2,6-Dimethyl-3,6,8-nonatrien-2-ol

Cactaceae [113, 115, 125]

Ethers and epoxides

(E)-2(3)-Epoxy-2,6-dimethyl-6,8-nonadiene

Cactaceae [113, 115, 125], Orchidaceae [117], Solanaceae [121]

C12

Hydrocarbons

Decahydro-1,6-dimethylnaphthalene

Asteraceae [213]

Alcohols

Dehydrogeosmin

Cactaceae [122, 125], Moraceae [121]

Geosmin

Cactaceae [122, 125], Moraceae [121, 125]

C13

Ketones

Geranylacetone

Actinidiaceae [237], Amaryllidaceae [21], Apocynaceae [141], Araceae [129, 229], Arecaceae [10], Asteraceae [214], Bignoniaceae [21], Cactaceae [122], Caprifoliaceae [102], Caryophyllaceae [5, 66, 141], Ericaceae [136], Hydnoraceae [48], Lecythidaceae [144], Malvaceae [67, 94], Myrsinaceae [103], Nyctaginaceae [151], Nymphaeaceae [14, 135], Oleaceae [33, 51], Orchidaceae [8, 30, 77, 201, 224], Orobanchaceae [141], Passifloraceae [157], Ranunculaceae [208], Rosaceae [60, 61, 176, 177, 178], Rubiaceae [5, 141], Rutaceae [119, 247], Sapotaceae [263], Scrophulariaceae [5], Solanaceae [94], Theophrastaceae [139], Verbenaceae [5], Winteraceae [239], Vitaceae [42, 44, 47]

(E)-Geranylacetone

Apiaceae [242], Berberidaceae [182, 215], Cactaceae [125], Commelinaceae [121], Oleaceae [119], Orchidaceae [77, 78, 117, 118, 120], Solanaceae [121]

(Z)-Geranylacetone

Orchidaceae [117]

Nerylacetone

Rubiaceae [141]

Alcohols

(E)-6-10-Dimethyl-5,9-tridecadien-2-ol

Solanaceae [121]

Esters

C14

Hydrocarbons

Decahydrotetramethylnaphthalene

Asteraceae [213]

Ethyldecahydrodimethylnaphthalene

Asteraceae [213]

(E)-2,6,10-Trimethyl-2,6-undecadiene

Rutaceae [119]

Esters

Methyl (E)-2,6,10-trimethyl-5,9-undecadienoate

Orchidaceae [116, 117]

C16

Hydrocarbons

(E,E)-4,8,12-Trimethyl-1,3,7,11-tridecatetraene

Apocynaceae [2, 115, 141], Arecaceae [68, 137, 143], Cactaceae [113, 115, 125], Caricaceae [141], Fabaceae [115, 138], Hydrangeaceae [115], Lecythidaceae [144], Liliaceae [115], Magnoliaceae [115], Oleaceae [115], Orchidaceae [115, 116, 117, 121, 141], Orobanchaceae [141], Passifloraceae [157], Polemoniaceae [5], Rubiaceae [5, 141], Ruscaceae [115], Thymelaeaceae [115], Valerianaceae [5], Verbenaceae [5, 101], Zingiberaceae [141, 174]

(Z,E)-4,8,12-Trimethyl-1,3,7,11-tridecatetraene

Arecaceae [137], Cactaceae [113, 115, 125], Caricaceae [141], Lecythidaceae [144], Orchidaceae [116, 117, 118], Polemoniaceae [5], Rubiaceae [5], Solanaceae [141], Valerianaceae [5], Zingiberaceae [174]

(Z,Z)-4,8,12-Trimethyl-1,3,7,11-tridecatetraene

Passifloraceae [157]

Alcohols

1,5,5,8-Tetramethyl-12-oxabicyclo[9.1.0]dodeca-3,7-diene

Nyctaginaceae [151]

Esters

Methyl (E,E)-4,8,12-trimethyl-3,7,11-tridecatrienoate

Orchidaceae [116, 117]

Ethers and epoxides

(E,E)-11(12)-Epoxy-4,8,12-trimethyl-1,3,7-tridecatriene

Orchidaceae [115, 117]

C18

Ketones

Hexahydrofarnesylacetone

Fabaceae [107]

NITROGEN CONTAINING COMPOUNDS

Ammonia

Ammonia

Ammonia

Araceae [230]

C1

Amines

Dimethylamine

Araceae [230]

N,N-Dimethylmethanamine

Rosaceae [220]

Methylamine

Araceae [230]

Trimethylamine

Araceae [134, 230]

Amides

N,N-Dimethyl formamide

Orchidaceae [117]

N-Methyl formamide

Orchidaceae [117]

C2

Amines

2-Aminoethanol

Araceae [230]

Diethylamine

Araceae [230]

Ethylamine

Araceae [230]

N-Methyleneethanamine

Brassicaceae [221], Oleaceae [39], Rosaceae [219, 220]

Amides

Acetamide

Araceae [26], Orchidaceae [117]

N,N-Dimethyl acetamide

Orchidaceae [117]

N-Methyl acetamide

Orchidaceae [117]

C3

Amines

2-Methylpropylamine

Araceae [230]

1,2-Propanediamine

Araceae [230]

Nitro compounds

1-Nitro-2-methylpropane

Zingiberaceae [124]

Nitriles

Isobutylnitrile, see 2-Methylpropylnitrile

2-Methylpropylnitrile (Isobutylnitrile)

Brassicaceae [142], Orchidaceae [191], Zingiberaceae [174]

Oximes

Isobutylalldoxime, see 2-Methylpropylalldoxime

2-Methylpropylalldoxime (Isobutylalldoxime)

Caprifoliaceae [115], Rubiaceae [115], Rutaceae [119], Solanaceae [115], Zingiberaceae [115, 124, 174]

C4

Amines

Agmatine

Araceae [230]

3-Methylbutylamine

Araceae [230]

Putrescine

Araceae [230]

Nitro compounds

2-Methyl-1-nitrobutane, see 1-Nitro-2-methylbutane

3-Methyl-1-nitrobutane, see 1-Nitro-3-methylbutane

1-Nitro-2-methylbutane (2-Methyl-1-nitrobutane)

Caprifoliaceae [124, 150], Rubiaceae [121]

1-Nitro-3-methylbutane (3-Methyl-1-nitrobutane)

Caprifoliaceae [124, 150], Orchidaceae [117], Zingiberaceae [124]

Nitriles

3-Methyl-2-butenylnitrile

Caprifoliaceae [124]

2-Methylbutylnitrile

Asteraceae [5], Brassicaceae [142], Caprifoliaceae [124, 150], Fabaceae [121, 179], Orchidaceae [117], Rubiaceae [5, 109], Rutaceae [124], Zingiberaceae [124, 141, 174]

3-Methylbutylnitrile

Caprifoliaceae [124], Fabaceae [179], Lecythidaceae [144], Malvaceae [142], Orchidaceae [116, 117], Polemoniaceae [5], Rubiaceae [5], Solanaceae [141], Zingiberaceae [124, 141, 174]

Oximes

Isobutyraldoxime-*O*-methyl ether

Fabaceae [121]

2-Methylbutylaldoxime

Caprifoliaceae [107, 115, 124, 150], Caryophyllaceae [112], Fabaceae [121], Magnoliaceae [115], Orchidaceae [115, 116, 117, 121], Rubiaceae [109, 115, 121], Rutaceae [115, 119], Solanaceae [115], Zingiberaceae [115, 124, 174]

(E)-2-Methylbutylaldoxime

Fabaceae [179]

(Z)-2-Methylbutylaldoxime

Fabaceae [179]

2-Methylbutylaldoxime-*O*-methyl ether

Apocynaceae [121], Cucurbitaceae [121], Fabaceae [121], Rubiaceae [118, 121], Salicaceae [118, 121], Solanaceae [118, 121], Zingiberaceae [118]

3-Methylbutylaldoxime

Apiaceae [31], Caprifoliaceae [115, 124], Caryophyllaceae [112, 141], Magnoliaceae [115], Malvaceae [142], Orchidaceae [115, 116, 117, 121, 141], Polemoniaceae [5], Rubiaceae [5, 115, 121], Solanaceae [115, 141], Zingiberaceae [115, 124, 141, 174]

(Z)-3-Methylbutylaldoxime

Fabaceae [179]

(E)-3-Methylbutylaldoxime

Fabaceae [179]

3-Methylbutylaldoxime-*O*-methyl ether

Caprifoliaceae [121], Cucurbitaceae [121], Rubiaceae [118], Salicaceae [118], Solanaceae [118, 121], Zingiberaceae [118, 121]

Esters

Methyl 2-amino-3-methylbutanoate (Valine methylester)

Adoxaceae [108]

Valine methylester, see Methyl 2-amino-3-methylbutanoate

C5

Amines

Cadaverine, see Pentamethylenediamine

Pentamethylenediamine (Cadaverine)

Araceae [230]

Nitriles

4-Methylpentynitrile

Caprifoliaceae [124]

Esters

Isoleucine methylester, see Methyl 2-amino-3-methylpentanoate

Methyl 2-amino-3-methylpentanoate (Isoleucine methylester)

Adoxaceae [108]

C6

Amines

1,6-Hexanediamine

Araceae [230]

Oximes

C6-Ketoxime

Rosaceae [220]

Cyclic

Imidazole

Histamine (4-Imidazole ethylamine)

Araceae [230]

1*H*-Imidazole

Araceae [152]

4-Imidazole ethylamine, see Histamine

Indole

Indole

Amaryllidaceae [6, 63, 177], Annonaceae [111], Apocynaceae [141], Araceae [26, 86, 129, 133, 134, 229], Arecaceae [137, 143, 145], Berberidaceae [215], Brassicaceae [36, 202, 217], Cactaceae [125], Calycanthaceae [120, 266], Caprifoliaceae [102, 107, 175, 178], Caryophyllaceae [141], Commelinaceae [121], Cucurbitaceae [3, 4, 71], Fabaceae [107,

119], Hemerocallidaceae [33], Hyacinthaceae [38], Hydrangeaceae [107, 108], Lecythidaceae [121, 144], Malvaceae [41], Nyctaginaceae [151], Oleaceae [33, 39, 52, 107, 114, 150, 176, 178, 267], Orchidaceae [13, 17, 75, 115, 116, 117, 118, 120, 141, 169, 180, 253, 258], Passifloraceae [157], Pittosporaceae [107], Rhizophoraceae [10], Rubiaceae [5, 141, 249], Ruscaceae [141, 171], Rutaceae [2, 119, 247], Solanaceae [131], Valerianaceae [5], Zingiberaceae [141, 174, 268]

N-Methylindole

Orchidaceae [23]

3-Methylindole, see Skatole

Skatole (3-Methylindole)

Araceae [26, 133, 230]

Pyrazine

2,5-Dimethylpyrazine

Apiaceae [31], Araceae [26]

Ethylldimethylpyrazine

Iridaceae [178]

2-Ethyl-3,5-dimethylpyrazine

Oleaceae [176, 178]

Ethyltrimethylpyrazine

Iridaceae [178]

2-Methoxy-3-*sec*-butylpyrazine (2-Methoxy-3-isobutylpyrazine)

Arecaceae [68, 143], Orchidaceae [117]

2-Methoxy-3-isobutylpyrazine, see 2-Methoxy-3-*sec*-butylpyrazine

Tetramethylpyrazine

Theophrastaceae [139]

Trimethylpyrazine

Theophrastaceae [139]

Pyrazole

Pyrazole

Araceae [86]

Pyridine

Actinidine

Araceae [26]

Nicotinaldehyde

Fumariaceae [56]

Methyl nicotinate

Cactaceae [113, 115, 125], Fabaceae [107], Magnoliaceae [11], Nyctaginaceae [151], Orchidaceae [117, 169], Rosaceae [220]

3-Pyridincarboxaldehyde

Apiaceae [242], Rosaceae [220]

Nicotine

Solanaceae [69]

Triazine

1,3,5-Tributylhexahydro-1,3,5-triazine

Calycanthaceae [266]

SULFUR-CONTAINING COMPOUNDS

Acyclic

Sulfoxides

Dimethyl sulfoxide

Onagraceae [265]

Isothiocyanate

S-Butyl isothiocyanate

Brassicaceae [142]

Cyclohexene isothiocyanate

Rosaceae [12]

Isopropyl isothiocyanate

Brassicaceae [142]

Sulfides

Carbon disulfide

Hydnoraceae [48], Orchidaceae [75]

- Dimethyl disulfide
 Araceae [26, 86, 130, 134, 231], Arecaceae [21, 137], Bignoniaceae [21, 142], Brassicaceae [142, 221], Bromeliaceae [21], Cactaceae [125], Fabaceae [21], Hydnoraceae [48], Malvaceae [94], Orchidaceae [117], Polemoniaceae [21], Solanaceae [94]
- Dimethyl disulfide monosulfone
 Cactaceae [125]
- Dimethyl sulfide
 Araceae [26, 229, 262]
- Dimethyl tetrasulfide
 Araceae [26, 134], Arecaceae [21], Bignoniaceae [21, 142], Cactaceae [125, 142], Fabaceae [21], Malvaceae [94], Polemoniaceae [21]
- Dimethyl trisulfide
 Agavaceae [142], Alliaceae [36], Araceae [26, 130, 134, 229, 231], Arecaceae [21, 137], Bignoniaceae [21, 142], Brassicaceae [142, 221], Cactaceae [125, 142], Fabaceae [21], Hydnoraceae [48], Malvaceae [94, 142], Polemoniaceae [21, 142], Solanaceae [94]
- (*E*)-Diprop-1-enyl disulfide
 Alliaceae [36]
- (*Z*)-Diprop-1-enyl disulfide
 Alliaceae [36]
- Diprop-2-enyl disulfide
 Alliaceae [36]
- Diprop-2-enyl sulfide
 Alliaceae [36]
- 2,4-Dithiapentane
 Arecaceae [21, 137], Bignoniaceae [21, 142], Brassicaceae [142], Fabaceae [21], Lythraceae [10], Malvaceae [94]
- Bis-(1-Methylethyl)-disulfide
 Araceae [262]
- Methyl (methylthio)methyl disulfide
 Hydnoraceae [48]
- Methyl (*E*)-prop-1-enyl disulfide
 Alliaceae [36]
- Methyl (*Z*)-prop-1-enyl disulfide
 Alliaceae [36]
- Methyl prop-2-enyl disulfide
 Alliaceae [36]
- Methyl prop-2-enyl trisulfide
 Alliaceae [36]
- Methyl propyl disulfide
 Alliaceae [36]
- 1,3,5,6,8-Pentathianonane
 Bignoniaceae [21]
- 2,3,5,6,8-Pentathianonane
 Arecaceae [21], Bignoniaceae [142]
- Propyl prop-2-enyl disulfide
 Alliaceae [36]
- 2,3,4,6-Tetrathiaheptane
 Arecaceae [21], Bignoniaceae [21, 142]
- 2,3,5,7-Tetrathiaoctane
 Bignoniaceae [142]
- 2,4,5,7-Tetrathiaoctane
 Arecaceae [137], Bignoniaceae [21], Cactaceae [125]
- 2,3,5-Trithiahexane
 Arecaceae [137], Bignoniaceae [21, 142], Cactaceae [125], Fabaceae [21]
- Thiocyanates
 Thiocyanate octadecanal
 Brassicaceae [217]
- Thioesters
 Dimethyltrithio carbonate
 Bignoniaceae [21]
- Methyl thioacetate

- Cactaceae [125]
- S-Methyl thioacetate
- Hydnoraceae [48], Orchidaceae [201]
- Thiols
 - 2-Aminobenzene thiol
 - Orchidaceae [13]
 - 4-Mercapto-4-methylpentan-2-one
 - Fabaceae [121]
 - Methanethiol
 - Araceae [86]
 - 2-Thiopropene-1-thiol
 - Arecaceae [21, 137]
- Cyclic
 - Thiofurans
 - Thiophene
 - Rosaceae [12]
 - 1,2,4-Trithiolane
 - Arecaceae [137]
- Thiazoles
 - Benzothiazole
 - Brassicaceae [36], Cactaceae [113], Hydrangeaceae [107], Myrsinaceae [103], Onagraceae [265], Rosaceae [12, 40]
 - Benzothiazolone
 - Amaryllidaceae [107]

MISCELLANEOUS CYCLIC COMPOUNDS

Carbocyclic

Monocyclic

Alkanes

- Butylcyclohexane
- Meliaceae [250]
- Cyclohexane
- Araceae [229], Rosaceae [12]
- Cyclotetradecane
- Rosaceae [51]
- Methylcyclohexane
- Rosaceae [12]
- Methylcyclopentane
- Asteraceae [213]

Alkenes

- 1,3,5-Cycloheptatriene
- Geraniaceae [32]
- 1,5-Cycloundecadiene
- Nyctaginaceae [151]
- 1-Formylcyclohexene
- Araceae [86]
- 3-Methylcyclopentene
- Oleaceae [267]

Ketones

- Cyclohexanone
- Araceae [86, 152, 229], Arecaceae [21], Brassicaceae [221], Bromeliaceae [21], Musaceae [21]
- 2-Cyclohexen-1-one
- Rosaceae [128]

Jasmone

- Cactaceae [125], Caprifoliaceae [107, 178], Hydrangeaceae [107], Magnoliaceae [240], Nelumbonaceae [178], Oleaceae [107, 114, 119, 176, 178], Orchidaceae [116, 117], Rutaceae [247], Solanaceae [121, 165, 166], Thymelaeaceae [234]

cis-Jasmone

- Actinidiaceae [237], Annonaceae [111], Arecaceae [137], Caprifoliaceae [102], Magnoliaceae [264], Nelumbonaceae [184], Nyctaginaceae [151], Thymelaeaceae [5], Zingiberaceae [141, 174]

- trans*-Jasmone
 Caprifoliaceae [102], Thymelaeaceae [5], Zingiberaceae [174]
- 4-Methylcyclohexanone
 Orchidaceae [201]
- 1-Methyl-1-cyclohexen-4-one
 Cactaceae [113, 125]
- 2-Methylcyclopentanone
 Fabaceae [104]
- 3-Methyl-2-cyclopenten-1-one
 Orchidaceae [117]
- Alcohols
 Cyclohexane ethanol
 Araceae [86]
- Cyclohexanol
 Araceae [86, 229]
- 3-Cyclohexene-1-methanol
 Araceae [86], Oleaceae [57, 58]
- 2-Ethylcyclobutanol
 Rosaceae [40]
- Esters
 Butyrolactone, see γ -Butyrolactone
- γ -Butyrolactone (Butyrolactone)
 Araceae [86], Nyctaginaceae [151], Oleaceae [177], Orchidaceae [117], Primulaceae [36], Sapindaceae [46]
- γ -Caprolactone, see γ -Decalactone
- (*Z,Z*)-2,6-Decadien-5-olide
 Orchidaceae [120, 121]
- δ -Decalactone (Decan-4-olide)
 Asteraceae [132], Caprifoliaceae [107], Orchidaceae [77, 117, 121, 169]
- γ -Decalactone (γ -Caprolactone)
 Nyctaginaceae [151], Oleaceae [57, 58, 177], Orchidaceae [117, 118]
- Decan-4-olide, see δ -Decalactone
- (*Z*)-2-Decen-5-olide
 Orchidaceae [120, 121]
- (*Z*)-7-Decen-4-olide (*cis*-Jasmin lactone, Jasmin lactone)
 Agavaceae [178], Caprifoliaceae [102, 107], Oleaceae [52, 177, 178], Orchidaceae [117, 118, 169], Rubiaceae [249], Zingiberaceae [174]
- 7-Decen-5-olide
 Nyctaginaceae [151]
- 4,4-Dimethylbut-2-enolide, see 4-Methyl-2-penten-4-olide
- (*Z*)-2,6-Dodecadien-5-olide
 Orchidaceae [120, 121]
- δ -Dodecalactone
 Agavaceae [178], Orchidaceae [117]
- γ -Dodecalactone
 Amaryllidaceae [6]
- (*Z*)-2-Dodecen-5-olide
 Orchidaceae [120, 121]
- γ -Hexalactone
 Orchidaceae [117]
- Jasmin lactone, see (*Z*)-7-Decen-4-olide
- cis*-Jasmin lactone, see (*Z*)-7-Decen-4-olide
- Methyl *cis*-(*Z*)-dehydrojasmonate
 Orchidaceae [116, 117]
- Methyl *trans*-(*Z*)-dehydrojasmonate
 Orchidaceae [116, 117]
- Methyl epijasmonate, see Methyl *cis*-(*Z*)-jasmonate
- (*Z*)-Methyl epijasmonate
 Rutaceae [172]
- Methyl jasmonate
 Araceae [86], Chloranthaceae [251], Oleaceae [114, 176, 178], Orchidaceae [176, 201], Zingiberaceae [174]

- cis*-Methyl jasmonate, see Methyl *cis*-jasmonate
Methyl *cis*-jasmonate (*cis*-Methyl jasmonate)
 Caprifoliaceae [102, 107], Chloranthaceae [251], Oleaceae [107], Orchidaceae [201], Rutaceae [172]
Methyl *cis*-(*Z*)-jasmonate (Methyl epijasmonate)
 Caprifoliaceae [102], Fabaceae [120], Oleaceae [119], Orchidaceae [77, 116, 117, 176], Zingiberaceae [174]
Methyl *trans*-(*Z*)-jasmonate
 Fabaceae [120], Oleaceae [119], Orchidaceae [77, 116, 117]
Methyl *trans*-jasmonate (*trans*-Methyl jasmonate)
 Caprifoliaceae [102, 107], Oleaceae [107]
trans-Methyl jasmonate, see Methyl *trans*-jasmonate
(*Z*)-3-Methyl-4-decanolide
 Orchidaceae [116, 117]
(*Z*)-4-Methyl-5-decanolide
 Orchidaceae [116, 117]
4-Methyl-5-hexen-4-olide (2-Methyl-2-vinyl-5-oxo-tetrahydro-lactone (furan), 4-Methyl-4-vinylbutyrolactone, 4-Methyl-5-hexen-1,4-olide, γ -Vinyl- γ -valerolactone)
 Arecaceae [143], Nyctaginaceae [151], Thymelaeaceae [5], Zamiaceae [211], Zingiberaceae [174]
4-Methyl-5-hexen-1,4-olide, see 4-Methyl-5-hexen-4-olide
4-Methyl-2-pentenolide, see 4-Methyl-2-penten-4-olide
4-Methyl-2-penten-4-olide (4,4-Dimethylbut-2-enolide, 4-Methyl-2-pentenolide)
 Cactaceae [125], Nyctaginaceae [151], Orchidaceae [117]
4-Methyl-4-vinylbutyrolactone, see 4-Methyl-5-hexen-4-olide
2-Methyl-2-vinyl-5-oxo-tetrahydro-lactone (furan), see 4-Methyl-5-hexen-4-olide
 δ -Nonalactone
 Nyctaginaceae [151]
 δ -Octalactone
 Nyctaginaceae [151], Orchidaceae [169]
 γ -Octalactone
 Orchidaceae [117]
(*Z*)-2,6-Undecadien-5-olide
 Orchidaceae [120]
(*Z,Z*)-2,6-Undecadien-5-olide
 Orchidaceae [121]
 γ -Vinyl- γ -valerolactone, see 4-Methyl-5-hexen-4-olide
Ethers
 1-Methyl-4-methoxy-1-cyclohexene
 Cactaceae [113, 115, 125]
Bicyclic
Bicyclo[3.1.0]
 Ketones
 1,5-di-*tert*-Butyl-3,3-dimethylbicyclo[3.1.0]hexan-2-one
 Onagraceae [265]
Bicyclo[3.1.1]
 Ketones
 6,6-Dimethylbicyclo[3.1.1]heptan-2-one
 Fabaceae [212]
Alcohols
 2-Methanolbicyclo[3.1.1]hept-2-ene
 Fabaceae [212]
Bicyclo[4.1.0]
 Hydrocarbons
 Bicyclo[4.1.0]heptane
 Orchidaceae [180]
Bicyclo[4.4.0] (Naphthalenes)
 Hydrocarbons
 Naphthalene
 Annonaceae [111], Berberidaceae [215], Calycanthaceae [266], Lauraceae [74], Orchidaceae [28], Violaceae [248]
 Ketones
 1,4-Dihydro-1-(2H)-naphthalenone

- Asteraceae [213]
- 3,4-Dihydro-1-(2H)-naphthalenone
- Asteraceae [213], Oleaceae [57, 58]
- Bicyclo[5.3.0] (Azulenes)
- Hydrocarbons
- Azulene
- Araceae [129]
- Heterocyclic
- Ethers
- 1,4-Diethylene dioxide (1,4-Dioxane)
- Rosaceae [40]
- 1,4-Dioxane, see 1,4-Diethylene dioxide
- 1,3-Dioxolane, see Glycol methylene ether
- Glycol methylene ether (1,3-Dioxolane)
- Calycanthaceae [266]
- Furan
- Furans
- (*E*)-2-Acetyl-5-methyl-5-ethenyl-tetrahydrofuran (*trans*-2-Acetyl-5-vinyl-5-methyltetrahydrofuran)
- Orchidaceae [117]
- (*Z*)-2-Acetyl-5-methyl-5-ethenyl-tetrahydrofuran
- Orchidaceae [117]
- trans*-2-Acetyl-5-vinyl-5-methyltetrahydrofuran, see (*E*)-2-Acetyl-5-methyl-5-ethenyl-tetrahydrofuran
- 2-Butylfuran
- Araceae [86]
- 2,5-Dihydrofuran
- Araceae [86]
- 3-(4,8-Dimethyl-3,7-nonadienyl)furan
- Asteraceae [213]
- (*E*)-2-Ethyl-3-acetoxy-tetrahydrofuran
- Oleaceae [114], Orchidaceae [118]
- 2-Ethylfuran
- Rosaceae [12, 219]
- 5-Methyl-5-ethenyl-2-isopropenyltetrahydrofuran (2-Methyl-5-isopropenyl-2-ethenyltetrahydrofuran)
- Rutaceae [247]
- 2-Methylfuran
- Araceae [229], Rosaceae [220]
- 3-Methylfuran
- Rosaceae [12], Zingiberaceae [174]
- 2-Methyl-5-isopropenyl-2-ethenyltetrahydrofuran, see 5-Methyl-5-ethenyl-2-isopropenyl-tetrahydrofuran
- 2-Pentylfuran
- Orchidaceae [227], Rosaceae [219]
- 2-Propylfuran
- Araceae [86]
- Aldehydes
- 2-Furancarboxyaldehyde
- Rosaceae [12]
- Furfural
- Orchidaceae [227]
- 5-Hydroxy-2-furancarboxaldehyde
- Orchidaceae [106]
- 5-Hydroxymethyl-2-furancarboxaldehyde (5-Hydroxymethyl-furfural)
- Orchidaceae [227], Sapindaceae [46]
- 5-Hydroxymethyl-furfural, see 5-Hydroxymethyl-2-furancarboxaldehyde
- 1-(5-Methyl-5-ethenyl-tetrahydrofuran-2-yl)ethanal
- Orchidaceae [243]
- 5-Methylfurfural
- Orchidaceae [227]

Ketones

2-Acetyldihydro-2(3H)furanone

Orchidaceae [106]

2,3-Dihydro-4-hydroxy-2,5-dimethyl-3-furanone, see 2,5-Dimethyl-4-hydroxy-2,3-dihydro-3-furanone

2,5-Dimethyl-4-hydroxy-2,3-dihydro-3-furanone (2,3-Dihydro-4-hydroxy-2,5-dimethyl-3-furanone)

Orchidaceae [227]

5-Ethenyldihydro-5-methyl-2(3H)-furanone, see 5-Methyl-5-ethenyldihydro-2(3H)-furanone

5-Ethenyl-5-methyl-2(3H)-furanone, see 5-Methyl-5-ethenyldihydro-2(3H)-furanone

(E)-2-Ethyl-3-benzoyloxo-tetrahydrofuran

Oleaceae [114]

5-(3-Furyl)-2-methyl-1-penten-3-one

Rutaceae [247]

5-Methylene-2(5H)-furanone (Protoanemonin)

Ranunculaceae [19]

5-Methyl-5-ethenyldihydro-2(3H)-furanone (5-Ethenyl-5-methyl-2(3H)-furanone, 5-Ethenyldihydro-5-methyl-2(3H)-furanone)

Arecaceae [137], Caryophyllaceae [141], Oleaceae [39, 51], Orchidaceae [243]

5-Methyl-2-furanone

Orchidaceae [227]

Protoanemonin, see 5-Methylene-2(5H)-furanone

Alcohols

2-Furanmethanol

Orchidaceae [106, 227]

5-Methyl-5-ethenyl-2-hydroxytetrahydrofuran

Apiaceae [246], Orchidaceae [241, 243]

Pyran

Pyrans

2-Ethenyltetrahydro-2,6,6-trimethyl-2H-pyran, see 2,2,6-Trimethyl-6-ethenyltetrahydro-2H-pyran

2,2,6-Trimethyl-6-ethenyltetrahydro-2H-pyran (2-Ethenyltetrahydro-2,6,6-trimethyl-2H-pyran)

Oleaceae [39]

Aldehydes

3,4-Dihydro-2,5-dimethyl-2H-pyran-2-carboxaldehyde, see 2,5-Dimethyl-3,4-dihydro-2H-pyran-2-carboxaldehyde

2,5-Dimethyl-3,4-dihydro-2H-pyran-2-carboxaldehyde (3,4-Dihydro-2,5-dimethyl-2H-pyran-2-carboxaldehyde)

Onagraceae [265]

Ketones

6-Ethenyltetrahydro-2,2,6-trimethyl-2H-pyran-3(4H)-one, see 2,2,6-Trimethyl-6-ethenyltetrahydro-2H-pyran-3(4H)-one

4-Methyl-6-isobutyl-2-pyranone

Cyclanthaceae [225]

3-Oxo-2,2,6-trimethyl-6-ethenyltetrahydropyran, see 2,2,6-Trimethyl-6-ethenyltetrahydro-3-pyranone

Pyranone

Verbenaceae [101]

2,2,6-Trimethyl-6-ethenyl-4,5-dihydro-3-pyranone

Arecaceae [137, 143], Orchidaceae [77, 117], Thymelaeaceae [24]

2,2,6-Trimethyl-6-ethenyltetrahydro-2H-pyran-3(4H)-one (6-Ethenyltetrahydro-2,2,6-trimethyl-2H-pyran-3(4H)-one)

Asteraceae [5], Polemoniaceae [5], Primulaceae [5], Rubiaceae [5], Verbenaceae [5]

2,2,6-Trimethyl-6-ethenyltetrahydro-3-pyranone (3-Oxo-2,2,6-trimethyl-6-ethenyltetrahydropyran)

Asteraceae [132], Fabaceae [121]

Alcohols

Pyranol

Verbenaceae [101]

Bicyclo[3.2.1]

(-)-endo-Brevicomine, see (-)-endo-7-Ethyl-5-methyl-6,8-dioxabicyclo[3.2.1]octane

- (-)-*endo*-7-Ethyl-5-methyl-6,8-dioxabicyclo[3.2.1]octane ((-)-*endo*-Brevicomine)
Orchidaceae [224]
1-Methyl-6-ethyl-2,7,8-trioxabicyclo[3.2.1]octane
Oleaceae [114]
1-Methyl-6(*exo*)-ethyl-2,7,8-trioxabicyclo[3.2.1]octane
Orchidaceae [118]
4,5,7-Trimethyl-6,8-dioxabicyclo[3.2.1]octane
Araceae [205]
Trimethyl-6,8-dioxabicyclo[3.2.1]octane
Araceae [205, 206]
Bicyclo[5.1.0]
8-Oxa-bicyclo[5.1.0]octane
Araceae [262]
Spiro
Chalcogran A
Orchidaceae [117]
Chalcogran B
Orchidaceae [117]
(*E*)-Chalcogran
Hydrangeaceae [115], Rubiaceae [121], Ruscaceae [115], Solanaceae [121]
(*Z*)-Chalcogran
Rubiaceae [121], Ruscaceae [115], Solanaceae [121]
8,8-Dimethyl-4-methylene-1-oxospiro[2.5]oct-5-ene
Oleaceae [57, 58]
(*E*)-7-Methyl-1,6-dioxaspiro[4.5]decane
Arecaceae [137, 143], Asteraceae [5], Fabaceae [212], Hydrangeaceae [115], Lecythidaceae [144], Malvaceae [142], Moraceae [121], Orchidaceae [117], Passifloraceae [157], Rubiaceae [5, 121, 141], Ruscaceae [115], Rutaceae [119], Solanaceae [121]
(*Z*)-7-Methyl-1,6-dioxaspiro[4.5]decane
Moraceae [121], Orchidaceae [117], Rubiaceae [121], Ruscaceae [115], Solanaceae [121]
Spiro[4.5]dec-1-ene
Zingiberaceae [268]

Appendix II: List of taxa

Taxonomic names are cited as they appear in the reviewed papers, and only obvious orthographic mistakes have been corrected. No taxonomic evaluations of the usage of names have been made, and no author names have been added if such are lacking in the original papers. Numbers within square brackets refer to reference numbers in Appendix III.

ALISMATALES

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- Amorphophallus albispatus* Hett. [133, 134]
A. albus Liu & Wei. [133, 134]
A. annulifer Hett. [133, 134]
A. arnautovii Hett. [133, 134]
A. brachyphyllus Hett. [133, 134]
A. bulbifer Blume [133, 134]
A. cicatricifer Hett. [133, 134]
A. eichleri (Engl.) N. E. Br. [133, 134]
A. elatus Hook. f. [133, 134]
A. fallax (syn. *Pseudodracontium fallax* Serebr.) [133, 134]
A. glossophyllus Hett. [133, 134]
A. haematospadix Hook. f. [133, 134]
A. konjac K. Koch [133, 134]
A. lacourii (Lind. & André) N. E. Br. (syn. *Pseudodracontium lacourii* (Lind. & André) N. E. Br.) [133, 134]
A. margaritifera (Roxb.) Kunth [133, 134]
A. maximus (Engl.) N. E. Br. [133, 134]
A. operatus Hett. [133]
A. paeoniifolius (Dennst.) Nicolson [133, 134]
A. prainii Hook. f. [133, 134]
A. sagittarius van Steenis [133, 134]
A. titanum (Becc.) Becc. ex Arcang. [133, 134]
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A. armenienne [146]
A. fragrantissimum [146]
A. lindenum [146]
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A. creticum Boiss. & Held. [133]
A. cyrenaicum Hruby [133]
A. dioscoridis Sibth. & Sm. [133, 230]
A. idaum Coust. & Gandoger [133]
A. italicum Mill. [133, 230]
A. maculatum L. [129, 133]
A. nigrum Schott [133]
A. palaestinum Boiss. [133]
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A. rupicola Boiss. [133]
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Sauromatum guttatum Schott [26, 86, 229, 230]
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Pittosporum tobira Ait. [107, 108]

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Aphandra natalia [68]
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Bactris gasipaes H.B.K. [143]
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C. ghiesbreghtiana (Linden & H. Wendl.) H. Wendl. [21, 94, 137]
Ceroxylon alpinum Bonpl. ex DC. [143]
Chamaedorea linearis (R. & P.) Mart. [143]
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G. cuneata H. Wendl. ex Spruce var. *cuneata* [137]
G. cuneata H. Wendl. ex Spruce var. *procumbens* (H. Wendl. ex Spruce) Skov [137]
G. cuneata H. Wendl. ex Spruce var. *sodiroidi* (Dammer ex Burret) Skov [137]
G. irena Borchs. [137]
G. longepedunculata Burret [137]
G. macrostachys Mart. var. *macrostachys* [137, 143, 145]
G. maxima (Poit.) Kunth [137]
G. orbignyana Mart. [137]
G. poeppigiana Mart. [137]
G. polyandra Skov [137, 143]
G. stricta (Poit.) Kunth var. *stricta* [137]
G. stricta (Poit.) Kunth var. *piscicauda* (Dammer) Hend. [137]
G. tenuissima H. E. Moore [137]
G. triglochis Burret [137, 143]
G. undata Klotsch. [137, 143]
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N. cuatrecasasii Fernandez-Casas, Lainz & Ruiz Rejon [63]
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N. papyraceus Ker-Gawler [63]
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N. serotinus L. [63]

- N. tazetta* [177]
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 Goday ex Fernandez-Casas [63]
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C. picturata Rchb. f. [77, 78, 117]
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Meliaceae

- Aglia odorata* Lour. [250]

Rutaceae

- Boronia megastigma* [115, 172]
Citrus amadurensis [118]
C. aurantium L. [115, 247]
C. bigaradia Risso [108]
C. depressa Hayata [159]
C. grandis Osbeck [247]
C. hassaku Hort. ex. Tanaka [247]
C. iyo Hort. ex. Tanaka [247]

- C. kawachinensis* Hort. [247]
C. limon Burm. f. [118, 119, 247]
C. madurensis [119]
C. medica L. [2, 118, 119, 173]
C. natsudaikai Hayata [247]
C. sinensis Osbeck [247]
C. sphaerocarpa Tanaka [247]
C. sudachi Hort. ex. Shirari [247]
C. sulcata Hort. ex. Tanaka [247]
C. tachibana Tanaka [247]
C. unshiu Marcovitch [247]
Citrus sp. [124]
Dictamnus albus [115]

Sapindaceae

- Aesculus hippocastanum* [46]

Simaroubaceae

- Ailanthus glandulosa* Desf. [108]

SAXIFRAGALES

Grossulariaceae

- Ribes nigrum* [89]

SOLANALES

Solanaceae

- Acnistus arborescens* [121]
Brugmansia suaveolens (Willd.) Bercht. & Presl. [141]
Brugmansia × *candida* Pers. [131]
Cestrum nocturnum L. [91, 154]
Cyphomandra diploconos Sendt. [222]
C. endopogon [79]
C. hartwegii (Miers) Dunal [222]
C. sciadostylis Sendt. [222]
Datura arborea [115]
Datura sp. [141]
Markea neurantha [94]
Nicotiana alata Link & Otto [113, 115, 118, 121, 165]
N. attenuata [69]
N. otophora Grisebach [166]
N. rustica L. [165]
N. suaveolens Lehmann [115, 118, 121, 165, 167, 168]
N. sylvestris Spegazzini & Cones [165, 166, 168]
N. tabacum L. [165]
N. tomentosiformis Goodspeed [165]
Solanum variabile [121]

VITALES

Vitaceae

- Vitis vinifera* [42, 44, 47]

ZINGIBERALES

Musaceae

- Musa* sp. [21, 94]

Zingiberaceae

- Hedychium coronarium* König [141, 174, 268]
H. flavum [113, 115, 124]
H. gardnerianum [115, 118, 121]

Appendix III: List of references from which the data included in appendices I and II are extracted.

1. Altenburger, R. & P. Matile. 1988. Circadian rhythmicity of fragrance emission in flowers of *Hoya carnosa* R. Br. *Planta* 174: 248–252.
2. ——— & ———. 1990. Further observations on rhythmic emission of fragrance in flowers. *Planta* 180: 194–197.
3. Andersen, J. F. & R. L. Metcalf. 1986. Identification of a volatile attractant for *Diabrotica* and *Acalymma* spp. from blossoms of *Cucurbita maxima* Duchesne. *J. Chem. Ecol.* 12: 687–699.
4. ——— & ———. 1987. Factors influencing distribution of *Diabrotica* spp. in blossoms of cultivated *Cucurbita* spp. *J. Chem. Ecol.* 13: 681–699.
5. Andersson, S., L. A. Nilsson, I. Groth & G. Bergström. 2002. Floral scents in butterfly-pollinated plants: possible convergence in chemical composition. *Bot. J. Linn. Soc.* 140: 129–153.
6. Arai, T. 1994. Volatile compounds of *Narcissus tazetta* var. *chinensis* flowers. *Nippon Koryo Kyokai* 184: 105–111.
7. Armbruster, W. S., S. Keller, M. Matsuki & T. P. Clausen. 1989. Pollination of *Dalechampia magnoliifolia* (Euphorbiaceae) by male euglossine bees. *Amer. J. Bot.* 76: 1279–1285.
8. Awano, K., T. Honda, T. Ogawa, S. Suzuki & Y. Matsunaga. 1997. Volatile components of *Phalaenopsis schilleriana* Rchb. f. *Flav. Fragr. J.* 12: 341–344.
9. Azuma, H., M. Toyota & Y. Asakawa. 2001. Intraspecific variation of floral scent chemistry in *Magnolia kobus* DC. (Magnoliaceae). *J. Pl. Res.* 114: 411–422.
10. ———, ———, ———, T. Takaso & H. Tobe. 2002. Floral scent chemistry of mangrove plants. *J. Pl. Res.* 115: 47–53.
11. ———, ———, ———, R. Yamaoka, J. G. Garcia-Franco, G. Dieringer, L. B. Thien & S. Kawano. 1997. Chemical divergence in floral scents of *Magnolia* and allied genera (Magnoliaceae). *Pl. Spec. Biol.* 12: 69–83.
12. Baraldi, R., F. Rapparini, F. Rossi, A. Latella & P. Ciccioli. 1999. Volatile organic compound emission from flowers of the most occurring and economically important species of fruit trees. *Phys. Chem. Earth* 24: 729–732.
13. Barkman, T. J., J. H. Beaman & D. A. Gage. 1997. Floral fragrance variation in *Cypripedium*: implications for evolutionary and ecological studies. *Phytochemistry* 44: 875–882.
14. Bartsch, A. 1997. Botanische Kuriositäten aus der Urwald. Der Kanonkugelbaum und die Königliche Wasserlilie. *Dragoco Rep.* 44: 63–71.
15. Bennett, D. E. Jr. & E. A. Christenson. 1994. New species of Peruvian Orchidaceae II. *Brittonia* 46: 228–259.
16. Bergström, G. 1973. Use of a pre-column tube for the quantitative isolation of natural, volatile compounds for gas chromatography/mass spectrometry. *Chem. Scripta* 4: 135–138.
17. ———, G. Birgersson, I. Groth & A. Nilsson. 1992. Floral fragrance disparity between three taxa of lady's slipper *Cypripedium calceolus* L. (Orchidaceae). *Phytochemistry* 31: 2315–2319.
18. ———, Groth, I., Pellmyr, O., Endress, P. K., Thien, L. B., Hübener, A. & W. Francke. 1991. Chemical basis of a highly specific mutualism: chiral esters attract pollinating beetles in Eupomatiaceae. *Phytochemistry* 30: 3221–3225.
19. ———, H. E. M. Dodson & I. Groth. 1995. Spatial fragrance patterns within the flowers of *Ranunculus acris* (Ranunculaceae). *Pl. Syst. Evol.* 195: 221–242.
20. Bergström, J. & G. Bergström. 1989. Floral scents of *Bartsia alpina* (Scrophulariaceae): chemical composition and variation between individual plants. *Nordic J. Bot.* 9: 363–365.
21. Bestmann, H. G., L. Winkler, & O. von Helversen. 1997. Headspace analysis of volatile flower scent constituents of bat-pollinated plants. *Phytochemistry* 46: 1169–1172.
22. Blight, M. M., M. Métayer, M.-H. Pham-Delegue, J. A. Pickett, F. Marion-Poll & L. J. Wadham. 1997. Identification of floral volatiles involved in recognition of oilseed rape flowers, *Brassica napus* by honeybees, *Apis mellifera*. *J. Chem. Ecol.* 23: 1715–1727.
23. Borg-Karlson, A.-K. 1987. Chemical basis for the relationship between *Ophrys* orchids and their pollinators. III. Volatile compounds of species in the *Ophrys* sections Fuciflorae and Bombyliflorae as insect mimetic attractants/excitants. *Chem. Scripta* 27: 313–325.
24. ———, C. K. Unelius, I. Valterova & A. Nilsson. 1996. Floral fragrance chemistry in the early flowering shrub *Daphne mezereum*. *Phytochemistry* 41: 1477–1483.
25. ———, H. H. Eidmann, M. Lindström, T. Norin & N. Wiersma. 1985. Odoriferous compounds from the flowers of the conifers *Picea abies*, *Pinus sylvestris* and *Larix sibirica*. *Phytochemistry* 24: 455–456.

26. ———, F.O. Englund & C.R. Unelius. 1994. Dimethyl oligosulphides, major volatiles released from *Sauromatum guttatum* and *Phallus impudicus*. *Phytochemistry* 35: 321–323.
27. ———, G. Bergström & B. Kullenberg. 1987. Chemical basis for the relationship between *Ophrys* orchids and their pollinators. II Volatile compounds of *O. insectifera* and *O. speculum* as insect mimetic attractants/excitants. *Chem. Scripta* 27: 303–311.
28. ———, ——— & I. Groth. 1985. Chemical basis for the relationship between *Ophrys* orchids and their pollinators. I. Volatile compounds of *Ophrys lutea* and *O. fusca* as insect mimetic attractants/excitants. *Chem. Scripta* 25: 283–294.
29. ——— & I. Groth. 1986. Volatiles from the flowers of four species in the sections Arachnitiformes and Araneiferae of the genus *Ophrys* as insect mimetic attractants. *Phytochemistry* 25: 1297–1299.
30. ———, ———, L. Ågren & B. Kullenberg. 1993. Form-specific fragrances from *Ophrys insectifera* L. (Orchidaceae) attract species of different pollinator genera. Evidence of sympatric speciation. *Chemoecology* 4: 39–45.
31. ———, I. Valterová & L. A. Nilsson. 1994. Volatile compounds from flowers of six species in the family Apiaceae: bouquets for different pollinators. *Phytochemistry* 35: 111–119.
32. Bozan, B., T. Ozek, M. Kurkuoglu, N. Kirimer & K. H. C. Baser. 1999. The analysis of essential oil and headspace volatiles of the flowers of *Pelargonium endlicherianum* used as an anthelmintic in folk medicine. *Pl. Med.* 65: 781–782.
33. Brunke, E.-J., F.-J. Hammerschmidt & G. Schmaus. 1992. L'analyse headspace des fragrances de fleurs. *Dragoco Rep.* 39: 3–31.
34. ———, ——— & ———. 1992. L'huile essentielle de *Santolina chamaecyparissus* L. *Dragoco Rep.* 39: 151–167.
35. ———, ——— & ———. 1992. Scent of roses—recent results. *Flav. Fragr. J.* 7: 195–198.
36. ———, ——— & ———. 1993. Analyses "headspace" de plantes médicinales. *Parfums Cosmet. Aromes* 110: 45–56.
37. ———, ——— & ———. 1993. Flower scent of some traditional medical plants. Pp. 282–295 in R. Teranishi, B. Buttery, R. G. & Sugisawa, H. (eds.), *Bioactive volatile compounds from plants*. American Chemical Soc. Symp. Ser. 525, Washington, DC.
38. ———, ——— & ———. 1994. Headspace analysis of hyacinth flowers. *Flav. Fragr. J.* 9: 59–69.
39. Bu, X., A. Huang & Y. Sun. 1987. Analysis of headspace volatile constituents jasmine (*Jasminum sambac* (L.) Aiton) flower. *Acta Sci. Nat. Univ. Pekin.* 6: 53–60.
40. ———, ———, ———, Z. Wu & M. Liu. 1987. Essence constituents of *Rosa chinensis* flower. *Acta Bot. Sin.* 29: 297–301.
41. Buchbauer, G., B. Remberg & L. Jirovetz. 1995. Comparative headspace analysis of living and fresh cut lime tree flowers (*Tiliae flores*). *Flav. Fragr. J.* 10: 221–224.
42. ———, L. Jirovetz, M. Wasicky, A. Herlitschka & A. Nikiforov. 1994. Aroma von Weissweibblüten: Korrelation sensorischer Daten mit Headspace-Inhaltsstoffen. *Z. Lebensm. Unters. Forsch.* 199: 1–4.
43. ———, ———, ——— & A. Nikiforov. 1993. Headspace and essential oil analysis of apple flowers. *J. Agric. Food Chem.* 41: 116–118.
44. ———, ———, ——— & ———. 1994. Headspace analysis of *Vitis vinifera* (Vitaceae) flowers. *J. Essential Oil Res.* 6: 313–314.
45. ———, ———, ——— & ———. 1994. Volatile constituents of *Exacum affine* Balf. f. flowers obtained by dynamic headspace sampling and as the essential oil. *Flav. Fragr. J.* 9: 55–58.
46. ———, ———, ——— & ———. 1994. Volatiles of common Horsechestnut (*Aesculus hippocastanum* L.) (Hippocastanaceae) blossoms. *J. Essential Oil Res.* 6: 93–95.
47. ———, ———, ——— & ———. 1995. Aroma von Rotweinblüten: Korrelation sensorischer daten mit Headspace-Inhaltsstoffen. *Z. Lebensm. Unters. Forsch.* 200: 443–446.
48. Burger, B. V., Z. M. Munro & J. H. Visser. 1988. Determination of plant volatiles 1: analysis of the insect-attracting allomone of the parasitic plant *Hydnora africana* using Grob-Habich activated charcoal traps. *J. High Resolution Chromatogr. Commun.* 11: 496–499.
49. Buttery, R. G., J. A. Kamm & L. C. Ling. 1982. Volatile components of alfalfa flowers and pods. *J. Agric. Food Chem.* 30: 739–742.
50. ———, ——— & ———. 1984. Volatile components of red clover leaves, flowers, and seed pods: possible insect attractants. *J. Agric. Food Chem.* 32: 254–256.
51. Chen, Y., Z. Li & H. Li. 1987. Analysis of fragrance volatiles of fresh flowers by pre-column absorption and GC/MS. *Chromatographia* 23: 502–506.
52. Christensen, L. P., H. B. Jakobsen, K. Kristiansen & J. Møller. 1997. Volatiles emitted from flowers of gamma-radiated and nonradiated *Jasminum polyanthum* Franch. in situ. *J. Agric. Food Chem.* 45: 2199–2203.

53. Christenson, E. A. & M. W. Whitten. 1995. *Phalaenopsis bellina* (Rchb.f) Christenson, a segregate from *P. violacea* Witte (Orchidaceae: Aeridinae). *Brittonia* 47: 57–60.
54. Clery, R. A., N. E. Owen & S. F. Chambers. 1999. An investigation into the scents of carnations. *J. Essential Oil Res.* 11: 355–359.
55. Connick, W. J. & R. C. French. 1991. Volatiles emitted during the sexual stage of the Canada thistle rust fungus and by thistle flowers. *J. Agric. Food Chem.* 39: 185–188.
56. Dahl, Å. E., A.-B. Wassgren & B. Bergström. 1990. Floral scents in *Hypocoum* sect. *Hypocoum* (Papaveraceae): chemical composition and relevance to taxonomy and mating system. *Biochem. Syst. Ecol.* 18: 157–168.
57. Ding, D., M. Zhu, Z. Huang, Z. Song & J. Zhao. 1989. Aroma constituents of different varieties of *Osmanthus fragrans* flowers. Pp. 27–35 in C. Bhattacharyas, N. Sen & K. L. Sethi (eds.), 11th Int. Congr. Essential Oils F & F Proceedings. Oxford & IBH Publishing Co., New Delhi.
58. ———, ——— & ———. 1989. Headspace analysis of *Osmanthus fragrans* Lour. *J. Essential Oil Res.* 1: 295–297.
59. Dobson, H. E. M. 1991. Pollen and flower fragrances in pollination. Pp. 313–320 in van Heemert, C. & A. de Ruijter (eds.), The sixth international symposium on pollination. *Acta Hort.* 288.
60. ———, G. Bergström & I. Groth. 1990. Differences in floral fragrance chemistry between different parts of *Rosa rugosa* Thunb. (Rosaceae). *Israel J. Bot.* 39: 143–156.
61. ———, J. Bergström, G. Bergström & I. Groth. 1987. Pollen and flower volatiles in two *Rosa* species. *Phytochemistry* 26: 3171–3173.
62. ———, I. Groth & G. Bergström. 1996. Pollen advertisement: chemical contrasts between whole-flower and pollen odors. *Amer. J. Bot.* 83: 877–885.
63. ———, J. Arroyo, G. Bergström & I. Groth. 1997. Interspecific variation in floral fragrances within the genus *Narcissus* (Amaryllidaceae). *Biochem. Syst. Ecol.* 25: 685–706.
64. Dudareva, N., L. M. Murfitt, C. J. Mann, N. Gorenstein, N. Kolosova, C. M. Kish, C. Bonham & K. Wood. 2000. Developmental regulation of methyl benzoate biosynthesis and emission in snapdragon flowers. *Pl. Cell* 12: 949–961.
65. Erhardt, A. 1993. Pollination of edelweiss, *Leontopodium alpinum*. *Bot. J. Linn. Soc.* 111: 229–240.
66. Erhardt, A. & B. Jäggi. 1995. From pollination by Lepidoptera to selfing: the case of *Dianthus glacialis* (Caryophyllaceae). *Pl. Syst. Evol.* 195: 67–76.
67. Erickson, B. J., A. M. Young, M. A. Strand & E. H. Erickson. 1987. Pollination biology of *Theobroma* and *Herrania* (Sterculiaceae). *Insect Sci. Applic.* 8: 301–310.
68. Ervik, F., L. Tollsten & J. T. Knudsen. 1999. Floral scent chemistry and pollination ecology in phytelephantoid palms (Arecaceae). *Pl. Syst. Evol.* 217: 279–297.
69. Euler, M. & I. T. Baldwin. 1996. The chemistry of defense and apparency in the corollas of *Nicotiana attenuata*. *Oecologia* 107: 102–112.
70. Evans, K. A. & L. J. Allen-Williams. 1992. Electroantennogram responses of the cabbage seed weevil, *Ceutorhynchus assimilis*, to oilseed rape, *Brassica napus* ssp. *oleifera*, volatiles. *J. Chem. Ecol.* 18: 1641–1659.
71. Fernando, L. N. & I. U. Grün. 2001. Headspace-SPME analysis of volatiles of the ridge gourd (*Luffa acutangula*) and bitter melon (*Momordica charantia*) flowers. *Flav. Fragr. J.* 16: 289–293.
72. Flament, I., C. Debonneville & A. Furrer. 1993. Volatile constituents of roses. Characterization of cultivars based on the headspace analysis of living flower emissions. Pp. 269–281 in R. Teranishi, R. G. Buttery & H. Sugisawa (eds.), Bioactive volatile compounds from plants. American Chemical Soc. Symp. Ser. 525, Washington, DC.
73. Flamini, G., P. L. Cioni & I. Morelli. 2002. Analysis of the essential oil of the aerial parts of *Viola etrusca* from Monte Labbro (South Tuscany, Italy) and in vivo analysis of flower volatiles using SPME. *Flav. Fragr. J.* 17: 147–149.
74. ———, ——— & ———. 2002. Differences in the fragrances of pollen and different floral parts of male and female flowers of *Laurus nobilis*. *J. Agric. Food Chem.* 50: 4647–4652.
75. Flath, R. A. & K. Ohinata. 1982. Volatile components of the orchid *Dendrobium superbum* Rchb. f. *J. Agric. Food Chem.* 30: 841–842.
76. Gerlach, G. & R. Schill. 1989. Fragrance analysis, an aid to taxonomic relationships of the genus *Coryanthes* (Orchidaceae). *Pl. Syst. Evol.* 168: 159–165.
77. ——— & ———. 1991. Composition of orchid scents attracting euglossine bees. *Bot. Acta* 104: 379–391.
78. ——— & ———. 1993. Die Gattung *Coryanthes* Hook. (Orchidaceae). Eine monographische Bearbeitung unter besonderer Berücksichtigung der Blütenduftstoffe. *Tropische und subtropische Pflanzenwelt* 83.
79. Gracie, C. 1993. Pollination of *Cyphomandra endopogon* var. *endopogon* (Solanaceae) by *Eufriesea* spp. (Euglossini) in French Guiana. *Brittonia* 45: 39–46.

80. Gregg, K.B. 1983. Variation in floral fragrances and morphology: incipient speciation in *Cynoches*? Bot. Gaz. 144: 566–576.
81. Griffiths, D.W., G.W. Robertson, T. Shepherd & G. Ramsay. 1999. Epicuticular waxes and volatiles from faba bean (*Vicia faba*) flowers. Phytochemistry 52: 607–612.
82. Grison, L., A.A. Edwards & M. Hossaert-McKey. 1999. Interspecies variation in floral fragrances emitted by tropical *Ficus* species. Phytochemistry 52: 1293–1299.
83. Grison-Pigé, L., J.-M. Bessiére & M. Hossaert-McKey. 2002. Specific attraction of fig-pollinating wasps: role of volatile compounds released by tropical figs. J. Chem. Ecol. 28: 283–295.
84. ———, ———, T.C.J. Turlings, F. Kjellberg, J. Roy & M.M. Hossaert-McKey. 2001. Limited intersex mimicry of floral odour in *Ficus carica*. Funct. Ecol. 15: 551–558.
85. Groth, I., G. Bergström & O. Pellmyr. 1987. Floral fragrances in *Cimicifuga*: chemical polymorphism and incipient speciation in *Cimicifuga simplex*. Biochem. Syst. Ecol. 15: 441–444.
86. Hadacek, F. & M. Weber. 2002. Club-shaped organs as additional osmophores within the *Sauromatum* inflorescence: odour analysis, ultrastructural changes and pollination aspects. Pl. Biol. 4: 367–383.
87. Hamilton-Kemp, T.R., J.H. Loughrin, R.A. Andersen & J.G. Rodriguez. 1993. Volatile compounds from strawberry foliage and flowers. Air versus nitrogen entrainment; quantitative changes and plant development. Pp. 229–239 in R. Teranishi, R.G. Buttery & H. Sugisawa (eds.), Bioactive volatile compounds from plants. American Chemical Soc. Symp. Ser. 525, Washington, DC.
88. ———, ——— & ———. 1990. Identification of some volatile compounds from strawberry flowers. Phytochemistry 29: 2847–2848.
89. Hansted, L., H.B. Jakobsen & C.E. Olsen. 1994. Influence of temperature on the rhythmic emission of volatiles from *Ribes nigrum* flowers in situ. Pl. Cell Environm. 17: 1069–1072.
90. Haynes, K.F., J.Z. Zhao & A. Latif. 1991. Identification of floral compounds from *Abelia grandiflora* that stimulate upwind flight in cabbage looper moths. J. Chem. Ecol. 17: 637–646.
91. Heath, R.R., P.J. Landolt, B. Dueben & B. Lenczewski. 1992. Identification of floral compounds of night-blooming jessamine attractive to cabbage looper moths. Environm. Entomol. 21: 854–859.
92. Hedström, I. 1983. Kungsängsliljans pollinationsekologi. Svensk Bot. Tidskr. 77: 313–325.
93. Helsper, J.P.F.G., J.A. Davies, H.J. Bouwmeester, A.F. Krol & M.H. van Kampen. 1998. Circadian rhythmicity in emission of volatile compounds by the flowers of *Rosa hybrida* L. cv. *Honesty*. Planta 207: 88–95.
94. Helversen, O. van, L. Winkler & H.J. Bestman. 2000. Sulphur containing “perfumes” attract flower-visiting bats. J. Comp. Physiol. A 186: 143–153.
95. Hills, H.G. 1989. Fragrance cycling in *Stanhopea pulla* (Orchidaceae, Stanhopeinae) and identification of trans-limonene oxide as a major fragrance components. Lindleyana 4: 61–67.
96. ——— & N.H. Williams. 1990. Fragrance cycle of *Clowesia rosea*. Orquídea (Méx.) 12: 19–22.
97. ———, ——— & C.H. Dodson. 1968. Identification of some orchid fragrance components. Amer. Orchid Soc. Bull. 37: 967–971.
98. ———, ——— & ———. 1972. Floral fragrances and isolating mechanisms in the genus *Catasetum* (Orchidaceae). Biotropica 4: 61–76.
99. Hiltunen, R., I. Laakso & J.J.C. Scheffer. 1983. Application of headspace gas chromatography in essential oil analysis. Acta Pharm. Fennica 92: 209–214.
100. Holman, R.T. & W.H. Heimermann. 1973. Identification of components of orchid fragrances by gas chromatography-mass spectrometry. Amer. Orchid Soc. Bull. 42: 678–682.
101. Hooper, A.M., J.A. Bennison, M.C. Luszniak, J.A. Pickett, E.M. Pow & L.J. Wadhams. 1999. *Verbena* × *hybrida* flower volatiles attractive to western flower thrips, *Frankliniella occidentalis*. Pestic. Sci. 55: 660–662.
102. Ikeda, N., M. Ishihara, T. Tseneya, M. Kawakita, M. Yoshihara, Y. Suzuki, R. Komaki & M. Inui. 1994. Volatile components of honeysuckle (*Lonicera japonica* Thunb.) flowers. Flav. Fragr. J. 9: 325–331.
103. Ishizaka, H., H. Yamada & K. Sasaki. 2002. Volatile compounds in the flowers of *Cyclamen persicum*, *C. purpurascens* and their hybrids. Sci. Hort. 94: 125–135.
104. Jakobsen, H.B. & C.E. Olsen. 1994. Influence of climatic factors on emission of flower volatiles in situ. Planta 192: 365–371.
105. ———, P. Friis, J.K. Nielsen & C.A. Olsen. 1994. Emission of volatiles from flowers and leaves of *Brassica napus* in situ. Phytochemistry 37: 695–699.
106. Jirovetz, L., J.E. Gonzalez, G. Silvera, A. Nikiforov & A. Woidich. 1992. Volatile constituents of *Peristeria elata* (Orchidaceae). J. Essential Oil Res. 4: 435–438.
107. Joulain, D. 1986. Study of the fragrance given off by certain springtime flowers. Pp. 57–67 in E.-J. Brunke (ed.), Progress in essential oil research. Walter de Gruyter & Co, Berlin.

108. ———. 1987. The composition of the headspace from fragrant flowers: further results. *Flav. Fragr. J.* 2: 149–155.
109. ——— & R. Laurent. 1988. The absolute from Karo-Karoude flowers. Pp. 607–625 in B.M. Lawrence, B.D. Mookherjee & R. Laurent (eds.), *Flavors and fragrances: a world perspective*. Proc. 10th Int. Congr. Essential Oils, F & F, Washington, DC, 16–20 November 1986. Elsevier Science Publ., Amsterdam.
110. ——— & R. Tabacchi. 1994. Two volatile β -chromenes from *Wistaria sinensis* flowers. *Phytochemistry* 37: 1769–1770.
111. Jürgens, A., A. C. Webber & G. Gottsberger. 2000. Floral scent compounds of Amazonian Annonaceae species pollinated by small beetles and thrips. *Phytochemistry* 55: 551–558.
112. ———, T. Witt & G. Gottsberger. 2002. Flower scent composition in night-flowering *Silene* species (Caryophyllaceae). *Biochem. Syst. Ecol.* 30: 383–397.
113. Kaiser, R. 1987. Night-scented flowers, not only attractive to moths. Oral presentation at Euechem 1987 meeting on “Semiocchemicals in the plant and animal kingdoms,” Angers (mimeographed).
114. ———. 1988. New volatile constituents of *Jasminum sambac* (L.) Aiton. Pp. 669–684 in B.M. Lawrence, B.D. Mookherjee & B.J. Willis (eds.), *Flavors and fragrances: a world perspective*. Proc. 10th Int. Congr. Essential Oils (Wash. DC. 1986). Elsevier Sci. Publ., Amsterdam.
115. ———. 1991. Trapping, investigation and reconstitution of flower scents. Pp. 213–250 in P.M. Müller & D. Lamparsky (eds.), *Perfumes: art, science and technology*. Elsevier Applied Science, London and New York.
116. ———. 1993. On the scents of orchids. Pp. 240–268 in R. Teranishi, R. G. Buttery & H. Sugisawa (eds.), *Bioactive volatile compounds from plants*, American Chemical Soc. Symp. Ser. 525, Washington, DC.
117. ———. 1993. *The scents of orchids*. Elsevier, Amsterdam.
118. ———. 1995. New and uncommon volatile compounds in floral scents. Pp. 135–168 in K.H.C. Baser (ed.), *Flavours, fragrances and essential oils*. Proc. 13th Int. Congr. Flavours, Fragrances and Essential Oils, Istanbul, Turkey, 15–19 Oct 1995, vol. 2.
119. ———. 1997. Environmental scents at the Ligurian coast. *Perfumer Flavorist* 22: 7–18.
120. ———. 1997. New or uncommon volatile compounds in the most diverse natural scents. *Revista Ital. Eppos* 18: 18–47.
121. ———. 2000. Scents from rain forests. *Chimia* 54: 346–363.
122. ——— & C. Nussbaumer. 1990. Dehydrogeosmin, a novel compound occurring in the flower scent of various species of Cactaceae. *Helv. Chim. Acta* 73: 133–139.
123. ——— & D. Lamparsky. 1977. Nouveaux constituants de l'absolue de jacinthe et leur compartement olfactif. *Parfumes Cosmet. Aromes* 17: 71–79.
124. ——— & D. Lamparsky. 1982. Constituants azotés en trace de quelques absolues de fleurs et leurs headspaces correspondants. Pp. 287–294 in Proc. 8th Int. Congr. of Essential Oils, Cannes 1980. FEDAROM, Grasse.
125. ——— & L. Tollsten. 1995. An introduction to the scent of Cacti. *Flav. Fragr. J.* 10: 153–164.
126. Kamdem, D. P., K. Gruber, T. Barkman & D. A. Gage. 1994. Characterization of black locust floral fragrance. *J. Essential Oil Res.* 6: 199–200.
127. Kawano, S., M. Odaki, R. Yamaoka, M. Oda-Tanabe, M. Takeuchi & N. Kawano. 1995. Pollination biology of *Oenothera* (Onagraceae). The interplay between floral UV-absorbancy patterns and floral volatiles as signals to nocturnal insects. *Pl. Spec. Biol.* 10: 31–38.
128. Kim, H.-J., K. Kim, N.-S. Kim & D.-S. Lee. 2000. Determination of floral fragrances of *Rosa hybrida* using solid-phase trapping-solvent extraction and gas chromatography-mass spectrometry. *J. Chromatogr. A* 902: 389–404.
129. Kite, G. C. 1995. The floral odour of *Arum maculatum*. *Biochem. Syst. Ecol.* 23: 343–354.
130. ———. 2000. Inflorescence odour of the foul-smelling aroid *Helicodiceros muscivorus*. *Kew Bull.* 55: 237–240.
131. ——— & C. Leon. 1995. Volatile compounds emitted from flowers and leaves of *Brugmansia* $\times$ *Candida* (Solanaceae). *Phytochemistry* 40: 1093–1095.
132. ——— & S.A.L. Smith. 1997. Inflorescence odour of *Senecio articulatus*: temporal variation in isovaleric acid levels. *Phytochemistry* 45: 1135–1138.
133. ———, W.L.A. Hettterscheid, M. J. Lewis, P.C. Boyce, J. Ollerton, E. Cocklin, A. Diaz & M.S.J. Simmonds. 1998. Inflorescence odours and pollinators of *Arum* and *Amorphophallus* (Araceae). Pp. 295–315 in S.J. Owens & P.J. Rudall (eds.), *Reproductive biology*. Royal Botanic Gardens, Kew.
134. ——— & ———. 1997. Inflorescence odours of *Amorphophallus* and *Pseudodracontium* (Araceae). *Phytochemistry* 46: 71–75.

135. ———, T. Reynolds & G. T. Prance. 1991. Potential pollinator-attracting chemicals from *Victoria* (Nymphaeaceae). *Biochem. Syst. Ecol.* 19: 535–540.
136. Knudsen, J. T. 1994. Floral scent variation in the *Pyrola rotundifolia* complex in Scandinavia and western Greenland. *Nordic J. Bot.* 14: 277–282.
137. ———. 1999. Floral scent chemistry in geonomoid palms (Palmae: Geonomeae) and its importance in maintaining reproductive isolation. *Mem. New York Bot. Gard.* 83: 141–168.
138. ——— & B. B. Klitgaard. 1998. Floral scent and pollination in *Browneopsis disepala* (Cae-salpinaceae). *Brittonia* 50: 174–182.
139. ——— & B. Ståhl. 1994. Floral scent in the Theophrastaceae. *Biochem. Syst. Ecol.* 22: 259–268.
140. ——— & L. Tollsten. 1991. Floral scent and intrafloral scent differentiation in *Moneses* and *Py-rola* (Pyrolaceae). *Pl. Syst. Evol.* 177: 81–91.
141. ——— & ———. 1993. Trends in floral scent chemistry in pollination syndromes: floral scent composition in moth-pollinated taxa. *Bot. J. Linn. Soc.* 113: 263–284.
142. ——— & ———. 1995. Floral scent in bat-pollinated plants: a case of convergent evolution. *Bot. J. Linn. Soc.* 119: 45–57.
143. ———, ——— & F. Ervik. 2001. Flower scent and pollination in selected neotropical palms. *Pl. Biol.* 3: 642–653.
144. ——— & S. A. Mori. 1996. Floral scents and pollination in neotropical Lecythidaceae. *Biotropica* 28: 42–60.
145. ———, S. Andersson & P. Bergman. 1999. Floral scent attraction in *Geonoma macrostachys*, an understory palm of the Amazonian rain forest. *Oikos* 85: 409–418.
146. Kuanprasert, N., A. R. Kuehnle & C. S. Tang. 1998. Floral fragrance compounds of *Anthurium* (Araceae) species and hybrids. *Phytochemistry* 49: 521–528.
147. Kullenberg, B., A.-K. Borg-Karlson & A.-L. Kullenberg. 1984. Field studies on the behaviour of the *Eucera nigrilabris* male in the odour flow from flower labellum extract of *Ophrys tenthre-dinifera*. *Nova Acta Regiae Soc. Sci. Upsal., Ser. V:C* 3: 79–110.
148. ——— & G. Bergström. 1974. The pollination of *Ophrys* orchids. Pp. 253–258 in G. Bendz & J. Santesson (eds.), *Chemistry in botanical classification*. Academic Press, New York and London.
149. ——— & ———. 1975. Chemical communication between living organisms. *Endeavor* 34: 59–66.
150. Lamparsky, D. 1985. Headspace technique as a versatile complementary tool to increase knowl-edge about constituents of domestic or exotic flowers and fruits. Pp. 79–92 in A. B. Svendsen & J. J. C. Scheffer (eds.), *Essential oils and aromatic plants*. Martinus Nijhoff/ Dr W. Junk Publishers, Dordrecht, The Netherlands.
151. Levin, R. A., R. A. Raguso & L. A. McDade. 2001. Fragrance chemistry and pollinator affinities in Nyctaginaceae. *Phytochemistry* 58: 429–440.
152. Lewis, J. A., Moore, C. J., Fletcher, M. T., Drew, R. A. & W. Kitching. 1988. Volatile compounds from flowers of *Spathiphyllum cannaefolium*. *Phytochemistry* 27: 2755–2757.
153. Li, C., Y. Zheng, Y. Sun, Z. Wu & M. Liu. 1988. Volatile constituents of the flower of *Albizia julib-rissin* Durazz. *Fenxi Huaxue* 16: 585–589. CA. 110:92059s (1989).
154. ———, ———, ———, ——— & ———. 1988. Studies on the odoriferous volatile constituents of the flower of *Cestrum nocturnum* L. *Youji Huaxue* 8: 357–361.
155. Li, Z., F. Zhao, N. Chen, D. Xue & Y. Chen. 1988. Chemical constituents of fragrance volatiles and essential oil of Lanzhou rose (*Rosa rugosa* Thunb.). *Sepu* 6: 18–23. CA. 109: 61290k (1988).
156. Light, D. M., J. A. Kamm & R. G. Buttery. 1992. Electroantennogram response of alfalfa seed chal-cid, *Bruchophagus roddi* (Hymenoptera: Eurytomidae) to host- and nonhost-plant volatiles. *J. Chem. Ecol.* 18: 333–352.
157. Lindberg, A. B., J. T. Knudsen & J. M. Olesen. 2000. Independence of floral morphology and scent chemistry as trait groups in a set of *Passiflora* species. Pp. 91–111 in Ö. Totland (ed.), *The Scandi-navian Association for Pollination Ecology honours Knut Fægri*. Det Norske Videnskaps-Akademi. I. Mat.-Naturv. Klasse, Ny Ser., 39.
158. Lindquist, N., M. A. Battiste, W. M. Whitten, N. H. Williams & L. Strekowski. 1985. *trans*-Carvone oxide, a monoterpene epoxide from the fragrance of *Catasetum*. *Phytochemistry* 24: 863–865.
159. Loper, G. M. 1972. *Medicago sativa* and *Citrus depressa* flower volatiles. *Phytochemistry* 11: 1865.
160. ———. 1976. Differences in alfalfa flower volatiles among parent and F1 plants. *Crop Sci.* 16: 107–110.
161. ——— & A. M. Lapiolo. 1971. Photoperiodic effects on the emanation of volatiles from alfalfa (*Medicago sativa* L.) florets. *Pl. Physiol.* 49: 729–732.
162. ——— & J. L. Webster. 1971. Gas sampling technique for the chromatography of alfalfa flower volatiles. *J. Chromatogr. Sci.* 9: 466–469.

163. ———, R.A. Flath & J.L. Webster. 1971. Identification of ocimene in alfalfa flower aroma by combined GC-mass spectrometry. *Crop Sci.* 11: 61–63.
164. ——— & R.L. Berdel. 1978. Seasonal emanation of ocimene from alfalfa flowers with three irrigation treatments. *Crop Sci.* 18: 447–452.
165. Loughrin, J.H., T.R. Hamilton-Kemp, R.A. Andersen & D.F. Hildebrand. 1990. Headspace compounds from flowers of *Nicotiana tabacum* and related species. *J. Agric. Food Chem.* 38: 455–460.
166. ———, ———, ——— & ———. 1990. Volatiles from flowers of *Nicotiana sylvestris*, *N. otophora*, and *Malus × domestica*: headspace components and day/night changes in their relative concentrations. *Phytochemistry* 29: 2473–2477.
167. ———, ———, H.R. Burton & R.A. Andersen. 1993. Effect of diurnal sampling on the headspace composition of detached *Nicotiana suaveolens* flowers. *Phytochemistry* 32: 1417–1419.
168. ———, ———, R.A. Andersen & D.F. Hildebrand. 1991. Circadian rhythm of volatile emission from flowers of *Nicotiana sylvestris* and *N. suaveolens*. *Physiol. Pl.* 83: 492–496.
169. Luyt, R. & S.D. Johnson. 2001. Hawkmoth pollination of the African epiphytic orchid *Mystacidium venosum*, with special reference to flower and pollen longevity. *Pl. Syst. Evol.* 228: 49–62.
170. Ma, L., Y. Zeng, Y. Sun, Z. Wu & M. Liu. 1988. Studies on the aroma volatile constituents of ylang-ylang flowers by gas chromatography and gas chromatography/mass spectrometry. *Sepu* 6: 11–18. CA. 109:98527u (1988).
171. Mack, H. & M. Köpsel. 1973. Die Blütenöle des Maiglöckchens und des Flieders. *Parfüm. Kosmetik* 54: 233–237.
172. MacTavish, H.S., N.W. Davies & R.C. Menary. 2000. Emission of volatiles from brown *Boronia* flowers: some comparative observations. *Ann. Bot.* 86: 347–354.
173. Matile, P. & R. Altenburger. 1988. Rhythms of fragrance emission in flowers. *Planta* 174: 242–247.
174. Matsumoto, F., H. Idetsuki, K. Harada, I. Nohara & T. Toyoda. 1993. Volatile components of *Hedychium coronarium* Koenig flowers. *J. Essential Oil Res.* 5: 123–133.
175. Miyake, T., R. Yamaoka & T. Yahara. 1998. Floral scents of hawkmoth-pollinated flowers in Japan. *J. Pl. Res.* 111: 199–205.
176. Mookherjee, B.D., R.W. Trenkle & R.A. Wilson. 1989. New Technology: IFF's Living flowers' technique. *Aerosol Age*, May 1989, pp. 20–23, 44–45.
177. ———, ——— & ———. 1989. Live vs. dead part II. A comparative analysis of the headspace volatiles of some important fragrance and flavour raw materials. *J. Essential Oil Res.* 2: 85–90.
178. ———, ——— & ———. 1990. The chemistry of flowers, fruits and spices: live vs. dead a new dimension in fragrance research. *Pure Appl. Chem.* 62: 1357–1364.
179. Mottram, H.R. & I.A. Flament. 1996. The volatile constituents of the flowers of the 'silk tree', *Albizia julibrissin*. *Cambridge Roy. Soc. Special Publ. Roy. Soc. Chemistry* (0260–6292) 197: 74–77.
180. Moya, S. & J.D. Ackerman. 1993. Variation in the floral fragrance of *Epidendrum ciliare* (Orchidaceae). *Nordic J. Bot.* 13: 41–47.
181. Murrell, J.T., N.H. Williams, A.M. Pridgeon & C.H. Dodson. 1981. Floral fragrances in *Angraecum* (Orchidaceae). *Selbyana* 5: 286–290.
182. Naef, A., B.A. Roy, R. Kaiser & R. Honegger. 2002. Insect-mediated reproduction of systemic infections by *Puccinia arrhenatheri* on *Berberis vulgaris*. *New Phytol.* 154: 717–730.
183. Nakamura, S. 1987. Scent and component analysis of the hybrid tea rose. *Perfumer Flavorist* 12: 43–45.
184. Nakanishi, H. & K. Yomogida. 1999. The scent of lotus (*Nelumbo nucifera* Gaertn.) flower. *Nippon Koryo Kyokai* 201: 101–110.
185. Naumann, C.M., P. Ockenfels, J. Schmitz, F. Schmidt & W. Francke. 1991. Reactions of *Zygaena* moths to volatile compounds of *Knautia arvensis* (Lepidoptera: Zygaenidae). *Entomol. Gen.* 15: 255–264.
186. Nielsen, J.K., B. Jakobsen, P. Friis, K. Hansen, J. Möller & C.A. Olsen. 1995. Asynchronous rhythms in the emission of volatiles from *Hesperis matronalis* flowers. *Phytochemistry* 38: 847–851.
187. Nilsson, L.A. 1978. Pollination ecology and adaptation in *Platanthera chlorantha* (Orchidaceae). *Bot. Not.* 131: 35–51.
188. ———. 1979. Anthecological studies on the Lady's Slipper, *Cypripedium calceolus* (Orchidaceae). *Bot. Not.* 132: 329–347.
189. ———. 1979. The pollination ecology of *Herminium monorchis* (Orchidaceae). *Bot. Not.* 132: 537–549.
190. ———. 1980. The pollination ecology of *Dactylorhiza sambucina* (Orchidaceae). *Bot. Not.* 133: 367–385.

191. ———. 1981. The pollination ecology of *Listera ovata* (Orchidaceae). *Nordic J. Bot.* 1: 461–480.
192. ———. 1983. Anthecology of *Orchis mascula* (Orchidaceae). *Nordic J. Bot.* 3: 157–179.
193. ———. 1983. Anthecology of *Orchis morio* (Orchidaceae) at its outpost in the north. *Nov. Acta Reg. Soc. Scient. Upsaliensis, Ser. V:C*, 3: 167–179.
194. ———. 1983. Processes of isolation and introgressive interplay between *Platanthera bifolia* (L.) Rich. and *P. chlorantha* (Custer) Reichb. (Orchidaceae). *Bot. J. Linn. Soc.* 87: 325–350.
195. ———. 1985. Characteristics and distribution of intermediate between *Platanthera bifolia* and *P. chlorantha* (Orchidaceae) in the Nordic countries. *Nordic J. Bot.* 5: 407–419.
196. Oka, N., H. Ohishi, T. Hatano, M. Hornberger, K. Sakata & N. Watanabe. 1999. Aroma evolution during flower opening in *Rosa damascena* Mill. *Z. Naturf.* 54c: 889–895.
197. Olesen, J. M. & J. T. Knudsen. 1994. Scent profiles of flower colour morphs of *Corydalis cava* (Fumariaceae) in relation to foraging behaviour of bumblebee queens (*Bombus terrestris*). *Biochem. Syst. Ecol.* 22: 231–237.
198. Omata, A., K. Yomogida, S. Nakamura, T. Ohta, Y. Izawa & S. Watanabe. 1989. The odour of *Lotus* (Nelumbonaceae) flower. Pp. 43–48 in C. Bhattacharyas, N. Sen & K. L. Sethi (eds.), 11th Int. Congr. Essential Oils F & F Proceedings, Oxford & IBH Publishing Co, New Delhi.
199. ———, ———, ———, ———, ——— & ———. 1991. The scent of *Lotus* flowers. *J. Essential Oil Res.* 3: 221–227.
200. ———, ———, ———, ——— & ———. 1989. Volatile compounds of *Camellia* flowers. *J. Jap. Soc. Hort. Sci.* 58: 429–434.
201. ———, S. Nakamura, K. Yomogida, K. Moriai, Y. Ichikawa & I. Watanabe. 1990. Volatile components of TO-YO-RAN flowers (*Cymbidium faberi* and *Cymbidium virens*). *Agric. Biol. Chem.* 54: 1029–1033.
202. Ômura, H., K. Honda & N. Hayashi. 1999. Chemical and chromatic bases for preferential visiting by the cabbage butterfly, *Pieris rapae*, to rape flowers. *J. Chem. Ecol.* 25: 1895–1906.
203. Ono, T. & M. Miyazawa. 1999. Headspace constituents of flowers of *Neofinetia falcata*. *Nat. Prod. Lett.* 13: 53–57.
204. Patt, J. M., D. F. Rhoades & J. A. Corkill. 1988. Analysis of the floral fragrance of *Platanthera stricta*. *Phytochemistry* 27: 91–95.
205. ———, J. C. French, C. Schal, J. Lech & T. G. Hartman. 1995. The pollination biology of Tuckahoe, *Peltandra virginica* (Araceae). *Amer. J. Bot.* 82: 1230–1240.
206. ———, T. G. Hartman, R. W. Creekmore, J. J. Elliott, C. Schal, J. Lech & R. T. Rosen. 1992. The floral odour of *Peltandra virginica* contains novel trimethyl-2,5-dioxabicyclo[3.2.1]nonanes. *Phytochemistry* 31: 487–491.
207. Pellmyr, O. 1985. The pollination biology of *Actaea pachypoda* and *A. rubra* (including *A. erythrocarpa*) in northern Michigan and Finland. *Bull. Torrey Bot. Club* 112: 265–273.
208. ———, G. Bergström & I. Groth. 1987. Floral fragrances in *Actaea*, using differential chromatograms to discern between floral and vegetative volatiles. *Phytochemistry* 26: 1603–1606.
209. ———, I. Groth & G. Bergström. 1984. Comparative analysis of the floral odors of *Actaea spicata* and *A. erythrocarpa* (Ranunculaceae). *Nova Acta Regiae Soc. Sci. Upsal. Ser. V:C*, 3: 157–160.
210. ———, L. B. Thien, G. Bergström & I. Groth. 1990. Pollination of New Caledonian Winteraceae: opportunistic shifts or parallel radiation with their pollinators. *Pl. Syst. Evol.* 173: 143–157.
211. ———, W. Tang, I. Groth, G. Bergström & L. B. Thien. 1991. Cycad cone and angiosperm floral volatiles: inferences for the evolution of insect pollination. *Biochem. Syst. Ecol.* 19: 623–627.
212. Pettersson, S. & J. T. Knudsen. 2001. Floral scent and nectar production in *Parkia biglobosa* Jacq. (Leguminosae: Mimosoideae). *Bot. J. Linn. Soc.* 135: 97–106.
213. Pham-Delegue, M. H., P. Etievant, E. Guichard & C. Masson. 1989. Sunflower volatiles involved in honeybee discrimination among genotypes and flowering stages. *J. Chem. Ecol.* 15: 329–343.
214. ———, ———, ———, R. Marilleau, P. Douault, J. Chauffaille & C. Masson. 1990. Chemicals involved in honeybee-sunflower relationship. *J. Chem. Ecol.* 16: 3053–3065.
215. Picone, J. M., H. S. MacTavish & R. A. Clery. 2002. Emission of floral volatiles from *Mahonia japonica* (Berberidaceae). *Phytochemistry* 60: 611–617.
216. Porter, A. E. A., D. W. Griffiths, G. W. Robertson & R. Sexton. 1999. Floral volatiles of sweet pea *Lathyrus odoratus*. *Phytochemistry* 51: 211–214.
217. Raguso, R. A. & B. A. Roy. 1998. "Floral" scent production by *Puccinia* rust fungi that mimic flowers. *Molec. Ecol.* 7: 1127–1136.
218. ——— & E. Pichersky. 1995. Floral volatiles from *Clarkia breweri* and *C. concinna* (Onagraceae): recent evolution of floral scent and moth pollination. *Pl. Syst. Evol.* 194: 55–67.
219. Robertson, G. W., D. W. Griffiths, J. A. T. Woodford & A. N. E. Birch. 1995. Changes in the chemical composition of volatiles released by the flowers and fruits of the red raspberry (*Rubus idaeus*) cultivar glen prosen. *Phytochemistry* 38: 1175–1179.

220. ———, ———, ———, ———, J. A. Picket & L. J. Wadhams. 1993. A comparison of the flower volatiles from hawthorn and four raspberry cultivars. *Phytochemistry* 33: 1047–1053.
221. ———, ———, W. M. Smith & R. D. Butcher. 1993. The application of thermal desorption-gas chromatography-mass spectrometry to the analyses of flower volatiles from five varieties of oilseed rape (*Brassica napus* spp. *oleifera*). *Phytochem. Analysis* 4: 152–157.
222. Sazima, M., S. Vogel, A. Cocucci & G. Hausner. 1993. The perfume flowers of *Cyphomandra* (Solanaceae): pollination by euglossine bees, bellows mechanism, osmophores, and volatiles. *Pl. Syst. Evol.* 187: 51–88.
223. Schiestl, F. P. & M. Ayasse. 2001. Post-pollination emission of a repellent compound in a sexually deceptive orchid: a new mechanism for maximising reproductive success. *Oecologia* 126: 531–534.
224. ———, ———, H. F. Paulus, D. Erdmann & W. Francke. 1997. Variation of floral scent emission and postpollination changes in individual flowers of *Ophrys sphegodes* subsp. *sphgodes*. *J. Chem. Ecol.* 23: 2881–2895.
225. Schultz, K., R. Kaiser & J. T. Knudsen. 1999. Cyclanthone and derivatives, new natural products in the flower scent of *Cyclanthus bipartitus* Poit. *Flav. Fragr. J.* 14: 185–190.
226. Schulz, S., W. Francke, J. Edgar & D. Schneider. 1988. Volatile compounds from androconial organs of Danaïne and Ithomiine butterflies. *Z. Naturf.* 43c: 99–104.
227. Silva, U. F. da, E. L. Borba, J. Semir & A. J. Marsaioli. 1999. A simple solid injection device for the analyses of *Bulbophyllum* (Orchidaceae) volatiles. *Phytochemistry* 50: 31–34.
228. Sirikulvadhana, S., W. G. Jennings & G. Vogel. 1975. Collection of flower aroma concentrates for gas chromatographic analysis. *Flav. Food Addit.* 6: 126–128.
229. Skubatz, H., D. D. Kunkel, W. N. Howald, R. Trenkle & B. Mookherjee. 1996. The *Sauromatum guttatum* appendix as an osmophore: excretory pathways, composition of volatiles and attractiveness to insects. *New Phytol.* 134: 631–640.
230. Smith, B. N. & B. J. D. Meeuse. 1966. Production of volatile amines and skatole at anthesis in some *Arum* lily species. *Pl. Physiol.* 41: 343–347.
231. Stránský, K. & I. Valterová. 1999. Release of volatiles during the flowering period of *Hydrosme riviera*. *Phytochemistry* 52: 1387–1390.
232. Surburg, H., M. Guentert & H. Harder. 1993. Investigation of volatiles from flowers. Analytical and olfactory aspects. Pp. 103–121 in R. Hopp, K. Mori & H. Surburg (eds.), *Recent development in flavor and fragrance chemistry VCH. Proceedings of the International Haarmann and Reimer Symposium*, Kyoto.
233. ———, ——— & ———. 1993. Volatile compounds from flowers: analytical and olfactory aspects. Pp. 168–186 in R. Teranishi, R. G. Buttery & H. Sugisawa (eds.), *Bioactive volatile compounds from plants. American Chemical Soc. Symp. Ser. 525*, Washington, DC.
234. ———, ——— & B. Schwarze. 1990. Volatile constituents of European bird cherry flowers (*Padus avium* Mill.). *J. Essential Oil Res.* 2: 307–316.
235. Sutton, C. J., S. J. Keegans, W. D. J. Kirk & E. D. Morgan. 1992. Floral volatiles of *Vicia faba*. *Phytochemistry* 31: 3427–3428.
236. Tatsuka, K., M. Kohama & S. Suekane. 1988. Floral fragrance of *Zygopetalum mackayi* (Orchidaceae). *Agric. Biol. Chem.* 52: 1599–1600.
237. ———, S. Suekane, S. Sakai & H. Sumitani. 1990. Volatile constituents of kiwi fruit flowers: simultaneous distillation and extraction versus headspace sampling. *J. Agric. Food Chem.* 38: 2176–2180.
238. ———, ——— & M. Kohama. 1986. Volatile components of the flower of *Michelia figo* Spreng. *Nippon Nogei Kagaku kaishi* 60: 457–461.
239. Thien, L. B., P. Bernhardt, G. W. Gibbs, O. Pellmyr, G. Bergström, I. Groth & G. McPherson. 1985. The pollination of *Zygogynum* (Winteraceae) by a moth, *Sabatinca* (Micropterigidae): an ancient association? *Science* 227: 540–543.
240. ———, W. H. Heimermann & R. T. Holman. 1975. Floral odors and quantitative taxonomy of *Magnolia* and *Liriodendron*. *Taxon* 24: 557–568.
241. Tollsten, L. 1993. A multivariate approach to post-pollination changes in the floral scent of *Platanthera bifolia* (Orchidaceae). *Nordic J. Bot.* 13: 495–499.
242. ——— & D. O. Øvstedal. 1994. Differentiation in floral scent chemistry among populations of *Conopodium majus* (Apiaceae). *Nordic J. Bot.* 14: 361–367.
243. ——— & G. Bergström. 1993. Fragrance chemotypes of *Platanthera* (Orchidaceae)—the result of adaptation to pollinating moths? *Nordic J. Bot.* 13: 607–613.
244. ——— & J. Bergström. 1989. Variation and post-pollination changes in floral odours released by *Platanthera bifolia* (Orchidaceae). *Nordic J. Bot.* 9: 359–362.
245. ——— & J. T. Knudsen. 1992. Floral scent in dioecious *Salix* (Salicaceae)—a cue determining the pollination system? *Pl. Syst. Evol.* 182: 229–237.

246. ———, J. T. Knudsen & L. G. Bergström. 1994. Floral scent in generalistic *Angelica* (Apiaceae)—an adaptive character? *Biochem. Syst. Ecol.* 22: 161–169.
247. Toyoda, T., I. Nohara & T. Sato. 1993. Headspace analysis of volatile compounds from various *Citrus* blossoms. Pp. 205–220 in R. Teranishi, R. G. Buttery & H. Sugisawa (eds.), *Bioactive volatile compounds from plants*, American Chemical Soc. Symp. Ser. 525, Washington, DC.
248. Tsuji, H. & K. Yomogida. 1997. The scent of *Viola odorata* L. flowers. *Nippon Koryo Kyokai* 193: 91–99.
249. Tsuneya, T., N. Ikeda, M. Shiga & N. Ichikawa. 1979. GC-MS analysis of *Gardenia* flower volatiles. Pp. 454–457 in *Proc. VII Int. Congr. Essential Oils* (Kyoto 1977).
250. Wang, T., A. Huang, Y. Sun, Z. Wu & M. Liu. 1986. Isolation and identification of the constituents of volatiles of dry flowers of *Aglaia odorata* Lour. *Acta Bot. Sin.* 28: 504–510.
251. ———, ———, ——— & ———. 1987. Isolation and identification of the constituents of volatiles of flowers of *Chloranthus spicatus* (Tunb.) Makino. *Acta Bot. Sin.* 29: 184–188.
252. Whitten, W. M., H. G. Hills & N. H. Williams. 1988. Occurrence of ipsdienol in floral fragrances. *Phytochemistry* 27: 2759–2760.
253. ——— & N. H. Williams. 1992. Floral fragrances of *Stanhopea* (Orchidaceae). *Lindleyana* 7: 130–153.
254. ———, ———, W. S. Armbruster, M. A. Battiste, L. Strekowski & N. Lindquist. 1986. Carvone oxide: an example of convergent evolution in euglossine pollinated plants. *Syst. Bot.* 11: 222–228.
255. Williams, N. H. 1981. Floral fragrance components of *Brassavola* (Orchidaceae: Laeliinae). *Selbyana* 5: 279–285.
256. ———, J. T. Atwood & C. H. Dodson. 1981. Floral fragrance analysis in *Anguloa*, *Lycaste* and *Mendoncella* (Orchidaceae). *Selbyana* 5: 291–295.
257. ——— & W. M. Whitten. 1982. Identification of floral fragrance components of *Stanhopea embreei* and attraction of its pollinators to synthetic fragrance compounds. *Amer. Orchid Soc. Bull.* 51: 1262–1266.
258. ——— & ———. 1983. Orchid floral fragrances and male euglossine bees: methods and advances in the last sesquidecade. *Biol. Bull.* 164: 355–395.
259. ———, ——— & C. H. Dodson. 1984. Preliminary analyses of the floral fragrances of species of *Acineta*, *Houlettia*, *Luddemannia*, *Lycomormium*, *Paphinia*, and *Sievekingia* (Orchidaceae). *Selbyana* 7: 315–317.
260. ———, ——— & L. F. Pedrosa. 1985. Crystalline production of fragrance in *Gongora quinquerervis*. *Amer. Orchid Soc. Bull.* 54: 598–603.
261. Wilson, L. A., N. P. Senéchal & M. P. Widrechner. 1992. Headspace analysis of the volatile oils of *Agastache*. *J. Agric. Food Chem.* 40: 1362–1366.
262. Vogel, S. & J. Martens. 2000. A survey of the function of the lethal kettle traps of *Arisaema* (Araceae), with records of pollinating fungus gnats from Nepal. *Bot. J. Linn. Soc.* 133: 61–100.
263. Wong, K. C. & Y. E. Teng. 1994. Volatile components of *Mimusops elengi* L. flowers. *J. Essential Oil Res.* 6: 453–458.
264. Yasukawa, S., H. Kato, R. Yamaoka, H. Tanaka, H. Arai & S. Kawano. 1992. Reproductive and poll. biology of *Magnolia* and its allied genera (Magnoliaceae) I. Floral volatiles of several *Magnolia* and *Michelia* species and their roles in attracting insects. *Pl. Spec. Biol.* 7: 121–140.
265. Zheng, Y., H. Xu, Y. Sun, Z. Wu & M. Liu. 1989. The analysis of aroma components for *Oenothera odorata* Jacq. fresh flowers. *Acta Bot. Sin.* 31: 69–72.
266. ———, Y. Zhu, R. Zhang, Y. Sun, Z. Wu & M. Liu. 1990. Studies on the natural flavor components of *Chimonanthus praecox* L. flower. *Acta Sci. Nat. Univ. Pekin.* 26: 667–673.
267. Zhu, L.-F., B.-Y. Lu & Y.-J. Luo. 1984. A preliminary study on the chemical constituents of head space of *Jasminum sambac* (L.) Ait. *Acta Bot. Sin.* 26: 189–194.
268. ———, ———, C.-S. Tang, K. Jiang & Z.-Q. Kang. 1983. Application of XAD-4 hydrophobic resin and GC/MS/DS to study the head space of *Hedychium coronarium* Koen. Pp. 127–129 in *IXth Int. Congr. Essential Oils*, 13–17 March 1983, Singapore. *Essential Oil Technical Paper Book* 1–5.